Functional Inequalities on the Quantum Boolean Cube with Applications to Quantum Learning Theory

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Abstract. This memoir develops a systematic theory of quantum Boolean analysis on the n-qubit matrix algebra $\mathbb{M}_{2^n}$, emphasizing the interplay among the Pauli expansion, Fourier degree, and the quantum Ornstein–Uhlenbeck semigroup. We first prove that this semigroup is holomorphic on noncommutative L_p-spaces for every $1 < p < \infty$, providing the analytic tool for degree-dependent estimates. For observables supported on high Fourier degrees, we establish heat-smoothing estimates and reverse Bernstein–Markov inequalities. We also obtain a strong contraction estimate for the first tail space. For low-degree observables, we obtain lower bounds for the decay of the heat semigroup and Bernstein–Markov inequalities for both the quantum Laplacian and gradient. Combined with hypercontractivity and semigroup representations of covariance, these results yield refined Talagrand-type and covariance inequalities in quantum tail spaces.

We next introduce a notion of monotone quantum observables and prove quantum versions of the Harris–Kleitman inequality, Talagrand's correlation inequality, and the Keller–Mossel–Sen correlation inequality. In the Talagrand correlation inequality, an additional similarity condition on the first-order Pauli coefficients is needed because each quantum coordinate has three Pauli directions. We also establish quantitative BKS and inverse BKS inequalities, relating quantum influences to noise sensitivity. The BKS estimate is proved by semigroup and noncommutative martingale methods, while the inverse inequality gives a converse estimate for monotone quantum observables.

Finally, we improve the quantum Bohnenblust–Hille inequality by showing that its constant is at most $3^{d/2}$ times the corresponding classical Boolean constant. A reduction to classical Boolean functions also yields quantum extensions of approximation results for Walsh tail spaces. Applying these estimates to quantum learning theory, we prove, for bounded self-adjoint observables of fixed degree, a lower bound matching the upper bound in its dependence on the number of qubits. Hence, this determines the sharp logarithmic order of the random-query complexity as $n \to \infty$. We also obtain results on exact learning, sparse approximation, and metric entropy.

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CHAPTER 1

Introduction

The analysis of Boolean functions lies at the intersection of functional and harmonic analysis, which is built on specific analytic structures: the Walsh expansion, the discrete gradient, the heat semigroup, and the decomposition of a function according to Fourier degree. These fundamental structures underlie many of the central inequalities on the discrete cube, including Poincaré and logarithmic Sobolev inequalities, hypercontractivity, Talagrand's influence inequality, KKL-type estimates, correlation inequalities, noise-sensitivity theorems, Bohnenblust–Hille inequalities, and learning theorems for low-degree functions. A common feature of these results is that analytic estimates for the heat semigroup convert information about Fourier degree into quantitative information about influences, covariance, correlation, approximation, and learnability.

In recent years, motivated by developments in quantum computing and quantum information theory, it has become natural to extend these classical results to the noncommutative setting. Analysis on the classical Boolean cube can be naturally viewed as the commutative algebra generated by n independent self-adjoint unitaries. In this interpretation, the coordinate functions are elementary two-valued observables, the Walsh characters are their products, and the Fourier degree measures how many coordinates an observable depends on. In the quantum Boolean cube, such commuting observables are replaced by observables on n-qubit systems, namely self-adjoint elements of $\mathbb{M}_2(\mathbb{C})^{\otimes n}$. Several central problems in quantum information—such as the design of quantum error-correcting codes, and the analysis of quantum learning algorithms—rely on quantitative estimates in quantum Boolean cubes, which promotes the development of *quantum* Boolean analysis.

The historical roots of non-commutative functional inequalities trace back to the fermionic setting, which serves as a fundamental tool in quantum field theory. Notably, Carlen and Lieb [**CL93**] established optimal hypercontractivity for Fermi fields by using the optimal uniform convexity estimate of noncommutative L_p spaces, while Lust-Piquard established Riesz transform estimates and the L_p Poincaré inequality in the fermionic setting in [**ELP08c, LP98**]. Building on the fundamental developments of noncommutative functional inequalities, Montanaro and Osborne [**MO10**] systematically formulated the modern framework for quantum Boolean functions. In their framework, a quantum Boolean function is a self-adjoint unitary $Q \in \mathbb{M}_2(\mathbb{C})^{\otimes n}$, and the Pauli expansion gives a Fourier analysis adapted to n-qubit observables. The associated quantum Ornstein–Uhlenbeck semigroup is the product depolarizing semigroup; on Pauli strings it acts by $\sigma_{\mathbf{s}} \mapsto e^{-t|\mathbf{s}|}\sigma_{\mathbf{s}}$ exactly as the classical heat semigroup acts on Walsh characters (we refer to Section 2 for the definitions). This structural correspondence ensures that concepts such as hypercontractivity, influence, noise sensitivity, and Fourier degree are mathematically well-defined in the quantum setting [**MO10, Kin14a**].

Applying the noncommutative semigroup theory and the Fourier analysis techniques, Montanaro and Osborne extended the Talagrand influence inequality and the Friedgut-Kalai-Naor theorem to the quantum Boolean cube setting.

It is worthwhile to mention here that extending classical Boolean analysis to the quantum setting invokes significant difficulties. The noncommutativity of $\mathbb{M}_2(\mathbb{C})^{\otimes n}$ causes certain fundamental classical identities (such as the exact equivalence between the L_1 and L_2 norms of gradients for Boolean functions) to fail. To overcome these essential obstacles, one must rely on the extensive developments in noncommutative harmonic analysis and noncommutative martingale theory achieved over the past few decades. Specifically, the noncommutative semigroup theory, the structures of noncommutative L_p spaces, completely bounded maps, and noncommutative martingale inequalities (such as the Burkholder-Gundy inequality and the Doob maximal-type inequalities) provide fundamental tools to address the difficulties arising from the noncommutative phenomena.

Recently, motivated by the study of the quantum KKL conjecture posed by Montanaro and Osborne [**MO10**], significant progress has been made in establishing noncommutative functional inequalities. In particular, Talagrand-type influence inequalities, quantum KKL-type inequalities for geometric influences, quantum Friedgut-type theorems has been extensively studied in [**BGX25, JLLZ24, RWZ24**], providing new insights into the complexity of quantum Boolean functions.

Despite these remarkable advances, a systematic and self-contained analytical framework capable of governing the finer, higher-order structural properties of the quantum Boolean cube remains visibly absent. Therefore, the main purpose of this monograph is to provide a systematic investigation of the theory of functional inequalities in quantum Boolean cubes. We focus on estimates which lie at the interaction between the quantum Ornstein–Uhlenbeck semigroup and Fourier degree. These include heat smoothing for quantum tail spaces, lower decay bounds and Bernstein–Markov inequalities for low degree observables, Talagrand-type inequalities, correlation inequalities for quantum monotone observables, BKS-type inequalities, Bohnenblust–Hille estimates, and applications to learning quantum observables. To highlight the obstacles imposed by the noncommutativity, it is natural to first trace the historical and conceptual evolution of these inequalities. In Sections 1.1–1.3, we review the relevant scalar and Banach space-valued theories on the classical Boolean cubes, alongside existing functional inequalities on the quantum Boolean cube. A detailed outline of the memoir is provided in Section 1.4.

1.1. Functional inequalities on the Boolean cube

Throughout this monograph, let $(\{-1,1\}^n,\mu)$ be the Boolean cube equipped with the uniform probability measure, which forms a basic model in Boolean analysis. Its Fourier–Walsh expansion, discrete gradient, and heat semigroup provide the standard analytic framework for Boolean function analysis and for many of its applications in probability, theoretical computer science, and social choice theory. We briefly recall the parts of this framework that will serve as the commutative model for the quantum results developed below.

For each $A \subseteq [n] \coloneqq \{1, \dots, n\}$, the *Walsh function* is given by

$$w_A(\varepsilon) = \prod_{j \in A} \varepsilon_j, \quad \varepsilon = (\varepsilon_1, \varepsilon_2, \cdots, \varepsilon_n) \in \{-1, 1\}^n.$$

The collection of Walsh functions $\{w_A : A \subset [n]\}$ is called the *Walsh system* in the sequel. Each function $f : \{-1, 1\}^n \to \mathbb{C}$ has the Walsh-Fourier expansion

$$f(\varepsilon) = \sum_{A \subseteq [n]} \widehat{f}(A) w_A(\varepsilon), \quad \varepsilon = (\varepsilon_1, \cdots, \varepsilon_n) \in \{-1, 1\}^n, \tag{1.1}$$

where $\widehat{f}(A) = \mathbb{E}(f w_A)$ is the Fourier coefficient. For each $\varepsilon \in \{-1, 1\}^n$ and $j \in [n]$, denote by $\varepsilon^{\oplus j}$ the flipping of the j-th variable of ε. For each $j \in [n]$, the discrete partial derivatives and the Laplacian operator on the Boolean cube as

$$D_j(f)(\varepsilon) = \frac{f(\varepsilon) - f(\varepsilon^{\oplus j})}{2}, \quad \Delta = \sum_{j=1}^{n} D_j. \tag{1.2}$$

The heat semigroup generated by $-\Delta$ is diagonal in the Walsh system:

$$e^{-t\Delta}(f) = \sum_{A \subseteq [n]} e^{-t|A|} \widehat{f}(A) w_A, \quad t \geq 0, \tag{1.3}$$

where $f : \{-1, 1\}^n \to \mathbb{C}$ and $|A| = \#\{j \in [n] : j \in A\}$ stands for the cardinality of the subset A. Thus, $e^{-t\Delta}$ is indeed a *Fourier multiplier* with symbol $A \mapsto e^{-t|A|}$ for each $A \subseteq [n]$. This spectral relation is fundamental for many of the inequalities recalled below. By the spectral gap of the operator $-\Delta$, it is clear that the following L_2-Poincaré inequality holds:

$$\|f - \mathbb{E}(f)\|_{L_2(\{-1,1\}^n)} \leq \left(\sum_{j=1}^{n} \|D_j(f)\|^2_{L_2(\{-1,1\}^n)} \right)^{1/2},$$

for every $f : \{-1, 1\}^n \to \mathbb{R}$.

A first nontrivial and essential inequality is the L_p-Poincaré inequality, which we now recall as follows. For $1 \leq p < \infty$, there exists a constant $K_p > 0$ depending only on p such that the following holds for each $f : \{-1, 1\}^n \to \mathbb{R}$,

$$\|f - \mathbb{E}(f)\|_{L_p(\{-1,1\}^n)} \leq K_p \left\| \left(\sum_{j=1}^{n} |D_j f|^2 \right)^{1/2} \right\|_{L_p(\{-1,1\}^n)}. \tag{1.4}$$

This inequality can be found in [**Pis86, Tal93**], and the best $K_p = O(\sqrt{p})$ (as $p \to \infty$) was proved in [**ELP08b**] via a noncommutative method. Extensions and refinements of the Poincaré inequality (1.4) have been a central object of study in the probability and analysis literature for decades.

In the seminal paper [**Gro75a**], Gross obtained the (gaussian) logarithmic Sobolev inequality, which can be viewed as an improvement of the L_2-Poincaré inequality. More precisely, for every $f : \{-1, 1\}^n \to \mathbb{R}$ the following *logarithmic Sobolev inequalities*

$$\mathbb{E}\left(|f|^2 \log(|f|)\right) - \|f\|^2_{L_2(\{-1,1\}^n)} \log \|f\|_{L_2(\{-1,1\}^n)} \leq \sum_{j=1}^{n} \|D_j(f)\|^2_{L_2(\{-1,1\}^n)}. \tag{1.5}$$

Furthermore, Gross showed that (1.5) is equivalent to the (optimal) hypercontractivity inequality of the corresponding semigroup $\{e^{-t\Delta}\}_{t\geq 0}$: for $1 < p \leq q < \infty$,

$$\|e^{-t\Delta}\|_{L_p(\{-1,1\}^n)\to L_q(\{-1,1\}^n)} = 1 \quad \text{if and only if} \quad e^{-2t} \leq \frac{p-1}{q-1}. \tag{1.6}$$

The estimate (1.6) was first proved by Bonami [**Bon70**]; Beckner [**Bec75**] subsequently extended it to complex-valued functions and related it to the sharp Hausdorff-Young inequality. We refer to [**BKS99, FKN02**] for the applications of hypercontractivity to Boolean analysis, and to [**IN22**] for a recent advance in this direction. A similar hypercontractive phenomenon also holds for other semigroups such as the Ornstein-Uhlenbeck semigroup on the Gaussian space $(\mathbb{R}^n, \gamma_n)$, which plays an essential role in Gaussian analysis; see e.g. [**Gro75a**].

In the remarkable paper, Talagrand [**Tal93**] extended (1.5) to general $1 \leq p < \infty$ in the following manner. For $1 \leq p < \infty$, there exists $K_p > 0$ such that

$$\|f - \mathbb{E}(f)\|_{L_p(\log L)^{\frac{p}{2}}(\{-1,1\}^n)} \leq K_p \Big\| \Big(\sum_{j=1}^n |D_j f|^2 \Big)^{\frac{1}{2}} \Big\|_{L_p(\{-1,1\}^n)}, \tag{1.7}$$

for every $f : \{-1,1\}^n \to \mathbb{R}$. The p-logarithmic Sobolev inequality is a refinement of the L_p-Poincaré inequality (1.4), providing an *optimal* Sobolev embedding result. Gaussian analogues were obtained in [**Ada79, Fei75**] for $1 < p < \infty$ and in [**Led88**] for $p = 1$.

Another major improvement of the L_2-Poincaré inequality is known as the *Talagrand influence inequality*, which states that for every $f : \{-1,1\}^n \to \mathbb{R}$, the following inequality holds ([**Tal94**])

$$\mathrm{Var} f \coloneqq \|f - \mathbb{E}(f)\|^2_{L_2(\{-1,1\}^n)} \leq K \sum_{j=1}^n \frac{\|D_j(f)\|^2_{L_2(\{-1,1\}^n)}}{1 + \log\big(\|D_j(f)\|_{L_2(\{-1,1\}^n)} / \|D_j(f)\|_{L_1(\{-1,1\}^n)}\big)}. \tag{1.8}$$

For Boolean function $f : \{-1,1\}^n \to \{0,1\}$, (1.8) yields the following Kahn-Kalai-Linial (KKL) inequality ([**KKL88**])(see also [**KKK+21, Tal94**]):

$$\mathrm{Var}(f) \leq K \frac{\sum_{j=1}^n \|D_j(f)\|^2_{L_2(\{-1,1\}^n)}}{\log\Big(1/\max_{j\in[n]} \|D_j(f)\|^2_{L_2(\{-1,1\}^n)}\Big)}, \tag{1.9}$$

where $K > 0$ is a universal constant. Consequently, we have

$$\max_{j\in[n]} \|D_j(f)\|^2_{L_2(\{-1,1\}^n)} \geq K \frac{\mathrm{Var}(f)\log(n)}{n}, \tag{1.10}$$

for every Boolean function $f : \{-1,1\}^n \to \{0,1\}$.

Interpreting the Boolean function f as a voting rule, (1.10) says that if a voting rule has all influences $\|D_j(f)\|^2_{L_2}$ are small, then f is necessarily an unfair system in the sense that its expectation is very close to either 0 or 1. Talagrand's inequality has since become a standard tool in the study of threshold phenomena, communication complexity, superconcentration, and metric nonembeddability; see for example, [**Cha14, CEE23a, EG20, KMS14, KR09, OW13**].

More recently, motivated by a conjecture of Talagrand [**Tal97**], Eldan and Gross [**EG20**] (see also [**EG22**]) applied stochastic analysis techniques to prove

the following result which is known as the *Eldan-Gross inequality*

(1.11)
$$\mathrm{Var}(f)\sqrt{\log\left(1+\frac{e}{\sum_{j=1}^{n}\|D_j(f)\|^2_{L_1(\{-1,1\}^n)}}\right)} \leq C\left\|\left(\sum_{j=1}^{n}|D_j(f)|^2\right)^{1/2}\right\|_{L_1(\{-1,1\}^n)},$$

for every Boolean function $f:\{-1,1\}^n\to\{0,1\}$. Notably, the Eldan-Gross inequality (1.11) unifies the KKL inequality and Talagrand's isoperimetric inequality [**Tal93**, Theorem 1.1], making (1.11) into one of the most efficient tools in Boolean analysis. An alternative proof invoking the Bakry–Émery theory is given by Ivanisvili and Zhang [**IZ26**]. Other recent proofs and related developments can be found in [**BIM23, EKLM22, Ros20**].

We conclude this section by recalling the *Bohnenblust–Hille (BH) inequality*, which was established by Bohnenblust and Hille in [**BH31**] to resolve Bohr's absolute convergence problem for Dirichlet series. In the Boolean cube setting, the Bohnenblust–Hille inequality was established by Defant, Mastyło, and Pérez [**DMoP19**] to explore the structures of low-degree polynomials: it asserts that for any polynomial $f:\{-1,1\}^n\to\mathbb{R}$ of degree at most d, its Fourier coefficients satisfy

$$\left(\sum_{A\subseteq[n],|A|\leq d}|\widehat{f}(A)|^{\frac{2d}{d+1}}\right)^{\frac{d+1}{2d}}\leq C_d\|f\|_{L_\infty(\{-1,1\}^n)},$$

where C_d is a constant independent of the dimension n. Such estimates connect the analytic property of a polynomial with the summability properties of its Fourier coefficients. In modern theoretical computer science and learning theory, the BH inequality in Boolean cubes serves as the mathematical foundation for analyzing and reconstructing low-degree Boolean functions. Most notably, the BH inequality in Boolean cubes provides optimal and dimension-free quantitative bounds that are intimately connected to the study of the Aaronson–Ambainis conjecture concerning the maximum influence of bounded polynomials [**AA14**]. Furthermore, within the Probably Approximately Correct (PAC) learning framework, the BH inequality provides a rigorous structural justification for the celebrated Linial–Mansour–Nisan (LMN) algorithm [**LMN93a**]. By guaranteeing that the tail of the Fourier spectrum of a bounded polynomial decays rapidly and concentrates in a highly summable regime, the BH inequality enables efficient reconstruction from low-frequency data. Consequently, it represents the fundamental link between analytic smoothness and sparsity in classical Boolean analysis. We refer interested readers to [**ADEGP25, Bha26, EI22, EZ24, SVZ24, VZ24a**] for more detail on the BH inequality and related topics.

1.2. Banach space-valued functional inequalities

The classical inequalities for scalar-valued functions recalled in the preceding section have several vector-valued counterparts. This direction is relevant here for two reasons. Firstly, it clarifies the connections between Boolean analysis and the non-linear geometry of Banach spaces which forms an essential part of the *Ribe program*. Secondly, the methods developed in the vector-valued theory, in particular Pisier's use of K-convexity and holomorphic semigroups, provide one of the main sources for the noncommutative semigroup estimates proved later in this paper.

We begin with Pisier's vector-valued Poincaré inequality. For a fixed Banach space $(X, \|\cdot\|)$, let $1 \leq p < \infty$ and ε, δ be independent random vectors that are uniformly distributed on the discrete cube $\{-1,1\}^n$. In the remarkable work [**Pis86**], Pisier discovered the following inequalities for vector-valued functions on the discrete cube:

$$\|f - \mathbb{E}(f)\|^p_{L_p(\{-1,1\}^n;X)} \leq C_n^p \mathbb{E}\left\|\sum_{j=1}^n \delta_j D_j f(\varepsilon)\right\|^p, \tag{1.12}$$

where $f : \{-1,1\}^n \to X$. It was proved by Pisier that $C_n \sim \log n$ for *arbitrary* Banach space X ([**Pis86**, Lemma 7.3]). Subsequently, it was shown by Talagrand [**Tal93**] that this order of growth is optimal for Banach X with trivial cotype.

Determining which Banach spaces admit a dimension-free Pisier inequality (1.12) is one of the fundamental problems in Banach space theory. This inequality serves as a key ingredient in establishing the coincidence of Rademacher type and Enflo type. Although the dimension-free form of (1.12) is false in general, Ivanisvili, van Handel and Volberg [**IvHV20**] resolved Enflo's problem by establishing the following new dimension-free *Poincaré-type inequality* invoking heat semigroup theory in the Boolean. More precisely, they proved that for each $1 \leq p < \infty$ there exists a constant $K_p > 0$ depending only on p such that for *arbitrary* Banach space X, $n \in \mathbb{N}$, and every $f : \{-1,1\}^n \to X$ we have

$$(\mathbb{E}\|f - \mathbb{E}(f)\|^p)^{\frac{1}{p}} \leq \int_0^\infty \left(\mathbb{E}\left\|\sum_{j=1}^n \delta_j(t) D_j(f)\right\|^p\right)^{\frac{1}{p}} dt, \tag{1.13}$$

where

$$\delta_j(t) := \frac{\xi_j(t) - e^{-t}}{\sqrt{1 - e^{-2t}}}$$

and $\{\xi_j(t)\}_{j=1}^n$ are i.i.d. biased Rademacher random variables, independent of ε, with

$$\mathbb{P}(\xi_j(t) = 1) = \frac{1 + e^{-t}}{2} \quad \text{and} \quad \mathbb{P}(\xi_j(t) = -1) = \frac{1 - e^{-t}}{2}.$$

Inspired by the remarkable paper [**IvHV20**], specific functional inequalities which refined the Poincaré inequalities have been extended to the Banach space-valued setting. For instance, in [**CEE23b, CEE24**], Cordero-Erausquin and Eskenazis extended Talagrand's influence inequality (1.8) and the logarithmic Sobolev inequality to Banach space-valued functions under natural type or cotype assumptions on the target space. We also refer the reader to [**IS24**] for the connection between Banach space-valued KKL inequality and the Rademacher type of the target space. More recently, Gordin [**Gor26**] extended the main results of [**IvHV20**] to biased Boolean cubes. These Banach space-valued inequalities were then applied to obtain lower bounds for the average distortion, with respect to biased product measures, of embeddings of the Boolean cube into Banach spaces.

Motivated by the main purpose of the present paper, one of the main focuses is the "heat smoothing estimate" and Bernstein-Markov type inequalities. For each $d \in [n]$, let

$$\mathcal{T}^{\geq d}(\{-1,1\}^n) = \left\{ T \in L_2(\{-1,1\}^n) : f = \sum_{|A| \geq d} \widehat{f}(A) w_A \right\}$$

and

$$\mathcal{P}^{\leq d}(\{-1,1\}^n) = \left\{ T \in L_2(\{-1,1\}^n) : f = \sum_{|A|\leq d} \widehat{f}(A) w_A \right\}.$$

The space $\mathcal{T}^{\geq d}(\{-1,1\}^n)$ is called the d-tail space. In [**MN14**, Remark 5.5], Mendel and Naor proposed the following problem which is known as the *heat smoothing conjecture*:

CONJECTURE 1.1. *Let $1 < p < \infty$ and $d \in [n]$. Then there exist $c_p, C_p \in (0,\infty)$ such that the following inequality holds for every $f \in \mathcal{T}^{\geq d}(\{-1,1\}^n)$,*

$$\|e^{-t\Delta} f\|_{L_p(\{-1,1\}^n)} \leq C_p e^{-c_p dt} \|f\|_{L_p(\{-1,1\}^n)}. \tag{1.14}$$

CONJECTURE 1.2. *Let $1 < p < \infty$ and $d \in [n]$. For each $f \in \mathcal{T}^{\geq d}(\{-1,1\}^n)$, there exists $c_p \in (0,\infty)$ such that*

$$\|\Delta f\|_{L_p(\{-1,1\}^n)} \geq c_p d \|f\|_{L_p(\{-1,1\}^n)}. \tag{1.15}$$

The original question of Mendel and Naor [**MN14**] was vector-valued, but the scalar form above is already open in this generality. Conjecture 1.2 is the corresponding reverse Bernstein–Markov inequality and would follow from Conjecture 1.1 by integrating the heat semigroup estimate. Related estimates of (1.14) date back to Meyer [**Mey84**] in the Gaussian setting (see also [**MN14**, Lemma 5.4] for the Boolean cube).

Heilman, Mossel and Oleszkiewicz [**HMO17**] resolved Conjecture 1.1 for $d = 1$. While Eskenazis and Ivanisvili [**Esk23**] considered a Gaussian analog of this conjecture and proved sharp heat smoothing estimates for holomorphic functions in the Gaussian setting.

There is also a complementary low-degree problem. Motivated by the heat smoothing conjecture, Eskenazis and Ivanisvili [**EI20b**] proposed the following dual estimates for functions of degree at most d.

CONJECTURE 1.3. *Let $1 < p < \infty$ and $d \in [n]$. Then there exist constants $c_p, C_p \in (0,\infty)$ such that the following inequality holds for $f \in \mathcal{P}^{\leq d}(\{-1,1\}^n)$,*

$$\|e^{-t\Delta} f\|_{L_p(\{-1,1\}^n)} \geq C_p e^{-c_p dt} \|f\|_{L_p(\{-1,1\}^n)}. \tag{1.16}$$

CONJECTURE 1.4. *Let $1 < p < \infty$ and $d \in [n]$. For each $f \in \mathcal{P}^{\leq d}(\{-1,1\}^n)$, there exists $c_p \in (0,\infty)$ such that*

$$\|\Delta f\|_{L_p} \leq c_p d \|f\|_{L_p}. \tag{1.17}$$

The relevance of these conjectures to the present paper is twofold. On the one hand, the heat smoothing estimates describe the decay of high Fourier levels under the semigroup; on the other hand, their dual low-degree counterparts control the stability of low-degree observables under the same semigroup. These are exactly the degree-sensitive results that will be developed on the quantum Boolean cube in Sections 4 and 5.

The best general results in this direction for Banach space-valued functions are due to Mendel and Naor [**MN14**]. For a K-convex Banach space X[1] and

[1] A Banach space $(X, \|\cdot\|_X)$ is said K-convex if each Rademacher projection $\mathrm{Rad}^{=k}(f) = \sum_{|A|=k} \widehat{f}(A) w_A$ is bounded on $L_p(\{-1,1\}^n; X)$, $1 < p < \infty$.

$1 < p < \infty$, there exist $c(p, X)$ and $C(p, X) \in (0, \infty)$, and $A(p, X) \geq 1$ such that the following holds for every $f \in \{-1, 1\}^n \to X$ lying in the d-tail space,

$$\|e^{-t\Delta} f\|_{L_p(\{-1,1\}^n;X)} \leq C(p, X) e^{-c(p,X) d \min\{t, t^{A(p,X)}\}} \|f\|_{L_p(\{-1,1\}^n;X)}, \tag{1.18}$$

and there exist $c(p, X) \in (0, \infty)$ and $\alpha \in (0, 1]$ such that

$$\|\Delta f\|_{L_p(\{-1,1\}^n;X)} \geq c(p, X) d^{\alpha} \|f\|_{L_p(\{-1,1\}^n;X)}. \tag{1.19}$$

Eskenazis and Ivanisvili later provided a short proof of (1.18) based on a duality argument developed in [**EI20b**], and Volberg proved (1.19) by analytic paraproduct methods [**Vol24**]. Partial results for the low-degree conjectures, including the equivalence between Conjectures 1.3 and 1.4, were obtained in [**DDF$^+$25**]; see also [**FHKL16, EI20b, Vol24**]. Pisier's holomorphic semigroup theorem for K-convex targets [**Pis82b**] is a fundamental result behind these developments, which is one of the prototypes for the quantum holomorphic semigroup theory derived in Section 3.

1.3. Functional inequalities on the Quantum Boolean cube

We next recall the noncommutative background most directly related to the present paper. This section is not meant to provide a comprehensive survey of noncommutative functional inequalities; its purpose is to isolate several foundational results that motivate the quantum Boolean analysis developed in the sequel.

Functional inequalities on the canonical anticommutation relations algebra, or CAR algebra for short, are fundamental tools in noncommutative analysis and quantum field theory. Motivated by Gross' work on logarithmic Sobolev inequalities and hypercontractivity [**Gro75a, Gro75b**], Carlen and Lieb proved optimal hypercontractivity for Fermi fields [**CL93**, Theorem 4]. Lust-Piquard established L_p estimates for the Riesz transforms associated with the fermionic number operator [**LP98**]. Efraim and Lust-Piquard subsequently obtained Poincaré type inequalities on the discrete cube and in the CAR algebra by a one-parameter group method [**ELP08c**, Theorem 6.1]; this approach also gives the correct order of the Poincaré constants. More recently, Jiao, Luo and Zhou proved Eldan–Gross and KKL-type inequalities in the CAR setting [**JLZ25**]. Heat smoothing estimates have also been studied in other noncommutative settings, for example, in holomorphic subalgebras of q-Gaussian algebras [**Zha23**].

The quantum Boolean cube shares structural features with the CAR algebra but differs from it in an essential way: it provides a finite-dimensional noncommutative model of the Boolean cube:

$$\mathbb{M}_{2^n}(\mathbb{C}) = \mathbb{M}_2(\mathbb{C})^{\otimes n}$$

with normalized trace, equipped with the Pauli basis. The Pauli strings play the role of Walsh characters, and their support size gives the Fourier degree. The corresponding quantum Ornstein–Uhlenbeck semigroup is the product depolarizing semigroup and acts diagonally by

$$\sigma_{\mathbf{s}} \mapsto e^{-t|\mathbf{s}|} \sigma_{\mathbf{s}}.$$

This is the structural reason why the semigroup and Fourier-degree questions recalled in Section 1.2 have natural quantum analogues.

Montanaro and Osborne introduced a systematic Fourier-analytic framework for quantum Boolean functions [**MO10**]. In their setting, a quantum Boolean function is a self-adjoint unitary in $\mathbb{M}_{2^n}(\mathbb{C})$. They proved, among other results, hypercontractivity for the quantum Ornstein–Uhlenbeck semigroup and quantum analogues of influence inequalities and Friedgut–Kalai–Naor type results; see also King's hypercontractivity theorem for unital qubit semigroups [**Kin14a**]. Montanaro and Osborne also formulated a quantum KKL problem. A direct transfer of the classical Talagrand-to-KKL argument is obstructed by a basic noncommutative phenomenon: for quantum Boolean functions one no longer has the classical identity relating the geometric and L_2 influences, namely $\|d_j(A)\|_{L_1(\mathbb{M}_{2^n})} = \|d_j(A)\|^2_{L_2(\mathbb{M}_{2^n})}$. This failure is one of the simplest indications that the quantum theory cannot be obtained by a formal replacement of Walsh characters with Pauli counterparts.

Subsequent work has clarified several aspects of the quantum influence theory. Rouzé, Wirth and Zhang [**RWZ24**] proved quantum Talagrand, KKL and Friedgut-type theorems using geometric influence. Blecher, Gao and Xu [**BGX25**], and independently Jiao, Luo, Lin and Zhou [**JLLZ24**], gave further approaches to quantum geometric Talagrand inequalities; the latter work also obtained a quantum Eldan–Gross inequality. These results show that many qualitative features of Boolean analysis persist on the quantum Boolean cube, but they also show that the appropriate formulations often involve theory from the geometry of noncommutative L_p spaces, noncommutative martingales, and noncommutative gradients.

The present paper continues this line of research in a degree-sensitive direction on the quantum Boolean cube. While the existing theory has focused mainly on quantum analogues of influence, KKL-type, Friedgut-type, and Eldan–Gross-type inequalities, the results developed below focus on the interaction between the quantum Ornstein–Uhlenbeck semigroup and the Pauli–Fourier degree. This perspective leads to heat smoothing estimates, Bernstein–Markov inequalities, tail Talagrand and covariance inequalities, correlation and BKS estimates, Bohnenblust–Hille inequalities, and applications to learning quantum observables. A recurring technical feature is that the classical pointwise and commutative semigroup arguments must be replaced by noncommutative L_p-methods, holomorphic functional calculus, noncommutative Khintchine inequalities, and martingale techniques.

Existing results in this direction have focused mainly on quantum analogues of influence, KKL-type, Friedgut-type, and Eldan–Gross-type phenomena. Here we take a degree-sensitive viewpoint, organizing the analysis around the interaction between the quantum Ornstein–Uhlenbeck semigroup and the Pauli–Fourier degree. This leads to heat smoothing estimates, Bernstein–Markov inequalities, tail Talagrand and covariance inequalities, correlation and BKS estimates, Bohnenblust–Hille inequalities, and applications to learning quantum observables. Several classical arguments must consequently be replaced by tools from noncommutative L_p space theory, functional calculus, noncommutative Khintchine inequalities, and noncommutative martingale theory.

1.4. Outline of the paper

The monograph begins with an analytic treatment of the quantum Boolean cube and then turns to degree-sensitive estimates. It concludes with applications to approximation and learning theory.

In Chapter 2, we collect basic notations and elementary analytic facts used throughout the monograph. We recall the Pauli-Walsh expansion of elements of $\mathbb{M}_{2^n}$, the quantum Ornstein–Uhlenbeck semigroup, the partial derivatives d_j, and the low-degree subspace $\mathcal{P}^{\leq d}(\mathbb{M}_{2^n})$ and the tail subspace $\mathcal{T}^{\geq d}(\mathbb{M}_{2^n})$, which are fundamental objects that will be investigated in detail through the monograph. Moreover, the (optimal) hypercontractivity of the quantum Ornstein–Uhlenbeck semigroup, a new anti-isomorphism of $\mathbb{M}_{2^n}$ (i.e., Lemma 2.4) used to handle sign multipliers are introduced. These preliminary results provide a basic bridge between the classical heat semigroup on the Boolean and the quantum Ornstein–Uhlenbeck semigroup on quantum Boolean cube.

Chapter 3 is devoted to the holomorphicity of the semigroup, which serves as a basic tool in subsequent chapters. We prove that the quantum Ornstein–Uhlenbeck semigroup is holomorphic on $L_p(\mathbb{M}_{2^n})$, uniformly in n, for every $1 < p < \infty$. This is the noncommutative counterpart of Pisier's holomorphic semigroup theorem for K-convex Banach spaces. Notably, the classical representation of the semigroup is no longer applicable in our framework; instead, the proof relies on a modified representation of the semigroup by commuting conditional expectations, together with the K-convexity of noncommutative L_p spaces; see Lemma 3.2 and Lemma 3.3. The holomorphic extension then gives us bounds for quantum Rademacher projections and for the functional calculus $w^{\mathcal{L}}$ on lens domains.

Chapters 4 and 5 form the foundation of the semigroup theory developed in this monograph. More precisely, Chapter 4 focuses on inequalities for observables lying in low-degree, or tail spaces. We prove quantum analogues of the Mendel–Naor heat smoothing estimate on $\mathcal{T}^{\geq d}(\mathbb{M}_{2^n})$, establish a quantum version of the Heilman–Mossel–Oleszkiewicz estimate in the first tail space, and derive reverse Bernstein–Markov inequalities for observables whose spectrum is localized between two degrees. Our arguments are inspired by [**MN14, EI20b, HMO17**] and make essential use of the holomorphic functional calculus developed in Chapter 3, together with conformal mapping arguments on lens domains. Substantial additional difficulties arise from noncommutativity and the lack of pointwise estimates in the quantum setting. To overcome these obstacles, we also derive probabilistic representation of noncommutative semigroups. Chapter 5 is devoted to observables lying in the low-degree regime. We first prove lower bounds for the decay of $e^{-t\mathcal{L}}$ on low-degree subspace $\mathcal{P}^{\leq d}(\mathbb{M}_{2^n})$, which can be viewed as a "dual" version of Theorem 4.1. The elementary Chebyshev bound gives a dimension-free estimate for all $1 \leq p \leq \infty$, and the holomorphic functional calculus gives a sharper estimate when $1 < p < \infty$. We then prove Bernstein–Markov inequalities for the quantum Laplacian and related operators. Our main estimates presented in this chapter use approximation-theoretic inequalities and a noncommutative Cowling-type theorem regarding the bound of imaginary Riesz potential. The gradient estimates require a substantially different approach. More precisely, we first develop a transference principle, with no direct classical analogue, that reduces the problem to a vector-valued setting. A further key ingredient is a tensor-product representation of the relevant operator in terms of completely positive and trace-preserving maps.

This representation allows us to combine noncommutative Khintchine inequalities, a quantum analogue of Naor's inequality, and Buser-type estimates to obtain the desired gradient bounds.

In Chapter 6, we focus on inequalities and covariance estimates in tail spaces. We prove a Talagrand inequality for observables in $\mathcal{T}^{\geq d}(\mathbb{M}_{2^n})$. In particular, the geometric influence estimate obtained from Theorem 6.1 carries a full logarithmic denominator and improves the dependence on small influences in the corresponding result of [**RWZ24**]. We then develop covariance Poincaré and covariance Talagrand inequalities for pairs of quantum observables. Our approach replies on the covariance representation in terms of the semigroup and the partial derivative, combined with hypercontractivity and the additional decay property in tail spaces. Thus Chapter 6 converts spectral localization into refined influence and covariance bounds. It is also worthwhile to mention that the covariance versions of the Poincaré and Talagrand inequalities are even new for the classical Boolean analysis theory.

Chapter 7 develops lower bounds for correlations on the quantum Boolean cube. We introduce a notion of monotonically increasing element in $\mathbb{M}_{2^n}$ by requiring positivity of the three directional derivatives $\partial_{j,\ell}$ associated with each coordinate and each Pauli direction. We then prove quantum analogues of the Harris–Kleitman, Talagrand [**Tal96**], and Keller–Mossel–Sen [**KMS14**] correlation inequalities. The quantitative Talagrand estimate is formulated in terms of the first-level quantity $\mathcal{W}_1(T_1, T_2)$, which records the interaction of matching Pauli directions, while the KMS-type estimate is expressed through the L_1-sizes of the coordinate derivatives. A key distinction from the classical setting is that each quantum coordinate carries three first-order Pauli coefficients rather than a single coefficient. This gives rise to an essential obstruction: the natural product of the L_1-influences need not vanish even when the covariance does, which prevents a direct extension of the classical Talagrand correlation inequality. To address this obstruction, we introduce a similarity condition requiring the first-order Pauli coefficient vectors of the two observables to be proportional coordinate by coordinate. This condition is automatic in the classical one-directional setting, but is genuinely additional in the quantum setting and is necessary for our Talagrand inequality, as demonstrated in Example 7.6 and Theorem 7.10. Chapter 7 thus extends monotone correlation inequalities to noncommutative observables while identifying the precise structural modification required by the three-directional Pauli structures.

Chapter 8 investigates quantum noise sensitivity through the quantum BKS inequality and its converse. We first establish a dimension-free quantitative BKS estimate, showing that the quantitative BKS theorem relates the decay of $\mathrm{Var}(e^{-t\mathcal{L}}T)$ to the L_1-sizes of the quantum influences. The main novelty is a noncommutative semigroup–martingale argument. The Pauli–Walsh expansion induces a dyadic filtration of matrix algebras, and the associated conditional expectations and martingale differences provide an orthogonal decomposition while contracting the coordinate derivatives. Combined with the commutation between the martingale differences, the quantum Ornstein–Uhlenbeck semigroup and hypercontractivity, this yields a direct proof that avoids the random-restriction method. We then prove an inverse quantum BKS inequality for monotone increasing observables, providing a lower bound for the variance after the action of quantum noise in terms of the same

influence quantities. We prove this converse estimate using the covariance representation from Chapter 7, together with two monotonicity facts: the semigroup preserves monotonicity, and the associated covariance terms are monotone. In the zero-noise limit, the estimate also gives a lower bound for the original variance. Hence, this chapter provides both upper and lower bounds relating quantum influence and noise sensitivity via noncommutative semigroup and martingale methods.

Chapter 9 develops coefficient estimates and approximation results for low-degree quantum observables. We reprove the quantum Bohnenblust–Hille inequality of Volberg and Zhang [**VZ24a**] with better constants. Our approach relies on the method of reducing a low-degree quantum observable to a classical polynomial on a Boolean cube with three copies of each coordinate and using a carefully chosen family of product density matrices. The same reduction principle is then combined with a quantum Figiel-type estimate for Rademacher projections, duality arguments, and K-functional methods to study the L_1-distance of low-degree observables from quantum tail spaces. In particular, the chapter extends approximation results for Walsh tail spaces to the quantum setting, establishes a quantum counterpart of the two-sided Oleszkiewicz estimate in the degree-one case, and derives corresponding bounds for quantum symmetric polynomials, including lower estimates that reveal the necessity of the Pauli-direction factor. These results serve as Fourier-analytic tools used to explore the learning theory in the next chapter.

Chapter 10 applies the preceding Fourier-analytic estimates to learning quantum observables. We formulate quantum deterministic and random query models in terms of expectations against density matrices. For bounded self-adjoint observables of degree at most d, we prove the sharp logarithmic asymptotic order of the random query complexity as $n \to \infty$. The upper bound uses the quantum Bohnenblust–Hille inequality and the reduction from Chapter 9. The lower bound is obtained from packing estimates for low-degree quantum Boolean observables, constructed through quantum analogues of decision trees. More precisely, the lower bound is based on two ingredients: a general counting principle that turns metric packings into query-complexity estimates, and explicit large packings built from quantum analogues of decision trees. The novelty on our construction of quantum decision trees use the three Pauli directions at each qubit to produce large families of pairwise separated, low-degree quantum Boolean observables. The packing argument also applies to all bounded self-adjoint contractions, not only to quantum Boolean observables, which gives a quantum counterpart of the classical sharpness result in [**EIS23**]. We also discuss exact learning, sparse recovery estimates, and metric entropy bounds. Thus, the final chapter shows that the analytic estimates developed earlier yield quantitative results for learning problems on the quantum Boolean cube.

1.5. List of notations

Basic notations used throughout the monograph are summarized below.

- Let $\mathbb{N}$, $\mathbb{R}$, and $\mathbb{C}$ denote the sets of natural, real, and complex numbers, respectively.
- $[n] \coloneqq \{1, 2, 3, \dots, n\}$.
- Let $(\{-1,1\}^n, \mu)$ be the n-dimensional Boolean cube equipped with the uniform probability measure μ.
- For each $A \subseteq [n]$ and $\varepsilon = \{\varepsilon_j\}_{j=1}^n \in \{-1,1\}^n$, define the Walsh function $w_A(\varepsilon) \coloneqq \prod_{j\in A} \varepsilon_j$.
- $\sigma_{\mathbf{s}} = \sigma_{s_1} \otimes \cdots \otimes \sigma_{s_n}$ for multi-index $\mathbf{s} \in \{0,1,2,3\}^n$.
- For $\mathbf{s} \in \{0,1,2,3\}^n$, define the support of $\mathbf{s}$ by $supp\,(\mathbf{s}) \coloneqq \{j \in [n] : s_j \neq 0\}$, and $|s| = \#supp\,(\mathbf{s})$ is the cardinality of the support.
- Let $\mathbb{M}_{2^n} \coloneqq \mathbb{M}_{2\times 2}^{\otimes n}(\mathbb{C})$ be the n-fold tensor product of 2×2 complex matrices, and tr be the normalized trace on $\mathbb{M}_{2^n}$.
- For $1 \leq p \leq \infty$, let $L_p(\mathbb{M}_{2^n})$ be the noncommutative L_p space generated by $(\mathbb{M}_{2^n}, \mathrm{tr})$.
- For $d \in [n]$, the bounded degree space is defined by
$$\mathcal{P}^{\leq d}(\mathbb{M}_{2^n}) = \left\{ T \in L_2(\mathbb{M}_{2^n}) : T = \sum_{|\mathbf{s}|\leq d} \widehat{T}(\mathbf{s})\sigma_{\mathbf{s}} \right\}.$$
- For $d \in [n]$, the tail space is defined by
$$\mathcal{T}^{\geq d}(\mathbb{M}_{2^n}) = \left\{ T \in L_2(\mathbb{M}_{2^n}) : T = \sum_{|\mathbf{s}|\geq d} \widehat{T}(\mathbf{s})\sigma_{\mathbf{s}} \right\}.$$
- For $k \in [n]$, the Rademacher projection: $\mathrm{Rad}^{=k}(T) = \sum_{\substack{\mathbf{s}\in\{0,1,2,3\}^n \\ |\mathbf{s}|=k}} \widehat{T}(\mathbf{s})\sigma_{\mathbf{s}}$, and $\mathrm{Rad}^{\leq k}(T) = \sum_{\substack{\mathbf{s}\in\{0,1,2,3\}^n \\ |\mathbf{s}|\leq k}} \widehat{T}(\mathbf{s})\sigma_{\mathbf{s}}$.

CHAPTER 2

Preliminaries

2.1. Quantum Boolean cube and Walsh system

Throughout this monograph, $[n] = \{1, \dots, n\}$. Let $\mathbb{M}_2(\mathbb{C})$ be the $*$-algebra of all 2×2 complex matrices, equipped with the normalized trace tr_2. We use the following Pauli matrices

$$\sigma_0 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad \sigma_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

The Pauli matrices are Hermitian unitaries with $\mathrm{tr}_2(\sigma_j) = 0$ for $j = 1, 2, 3$, and

$$\sigma_j \sigma_k + \sigma_k \sigma_j = 0, \quad j \neq k \in \{1, 2, 3\}. \tag{2.1}$$

The quantum analogue of functions on the Boolean cubes is $\mathbb{M}_2(\mathbb{C})^{\otimes n} \cong \mathbb{M}_{2^n}(\mathbb{C})$ endowed with the normalized trace $\mathrm{tr} := \mathrm{tr}_2^{\otimes n}$ and the unit $\mathbf{1} := \sigma_0^{\otimes n}$. Following [**MO10**, Definition 3.1], an observable $A \in \mathbb{M}_{2^n}(\mathbb{C})$ is said to be a *quantum Boolean function* if $A = A^*$ and $A^2 = \mathbf{1}$.

In the sequel, we may shortly write $\mathbb{M}_2$ and $\mathbb{M}_{2^n}$ for $\mathbb{M}_2(\mathbb{C})$ and $\mathbb{M}_{2^n}(\mathbb{C})$, respectively. The identity maps on $\mathbb{M}_2$ and $\mathbb{M}_{2^n}$ will be denoted by $\mathbb{I}_2$ and $\mathbb{I}$, respectively. For $1 \leq p \leq \infty$, let $L_p(\mathbb{M}_{2^n})$ be the noncommutative L_p-space associated with $(\mathbb{M}_{2^n}, \mathrm{tr})$. Thus

$$\|x\|_{L_\infty(\mathbb{M}_{2^n})} = \|x\|_{\mathbb{M}_{2^n}}$$

for $p = \infty$, while

$$\|x\|_{L_p(\mathbb{M}_{2^n})} = \big(\mathrm{tr}(|x|^p)\big)^{1/p}, \qquad 1 \leq p < \infty,$$

where $|x| = (x^* x)^{1/2}$.

For each multi-index $\mathbf{s} = (s_1, \dots, s_n) \in \{0, 1, 2, 3\}^n$, set

$$\sigma_{\mathbf{s}} = \sigma_{s_1} \otimes \cdots \otimes \sigma_{s_n}.$$

The family $\{\sigma_{\mathbf{s}} : \mathbf{s} \in \{0, 1, 2, 3\}^n\}$ forms an orthonormal basis of $L_2(\mathbb{M}_{2^n})$. Consequently every $T \in \mathbb{M}_{2^n}$ has the Pauli–Walsh expansion

$$T = \sum_{\mathbf{s} \in \{0,1,2,3\}^n} \widehat{T}(\mathbf{s}) \sigma_{\mathbf{s}},$$

where $\widehat{T}(\mathbf{s})$ is the Fourier coefficient defined by $\widehat{T}(\mathbf{s}) = \mathrm{tr}(\sigma_{\mathbf{s}} T)$. For $\mathbf{s} \in \{0, 1, 2, 3\}^n$, define the *support* of $\mathbf{s}$ by

$$supp\,(\mathbf{s}) := \{j \in [n] : s_j \neq 0\},$$

and $|\mathbf{s}| = \#supp\,(\mathbf{s})$ is the cardinality of the support.

This is already explained in the notation section. Should we delete it here)For each $d \in [n]$, we define

$$\mathcal{T}^{\geq d}(\mathbb{M}_{2^n}) = \left\{ T \in L_2(\mathbb{M}_{2^n}) : T = \sum_{|\mathbf{s}| \geq d} \widehat{T}(\mathbf{s})\sigma_{\mathbf{s}} \right\}$$

and

$$\mathcal{P}^{\leq d}(\mathbb{M}_{2^n}) = \left\{ T \in L_2(\mathbb{M}_{2^n}) : T = \sum_{|\mathbf{s}| \leq d} \widehat{T}(\mathbf{s})\sigma_{\mathbf{s}} \right\}.$$

The space $\mathcal{T}^{\geq d}(\mathbb{M}_{2^n})$ is called the d-tail space in $\mathbb{M}_{2^n}$. When no confusion arises, for convenience, we may write $\mathcal{T}^{\geq d}$ and $\mathcal{P}^{\leq d}$ instead of $\mathcal{T}^{\geq d}(\mathbb{M}_{2^n})$ and $\mathcal{P}^{\leq d}(\mathbb{M}_{2^n})$, respectively.

2.2. The quantum Ornstein-Uhlenbeck semigroup

We next recall the product depolarizing semigroup, which will be denoted by the quantum Ornstein–Uhlenbeck semigroup. Let $L : \mathbb{M}_2 \to \mathbb{M}_2$ defined by

$$L(A) \coloneqq A - \mathrm{tr}_2[A]\mathbf{1}_2, \quad A \in \mathbb{M}_2. \tag{2.2}$$

Since the operator L is indeed a projection on $\mathbb{M}_2$, it follows from the Taylor expansion that

$$e^{-tL}(A) = e^{-t}A + (1 - e^{-t})\mathrm{tr}_2[A]\mathbf{1}_2, \quad A \in \mathbb{M}_2. \tag{2.3}$$

In view of (2.2) and (2.3), for each $j \in \{0, 1, 2, 3\}$, the following identities are obvious

$$L(\sigma_j) = \begin{cases} \sigma_j, & j \neq 0; \\ 0, & \text{otherwise}, \end{cases}$$

and

$$e^{-tL}(\sigma_j) = \begin{cases} e^{-t}\sigma_j, & j \neq 0; \\ \sigma_0, & \text{otherwise}. \end{cases}$$

For each stabilizer operator $\sigma_{\mathbf{s}}$, let

$$P_t(\sigma_{\mathbf{s}}) = e^{-tL}(\sigma_{\mathbf{s}_1}) \otimes \cdots \otimes e^{-tL}(\sigma_{\mathbf{s}_n}).$$

Then $\{P_t\}_{t \geq 0}$ forms a semigroup on $L_2(\mathbb{M}_{2^n})$ with the *infinitesimal generator* given by

$$\mathcal{L}(\sigma_{\mathbf{s}}) \coloneqq \lim_{t \to 0} \frac{\sigma_{\mathbf{s}} - (e^{-tL} \otimes \cdots \otimes e^{-tL})(\sigma_{\mathbf{s}})}{t} = |\mathbf{s}|\sigma_{\mathbf{s}},$$

for each stabilizer operator $\sigma_{\mathbf{s}}$. In the sequel, we denote $\{P_t\}_{t \geq 0}$ by $\left\{e^{-t\mathcal{L}}\right\}_{t \geq 0}$ to emphasize the generator $\mathcal{L}$. Equivalently, for $T \in \mathbb{M}_{2^n}$ we have

$$\mathcal{L}(T) = \sum_{\mathbf{s} \in \{0,1,2,3\}^n} |\mathbf{s}|\widehat{T}(\mathbf{s})\sigma_{\mathbf{s}} \tag{2.4}$$

and

$$e^{-t\mathcal{L}}(T) = \sum_{\mathbf{s} \in \{0,1,2,3\}^n} e^{-t|\mathbf{s}|}\widehat{T}(\mathbf{s})\sigma_{\mathbf{s}}. \tag{2.5}$$

For $j \in [n]$, the j-th partial differential operator (or, quantum bit-flip map) is given by

$$\mathbf{d}_j \coloneqq \mathbb{I}_2^{\otimes(j-1)} \otimes (\mathbb{I}_2 - \mathrm{tr}_2) \otimes \mathbb{I}_2^{\otimes(n-j)}.$$

Due to the Pauli-Walsh expansion of $T \in \mathbb{M}_{2^n}$, it follows that each $\mathbf{d}_j$ is a projection on $L_2(\mathbb{M}_{2^n})$ such that

$$\mathbf{d}_j(T) = \sum_{\mathbf{s}\in\{0,1,2,3\}^n} \widehat{T}(\mathbf{s}) d_j(\sigma_{\mathbf{s}}) = \sum_{\substack{\mathbf{s}\in\{0,1,2,3\}^n \\ \mathbf{s}_j\neq 0}} \widehat{T}(\mathbf{s})\sigma_{\mathbf{s}}, \tag{2.6}$$

and $\mathcal{L} = \sum_{j=1}^n \mathbf{d}_j$.

We recall the following *optimal hypercontractivity* for the quantum Ornstein-Uhlenbeck semigroup $\{e^{-t\mathcal{L}}\}_{t\geq 0}$ as follows and refer to [**Kin14b**] (or, [**MO10**, Theorem 8.4]) for more details.

THEOREM 2.1. *For each $1 < p < q < \infty$, we have*

$$\|e^{-t\mathcal{L}}\|_{L_p\to L_q} = 1 \quad \textit{if and only if } t \geq \frac{1}{2}\log(\frac{q-1}{p-1}).$$

The rest of this chapter is devoted to deriving and recalling technical lemmas that will be used throughout this monograph.

LEMMA 2.2. *For all $j \in [n]$ and $t \geq 0$, we have*

$$\mathbf{d}_j e^{-t\mathcal{L}} = e^{-t\mathcal{L}}\mathbf{d}_j.$$

PROOF. For each $j \in [n]$, by the definitions of (2.5) and (2.6), we have

$$\mathbf{d}_j e^{-t\mathcal{L}}(T) = e^{-t\mathcal{L}}\mathbf{d}_j(T) = \sum_{\substack{\mathbf{s}\in\{0,1,2,3\}^n \\ \mathbf{s}_j\neq 0}} e^{-t|\mathbf{s}|}\widehat{T}(\mathbf{s})\sigma_{\mathbf{s}}, \quad \forall T \in \mathbb{M}_{2^n}.$$

□

The following lemma is taken from [**JLLZ24**, Proposition 2.7 (i)].

LEMMA 2.3. *For $T \in \mathbb{M}_{2^n}$ with $\|T\|_{L_\infty} \leq 1$ and each $j \in [n]$, we have*

$$\|\mathbf{d}_j(T)\|_{L_\infty} \leq 1.$$

2.3. Auxiliary lemmas

Set

$$R_\rho := \begin{pmatrix} 0 & i \\ -i & 0 \end{pmatrix}.$$

Then, $R_\rho = R_\rho^* = R_\rho^{-1}$ and for each $j \in \{0,1,2,3\}$, we have

$$R_\rho \sigma_j^{\mathsf{T}} R_\rho = \begin{cases} -\sigma_j, & j \neq 0; \\ \sigma_j, & \text{otherwise,} \end{cases}$$

where σ_j^{T} is the transpose of σ_j. The following map is inspired by Efraim and Lust-Piquard [**ELP08c**]. Define the map $\Phi_\rho : \mathbb{M}_{2^n} \to \mathbb{M}_{2^n}$ by setting

$$\Phi_\rho(\sigma_{\mathbf{s}}) := R_\rho^{\otimes n}\sigma_{\mathbf{s}}^{\mathsf{T}} R_\rho^{\otimes n} = \otimes_{j=1}^n R_\rho \sigma_{s_j}^{\mathsf{T}} R_\rho. \tag{2.7}$$

Equivalently,

$$\Phi_\rho(\sigma_{\mathbf{s}}) = R_\rho^{\otimes n}\sigma_{\mathbf{s}}^{\mathsf{T}} R_\rho^{\otimes n} = \bigotimes_{j=1}^n R_\rho \sigma_{s_j}^{\mathsf{T}} R_\rho = (-1)^{|\mathbf{s}|}\sigma_{\mathbf{s}}. \tag{2.8}$$

Thus, Φ_ρ is a linear bijection on $\mathbb{M}_{2^n}$.

LEMMA 2.4. *For $1 \le p \le \infty$ and $T \in \mathbb{M}_{2^n}$, the mapping Φ_ρ given as in (2.7) satisfies the following properties:*

(i) *Φ_ρ is $*$-preserving, that is, $\Phi_\rho(T^*) = \Phi_\rho(T)^*$, for any $T \in \mathbb{M}_{2^n}$;*
(ii) *Φ_ρ is an anti-isomorphism, that is, $\Phi_\rho(T_1T_2) = \Phi_\rho(T_2)\Phi_\rho(T_1)$, for any $T_1, T_2 \in \mathbb{M}_{2^n}$;*
(iii) *$\|\Phi_\rho(T)\|_{L_p} = \|T\|_{L_p}$, for any $T \in \mathbb{M}_{2^n}$, and $1 \le p \le \infty$.*

PROOF. Since Pauli matrix σ_j is a self adjoint matrix, we obtain

$$R_\rho(\sigma_j^*)R_\rho = [R_\rho\sigma_j R_\rho]^* = \begin{cases} -\sigma_j, & j \neq 0; \\ \sigma_j, & \text{otherwise.} \end{cases}$$

Therefore, for any $T \in \mathbb{M}_{2^n}$, by the definition and linearity of Φ_ρ, we have $\Phi_\rho(T^*) = \Phi_\rho(T)^*$.

We now prove item (ii). For any $T_1, T_2 \in \mathbb{M}_{2^n}$, we obtain

$$\begin{aligned} \Phi_\rho(T_1T_2) &= \Phi_\rho\left(\left(\sum_{\mathbf{s}\in\{0,1,2,3\}^n} \widehat{T_1}(\mathbf{s})\sigma_{\mathbf{s}}\right)\left(\sum_{\ell\in\{0,1,2,3\}^n} \widehat{T_2}(\ell)\sigma_\ell\right)\right) \\ &= \sum_{\mathbf{s}\in\{0,1,2,3\}^n}\sum_{\ell\in\{0,1,2,3\}^n} \widehat{T_1}(\mathbf{s})\widehat{T_2}(\ell)\Phi_\rho(\sigma_{\mathbf{s}}\sigma_\ell) \\ &= \sum_{\mathbf{s}\in\{0,1,2,3\}^n}\sum_{\ell\in\{0,1,2,3\}^n} \widehat{T_1}(\mathbf{s})\widehat{T_2}(\ell)\Phi_\rho(\otimes_{j=1}^n\sigma_{\mathbf{s}_j}\sigma_{\ell_j}). \end{aligned} \tag{2.9}$$

By the definition of the operator Φ_ρ, it is clear that

$$\begin{aligned} \Phi_\rho\left(\otimes_{j=1}^n\sigma_{\mathbf{s}_j}\sigma_{\ell_j}\right) &= \otimes_{j=1}^n R_\rho(\sigma_{\mathbf{s}_j}\sigma_{\ell_j})^{\mathsf{T}}R_\rho \\ &= \otimes_{j=1}^n R_\rho\sigma_{\ell_j}^{\mathsf{T}}\sigma_{\mathbf{s}_j}^{\mathsf{T}}R_\rho \\ &= \otimes_{j=1}^n (R_\rho\sigma_{\ell_j}^{\mathsf{T}}R_\rho)(R_\rho\sigma_{\mathbf{s}_j}^{\mathsf{T}}R_\rho) \\ &= \Phi_\rho(\sigma_\ell)\Phi_\rho(\sigma_{\mathbf{s}}). \end{aligned}$$

Then we get $\Phi_\rho(T_1T_2) = \Phi_\rho(T_2)\Phi_\rho(T_1)$ by (2.9).

Since $\|\cdot\|_{L_p}$ is unitary invariant, the following holds

$$\begin{aligned} \|\Phi_\rho(T)\|_{L_p} &= \left\|\sum_{\mathbf{s}\in\{0,1,2,3\}^n} \widehat{T}(\mathbf{s})R_\rho^{\otimes n}\sigma_{\mathbf{s}}^{\mathsf{T}}R_\rho^{\otimes n}\right\|_{L_p} \\ &= \left\|R_\rho^{\otimes n}\Big(\sum_{\mathbf{s}\in\{0,1,2,3\}^n} \widehat{T}(\mathbf{s})\sigma_{\mathbf{s}}\Big)^{\mathsf{T}}R_\rho^{\otimes n}\right\|_{L_p} \\ &= \left\|\left(\sum_{\mathbf{s}\in\{0,1,2,3\}^n} \widehat{T}(\mathbf{s})\sigma_{\mathbf{s}}\right)^{\mathsf{T}}\right\|_{L_p} \\ &= \|T\|_{L_p}. \end{aligned}$$

The proof is complete. □

In the sequel, we also need to apply the well-known *noncommutative Khintchine inequality*, which was first proved by Lust-Piquard [**LP86**] for the case $1 < p < \infty$, and then extended to $0 < p \leq 1$ by Lust-Piquard, Pisier and Ricard [**LPP91, PR17**]. For each finite sequence $\{T_j\}_{j=1}^N \subseteq L_p(\mathbb{M}_{2^n})$ $(0 < p \leq \infty)$, define

$$\|(T_j)\|_{L_p(\mathbb{M}_{2^n},\ell_2^c)} := \Big\| \Big(\sum_{j=1}^N |T_j|^2 \Big)^{1/2} \Big\|_{L_p}, \quad \|(T_j)\|_{L_p(\mathbb{M}_{2^n},\ell_2^r)} := \Big\| \Big(\sum_{j=1}^N |T_j^*|^2 \Big)^{1/2} \Big\|_{L_p}.$$

For $2 \leq p \leq \infty$, define

$$\|(T_j)\|_{L_p(\mathbb{M}_{2^n},\ell_2^{cr})} \coloneqq \max\{\|(T_j)\|_{L_p(\mathbb{M}_{2^n},\ell_2^c)}, \|(T_j)\|_{L_p(\mathbb{M}_{2^n},\ell_2^r)}\},$$

and for $0 < p \leq 2$, define

$$\|(T_j)\|_{L_p(\mathbb{M}_{2^n},\ell_2^{cr})} \coloneqq \inf_{T_j = A_j + B_j} \{\|(A_j)\|_{L_p(\mathbb{M}_{2^n},\ell_2^c)} + \|(B_j)\|_{L_p(\mathbb{M}_{2^n},\ell_2^r)}\}.$$

We only state the following special form of the noncommutative Khintchine inequality which will be used in the sequel.

LEMMA 2.5 (Lust-Piquard-Pisier). *There exists a universal constant $C > 0$ such that for each $2 \leq p < \infty$, and any finite sequence $\{T_j\}_{j=1}^N \subseteq L_p(\mathbb{M}_{2^n})$ the following holds*

$$\big\|\{T_j\}_{j=1}^N\big\|_{L_p(\mathbb{M}_{2^n},\ell_2^{cr})} \leq \left(\mathbb{E}_\delta \left\| \sum_{j=1}^N \delta_j T_j \right\|_{L_p}^p \right)^{1/p} \leq C\sqrt{p}\, \big\|\{T_j\}_{j=1}^N\big\|_{L_p(\mathbb{M}_{2^n},\ell_2^{cr})},$$

where $\delta = \{\delta_j\}_{j=1}^N$ is a sequence of i.i.d. (unbiased) Rademacher random variables.

CHAPTER 3

Holomorphic quantum semigroups

Recall that a bounded semigroup $\{S_t\}_{t\geq 0}$ on a real or complex Banach space X is called holomorphic if there exists $\theta > 0$ such that, for each $x \in X$ and $x^* \in X^*$ the function $t \to \langle x^*, S_t(x)\rangle$ has holomorphic extension in the sector $\{z \in \mathbb{C} : \mathrm{Re} z \geq 0, |\arg z| < \theta\}$. The purpose of this chapter is to prove that, for $1 < p < \infty$, the quantum Ornstein-Uhlenbeck semigroup is holomorphic on $L_p(\mathbb{M}_{2^n})$ with constants independent of n; see Theorem 3.1.

3.1. Holomorphic quantum Ornstein-Uhlenbeck semigroup

For $z \in \mathbb{C}$, define the Fourier multiplier

$$e^{-z\mathcal{L}}(T) \coloneqq \sum_{\mathbf{s}\in\{0,1,2,3\}^n} \widehat{T}(\mathbf{s}) e^{-z|\mathbf{s}|}\sigma_{\mathbf{s}}, \tag{3.1}$$

where $T = \sum_{\mathbf{s}\in\{0,1,2,3\}^n} \widehat{T}(\mathbf{s})\sigma_{\mathbf{s}} \in \mathbb{M}_{2^n}$. When $z = t \geq 0$, this agrees with the quantum Ornstein–Uhlenbeck semigroup defined in (2.5). The main result in this section is the following holomorphic extension theorem for quantum Ornstein-Uhlenbeck semigroups.

THEOREM 3.1. *For every $p \in (1,\infty)$, there exist constants $\theta = \theta_p \in (0, \frac{\pi}{2}]$ and $M_p \in [1,\infty)$ such that for every $n \in \mathbb{N}$ and for each $z \in \mathbb{C}$ with $|\arg z| \leq \theta$, one has*

$$\|e^{-z\mathcal{L}}\|_{L_p\to L_p} \leq M_p, \quad |\arg(z)| \leq \theta, \tag{3.2}$$

where the operator $e^{-z\mathcal{L}}$ is defined in (3.1).

Our proof follows Pisier's holomorphic-semigroup argument for K-convex Banach spaces (see [**Pis82b**]), adapted to the product depolarizing semigroup. The key point is that the semigroup can be represented through commuting conditional expectations. More precisely, let $L : \mathbb{M}_2 \to \mathbb{M}_2$ be as in (2.2). For each $j \in [n]$ and $t > 0$, define the map $L_{j,t} : \mathbb{M}_{2^n} \to \mathbb{M}_{2^n}$ by setting

$$L_{j,t} = \mathbb{I}_2^{\otimes(j-1)} \otimes ((\mathbb{I}_2 - L) + e^{-t}L) \otimes \mathbb{I}_2^{\otimes(n-j)}$$

with $\mathbb{I}_2$ being the identity map over $\mathbb{M}_2$. It is not hard to see that $(L_{j,t})_{j=1}^n$ is a family of commuting projections on $L_2(\mathbb{M}_{2^n})$. Set

$$S_t \coloneqq \prod_{j=1}^n L_{j,t}.$$

Moreover, the family of operators $\{S_t\}_{t>0}$ is exactly the quantum Ornstein-Uhlenbeck semigroup defined in (2.5).

LEMMA 3.2. *For each $t > 0$, we have*

$$S_t = e^{-t\mathcal{L}}.$$

PROOF. Fix $t>0$, by the orthonormality of $\{\sigma_{\mathbf{s}}\}_{\mathbf{s}\in\{0,1,2,3\}^n}$, it suffices to verify that $S_t(\sigma_{\mathbf{s}})=e^{-t\mathcal{L}}(\sigma_{\mathbf{s}})$ for each $\mathbf{s}\in\{0,1,2,3\}^n$. Note that

$$S_t(\sigma_{\mathbf{s}})=\prod_{j=1}^n L_{j,t}(\sigma_{\mathbf{s}})=\prod_{j=1}^n L_{j,t}(\otimes_{j=1}^n\sigma_{\mathbf{s}_j})=\otimes_{j=1}^n((\mathbb{I}_2-L)+e^{-t}L)(\sigma_{\mathbf{s}_j})=e^{-t|\mathbf{s}|}\sigma_{\mathbf{s}},$$

where we used the definition of L as in (2.2). The assertion follows. □

We shall also use the following equivalent formula of $\{S_t\}_{t>0}$. For each $j\in[n]$, define $Q_j:\mathbb{M}_{2^n}\to\mathbb{M}_{2^n}$ by

$$Q_j:=\mathbb{I}_2^{\otimes(j-1)}\otimes\mathcal{E}_2\otimes\mathbb{I}_2^{\otimes(n-j)},$$

where $\mathcal{E}_2:\mathbb{M}_2\to\mathbb{M}_2$ is a conditional expectation given by $\mathcal{E}_2(A)=\mathrm{tr}_2(A)\mathbf{1}_2$, for all $A\in\mathbb{M}_2$. By the construction, it follows that $\{Q_j\}_{j=1}^n$ forms a commuting family of conditional expectations. Thus $\{Q_j\}_{j=1}^n$ is a commuting family of norm-one projections on $L_p(\mathbb{M}_{2^n})$, $1\le p\le\infty$.

LEMMA 3.3. *For each $t>0$, we have*

$$S_t=\prod_{j=1}^n\left(Q_j+e^{-t}\left(\mathbb{I}_2^{\otimes_n}-Q_j\right)\right).$$

PROOF. For any $t>0$ and $\mathbf{s}\in\{0,1,2,3\}^n$, it is not hard to check that for each $j\in[n]$,

$$\big(Q_j+e^{-t}(\mathbb{I}_2^{\otimes_n}-Q_j)\big)(\sigma_{\mathbf{s}})=e^{-t}\sigma_{\mathbf{s}}$$

whenever $\mathbf{s}_j\neq 0$, and $\big(Q_j+e^{-t}(\mathbb{I}_2^{\otimes_n}-Q_j)\big)(\sigma_{\mathbf{s}})=\sigma_{\mathbf{s}}$ whenever $\mathbf{s}_j=0$. It follows directly that

$$\prod_{j=1}^n(Q_j+e^{-t}(\mathbb{I}_2^{\otimes_n}-Q_j))(\sigma_{\mathbf{s}})=e^{-t|\mathbf{s}|}\sigma_{\mathbf{s}}=e^{-t\mathcal{L}}(\sigma_{\mathbf{s}}).$$

Since $t>0$ and $\mathbf{s}\in\{0,1,2,3\}^n$ are arbitrary, the desired assertion follows from Lemma 3.2. □

We next recall two standard facts from semigroup theory in Banach space-valued setting. Given a Banach space X, recall that for each $k\in[n]$ and $f\in L_p(\{-1,1\}^n;X)$, the *Rademacher projection* $\mathrm{Rad}^{=k}$ is defined by

$$\mathrm{Rad}^{=k}(f)=\sum_{|A|=k}\widehat{f}(A)w_A. \tag{3.3}$$

We say a Banach space X is K-convex if

$$\sup_{k\ge0}\|\mathrm{Rad}^{=k}\|_{L_p(\{-1,1\}^n;X)\to L_p(\{-1,1\}^n;X)}<\infty.$$

According to [**Pis82b**, Theorem 2.1], a Banach space X is K-convex is equivalent to that X is non-trivial type. Hence, each noncommutative Lebesgue space $L_p(\mathcal{M})$ with $1<p<\infty$ is K-convex.

The following result is a restatement of the fundamental theorem of Pisier [**Pis82b**].

LEMMA 3.4 ([**Pis82b**, Lemma 1.5]). *Suppose that X is a K-convex Banach space. For any commuting norm-one projections $P_1, \dots, P_N \in B(X)$[1], there exists a real number $0 < \rho < 2$ (depending on N and X) such that*

$$\left\| \prod_{j=1}^{N} (\mathbb{I} - P_j) \right\| \leq \rho^N$$

where $\mathbb{I}$ denotes the identity map on X.

Next lemma is due to Kato and Beurling [**Beu70, Kat70**] which is known as the *Kato–Beurling criterion*.

LEMMA 3.5 (Kato–Beurling). *Let $\{T_t\}_{t\geq 0}$ be a strongly continuous semigroup on some Banach space X. Suppose that there exists a natural number N such that*

$$\limsup_{t\to 0^+} \left\| (\mathbb{I} - T_t)^N \right\|_{X\to X}^{1/N} = \rho < 2,$$

where $\mathbb{I}$ is the identity operator in $B(X)$. Then there exists $\theta > 0$ such that $\{T_t\}_{t\geq 0}$ admits a holomorphic semigroup extension $z \mapsto T_z$ to the sector

$$\Sigma_\theta := \{z \in \mathbb{C} \setminus \{0\} : |\arg z| < \theta\}.$$

Furthermore, after decreasing the angle if necessary, the extension is uniformly bounded on the closed smaller sector.

We now are ready to prove Theorem 3.1.

PROOF OF THEOREM 3.1. For any $1 < p < \infty$, by Lemma 3.2 and Lemma 3.5, it suffices to show that there exist $0 < \rho < 2$ and $N \in \mathbb{N}$ such that

$$\sup_{t>0} \|(\mathbb{I} - S_t)^N\|_{L_p\to L_p} \leq \rho^N. \tag{3.4}$$

For a fixed $t > 0$, let $\{\xi_j\}_{j=1}^n$ be a family of independent and identically distributed functions defined on $(\Omega, \mathcal{F}, \mathbb{P}) := (\{0,1\}^n, \otimes_{i=1}^n \mu)$ satisfying

$$\mathbb{P}(\{\xi_j = 0\}) = 1 - e^{-t} \quad \text{and} \quad \mathbb{P}(\{\xi_j = 1\}) = e^{-t}.$$

Define $\pi : (\Omega, \mathcal{F}, \mathbb{P}) \to B(L_p(\mathbb{M}_{2^n}))$ by setting

$$\omega \mapsto \prod_{j=1}^{n} (Q_j + \xi_j(\omega)(\mathbb{I}_2^{\otimes_n} - Q_j)). \tag{3.5}$$

[1]For a Banach space E, $B(E)$ stands for the space of all bounded linear operators from E to itself.

By the independence of $\{\xi_j\}_{j=1}^n$, we set that

$$\begin{aligned}\int_\Omega \pi(\omega)d\mathbb{P}(\omega) &= \int_\Omega \prod_{j=1}^n (Q_j + \xi_j(\omega)(\mathbb{I}_2^{\otimes_n} - Q_j))d\mathbb{P}(\omega)\\ &= \prod_{j=1}^n \int_\Omega (Q_j + \xi_j(\omega)(\mathbb{I}_2^{\otimes_n} - Q_j))d\mathbb{P}(\omega)\\ &= \prod_{j=1}^n [(1-e^{-t})Q_j + e^{-t}\mathbb{I}_2^{\otimes_n}] \qquad (3.6)\\ &= \prod_{j=1}^n [Q_j + e^{-t}(\mathbb{I}_2^{\otimes_n} - Q_j)] = S_t,\end{aligned}$$

where the last equality is due to Lemma 3.3.

For $N \in \mathbb{N}$, let $(\Omega^N, \mathcal{F}^{\otimes_N}, \mathbb{P}^{\otimes_N})$ be the N-Cartesian product of the probability space $(\Omega, \mathcal{F}, \mathbb{P})$. For each $1 \le k \le N$, let

$$P_k : (\Omega^N, \mathcal{F}^{\otimes_N}, \mathbb{P}^{\otimes_N}) \to (\Omega, \mathcal{F}, \mathbb{P}),$$

that is, $P_k(\omega) = \omega_k$. We claim that, for each given $\omega \in \Omega^N$, $\{\pi \circ P_\ell(\omega)\}_{\ell=1}^N$ forms a commuting norm-one projections family of $B(L_p(\mathbb{M}_{2^n}))$. Once this claim is proved, applying Lemma 3.4 and the K-convexity of $L_p(\mathbb{M}_{2^n})$ for all $1 < p < \infty$, there exists $0 < \rho < 2$ such that

$$\left\| \prod_{\ell=1}^N (\mathbb{I} - \pi \circ P_\ell(\omega)) \right\|_{L_p \to L_p} \le \rho^N. \qquad (3.7)$$

Furthermore, by (3.6), we have

$$\begin{aligned}(\mathbb{I} - S_t)^N &= \left(\int_\Omega \mathbb{I} - \pi(\omega)d\mathbb{P}(\omega) \right)^N\\ &= \prod_{\ell=1}^N \int_{\Omega^N} (\mathbb{I} - \pi \circ P_\ell(\omega))d\mathbb{P}^{\otimes_N}(\omega)\\ &= \int_{\Omega^N} \prod_{\ell=1}^N (\mathbb{I} - \pi \circ P_\ell(\omega))d\mathbb{P}^{\otimes_N}(\omega).\end{aligned}$$

Then the Jensen inequality entails that

$$\|(\mathbb{I} - S_t)^N\|_{L_p \to L_p} \le \int_{\Omega^N} \| \prod_{\ell=1}^N (\mathbb{I} - \pi \circ P_\ell(\omega))\|_{L_p \to L_p} d\mathbb{P}^{\otimes_N}.$$

which, together with (3.7), gives

$$\|(\mathbb{I} - S_t)^N\|_{L_p \to L_p} \le \int_{\Omega^N} \rho^N d\mathbb{P}^{\otimes_N} = \rho^N.$$

This proves (3.4).

It now remains to verify the claim. Indeed, for any $\omega \in \Omega^N$ and $1 \leq \ell \leq N$, by the definition of the map π, we have

$$\pi \circ P_\ell(\omega) = \pi(\omega_l) = \prod_{j=1}^{n} (Q_j + \xi_j(\omega_\ell)(\mathbb{I}_2^{\otimes_n} - Q_j)) = \prod_{\substack{\xi_j(\omega_\ell)=0 \\ j\in[n]}} Q_j.$$

Since $(Q_j)_{j=1}^n$ is a family norm-one projections on $L_p(\mathbb{M}_{2^n})$ with $1 \leq p \leq \infty$, it follows that $\pi \circ P_l(\omega)$ is also a norm-one projection. Since the family $(Q_j)_{j=1}^n$ are commuting, for any $\ell_1, \ell_2 \in [N]$ and $\ell_1 \neq \ell_2$, it follows that

$$\begin{aligned}
\pi \circ P_{\ell_1}(\omega) \cdot \pi \circ P_{\ell_2}(\omega) &= \left(\prod_{\substack{\xi_j(\omega_{\ell_1})=0 \\ j\in[n]}} Q_j \right) \left(\prod_{\substack{\xi_k(\omega_{\ell_2})=0 \\ k\in[n]}} Q_k \right) \\
&= \prod_{\substack{\xi_j(\omega_{\ell_1})=0,\ \xi_k(\omega_{\ell_2})=0 \\ j,k\in[n]}} Q_j \cdot Q_k \\
&= \prod_{\substack{\xi_j(\omega_{\ell_1})=0,\ \xi_k(\omega_{\ell_2})=0 \\ j,k\in[n]}} Q_k \cdot Q_j \\
&= \pi \circ P_{\ell_2}(\omega) \cdot \pi \circ P_{\ell_1}(\omega).
\end{aligned}$$

This verifies the claim, and the proof is complete. □

3.2. Quantum Rademacher projection

The holomorphic estimate of Theorem 3.1 has two immediate consequences that will be used repeatedly in the sequel. First is a dimension-free bound for the spectral projections onto fixed Pauli–Fourier levels. The second is a bounded functional calculus for the multiplier $w^{\mathcal{L}}$ on suitable lens domains inside the unit disc. These two estimates are the analytic bridge between holomorphicity of the quantum Ornstein–Uhlenbeck semigroup and the degree sensitive estimates in the next two chapters.

For any $k \in [n]$, define the *quantum Rademacher projection* $\mathrm{Rad}_{\mathbb{M}_2}^{=k} : \mathbb{M}_{2^n} \to \mathbb{M}_{2^n}$ of level k is defined by

$$\mathrm{Rad}_{\mathbb{M}_2}^{=k}(T) = \sum_{\substack{\mathbf{s}\in\{0,1,2,3\}^n \\ |\mathbf{s}|=k}} \widehat{T}(\mathbf{s})\sigma_{\mathbf{s}}, \tag{3.8}$$

where $T = \sum_{\mathbf{s}\in\{0,1,2,3\}^n} \widehat{T}(\mathbf{s})\sigma_{\mathbf{s}}$. If the context is clear, we simply use $\mathrm{Rad}^{=k}$ instead of $\mathrm{Rad}_{\mathbb{M}_2}^{=k}$, and simply denote $\mathrm{Rad}^{=1}$ by Rad. Notice that $\mathrm{Rad}^{=k}$ is the spectral projection of the number operator $\mathcal{L}$ corresponding to the eigenvalue k.

LEMMA 3.6. *Let $1 < p < \infty$, and let $\theta = \theta_p \in (0, \pi/2]$ and M_p be as in Theorem 3.1. Then, for any $k \in [n]$, we have*

$$\|\mathrm{Rad}^{=k}\|_{L_p \to L_p} \leq M_p e^{ak}, \quad a = \frac{\pi}{\tan\theta}.$$

PROOF. Let $a = \frac{\pi}{\tan\theta}$. For each $k \in [n]$ and every $-\pi \leq t \leq \pi$ (which ensures that $|\arg(a+it)| \leq \theta$), applying the orthogonality of trigonometric series, we obtain

$$\frac{1}{2\pi}\int_{-\pi}^{\pi} e^{ikt}e^{-(a+it)\mathcal{L}}\mathrm{d}t = \frac{1}{2\pi}\int_{-\pi}^{\pi}\sum_{j=0}^{n} e^{ikt-(a+it)j}\,\mathrm{Rad}^{=j}\;\mathrm{d}t = e^{-ka}\mathrm{Rad}^{=k}.$$

Then, we have

$$\begin{aligned}\|\mathrm{Rad}^{=k}\|_{L_p\to L_p} &= e^{ka}\|\frac{1}{2\pi}\int_{-\pi}^{\pi} e^{ikt}e^{-(a+it)\mathcal{L}}\;\mathrm{d}t\|_{L_p\to L_p}\\ &\leq \frac{e^{ka}}{2\pi}\int_{-\pi}^{\pi}\|e^{-(a+it)\mathcal{L}}\|_{L_p\to L_p}\;\mathrm{d}t\\ &\leq M_p e^{ka},\end{aligned}$$

where the last inequality is due to Theorem 3.1. The proof is complete. □

We next explain why the following lens domains (3.9) naturally enter the argument. Estimates for heat smoothing and for low-degree lower bounds are not obtained from the positive semigroup parameter $w = e^{-t}$ alone. They require applying scalar polynomial inequalities to the spectral variable w and then transferring those polynomial estimates to the operator-valued multiplier $w^{\mathcal{L}}$. In particular, the Hahn–Banach and conformal-mapping arguments used later require uniform bounds for $w^{\mathcal{L}}$ on domains that touch the unit circle at both 1 and -1. The point 1 corresponds to the usual heat semigroup, while the point -1 is needed for Chebyshev-type and low-degree interpolation arguments. The anti-isomorphism Φ_ρ from Lemma 2.4 supplies the required control of negative multipliers, since it implements the sign $(-1)^{|\mathbf{s}|}$ on each Pauli level.

Let $\mathbb{D} = \{z \in \mathbb{C} : |z| < 1\}$ the open unit disc. For $r \in [1,\infty)$, define the lens domain

$$\Omega(r) \stackrel{\mathrm{def}}{=} \left\{z \in \mathbb{C} : \max\left\{\left|z - i\sqrt{r^2-1}\right|, \left|z + i\sqrt{r^2-1}\right|\right\} < r\right\} \subseteq \mathbb{D}. \tag{3.9}$$

Note that $\partial\Omega(r)$ is a Jordan curve with an interior angle

$$\theta(r) = 2\arcsin(1/r) \tag{3.10}$$

at $z = \pm 1$.

LEMMA 3.7 ([**EI20b**, Lemma 27]). *For $\alpha \in (0,\infty)$, let $\phi_\alpha : \mathbb{C}\backslash(-\infty,0] \to \mathbb{C}$ be a holomorphic branch of $z \mapsto z^\alpha$. Also, consider the Möbius transformations $\psi_1(z) = \frac{1-z}{1+z}$ and $\psi_2(z) = \frac{z+1}{z-1}$. Then, for every $r \in [1,\infty)$, the map $\psi_1 \circ \phi_{\pi/\theta(r)} \circ \psi_1$ is a conformal equivalence between $\Omega(r)$ and the unit disc $\mathbb{D}$. Furthermore, the map $\psi_2 \circ \phi_{\pi/(2\pi-\theta(r))} \circ \psi_2$ is a conformal equivalence between the complement $\overline{\Omega(r)}^c$ of $\overline{\Omega(r)}$ and the complement $\overline{\mathbb{D}}^c$ of the closed unit disc.*

LEMMA 3.8. *For any $p \in (1,\infty)$, there exist constants $r_p \in [1,\infty)$ and $K_p \in [1,\infty)$ such that*

$$\sup_{w\in\Omega(r_p)} \|w^{\mathcal{L}}\|_{L_p\to L_p} \leq K_p.$$

PROOF. Let $\theta \in \big(0, \frac{\pi}{2}\big]$, and let $a = \frac{\pi}{\tan\theta}$. Take $w \in \mathbb{D}$ with $|w| < e^{-2a}$. According to Lemma 3.6, we have

$$\begin{aligned}\|w^{\mathcal{L}}\|_{L_p\to L_p} &\le \sum_{k=0}^{n} |w|^k \left\|\mathrm{Rad}^{=k}\right\|_{L_p\to L_p} \\ &\le M_p \sum_{k=0}^{n} (|w|e^a)^k,\end{aligned}$$

which implies that for $|w| < e^{-2a}$, we also have the estimate

$$\|w^{\mathcal{L}}\|_{L_p\to L_p} \le \frac{M_p}{1-e^{-a}}.$$

For every $w \in \{e^{-z} : |\arg z| < \theta\}$, applying Lemma 2.4 (iii) and Theorem 3.1, we have

$$\left\|w^{\mathcal{L}}\right\|_{L_p\to L_p} = \left\|(-w)^{\mathcal{L}}\right\|_{L_p\to L_p} \le M_p.$$

By the above estimates, one can easily deduce that there exists $r_p \in [1,\infty)$ such that

$$\Omega(r_p) \subseteq \left\{e^{-z} : |\arg z| < \theta\right\} \cup \left\{-e^{-z} : |\arg z| < \theta\right\} \cup \{w \in \mathbb{D} : |w| < e^{-2a}\}. \tag{3.11}$$

and the desired result follows with $K_p = \frac{M_p}{1-e^{-a}}$. □

LEMMA 3.9. *For each $k \in [n]$ and $T \in \mathbb{M}_{2^n}$, we have*

$$\sup_{k\ge 0} \|\mathrm{Rad}^{=k}\|_{L_p\to L_p}^{\frac{1}{k}} \le \begin{cases} \sqrt{p-1}, & 2 \le p < \infty; \\ \sqrt{\frac{1}{p-1}}, & 1 < p \le 2, \end{cases}.$$

PROOF. We first prove the case $2 \le p < \infty$ and use the duality argument to derive the case $1 < p \le 2$. It is clear that for $T \in \mathbb{M}_{2^n}$ we have

$$\mathrm{Rad}^{=k}(T) = e^{tk} e^{-t\mathcal{L}}(\mathrm{Rad}^{=k}(T)).$$

By the hypercontractivity (i.e., Theorem 2.1), we obtain

$$\begin{aligned}\|\mathrm{Rad}^{=k}(T)\|_{L_p} &= \|e^{tk} e^{-t\mathcal{L}}(\mathrm{Rad}^{=k}(T))\|_{L_p} \\ &\le e^{tk} \|\mathrm{Rad}^{=k}(T)\|_{L_2} \\ &\le e^{tk} \|T\|_{L_2} \\ &\le e^{tk} \|T\|_{L_p},\end{aligned}$$

where we used the fact $2 \le p < \infty$ and $(\mathbb{M}_{2^n}, \mathrm{tr})$ is a probability space in the last inequality. Taking $t = \frac{1}{2}\log(p-1)$, it follows that

$$\sup_{k\ge 0} \|\mathrm{Rad}^{=k}\|_{L_p\to L_p} \le (p-1)^{k/2}, \quad 2 \le p < \infty.$$

By the duality argument, we obtain the estimate for the case $1 < p \le 2$, which completes our proof. □

CHAPTER 4

Quantum heat smoothing estimates

This chapter develops the high-frequency side of the degree-sensitive semigroup theory on the quantum Boolean cube. Throughout the chapter, $\mathcal{T}^{\geq d}(\mathbb{M}_{2^n})$ and $\mathcal{P}^{\leq d}(\mathbb{M}_{2^n})$ denote the corresponding Pauli–Walsh tail and low-degree subspaces, respectively.

The first section proves a quantum analogue of the Mendel–Naor heat smoothing estimate [**MN14**] for observables in $\mathcal{T}^{\geq d}(\mathbb{M}_{2^n})$. The proof combines the holomorphic functional calculus from Chapter 3 with a conformal mapping argument on lens domains. The second section gives a quantum analogy of the first-tail (i.e., $d = 1$) Heilman–Mossel–Oleszkiewicz type strong contraction estimate [**HMO17**]. Our approach is inspired by [**HMO17**], but the proof replaces the pointwise order arguments used in the commutative setting by a probabilistic representation for noncommutative Markov maps. We conclude the chapter with reverse Bernstein–Markov inequalities for observables whose Pauli-Walsh spectrum is localized in the finite band $[d, d+m]$.

4.1. Quantum Mendel-Naor heat smoothing estimates

Heat smoothing estimates quantify how fast the heat semigroup damps the part of a function supported on high frequencies. In the classical framework, this idea goes back to Meyer [**Mey84**] in the Gaussian setting and was developed systematically by Mendel and Naor [**MN14**] in the Boolean cube setting. The following theorem is the quantum counterpart needed later for the tail Talagrand and covariance estimates.

THEOREM 4.1. *For $n \geq 1, d \in [n]$ and $1 < p < \infty$, there exist $C_p, c_p \in (0, \infty)$ and $A_p \in [1, \infty)$ depending only p such that, for every $T \in \mathcal{T}^{\geq d}(\mathbb{M}_{2^n})$, we have*

$$\left\| e^{-t\mathcal{L}}(T) \right\|_{L_p} \leq C_p e^{-c_p d \min\{t, t^{A_p}\}} \|T\|_{L_p}, \quad \forall t \geq 0. \tag{4.1}$$

The proof is postponed until after Proposition 4.2. Recall that the d-th Chebyshev polynomial of the first kind, denoted by $T_d(x)$, is the unique polynomial of degree d satisfying $T_d(\cos\theta) = \cos(d\theta)$ for all $\theta \in \mathbb{R}$. Chebyshev's inequality [**BE95**, p.235] asserts that the d-th Chebyshev polynomial of the first kind is characterized by the extremal property

$$|T_d(x)| = \max\left\{ |P(x)| : \deg(P) \leq d \text{ and } \|P\|_{\mathcal{C}([-1,1])} = 1 \right\}, \ \forall x \in \mathbb{R}\backslash[-1,1]. \tag{4.2}$$

Here and what follows, for each polynomial $P \in \mathcal{P}(\mathbb{K})$, defined on a compact $\mathbb{K} \subseteq \mathbb{C}$, we denote $\|P\|_{\mathcal{C}(\mathbb{K})} = \sup_{z \in \mathbb{K}} |P(z)|$ its supremum norm in $\mathbb{K}$. For each compact $\mathbb{K} \subseteq \mathbb{C}$ and $d \in \mathbb{N}$, denote

$$\mathcal{P}^{\leq d}(\mathbb{K}) = \{P : \mathbb{K} \to \mathbb{C} : \ P \text{ is a polynomial with } \deg(P) \leq d\}$$

and

$$\mathcal{T}^{\geq d}(\mathbb{K}) = \left\{ P : \mathbb{K} \to \mathbb{C} : P(w) = \sum_{n=d}^{N} a_n w^n,\ N \in \mathbb{N} \right\}.$$

PROPOSITION 4.2. *Let $n \geq 1, d \in [n]$, $1 < p < \infty$, and $\Omega \subseteq \mathbb{D}$ be a simply connected domain bounded by a Jordan curve γ such that $(-1,1) \subseteq \Omega$ and suppose that*

$$\sup_{w \in \Omega} \|w^{\mathcal{L}}\|_{L_p \to L_p} \leq K_p. \tag{4.3}$$

Let $\phi_\Omega : \Omega \to \mathbb{D}$ be a conformal mapping of Ω onto $\mathbb{D}$ with $\phi_\Omega(0) = 0$ which extends continuously as a homeomorphism between $\Omega \cup \gamma$ and $\overline{\mathbb{D}}$. Then, for every $T \in \mathcal{T}^{\geq d}(\mathbb{M}_{2^n})$, we have

$$\|e^{-t\mathcal{L}}(T)\|_{L_p} \leq K_p |\phi_\Omega(e^{-t})|^d \|T\|_{L_p}, \quad \forall t > 0.$$

PROOF. Consider the linear functional $\zeta_t : \left(\mathcal{T}^{\geq d}(\overline{\Omega}), \|\cdot\|_{\mathcal{C}(\overline{\Omega})}\right) \to \mathbb{C}$ defined by

$$\zeta_t(P) = P(e^{-t}), \quad t \geq 0.$$

For each $P \in \mathcal{T}^{\geq d}(\overline{\Omega})$, consider the function $h : \Omega \to \mathbb{C}$ given by

$$h(w) \coloneqq \frac{P(w)}{\phi_\Omega(w)^d}, \quad \forall w \in \Omega.$$

Note that ϕ_Ω does not vanish on $\Omega \backslash \{0\}$ and has a single root at 0. Therefore, the multiplicity of the root 0 in the numerator is at least that in the denominator, ensuring that h is a holomorphic function on Ω. Furthermore, we have

$$|\phi_\Omega(w)| = 1, \quad w \in \gamma,$$

since ϕ_Ω is a homeomorphism between γ and $\partial\mathbb{D}$ (such a conformal map ϕ_Ω always exists by Caratheodory's theorem, see [**Kra06**, Theorem 5.1.1]). Therefore, by the maximum principle,

$$\sup_{w \in \Omega} |h(w)| = \max_{w \in \gamma} |h(w)| = \max_{w \in \gamma} \frac{|P(w)|}{|\phi_\Omega(w)|^d} = \|P\|_{\mathcal{C}(\gamma)} = \|P\|_{\mathcal{C}(\overline{\Omega})},$$

which implies

$$|P(w)| \leq |\phi_\Omega(w)|^d \|P\|_{\mathcal{C}(\overline{\Omega})}, \quad w \in \overline{\Omega}.$$

Applying the above inequality for $w = e^{-t} \in \Omega$, we deduce that the linear functional ζ_t satisfies

$$|\zeta_t(P)| \leq \left|\phi_\Omega\left(e^{-t}\right)\right|^d \|P\|_{\mathbb{C}(\overline{\Omega})}, \quad \forall P \in \mathcal{T}^{\geq d}(\overline{\Omega}). \tag{4.4}$$

Therefore, by the Hahn-Banach theorem and the Riesz representation theorem, there exists a complex measure τ_t on $\overline{\Omega}$ such that

$$\|\tau_t\|_{V(\overline{\Omega})} \leq \left|\phi_\Omega\left(e^{-t}\right)\right|^d$$

and

$$\int_{\overline{\Omega}} P(w) \mathrm{d}\tau_t(w) = \zeta_t(P) = P\left(e^{-t}\right), \quad P \in \mathcal{P}(\overline{\Omega}). \tag{4.5}$$

Since $\widehat{T}(\mathbf{s}) = 0$ when $|\mathbf{s}| < d$, by the Jensen inequality, we obtain

$$\begin{aligned}
\|e^{-t\mathcal{L}}(T)\|_{L_p} &= \Big\| \sum_{\mathbf{s}\in\{0,1,2,3\}^n, |\mathbf{s}|\geq d} e^{-t|\mathbf{s}|}\widehat{T}(\mathbf{s})\sigma_{\mathbf{s}} \Big\|_{L_p} \\
&\overset{(4.5)}{=} \Big\| \int_{\overline{\Omega}} \sum_{\mathbf{s}\in\{0,1,2,3\}^n, |\mathbf{s}|\geq d} w^{|\mathbf{s}|}\widehat{T}(\mathbf{s})\sigma_{\mathbf{s}} \, \mathrm{d}\tau_t(w) \Big\|_{L_p} \\
&= \Big\| \int_{\overline{\Omega}} w^{\mathcal{L}}(T) \, \mathrm{d}\tau_t(w) \Big\|_{L_p} \leq \int_{\overline{\Omega}} \|w^{\mathcal{L}}(T)\|_{L_p} \mathrm{d}|\tau_t|(w) \\
&\overset{(4.3)}{\leq} K_p \int_{\overline{\Omega}} \|T\|_{L_p} \mathrm{d}|\tau_t|(w) = K_p \, \|\tau_t\|_{V(\overline{\Omega})} \, \|T\|_{L_p} \\
&\leq K_p |\phi_\Omega(e^{-t})|^d \|T\|_{L_p},
\end{aligned}$$

which concludes the proof. □

Proof of Theorem 4.1. Let $T \in \mathcal{T}^{\geq d}(\mathbb{M}_{2^n})$ and fix $1 < p < \infty$. Choose a sectorial angle $0 < \theta < \theta_p$, where θ_p is supplied by the holomorphic semigroup theorem of Chapter 3, and put

$$a \coloneqq \frac{\pi}{\tan\theta} > 0.$$

We divide the proof into two cases.

Case I: For $t \geq 2a$, by Lemma 3.6, we have the following estimate

$$\begin{aligned}
\|e^{-t\mathcal{L}}(T)\|_{L_p} &\leq \sum_{k=d}^{n} e^{-tk} \|\mathrm{Rad}^{=k}(T)\|_{L_p} \\
&\leq M_p \sum_{k=d}^{n} e^{-tk} e^{ak} \|T\|_{L_p} \leq K_p e^{-\frac{1}{2}td} \|T\|_{L_p},
\end{aligned}$$

where $K_p = \frac{M_p}{1-e^{-a}}$.

Case II: For $t \leq 2a$, by Lemma 3.8, there exist constants $r_p \in [1,\infty)$ and $K_p \in [1,\infty)$ such that

$$\sup_{\omega\in\Omega(r_p)} \sup_{n\in\mathbb{N}} \|\omega^{\mathcal{L}}\|_{L_p\to L_p} \leq K_p,$$

where $\Omega(r_p)$ is the domain (3.11). By Proposition 4.2, we have

$$\|e^{-t\mathcal{L}}(T)\|_{L_p} \leq K_p |\phi_{\Omega(r_p)}(e^{-t})|^d \|T\|_{L_p}, \quad \forall t \geq 0.$$

By Lemma 3.7, $\theta = \theta(r_p) = 2\arcsin\left(\frac{1}{r_p}\right) \in (0,\pi)$, and the function $\phi_{\Omega(r_p)}$ can be expressed as

$$\phi_{\Omega(r_p)}(e^{-t}) = \psi_1 \circ \phi_{\pi/\theta} \circ \psi_1 = \frac{(1+e^{-t})^{\pi/\theta} - (1-e^{-t})^{\pi/\theta}}{(1+e^{-t})^{\pi/\theta} + (1-e^{-t})^{\pi/\theta}}.$$

To analyze the decay of $\phi_{\Omega(r_p)}$, we take the logarithm and use the inequality $\log(1-x) \leq -x$ on $(0,1)$:

$$\begin{aligned}
\log|\phi_{\Omega(r_p)}(e^{-t})| &= \log\Big(1 - \frac{2(1-e^{-t})^{\pi/\theta}}{(1+e^{-t})^{\pi/\theta} + (1-e^{-t})^{\pi/\theta}}\Big) \\
&\leq -\frac{2(1-e^{-t})^{\pi/\theta}}{(1+e^{-t})^{\pi/\theta} + (1-e^{-t})^{\pi/\theta}}.
\end{aligned}$$

For all $t \le 2a$, we further simplify the bound

$$-\frac{2(1-e^{-t})^{\pi/\theta}}{(1+e^{-t})^{\pi/\theta}+(1-e^{-t})^{\pi/\theta}} \le -2^{1-\frac{\pi}{\theta}}(1-e^{-t})^{\pi/\theta} \le \frac{-2^{1-\frac{\pi}{\theta}}(1-e^{-2a})^{\pi/\theta}}{(2a)^{\pi/\theta}} t^{\pi/\theta},$$

where the last inequality is because the function $f(x) = \frac{1-e^{-x}}{x}$ is monotonically decreasing on $x \in (0,\infty)$. Thus, for all $0 \le t \le 2a$, we have

$$|\phi_{\Omega(r)}\left(e^{-t}\right)| \le e^{-ct^{\pi/\theta}},$$

where $c = \frac{2^{1-\frac{\pi}{\theta}}\left(1-e^{-2a}\right)^{\pi/\theta}}{(2a)^{\pi/\theta}}$. The proof is complete. □

4.2. Quantum Heilman-Mossel-Oleszkiewicz heat smoothing estimate

The preceding estimate gives heat smoothing on d-tail spaces for arbitrary d. We next prove a first-tail strong contraction estimate in the spirit of Heilman–Mossel–Oleszkiewicz [**HMO17**]. In the hilbertian case the spectral gap of the Laplacian operator is equivalent to the following exponential ergodicity of the semigroup, that is, for every $T \in \mathbb{M}_{2^n}$ with $\mathrm{tr}(T) = 0$, we have

$$\|e^{-t\mathcal{L}}(T)\|_{L_2} \le e^{-t}\|T\|_{L_2}. \tag{4.6}$$

The point of this section is to derive a general case of (4.6). After extensive investigations such as [**EI20b, Esk23, HMO17, Vol24**], the heat smoothing conjecture, that is, Conjecture 1.1, remains open (even for $X = \mathbb{R}$ there). Particularly, Heilman, Mossel and Oleszkiewicz [**HMO17**] resolved Conjecture 1.1 with $X = \mathbb{R}$ and $d = 1$. The next theorem is a quantum counterpart of the main estimate of Heilman, Mossel and Oleszkiewicz [**HMO17**].

For $p \in (1,\infty) \setminus \{2\}$, set

$$\kappa(p) \coloneqq \inf_{x>1} \frac{x^{p/(p-2)} + x^{-p/(p-2)}}{x - x^{-1}}.$$

THEOREM 4.3. *For each $p \in (1,\infty)\setminus\{2\}$, the following heat smoothing inequality holds for each self-adjoint $T \in \mathbb{M}_{2^n}$ with $\mathrm{tr}(T) = 0$,*

$$\|e^{-t\mathcal{L}}(T)\|_{L_p} \le \exp\left(-\frac{(4p-4)t}{p^2(1+\kappa(p)^{-2})}\right)\|T\|_{L_p}.$$

We first isolate a spectral approximation bound for the heat semigroup.

LEMMA 4.4. *For any $\epsilon, t \in (0,\infty)$, we have*

$$\|e^{-t\mathcal{L}} - e^{-(\epsilon+t)\mathcal{L}}\|_{L_2 \to L_2} \le \frac{\epsilon}{t}.$$

PROOF. Without loss of generality, we assume that

$$T = \sum_{\mathbf{s}\in\{0,1,2,3\}^n} \widehat{T}(\mathbf{s})\sigma_{\mathbf{s}}.$$

We have

$$e^{-ts} - e^{-(t+\epsilon)s} \le \frac{\epsilon}{t}, \quad \text{for all } s \ge 0. \tag{4.7}$$

Indeed, let $F(s) \coloneqq e^{-ts} - e^{-(\epsilon+t)s}$ for all $s \geq 1$. It is easy to see that

$$F'(s) = e^{-ts}\left((t+\epsilon)e^{-\epsilon s} - t\right), \quad \text{for } s \geq 1.$$

The function $(t+\epsilon)e^{-\epsilon s} - t$ is strictly decreasing and vanishes at $s_0 = \frac{1}{\epsilon}\log\left(1+\frac{\epsilon}{t}\right)$. Hence, F is increasing on $[0, s_0]$ and decreasing on $[s_0, \infty)$, so it attains its maximum at s_0. Evaluating F at s_0, we obtain

$$F(s_0) = \left(1+\frac{\epsilon}{t}\right)^{-\frac{t}{\epsilon}}\left(1 - \frac{t}{t+\epsilon}\right) \leq \frac{\epsilon}{t+\epsilon} \leq \frac{\epsilon}{t}.$$

Hence, we apply (4.7) to get that

$$\begin{aligned}\left\|\left(e^{-t\mathcal{L}} - e^{-(\epsilon+t)\mathcal{L}}\right)(T)\right\|_{L_2}^2 &= \sum_{\mathbf{s}\in\{0,1,2,3,\}^n, \mathbf{s}\neq 0} (e^{-t|\mathbf{s}|} - e^{-(\epsilon+t)|\mathbf{s}|})^2 \left|\widehat{T}(\mathbf{s})\right|^2 \\ &\leq \left(\frac{\epsilon}{t}\right)^2 \|T - \mathrm{tr}(T)\|_{L_2}^2 \leq \frac{\epsilon^2}{t^2}\|T\|_{L_2}^2,\end{aligned}$$

which is the desired inequality. □

The next proposition is a probabilistic representation of positive operators, established in [**Xio17**, Lemma 4.5], which enables us to bypass pointwise estimate heavily relied upon in the approach of [**HMO17**].

PROPOSITION 4.5. *Suppose that $\mathcal{N}$ is a σ-weakly dense C^*-subalgebra of $(\mathcal{M}, \tau)$. Let $x \in \mathcal{N}$ be a normal element and $P : \mathcal{N} \to \mathcal{N}$ be a positive map. Then, there exists a positive measure μ on $\sigma(x) \times \sigma(x)$ with $\|\mu\| \leq \|P\|$ such that for all $\varphi, \psi \in C(\sigma(x))$,*

$$\tau\big(\varphi(x)P(\psi(x))\big) = \int_{\sigma(x)\times\sigma(x)} \varphi(t)\psi(s)\ \mathrm{d}\mu(t,s).$$

For $a \in \mathbb{R}$ and $q \in [1, \infty)$, we use the shorthand notation

$$\phi_q(a) \coloneqq |a|^{q-1}\mathrm{sign}(a) \tag{4.8}$$

in the sequel. The well-known Stroock-Varopulos inequality is recalled as follows; see, for instance, [**HMO17**, Lemma 2.4].

LEMMA 4.6 (Stroock–Varopoulos). *Let $a, b \in \mathbb{R}$ and $p > 1$. Then*

$$\big(\phi_{p-1}(a) - \phi_{p-1}(b)\big)(a-b) \geq \frac{4(p-1)}{p^2}\big(\phi_{\frac{p}{2}}(a) - \phi_{\frac{p}{2}}(b)\big)^2.$$

Combining Proposition 4.5 with the Stroock-Varopoulos inequality, we get the following inequality.

LEMMA 4.7. *For every $1 < p < \infty$ and $T \in L_p(\mathbb{M}_{2^n})$ with $T = T^*$, we have*

$$(p-2)^2\|T\|_{L_p}^p + 4(p-1)\mathrm{tr}(\phi_{\frac{p}{2}}(T)\cdot e^{-t\mathcal{L}}(\phi_{\frac{p}{2}}(T))) \geq p^2\mathrm{tr}(\phi_{p-1}(T)\cdot e^{-t\mathcal{L}}(T))$$

PROOF. By Lemma 4.6, the following inequality holds for all a, $b \in \mathbb{R}$

$$p^2\big(\phi_{p-1}(a) - \phi_{p-1}(b)\big)(a-b) \geq 4(p-1)\big(\phi_{\frac{p}{2}}(a) - \phi_{\frac{p}{2}}(b)\big)^2,$$

which can be rearranged to the follows

$$(p-2)^2(|a|^p + |b|^p) + 8(p-1)\phi_{\frac{p}{2}}(a)\phi_{\frac{p}{2}}(b) \geq p^2(a\phi_{p-1}(b) + \phi_{p-1}(a)b). \tag{4.9}$$

Applying Proposition 4.5 to (4.9) and taking the integration with respect to the positive measure μ on $\sigma(T)\times\sigma(T)$ yields the desired inequality. □

LEMMA 4.8. *Let a, $b \in \mathbb{R}$. For each $2 < p < \infty$, we have*

$$|a|^p + (p-1)|b|^p - pa|b|^{p-1}\mathrm{sign}(b) \leq p(p-1)|a-b|^2(|a|^{p-2} + |b|^{p-2}), \tag{4.10}$$

and for each $1 < p \leq 2$, we have

$$|a|^p + (p-1)|b|^p - pa|b|^{p-1}\mathrm{sign}(b) \leq 2p|a-b|^p. \tag{4.11}$$

PROOF. The proofs of both inequalities are similar to each other which relies on specific convexity of the function

$$F(t) \coloneqq |b + t(a-b)|^p, \quad \text{for } t \in [0,1].$$

It is easy to see that F is a differentiable convex function on $[0,1]$ such that $F(0) = |b|^p$, $F(1) = |a|^p$ and

$$F'(t) = p(a-b)|b + t(a-b)|^{p-1}\mathrm{sign}(b + t(a-b)),$$

and

$$F''(t) = p(p-1)(a-b)^2|b + t(a-b)|^{p-2},$$

for all $t \in [0,1]$. Thanks to the Newton-Leibniz formula, we have

(4.12)

$$\begin{aligned}
|a|^p - |b|^p = F(1) - F(0) &= \int_0^1 F'(t)dt \\
&= p(a-b)\int_0^1 |b + t(a-b)|^{p-1}\mathrm{sign}(b + t(a-b))dt = \int_0^1\int_0^t F''(u)dudt + F'(0) \\
&= p(p-1)|a-b|^2\int_0^1\int_0^t |b + u(a-b)|^{p-2}dudt + p(a-b)|b|^{p-1}\mathrm{sign}(b).
\end{aligned}$$

Rearranging (4.12) yields the following

(4.13)

$$|a|^p + (p-1)|b|^p - pa|b|^{p-1}\mathrm{sign}(b) = p(p-1)|a-b|^2\int_0^1\int_0^t |b + u(a-b)|^{p-2}dudt,$$

where $t \in [0,1]$.

For the inequality (4.10), we note that, for every $u \in [0,1]$,

$$|b + u(a-b)| = |(1-u)b + ua| \leq (1-u)|b| + u|a| \leq \max\{|a|, |b|\}.$$

Since $p > 2$, it follows that

$$|b + u(a-b)|^{p-2} \leq \max\{|a|^{p-2}, |b|^{p-2}\} \leq |a|^{p-2} + |b|^{p-2}.$$

Therefore, by (4.13) we have

$$|a|^p + (p-1)|b|^p - pa|b|^{p-1}\mathrm{sign}(b) \leq p(p-1)|a-b|^2(|a|^{p-2} + |b|^{p-2}),$$

which completes the proof of (4.10).

If $1 < p \leq 2$, we note that the convexity of F implies that F' is an increasing function. Hence,

$$\begin{aligned}
&|a|^p + (p-1)|b|^p - pa|b|^{p-1}\mathrm{sign}(b) = |a|^p - |b|^p + p(b-a)|b|^{p-1}\mathrm{sign}(b) \\
=&F(1) - F(0) - F'(0) = \int_0^1 F'(t)dt - F'(0) \leq F'(1) - F'(0) \\
\leq&p|a-b|\left||a|^{p-1}\mathrm{sign}(a) - |b|^{p-1}\mathrm{sign}(b)\right|.
\end{aligned} \tag{4.14}$$

Since $1 < p \leq 2$, it follows that $\big||a|^{p-1} - |b|^{p-1}\big| \leq |a-b|^{p-1}$ for all a, $b \in \mathbb{R}$ and $\mathrm{sign}(a)\mathrm{sign}(b) \geq 0$. Thus, if $\mathrm{sign}(a)\mathrm{sign}(b) \geq 0$, (4.14) can be estimated by

$$|a|^p + (p-1)|b|^p - pa|b|^{p-1}\mathrm{sign}(b) \leq p|a-b|^p. \tag{4.15}$$

If $\mathrm{sign}(a)\mathrm{sign}(b) < 0$, we have

$$\big||a|^{p-1} + |b|^{p-1}\big| \leq 2||a| + |b||^{p-1} = 2|a-b|^{p-1}.$$

Thus, (4.14) can be estimated by

$$|a|^p + (p-1)|b|^p - pa|b|^{p-1}\mathrm{sign}(b) \leq 2p|a-b|^p. \tag{4.16}$$

Combining (4.15) and (4.16) yields the desired inequality (4.11). □

Definition 4.9. For M, $\epsilon > 0$, define

$$\mathcal{C}(M,\epsilon) \coloneqq \{T : T = e^{-\epsilon\mathcal{L}}(S), S = S^*, \|S\|_{L_\infty} < M\}.$$

For every $1 < p < \infty$, we define the parameter

$$\alpha_{p,M,\epsilon}(t) \coloneqq \sup_{T\in\mathcal{C}(M,\epsilon)} (p-1)\|T\|_{L_p}^p + \|e^{-t\mathcal{L}}(T)\|_{L_p}^p - p\mathrm{tr}(\phi_{p-1}(T)\cdot e^{-t\mathcal{L}}(T)).$$

Lemma 4.10. *For any $1 < p < \infty$ and $M, \epsilon > 0$, we have*

$$\alpha_{p,M,\epsilon}(t)/t \to 0, \quad \textit{as} \quad t \to 0^+.$$

Proof. Combining Lemma 4.8 and Proposition 4.5, we get that

$$\alpha_{p,M,\epsilon}(t) \leq p(p-1)\sup_{T\in\mathcal{C}(M,\epsilon)} \left(\|T\|_{L_\infty}^{p-2} + \|e^{-t\mathcal{L}}(T)\|_{L_\infty}^{p-2}\right)\int_{\sigma(T)\times\sigma(T)} |a-b|^2 d\mu(a,b),$$

if $2 < p < \infty$, and

$$\alpha_{p,M,\epsilon}(t) \leq 2p\sup_{T\in\mathcal{C}(M,\epsilon)} \int_{\sigma(T)\times\sigma(T)} |a-b|^p d\mu(a,b),$$

if $1 < p \leq 2$. It is clear that

$$\int_{\sigma(T)\times\sigma(T)} |a-b|^2 d\mu(a,b) = \|e^{-t\mathcal{L}}(T)\|_{L_2}^2 - 2\mathrm{tr}\left(e^{-t\mathcal{L}}(T)T\right) + \|T\|_{L_2}^2 = \|e^{-t\mathcal{L}}(T) - T\|_{L_2}^2.$$

Hence, for the case $2 < p < \infty$, we have

$$\begin{aligned}\alpha_{p,M,\epsilon}(t) &\leq p(p-1)\sup_{T\in\mathcal{C}(M,\epsilon)} (\|T\|_{L_\infty}^{p-2} + \|e^{-t\mathcal{L}}(T)\|_{L_\infty}^{p-2})\|e^{-t\mathcal{L}}(T) - T\|_{L_2}^2 \\ &\leq 2p(p-1)M^{p-2}\sup_{T\in\mathcal{C}(M,\epsilon)} \|e^{-t\mathcal{L}}(T) - T\|_{L_2}^2,\end{aligned}$$

where we used $T \in \mathcal{C}(\mathcal{M},\epsilon)$ and the fact $e^{-t\mathcal{L}}$ is a contraction in the last inequality. Applying Lemma 4.4 to the above inequality yields that

$$\begin{aligned}\alpha_{p,M,\varepsilon}(t) &\leq 2p(p-1)M^{p-2}\sup_{\|\varphi\|_{L_\infty}\leq M} \|e^{-(t+\epsilon)\mathcal{L}}(\varphi) - e^{-\epsilon\mathcal{L}}(\varphi)\|_{L_2}^2 \\ &\leq 2p(p-1)M^p\left(\frac{t^2}{\epsilon^2}\right),\end{aligned}$$

which further entails that $\alpha_{p,M,\epsilon}(t)/t \to 0^+$ as $t \to 0^+$.

For the case $1 < p \leq 2$, Jensen's inequality gives

$$\begin{aligned}\int_{\sigma(T)\times\sigma(T)} |a-b|^p d\mu(a,b) &= \int_{\sigma(T)\times\sigma(T)} \big(|a-b|^2\big)^{p/2} d\mu(a,b) \\ &\leq \left(\int |a-b|^2 \,\mathrm{d}\mu(a,b)\right)^{p/2} \\ &= \|e^{-t\mathcal{L}}(T) - T\|_{L_2}^p.\end{aligned}$$

It follows that

$$\alpha_{p,M,\epsilon}(t) \leq 2pM^p \left(\frac{t}{\epsilon}\right)^p,$$

and hence

$$0 \leq \frac{\alpha_{p,M,\epsilon}(t)}{t} \leq \frac{2pM^p}{\epsilon^p} t^{p-1} \longrightarrow 0 \qquad \text{as } t \to 0^+.$$

The proof is complete. □

LEMMA 4.11 ([**HMO17**, Lemma 2.2 and Remark 3.2]). *Let $p \in (1,\infty) \setminus \{2\}$. For any $a\ , b > 0$, we have*

$$\big(1+\kappa(p)^2\big)(a-b)^2 \leq a^2 + b^2 + a^{\frac{2}{2-p}} b^{\frac{2p-2}{p-2}} + b^{\frac{2}{2-p}} a^{\frac{2p-2}{p-2}}.$$

LEMMA 4.12. *For each $p \in (1,\infty)\setminus\{2\}$ and a non-zero self-adjoint $T \in L_p(\mathbb{M}_{2^n})$ with $\mathrm{tr}(T) = 0$, we have*

$$\begin{aligned}(\mathrm{tr}(T_+^{p/2}))^2 + (\mathrm{tr}(T_-^{p/2}))^2 + \big(\mathrm{tr}(T_+^{p/2})\big)^{-\frac{2}{p-2}} \big(\mathrm{tr}(T_-^{p/2})\big)^{\frac{2p-2}{p-2}} \\ + \big(\mathrm{tr}(T_-^{p/2})\big)^{-\frac{2}{p-2}} \big(\mathrm{tr}(T_+^{p/2})\big)^{\frac{2p-2}{p-2}} \leq \mathrm{tr}(|T|^p).\end{aligned}$$

PROOF. For each $T \in L_p(\mathbb{M}_{2^n})$, we write $T = T_+ - T_-$. Then $|T| = T_+ + T_-$ and $|T|^p = T_+^p + T_-^p$. By the fact $\mathrm{tr}(T) = 0$, we have

$$\mathrm{tr}(T_+) = \mathrm{tr}(T_-) = \frac{\mathrm{tr}(|T|)}{2}. \tag{4.17}$$

If we prove the following two inequalities,

$$\mathrm{tr}\big(T_+^{\frac{p}{2}}\big)^{\frac{2p-2}{p-2}} + \mathrm{tr}\big(T_-^{\frac{p}{2}}\big)^{\frac{2p-2}{p-2}} \leq 2^{-\frac{p}{p-2}} \mathrm{tr}(|T|)^{\frac{p}{p-2}} \mathrm{tr}(|T|^p) \tag{4.18}$$

and

$$\mathrm{tr}\big(T_+^{\frac{p}{2}}\big)^{-\frac{2}{p-2}} + \mathrm{tr}\big(T_-^{\frac{p}{2}}\big)^{-\frac{2}{p-2}} \leq 2^{\frac{p}{p-2}} \mathrm{tr}(|T|)^{-\frac{p}{p-2}}, \tag{4.19}$$

multiplying them together yields the desired estimate.

For $p > 2$, by the Hölder inequality with indexes $\left(\frac{2p-2}{p-2}, \frac{2p-2}{p}\right)$ and (4.17), we have

$$\mathrm{tr}\left(T_+^{\frac{p}{2}}\right) = \mathrm{tr}\left(T_+^{\frac{p^2-2p}{2p-2}} T_+^{\frac{p}{2p-2}}\right) \leq \mathrm{tr}\left(T_+^p\right)^{\frac{p-2}{2p-2}} \mathrm{tr}\left(\frac{|T|}{2}\right)^{\frac{p}{2p-2}}, \tag{4.20}$$

and

$$\mathrm{tr}\left(T_-^{\frac{p}{2}}\right) \leq \mathrm{tr}(T_-^p)^{\frac{p-2}{2p-2}} \mathrm{tr}\big(|T|/2\big)^{\frac{p}{2p-2}}. \tag{4.21}$$

Taking the power of $(2p-2)/(p-2)$ to both (4.20) and (4.21), and summing the resulting inequalities together, we obtain (4.18). In addition, by the Hölder inequality with $(\frac{p}{2}, \frac{p}{p-2})$ and (4.17),

$$\mathrm{tr}(|T|/2) = \mathrm{tr}(T_+) \leq \mathrm{tr}\big(T_+^{\frac{p}{2}}\big)^{\frac{2}{p}} \mathrm{tr}(\chi_{(0,\infty)}(T))^{\frac{p-2}{p}}. \tag{4.22}$$

Then, taking the power $p/(p-2)$ to the both sides of (4.22) yields the inequality

$$\mathrm{tr}\left(T_+^{\frac{p}{2}}\right)^{-\frac{2}{p-2}} \leq 2^{\frac{p}{p-2}}\mathrm{tr}(|T|)^{-\frac{p}{p-2}} \cdot \mathrm{tr}(\chi_{(0,\infty)}(T)).$$

Similarly, we have

$$\mathrm{tr}\left(T_-^{\frac{p}{2}}\right)^{-\frac{2}{p-2}} \leq 2^{\frac{p}{p-2}}\mathrm{tr}(|T|)^{-\frac{p}{p-2}} \cdot \mathrm{tr}(\chi_{(-\infty,0)}(T)).$$

Summing the two inequalities above gives (4.19).

For $1 < p < 2$, by the Hölder inequality with $(\frac{2}{p}, \frac{2}{2-p})$ and (4.17),

$$\mathrm{tr}\big(T_+^{\frac{p}{2}}\big) \leq \mathrm{tr}(T_+)^{\frac{p}{2}}\tau(\chi_{(0,\infty)}(T))^{\frac{2-p}{2}} = 2^{-\frac{p}{2}}\mathrm{tr}(|T|)^{\frac{p}{2}}\mathrm{tr}\left(\chi_{(0,\infty)}(T)\right)^{\frac{2-p}{2}},$$

and

$$\mathrm{tr}\big(T_-^{\frac{p}{2}}\big) \leq 2^{-\frac{p}{2}}\mathrm{tr}(|T|)^{\frac{p}{2}}\mathrm{tr}(\chi_{(-\infty,0)}(T))^{\frac{2-p}{2}}.$$

Exponentiating both sides of the two inequalities above to the power $2/(2-p)$ and summing the resulting relations gives inequality (4.19). The Hölder inequality with indexes $\left(\frac{p}{2p-2}, \frac{p}{2-p}\right)$ and (4.17) give

$$\mathrm{tr}(|T|/2) = \mathrm{tr}(T_+) = \mathrm{tr}\left(T_+^{p-1}T_+^{2-p}\right) \leq \mathrm{tr}\big(T_+^{\frac{p}{2}}\big)^{\frac{2p-2}{p}}\mathrm{tr}\big(T_+^{p}\big)^{\frac{2-p}{p}}.$$

Then, taking both sides to the $p/(2-p)$-th power, we obtain

$$\mathrm{tr}\big(T_+^{\frac{p}{2}}\big)^{-\frac{2p-2}{2-p}} \leq 2^{\frac{p}{2-p}}\mathrm{tr}(|T|)^{-\frac{p}{2-p}}\mathrm{tr}\big(T_+^{p}\big).$$

Similarly,

$$\mathrm{tr}\big(T_-^{\frac{p}{2}}\big)^{-\frac{2p-2}{2-p}} \leq 2^{\frac{p}{2-p}}\mathrm{tr}(|T|)^{-\frac{p}{2-p}}\mathrm{tr}\big(T_-^{p}\big).$$

Adding the two inequalities together yields (4.18), which completes the proof. □

LEMMA 4.13. *For each $p \in (1,\infty) \setminus \{2\}$ and self-adjoint $T \in L_p(\mathbb{M}_{2^n})$ with $\mathrm{tr}(T) = 0$,*

$$\big(1+\kappa(p)^2\big)\big(\mathrm{tr}(T_+^{\frac{p}{2}}) - \mathrm{tr}(T_-^{\frac{p}{2}})\big)^2 \leq \mathrm{tr}(|T|^p).$$

PROOF. Lemma 4.11 applies only to two positive real numbers $a = \mathrm{tr}\big(T_+^{\frac{p}{2}}\big)$ and $b = \mathrm{tr}\big(T_-^{\frac{p}{2}}\big)$. Combining this with Lemma 4.12, we complete the proof. □

Before proving Theorem 4.3, we note that for self-adjoint T, $\phi_q(T) = u_T|T|^{q-1}$, where ϕ_q is as in (4.8), $q \geq 1$, and u_T is a partial isometry coming from the polar decomposition $T = u_T|T|$.

PROOF OF THEOREM 4.3. Denote for simplicity that

$$\mathcal{C}_0(M,\epsilon) = \{T \in \mathcal{C}(M,\epsilon) : \mathrm{tr}(T) = 0\},$$

and

$$L_{p,0}(\mathbb{M}_{2^n}) = \{T \in L_p(\mathbb{M}_{2^n}) : \mathrm{tr}(T) = 0\}.$$

It is clear that

$$\bigcup_{M,\varepsilon>0} \mathcal{C}_0(M,\varepsilon) = L_{p,0}(\mathbb{M}_{2^n}).$$

Indeed, for each $T \in L_{p,0}(\mathbb{M}_{2^n})$, we have $T = e^{-\epsilon\mathcal{L}}\left(e^{\epsilon\mathcal{L}}(T)\right)$ with

$$\left\|e^{\epsilon\mathcal{L}}(T)\right\|_{L_\infty} \leq 2^{n/2}\|e^{\epsilon\mathcal{L}}(T)\|_{L_2} \leq 2^{n/2}e^{\epsilon n}\|T\|_{L_2} \leq 2^n e^{\varepsilon n}\|T\|_{L_\infty}.$$

For each $T \in \mathcal{C}_0(M,\epsilon)$ and $M,\epsilon>0$, it follows from Definition 4.9 and Lemma 4.7 that

$$\begin{aligned}\|e^{-t\mathcal{L}}(T)\|_{L_p}^p &\le \alpha_{p,M,\epsilon}(t)+p\mathrm{tr}(\phi_{p-1}(T)\cdot e^{-t\mathcal{L}}(T))-(p-1)\|T\|_{L_p}^p\\ &\le \alpha_{p,M,\epsilon}(t)+\frac{1}{p}\left[(p-2)^2\|T\|_{L_p}^p+4(p-1)\mathrm{tr}(\phi_{\frac{p}{2}}(T)\cdot e^{-t\mathcal{L}}(\phi_{\frac{p}{2}}(T)))\right]\\ &\quad-(p-1)\|T\|_{L_p}^p.\end{aligned}$$

By the exponential ergodicity of $\{e^{-t\mathcal{L}}\}_{t\ge0}$, we have

$$\begin{aligned}\mathrm{tr}(\phi_{\frac{p}{2}}(T)e^{-t\mathcal{L}}(\phi_{\frac{p}{2}}(T)))-(\mathrm{tr}(\phi_{\frac{p}{2}}(T)))^2&=\mathrm{Var}\left(e^{-t\mathcal{L}/2}(\phi_{\frac{p}{2}}(T))\right)\\ &\le e^{-t}\mathrm{Var}(\phi_{\frac{p}{2}}(T))=e^{-t}\left(\|T\|_{L_p}^p-\mathrm{tr}(\phi_{\frac{p}{2}}(T))^2\right).\end{aligned}$$

By Lemma 4.13, it follows that

$$\mathrm{tr}(\phi_{\frac{p}{2}}(T))^2=\left(\mathrm{tr}(T_+^{\frac{p}{2}})-\mathrm{tr}(T_-^{\frac{p}{2}})\right)^2\le\frac{1}{\left(1+\kappa(p)^2\right)}\mathrm{tr}(|T|^p)$$

Combining the above estimates, we have

$$\begin{aligned}\|e^{-t\mathcal{L}}(T)\|_{L_p}^p &\le \alpha_{p,M,\epsilon}(t)+\frac{1}{p}\big[\left((p-2)^2+4(p-1)e^{-t}\right)\|T\|_{L_p}^p\\ &\quad+4(p-1)(1-e^{-t})(\mathrm{tr}(\phi_{\frac{p}{2}}(T)))^2\big]-(p-1)\|T\|_{L_p}^p\\ &\le \alpha_{p,M,\epsilon}(t)+\Big[\frac{(p-2)^2+4(p-1)e^{-t}}{p}+\frac{4(p-1)(1-e^{-t})}{(1+\kappa(p)^2)p}-(p-1)\Big]\|T\|_{L_p}^p\\ &= \alpha_{p,M,\epsilon}(t)+\Big[1-\frac{4(p-1)(1-e^{-t})}{p}+\frac{4(p-1)(1-e^{-t})}{(1+\kappa(p)^2)p}\Big]\|T\|_{L_p}^p\\ &= \alpha_{p,M,\epsilon}(t)+\Big[1-\frac{4(p-1)(1-e^{-t})}{(1+\kappa(p)^{-2})p}\Big]\|T\|_{L_p}^p.\end{aligned}$$

For arbitrary $m\in\mathbb{N}$ and $k\in\mathbb{N}$, standard induction argument gives

$$\left\|e^{-kt\mathcal{L}/m}(T)\right\|_{L_p}^p\le k\alpha_{p,M,\epsilon}\left(\frac{t}{m}\right)+\left(1-\frac{4(p-1)(1-e^{-\frac{t}{m}})}{(1+\kappa(p)^{-2})p}\right)^k\|T\|_{L_p}^p. \tag{4.23}$$

It follows from Lemma 4.10 that

$$m\alpha_{p,M,\varepsilon}(t/m)\to0,\quad\text{as } m\to\infty.$$

Also note that

$$\lim_{m\to\infty}\left(1-\frac{4(p-1)(1-e^{-\frac{t}{m}})}{(1+\kappa(p)^{-2})p}\right)^m=\exp\left(-\frac{(4p-4)t}{p(1+\kappa(p)^{-2})}\right)$$

for some constant $c>0$. The desired inequality follows by taking $k=m$ and $m\to\infty$ in (4.23). □

Remark 4.14. In the classical treatments of establishing the Heilman–Mossel–Oleszkiewicz inequality [**HMO17**], the proof of Lemmas 4.7 and 4.10 relies on pointwise estimates for the heat semigroup. However, such pointwise arguments are not available in the quantum Boolean cube setting. To overcome this difficulty, we applied the probabilistic representation of the noncommutative semigroups (see Proposition 4.5) to prove the desired results.

We now provide a simpler, although weaker, contraction estimate. It will be useful when only an explicit dimension-free contraction factor is needed. The proof presented below is motivated by [**HMO17**, Proposition 4.1].

THEOREM 4.15. *For each $1 < p < \infty$, let $T \in L_p(\mathbb{M}_{2^n})$ with $\mathrm{tr}(T) = 0$. Then*

$$\|e^{-t\mathcal{L}}(T)\|_{L_p} \leq \left(1 - 2^{2-p^*}(1 - e^{-2t})\right)^{\frac{1}{p^*}} \|T\|_{L_p},$$

where $p^* = \max\{p, p'\}$.

PROOF. Let $c > 0$ be the constant that will be specified later. We define a positive measure on the two-point set $\{0, 1\}$ as follows

$$\mu_c(\{0\}) = 1 \quad \text{and} \quad \mu_c(\{1\}) = c^2(1 - e^{-2t}).$$

Let $\mathcal{N} \coloneqq L_\infty(\{0,1\}, \mu_c) \bar{\otimes} \mathbb{M}_{2^n}$ be the tensor product of two von Neumann algebras, equipped with the trace $\tau \coloneqq \mu_c \otimes \mathrm{tr}$. Let $F : L_1(\mathbb{M}_{2^n}) \to L_1(\mathcal{N})$ be the linear operator defined by

$$F(T)(0) \coloneqq ce^{-t\mathcal{L}}(T) \quad \text{and} \quad F(T)(1) \coloneqq T - \mathrm{tr}(T),$$

for each $T \in L_1(\mathbb{M}_{2^n})$. We now verify the boundedness of the operator F as follows. By the contraction of the semigroup, it follows that

$$\begin{aligned}
\|F(T)\|_{L_1(\mathcal{N})} &= \|F(T)(0)\|_{L_1(\mathbb{M}_{2^n})} + c^2(1 - e^{-2t})\|F(T)(1)\|_{L_1(\mathbb{M}_{2^n})} \\
&= c\|e^{-t\mathcal{L}}(T)\|_{L_1(\mathbb{M}_{2^n})} + c^2(1 - e^{-2t})\|T - \mathrm{tr}(T)\|_{L_1(\mathbb{M}_{2^n})} \\
&\leq c\|T\|_{L_1(\mathbb{M}_{2^n})} + 2c^2(1 - e^{-2t}) \cdot \|T\|_{L_1(\mathbb{M}_{2^n})}.
\end{aligned}$$

Hence, we have

$$\|F\|_{L_1(\mathbb{M}_{2^n}) \to L_1(\mathcal{N})} \leq c + 2c^2(1 - e^{-2t}). \tag{4.24}$$

Given $T \in L_2(\mathbb{M}_{2^n})$, set $T_0 \coloneqq T - \mathrm{tr}(T)$. Then

$$\begin{aligned}
\|F(T)\|^2_{L_2(\mathcal{N})} &= c^2\|e^{-t\mathcal{L}}(T)\|^2_{L_2(\mathbb{M}_{2^n})} + c^2(1 - e^{-2t}) \cdot \|T_0\|^2_{L_2(\mathbb{M}_{2^n})} \\
&= c^2\|e^{-t\mathcal{L}}(T_0)\|^2_{L_2} + c^2|\mathrm{tr}(T)|^2 + c^2(1 - e^{-2t})\|T_0\|^2_{L_2} \\
&\leq c^2e^{-2t}\|T_0\|^2_{L_2} + c^2|\mathrm{tr}(T)|^2 + c^2(1 - e^{-2t})\|T_0\|^2_{L_2} \\
&= c^2\left(|\mathrm{tr}(T)|^2 + \|T_0\|^2_{L_2}\right) = c^2\|T\|^2_{L_2},
\end{aligned}$$

where we used the exponential ergodic of the semigroup in the inequality. Thus,

$$\|F\|_{L_2(\mathbb{M}_{2^n}) \to L_2(\mathcal{N})} \leq c. \tag{4.25}$$

Moreover, it is clear that

$$\begin{aligned}
\|F(T)\|_{L_\infty(\mathcal{N})} &= \max\{c\|e^{-t\mathcal{L}}(T)\|_{L_\infty(\mathbb{M}_{2^n})}, \|T - \mathrm{tr}(T)\|_{L_\infty(\mathbb{M}_{2^n})}\} \\
&\leq \max\{c, 2\}\|T\|_{L_\infty(\mathbb{M}_{2^n})}.
\end{aligned}$$

Hence,

$$\|F\|_{L_\infty(\mathbb{M}_{2^n}) \to L_\infty(\mathcal{N})} \leq \max\{c, 2\}. \tag{4.26}$$

We now apply the Riesz-Thorin interpolation theorem to conclude the proof. For the case $1 < p \leq 2$, applying the Riesz-Thorin interpolation theorem to (4.24) and (4.25) yields

$$\|F\|_{L_p(\mathbb{M}_{2^n}) \to L_p(\mathcal{N})} \leq \left(c + 2c^2(1 - e^{-2t})\right)^{\frac{2-p}{p}} \cdot c^{\frac{2p-2}{p}} = c\left(1 + 2c(1 - e^{-2t})\right)^{\frac{2-p}{p}}.$$

Putting $c = \left(2^{\frac{1}{p-1}} - 2(1-e^{-2t})\right)^{-1}$, it follows that

$$\begin{aligned}\|e^{-t\mathcal{L}}(T)\|_{L_p} &\leq \left(\left(1+2c(1-e^{-2t})\right)^{2-p} - c^{2-p}(1-e^{-2t})\right)^{\frac{1}{p}}\|T\|_{L_p}\\ &= \left(\frac{c^{1-p}}{2}\right)^{\frac{1}{p}}\|T\|_{L_p}\\ &= \left(\frac{(2^{\frac{1}{p-1}} - 2(1-e^{-2t}))^{\frac{1}{p'}}}{2^{\frac{1}{p}}}\right)\|T\|_{L_p}\\ &= (1-2^{2-p'}(1-e^{-2t}))^{\frac{1}{p'}}\|T\|_{L_p}.\end{aligned}$$

For the case $2 \leq p < \infty$, we put $c = 2$. Then, combining (4.25), (4.26) with interpolation theorem, we have $\|F\|_{L_p \to L_p(\mathcal{N})} \leq 2$. Therefore, for each $T \in L_p(\mathbb{M}_{2^n})$ with $\mathrm{tr}(T) = 0$, it follows from the definition of F that

$$\|e^{-t\mathcal{L}}(T)\|_{L_p} \leq (1-2^{2-p}(1-e^{-2t}))^{\frac{1}{p}}\|T\|_{L_p},$$

which completes our proof. □

4.3. A reverse Bernstein-Markov type inequality

The previous results control the decay property of observables under the heat semigroup. We now convert this heat control into reverse Bernstein–Markov inequalities for observables whose spectrum lies in the band $[d, d+m]$. Such estimates are the high-frequency counterpart of the low-degree Bernstein–Markov inequalities proved in Chapter 5.

THEOREM 4.16. *Let $1 < p < \infty$, and let $n, d, m \in \mathbb{N}$ satisfy $d + m \leq n$. For every $T \in \mathcal{P}^{\leq d+m}(\mathbb{M}_{2^n}) \cap \mathcal{T}^{\geq d}(\mathbb{M}_{2^n})$, there exists $c_p \in (0,\infty)$ such that*

$$\|\mathcal{L}^{1/2}(T)\|_{L_p} \geq c_p\sqrt{\frac{d}{m}}\|T\|_{L_p}.$$

THEOREM 4.17. *Let $1 < p < \infty$, and let $n, d, m \in \mathbb{N}$ satisfy $d + m \leq n$. For every $T \in \mathcal{P}^{\leq d+m}(\mathbb{M}_{2^n}) \cap \mathcal{T}^{\geq d}(\mathbb{M}_{2^n})$, there exists $c_p \in (0,\infty)$ such that*

$$\|\mathcal{L}(T)\|_{L_p} \geq c_p\frac{d}{m}\|T\|_{L_p}.$$

We now derive Theorem 4.17 from Theorem 4.16 as follows.

PROOF OF THEOREM 4.17. By Theorem 4.16 twice, we have

$$\begin{aligned}\|\mathcal{L}(T)\|_{L_p} &= \|\mathcal{L}^{1/2}\mathcal{L}^{1/2}(T)\|_{L_p}\\ &\geq c_p\sqrt{\frac{d}{m}}\|\mathcal{L}^{1/2}(T)\|_{L_p} \geq c_p^2\frac{d}{m}\|T\|_{L_p}.\end{aligned}$$

□

To prove Theorem 4.16, we shall use the following standard fractional-power estimate for generators of contraction semigroups which can be found from [**ELP08b**].

LEMMA 4.18 ([**ELP08b**, Lemma 5.6]). *Let E be a Banach space, $\{T_t\}_{t\geq 0}$ be a strongly continuous semigroup of contractions of E, and let A be its infinitesimal generator. Then for $0 < \beta < 1$ and every f in the domain of A*

$$\|A^{\beta}(f)\|_E \leq 4\|f\|_E^{1-\beta}\|A(f)\|_E^{\beta}.$$

PROPOSITION 4.19. *Let $1 < p < \infty$. For every $T \in \mathbb{M}_{2^n}$, there exists $K_p \in (0, \infty)$ such that*

$$\|\mathcal{L}^{1/2} e^{-t\mathcal{L}}(T)\|_{L_p} \leq \frac{K_p}{\sqrt{e^{2t}-1}} \|T\|_{L_p}, \quad \forall t > 0.$$

PROOF. For $1 < p < \infty$, we take $\theta = \theta_p \in (0, \frac{\pi}{2}]$ as in Theorem 3.1. Set $a = \frac{\pi}{\tan(\theta)} > 0$. We divide the proof into following two cases.

Case I: For $t \geq 2a$, we apply Lemma 3.6 to obtain the following estimate

$$\begin{aligned}
\|\mathcal{L}^{1/2} e^{-t\mathcal{L}}(T)\|_{L_p} &= \left\| \sum_{k=1}^{n} \sqrt{k} e^{-tk} \mathrm{Rad}^{=k}(T) \right\|_{L_p} \\
&\leq \sum_{k=1}^{n} \sqrt{k} e^{-tk} \|\mathrm{Rad}^{=k}(T)\|_{L_p} \\
&\leq M_p \sum_{k=1}^{n} \sqrt{k} e^{-(t-a)k} \|T\|_{L_p} \\
&\leq \left(M_p e^{2a} \sum_{k=1}^{\infty} \sqrt{k} e^{-ak} \right) e^{-t} \|T\|_{L_p},
\end{aligned}$$

where the last inequality follows from the assumption $t \geq 2a$. Indeed, for every $k \geq 1$ we have

$$e^{-(t-a)k} = e^{-ak} e^{-(t-2a)k} \leq e^{-ak} e^{-(t-2a)} = e^{2a} e^{-t} e^{-ak}.$$

Moreover, $e^{-t} \leq \frac{1}{\sqrt{e^{2t}-1}}$ for $t > 0$. Consequently,

$$\|\mathcal{L}^{1/2} e^{-t\mathcal{L}}(T)\|_{L_p} \leq \frac{K_p}{\sqrt{e^{2t}-1}} \|T\|_{L_p},$$

where $K_p = M_p e^{2a} \sum_{k=1}^{\infty} \sqrt{k} e^{-ak}$.

Case II: For $0 < t < 2a$, let $r = t \sin\theta$. Notice that the closed disc $\overline{D}(t, r)$ of radius r centered at t is contained in $\{z \in \mathbb{C} : |\arg z| \leq \theta\}$. By the Cauchy integral formula for the derivative, we have

$$-\mathcal{L} e^{-t\mathcal{L}} = \frac{1}{2\pi i} \int_{\partial D(t,r)} \frac{e^{-\zeta\mathcal{L}}}{(\zeta - t)^2} \mathrm{d}\zeta,$$

Hence,

$$\begin{aligned}
\|\mathcal{L} e^{-t\mathcal{L}}(T)\|_{L_p} &= \left\| \frac{1}{2\pi i} \int_{\partial D(t,r)} \frac{e^{-\zeta\mathcal{L}} T}{(\zeta - t)^2} \, \mathrm{d}\zeta \right\|_{L_p} \\
&\leq \left| \frac{1}{2\pi i} \int_{\partial D(t,r)} \frac{1}{(\zeta - t)^2} \, \mathrm{d}\zeta \right| \sup_{\zeta \in \partial D(t,r)} \|e^{-\zeta\mathcal{L}}(T)\|_{L_p} \\
&\leq \frac{1}{r} M_p \|T\|_{L_p} \\
&= \frac{M_p}{t \sin\theta} \|T\|_{L_p},
\end{aligned}$$

where the second inequality is due to Theorem 3.1. Applying Lemma 4.18 (with $f = e^{-t\mathcal{L}}(T)$ and $\beta = \frac{1}{2}$ there) and Lemma 2.1 (with $p = q$), we have

$$\|\mathcal{L}^{1/2}e^{-t\mathcal{L}}(T)\|_{L_p} \leq 4\|\mathcal{L}e^{-t\mathcal{L}}(T)\|_{L_p}^{\frac{1}{2}}\|e^{-t\mathcal{L}}(T)\|_{L_p}^{\frac{1}{2}} \leq \frac{4\sqrt{M_p}}{\sqrt{t\sin\theta}}\|T\|_{L_p}.$$

Since for all $t \in (0, 2a)$, we have $\sqrt{e^{2t}-1} \leq c_a\sqrt{t}$, where the constant $c_a > 0$ depends only on p, because $a = \pi/\tan(\theta_p)$, it gives the desired estimate. □

We shall also use the following reverse Markov inequality in the proof of Theorem 4.16.

LEMMA 4.20 ([**Erd20**, Theorem 2.2]). *Let $d, m \in \mathbb{N}$, and let $P(x)$ be a polynomial of the form*

$$P(x) = a_d x^d + a_{d+1}x^{d+1} + \cdots + a_{d+m}x^{d+m}.$$

Then, the following estimate holds

$$|P(1)| \leq 6\sqrt{\frac{m}{d}}\left\|\sqrt{1-x^2}P'(x)\right\|_{\mathcal{C}([0,1])}.$$

The order of this estimate is sharp up to the universal constant.

PROOF OF THEOREM 4.16. Let $x = e^{-t} \in [0, 1]$. Rewrite Proposition 4.19 as follows

(4.27)
$$\left\|\sqrt{1-x^2}\mathcal{L}x^{\mathcal{L}-1}(T)\right\|_{L_p} = \left\|\sqrt{x^{-2}-1}\mathcal{L}^{1/2}x^{\mathcal{L}}(\mathcal{L}^{1/2}T)\right\|_{L_p} \leq K_p\|\mathcal{L}^{1/2}(T)\|_{L_p}.$$

Consider the linear functional

$$\xi : \left(\operatorname{span}\left\{\sqrt{1-x^2}x^{d-1}, \ldots, \sqrt{1-x^2}x^{d+m-1}\right\}, \|\cdot\|_{\mathcal{C}([0,1])}\right) \to \mathbb{C}$$

given by

$$\xi\left(\sqrt{1-x^2}\sum_{k=d}^{d+m} ka_k x^{k-1}\right) \coloneqq \sum_{k=d}^{d+m} a_k = P(1),$$

where $P(x) = \sum_{k=d}^{d+m} a_k x^k$ is the polynomial defined as in Lemma 4.20. Thus, by lemma 4.20, it follows that for each

$$Q(x) = \sqrt{1-x^2}P'(x) = \sqrt{1-x^2}\sum_{k=d}^{d+m} ka_k x^{k-1} \in \operatorname{span}\left\{\sqrt{1-x^2}x^{d-1}, \ldots, \sqrt{1-x^2}x^{d+m-1}\right\},$$

we have

$$|\xi(Q)| = |P(1)| \leq 6\sqrt{\frac{m}{d}}\|\sqrt{1-x^2}P'(x)\|_{\mathcal{C}([0,1])} = 6\sqrt{\frac{m}{d}}\|Q\|_{\mathcal{C}([0,1])}.$$

Therefore, by the Hahn-Banach theorem and the Riesz representation theorem, there exists a complex measure σ on $[0, 1]$ such that

$$\|\sigma\|_{\mathcal{M}([0,1])} \leq 6\sqrt{\frac{m}{d}}$$

and

$$\int_0^1 Q(x)\,\mathrm{d}\,\sigma(x) = \xi(Q), \quad \forall Q \in \operatorname{span}\left\{\sqrt{1-x^2}x^{d-1}, \ldots, \sqrt{1-x^2}x^{d+m-1}\right\}.$$

Since $\widehat{T}(\mathbf{s}) = 0$ when $|\mathbf{s}| \notin \{d, \dots, d+m\}$, by the above equation, Jensen inequality and (4.27), we obtain that

$$\begin{aligned}
\|T\|_{L_p} &= \left\| \sum_{\substack{\mathbf{s}\in\{0,1,2,3\}^n \\ d\le|\mathbf{s}|\le d+m}} \widehat{T}(\mathbf{s})\sigma_{\mathbf{s}} \right\|_{L_p} \\
&= \left\| \int_0^1 \sqrt{1-x^2}\mathcal{L}x^{\mathcal{L}-1}(T)\ \mathrm{d}\sigma(x) \right\|_{L_p} \\
&\le \int_0^1 \left\| \sqrt{1-x^2}\mathcal{L}x^{\mathcal{L}-1}(T) \right\|_{L_p} \mathrm{d}|\sigma|(x) \\
&\le K_p \int_0^1 \left\| \mathcal{L}^{1/2}(T) \right\|_{L_p} \mathrm{d}|\sigma|(x) \\
&= K_p \|\sigma\|_{\mathcal{M}([0,1])} \left\| \mathcal{L}^{1/2}(T) \right\|_{L_p} \le 6K_p\sqrt{\frac{m}{d}} \left\| \mathcal{L}^{1/2}(T) \right\|_{L_p}.
\end{aligned}$$

□

CHAPTER 5

Bernstein-Markov type inequalities for low degree observables

In this chapter, the observables of interest have a degree of at most $d \in [n]$, which shifts our focus from the preceding chapter. We first investigate a lower bound on the decay of the quantum heat semigroup, which may be viewed as the "dual" formulation of Theorem 4.1. Furthermore, we establish Bernstein-Markov-type inequalities for low-degree quantum functions with respect to both the quantum Laplacian and its associated gradient.

5.1. Lower bound on the decay of the quantum heat semigroup

In the classical setting, the study of lower bounds on the decay of the heat semigroup acting on functions of low degree goes back at least to [**FHKL16**]. Later, Eskenazis and Ivanisvili [**EI20b**] extended it to the vector-valued setting and found interesting applications to moment comparison for functions of low degree and Pisier's remarkable inequality (1.12). The main results here, namely Theorems 5.1 and 5.2, serve as the quantum counterparts of theirs. The Chebyshev polynomial was defined above (see the explanation after Theorem 4.1).

THEOREM 5.1. *For every $d \in [n]$, $1 \le p \le \infty$, and $T \in \mathcal{P}^{\le d}(\mathbb{M}_{2^n})$, we have*

$$\|e^{-t\mathcal{L}}(T)\|_{L_p} \ge \frac{1}{T_d(e^t)}\|T\|_{L_p}, \quad \forall t \ge 0, \tag{5.1}$$

where T_d is the d-th Chebyshev polynomial of the first kind.

PROOF. Let $1 \le p \le \infty$. The contractivity of the semigroup can be expressed as:

$$\left\|x^{\mathcal{L}}(T)\right\|_{L_p} \le \|T\|_{L_p}, \quad \forall x \in [0,1], T \in \mathbb{M}_{2^n}.$$

As for $x \in [-1,0]$, we have

$$\begin{aligned}
\left\|(-x)^{\mathcal{L}}(T)\right\|_{L_p} &= \left\|\sum_{\mathbf{s}\in\{0,1,2,3\}^n} (-x)^{|\mathbf{s}|}\widehat{T}(\mathbf{s})\sigma_{\mathbf{s}}\right\|_{L_p} \\
&= \left\|\sum_{\mathbf{s}\in\{0,1,2,3\}^n} x^{|\mathbf{s}|}\widehat{T}(\mathbf{s})\Phi_\rho(\sigma_{\mathbf{s}})\right\|_{L_p} \\
&= \left\|x^{\mathcal{L}}\Phi_\rho(T)\right\|_{L_p} \\
&\le \left\|\Phi_\rho(T)\right\|_{L_p} = \|T\|_{L_p},
\end{aligned}$$

where we used (2.8) and Lemma 2.4 (iii). Consequently, we conclude from the above argument that

$$\left\|x^{\mathcal{L}}(T)\right\|_{L_p} \leq \|T\|_{L_p}, \quad \forall x \in [-1,1]. \tag{5.2}$$

Let μ be any complex measure on $[-1,1]$. Then, averaging over (5.2), and applying the Jensen inequality, we get

$$\begin{aligned}\left\|\int_{-1}^{1} x^{\mathcal{L}}(T)d\mu(x)\right\|_{L_p} &\leq \int_{-1}^{1} \|x^{\mathcal{L}}(T)\|_{L_p}\mathrm{d}|\mu|(x)\\ &\overset{(5.2)}{\leq} \int_{-1}^{1} \|T\|_{L_p}\mathrm{d}|\mu|(x)\\ &= \|\mu\|_{V([-1,1])}\|T\|_{L_p},\end{aligned} \tag{5.3}$$

where $\|\mu\|_{V([-1,1])}$ denotes the total variation of μ. Consider the linear functional

$$\varphi_t : (\mathcal{P}^{\leq d}[-1,1], \|\cdot\|_{\mathcal{C}([-1,1])}) \to \mathbb{C}, \quad \text{for every } t \geq 0$$

given by

$$\varphi_t(P) = P(e^t), \quad \forall P \in \mathcal{P}^{\leq d}[-1,1].$$

Then, Chebyshev's inequality (4.2) can be rewritten as

$$|\varphi_t(P)| = |P(e^t)| \leq T_d(e^t)\|P\|_{\mathcal{C}([-1,1])}, \quad \forall P \in \mathcal{P}^{\leq d}[-1,1].$$

Therefore, by the Hahn-Banach theorem and the Riesz representation theorem, there exists a complex measure μ_t on $[-1,1]$ such that

$$\|\mu_t\|_{V([-1,1])} = \|\varphi_t\| \leq T_d(e^t), \tag{5.4}$$

and

$$\int_{-1}^{1} P(x)\mathrm{d}\mu_t(x) = \varphi_t(P) = P(e^t), \quad \forall P \in \mathcal{P}^{\leq d}[-1,1].$$

Since $\widehat{T}(\mathbf{s}) = 0$ when $|\mathbf{s}| > d$, we deduce that for all $t \geq 0$,

$$\begin{aligned}\|e^{t\mathcal{L}}(T)\|_{L_p} &= \left\|\sum_{\substack{\mathbf{s}\in\{0,1,2,3\}^n\\ |\mathbf{s}|\leq d}} e^{t|\mathbf{s}|}\widehat{T}(\mathbf{s})\sigma_{\mathbf{s}}\right\|_{L_p}\\ &\overset{(5.1)}{=} \left\|\int_{-1}^{1}\sum_{\substack{\mathbf{s}\in\{0,1,2,3\}^n\\ |\mathbf{s}|\leq d}} x^{|\mathbf{s}|}\widehat{T}(\mathbf{s})\sigma_{\mathbf{s}}d\mu_t(x)\right\|_{L_p}\\ &= \left\|\int_{-1}^{1} x^{\mathcal{L}}(T)\ \mathrm{d}\mu_t(x)\right\|_{L_p}\\ &\overset{(5.3),(5.4)}{\leq} T_d(e^t)\|T\|_{L_p}.\end{aligned}$$

We can get the desired estimate (5.1) by taking $T = e^{-t\mathcal{L}}(T)$. □

We should note that even though Theorem 5.2 can be understood as the "dual" of Theorem 4.1, it can not be deduced from Theorem 4.1 via duality argument. This is merely a formal duality relation.

We now establish an improvement of Theorem 5.1 whenever $1 < p < \infty$.

THEOREM 5.2. *Let $d \in [n]$ and $1 < p < \infty$. for every $T \in \mathcal{P}^{\leq d}(\mathbb{M}_{2^n})$, there exist $C_p, c_p \in (0,\infty)$ and $\eta_p \in (\frac{1}{2}, 1]$ such that*

$$\left\|e^{-t\mathcal{L}}(T)\right\|_{L_p} \geq C_p e^{-c_p d \min\{t, t^{\eta_p}\}} \|T\|_{L_p}, \quad \forall t \geq 0. \tag{5.5}$$

To prove this theorem, we begin by stating a fundamental result from complex analysis, which will be instrumental in proving the subsequent proposition.

LEMMA 5.3 ([**EI20b**, Lemma 29]). *Let $\Omega \subseteq \mathbb{C}$ be a bounded simply connected domain and let $\phi_{\Omega^c} : \overline{\Omega}^c \to \overline{\mathbb{D}}^c$ be a conformal mapping of the complement of $\overline{\Omega}$ onto the complement of the closed unit disc, satisfying $\phi_{\Omega^c}(\infty) = \infty$. Then*

$$\lim_{z\to\infty} \frac{\phi_{\Omega^c}(z)}{z} = \beta \in \mathbb{C}\backslash\{0\}.$$

PROPOSITION 5.4. *Let $d \in [n]$, $1 < p < \infty$, $K_p \in (0,\infty)$, and let $\Omega \subseteq \mathbb{D}$ be a simply connected domain bounded by a Jordan curve γ. Suppose that*

$$\sup_{w\in\Omega} \|w^{\mathcal{L}}\|_{L_p \to L_p} \leq K_p. \tag{5.6}$$

Also, let $\phi_{\Omega^c} : \overline{\Omega}^c \to \overline{\mathbb{D}}^c$ be a conformal mapping of the complement of $\overline{\Omega}$ onto the complement of the closed unit disc, which extends continuously as a homeomorphism between Ω^c and $\mathbb{D}^c$. Then, for $T \in \mathcal{P}^{\leq d}(\mathbb{M}_{2^n})$, we have

$$\|e^{-t\mathcal{L}}(T)\|_{L_p} \geq \frac{1}{K_p |\phi_{\Omega^c}(e^t)|^d} \|T\|_{L_p}, \quad \forall t \geq 0.$$

PROOF. Consider the linear functional

$$\varphi_t : (\mathcal{P}(\overline{\Omega}), \|\cdot\|_{\mathcal{C}(\overline{\Omega})}) \to \mathbb{C}$$

defined by

$$\varphi_t(P) = \varphi_t(\sum_{k=0}^{d} a_k w^k) \overset{\text{def}}{=} \sum_{k=0}^{d} a_k e^{tk} = P(e^t), \quad P \in \mathcal{P}^{\leq d}(\overline{\Omega}).$$

For such a polynomial $P \in \mathcal{P}^{\leq d}(\overline{\Omega})$, define the function $h : \overline{\Omega}^c \to \mathbb{C}$ by

$$h(z) = \frac{P(z)}{(\phi_{\Omega^c})^d}, \quad z \in \overline{\Omega}^c,$$

where ϕ_{Ω^c} is a conformal map that extends to a homeomorphism between Ω^c and $\mathbb{D}^c$, satisfying $\phi_{\Omega^c}(\infty) = \infty$. By Lemma 5.3, it follows that $\lim_{z\to\infty} h(z) \in \mathbb{C}$, and thus h is a holomorphic function on $\overline{\Omega}^c \cup \{\infty\}$. For $z \in \gamma$, we have $|\phi_{\Omega^c}(z)| = 1$ since ϕ_{Ω^c} is a homeomorphism between γ and $\partial\mathbb{D}$. By the maximum principle,

$$\sup_{z\in\Omega^c} |h(z)| = \max_{z\in\gamma} |h(z)| = \max_{z\in\gamma} \frac{|P(z)|}{|\phi_{\Omega^c}(z)|^d} = \|P\|_{\mathcal{C}(\gamma)} = \|P\|_{\mathcal{C}(\overline{\Omega})},$$

which implies that

$$|P(z)| \leq |\phi_{\Omega^c}(z)|^d \|P\|_{\mathcal{C}(\overline{\Omega})}, \quad z \in \Omega^c.$$

For $z = e^t \in \Omega^c$, we obtain the following estimate

$$|\varphi_t(P)| \leq |\phi_{\Omega^c}(e^t)|^d \|P\|_{\mathcal{C}(\overline{\Omega})}, \quad \forall P \in \mathcal{P}^{\leq d}(\overline{\Omega}).$$

Therefore, by the Hahn-Banach theorem and the Riesz representation theorem, there exists a complex measure ρ_t on $\overline{\Omega}$ such that $\|\rho_t\|_{V(\overline{\Omega})} \le |\phi_{\Omega^c}(e^t)|^d$ and

$$\int_{\overline{\Omega}} P(w)\mathrm{d}\rho_t(w) = P(e^t), \quad P \in \mathcal{P}^{\le d}(\overline{\Omega}). \tag{5.7}$$

Since $\widehat{T}(\mathbf{s}) = 0$ when $|\mathbf{s}| > d$, combining the above estimates, the Jensen inequality, and the given assumptions, it follows that

$$\begin{aligned}
\|e^{t\mathcal{L}}(T)\|_{L_p} &= \Big\| \sum_{\mathbf{s}\in\{0,1,2,3\}^n, |\mathbf{s}|\le d} e^{t|\mathbf{s}|}\widehat{T}(\mathbf{s})\sigma_{\mathbf{s}} \Big\|_{L_p} \\
&\overset{(5.7)}{=} \Big\| \int_{\overline{\Omega}} \sum_{\mathbf{s}\in\{0,1,2,3\}^n, |\mathbf{s}|\le d} w^{|\mathbf{s}|}\widehat{T}(\mathbf{s})\sigma_{\mathbf{s}} \, \mathrm{d}\rho_t(w) \Big\|_{L_p} \\
&= \Big\| \int_{\overline{\Omega}} w^{\mathcal{L}}(T) \, \mathrm{d}\rho_t(\omega) \Big\|_{L_p} \\
&\le \int_{\overline{\Omega}} \|w^{\mathcal{L}}(T)\|_{L_p} \mathrm{d}|\rho_t|(w) \\
&\overset{(5.6)}{\le} K_p \int_{\overline{\Omega}} \|T\|_{L_p} \mathrm{d}\,|\rho_t|\,(w) = K_p \|\rho_t\|_{V(\overline{\Omega})} \|T\|_{L_p} \\
&\le K_p |\phi_{\Omega^c}(e^t)|^d \|T\|_{L_p},
\end{aligned}$$

which is equivalent to the desired estimate by taking $T = e^{-t\mathcal{L}}(T)$. □

Now we are ready to prove Theorem 5.2.

Proof of Theorem 5.2. Let $d \in [n]$, $1 < p < \infty$, and $T \in \mathcal{P}^{\le d}(\mathbb{M}_{2^n})$. By Theorem 5.1, for every $t \ge 0$,

$$\|e^{-t\mathcal{L}}(T)\|_{L_p} \ge \frac{1}{T_d(e^t)} \|T\|_{L_p},$$

where

$$T_d(e^t) = \frac{(e^t + \sqrt{e^{2t}-1})^d + (e^t - \sqrt{e^{2t}-1})^d}{2} \le (2e^t)^d \le e^{2td}.$$

The desired assertion for $1 < t < \infty$ follows.

It remains to consider the case $0 < t \le 1$. By Lemma 3.8, there exist $r_p \in [1,\infty)$ and $K_p \in [1,\infty)$ such that

$$\sup_{w\in\Omega(r_p)} \|w^{\mathcal{L}}\|_{L_p\to L_p} \le K_p,$$

where $\Omega(r_p)$ is the domain (3.11). Therefore, by Proposition 5.4, we have

$$\|e^{-t\mathcal{L}}T\|_{L_p} \ge \frac{1}{K_p |\phi_{\Omega(r_p)^c}(e^t)|^d} \|T\|_{L_p}, \quad \forall t \ge 0. \tag{5.8}$$

Let $\theta = \theta(r_p) = 2\arcsin(1/r_p) \in (0,\pi]$. By Lemma 3.7, the conformal map $\phi_{\Omega(r_p)^c}$ is given by

$$\phi_{\Omega(r_p)^c}(e^t) = \psi_2 \circ \phi_{\frac{\pi}{2\pi-\theta}} \circ \psi_2(e^t) = \frac{(e^t+1)^{\frac{\pi}{2\pi-\theta}} + (e^t-1)^{\frac{\pi}{2\pi-\theta}}}{(e^t+1)^{\frac{\pi}{2\pi-\theta}} - (e^t-1)^{\frac{\pi}{2\pi-\theta}}}.$$

Taking the logarithm of its modulus, we obtain

$$\log\left|\phi_{\Omega(r_p)^c}\left(e^t\right)\right| = \log\left(1+\frac{2\left(e^t-1\right)^{\frac{\pi}{2\pi-\theta}}}{\left(e^t+1\right)^{\frac{\pi}{2\pi-\theta}}-\left(e^t-1\right)^{\frac{\pi}{2\pi-\theta}}}\right) \leq \frac{2\left(e^t-1\right)^{\frac{\pi}{2\pi-\theta}}}{\left(e^t+1\right)^{\frac{\pi}{2\pi-\theta}}-\left(e^t-1\right)^{\frac{\pi}{2\pi-\theta}}},$$

where the inequality is due to the fact $\log(1+x)\leq x$ for any $x\in(0,\infty)$. We further have

$$\frac{2(e^t-1)^{\frac{\pi}{2\pi-\theta}}}{(e^t+1)^{\frac{\pi}{2\pi-\theta}}-(e^t-1)^{\frac{\pi}{2\pi-\theta}}} \leq \frac{2(e^t-1)^{\frac{\pi}{2\pi-\theta}}}{(e+1)^{\frac{\pi}{2\pi-\theta}}-(e-1)^{\frac{\pi}{2\pi-\theta}}} \leq \frac{2(e-1)^{\frac{\pi}{2\pi-\theta}}}{(e+1)^{\frac{\pi}{2\pi-\theta}}-(e-1)^{\frac{\pi}{2\pi-\theta}}}t^{\frac{\pi}{2\pi-\theta}},$$

where the first inequality is because that the denominator is monotonically decreasing in t, and the second inequality is due to the fact $e^t-1\leq(e-1)t$ for $t\leq 1$. By (5.8), we obtain the desired estimate with $C=\frac{2(e-1)^{\frac{\pi}{2\pi-\theta}}}{(e+1)^{\frac{\pi}{2\pi-\theta}}-(e-1)^{\frac{\pi}{2\pi-\theta}}}$ and $\eta_p=\frac{\pi}{2\pi-\theta}\in(\frac{1}{2},1]$. The proof is complete. □

5.2. Bernstein-Markov type inequalities for Laplacian

Recall that partial answers to Conjecture 1.4 have been provided in [**EI20b**] and [**DDF$^+$25**]. In this section, we extend the corresponding Bernstein-Markov type inequalities in [**EI20b**] and [**DDF$^+$25**] to the quantum setting. Although we adopt similar ideas to [**EI20b**] and [**DDF$^+$25**], we still need to apply the newly developed tools in noncommutative analysis, such as the noncommutative Cowling theorem. By the definition of (2.4), when $p=2$, it follows directly from Parseval's identity

$$\|\mathcal{L}(T)\|_{L_2}\leq d\|T\|_{L_2}, \tag{5.9}$$

whenever $T\in\mathcal{P}^{\leq d}(\mathbb{M}_{2^n})$. The first purpose of this section is to establish a variant of (5.9) on L_p spaces when $p\neq 2$.

THEOREM 5.5. *Let $d\in[n]$ and $1<p<\infty$. For $T\in\mathcal{P}^{\leq d}(\mathbb{M}_{2^n})$, there exist $\alpha_p\in[1,2)$ and $C_p\in(0,\infty)$ such that*

$$\|\mathcal{L}(T)\|_{L_p}\leq C_p d^{\alpha_p}\|T\|_{L_p}.$$

We shall use the classical approximation theoretic result of Szegö [**Sze25**], indeed, we just apply the following Szegö theorem specifically for lens domains of the form (3.9).

LEMMA 5.6 ([**EI20a**, Proposition 15]). *Let $r\in[1,\infty)$, and let lens domain $\Omega(r)\subseteq\mathbb{C}$ (defined in (3.9)) be a domain bounded by a Jordan curve γ. The exterior angle of γ at ±1 is $2\pi-\theta(r)$ ($\theta(r)$ is given in (3.10)). Then, for every $d\in\mathbb{N}$ and every polynomial $P\in\mathcal{P}^{\leq d}(\overline{\Omega})$, we have*

$$|P'(1)|\leq 10d^{2\pi-\theta(r)/\pi}\|P\|_{\mathcal{C}(\overline{\Omega})}.$$

PROOF OF THEOREM 5.5. Let $T\in\mathcal{P}^{\leq d}(\mathbb{M}_{2^n})$. Consider the linear functional

$$\psi:(\mathcal{P}^{\leq d}(\overline{\Omega}),\|\cdot\|_{\mathcal{C}(\overline{\Omega})})\to\mathbb{C}$$

given by

$$\psi(P) \stackrel{\text{def}}{=} P'(1),$$

where

$$\Omega = \{e^{-z} : |\arg z| < \theta\} \subseteq \mathbb{D},$$

where θ is given by Theorem 3.1. Notice that Ω is a domain bounded by a Jordan curve γ with $1 \in \gamma$ which forms an exterior angle $2\pi - 2\theta$ at 1. Then, Lemma 5.6 implies that

$$|\psi(P)| \leq 10d^{2-\frac{2\theta}{\pi}} \|p\|_{\mathcal{C}(\overline{\Omega})}, \quad \forall P \in \mathcal{P}^{\leq d}(\overline{\Omega}).$$

Therefore, by the Hahn-Banach theorem and the Riesz representation theorem, there exists a complex measure v on $\overline{\Omega}$ such that

$$\|v\|_{\mathcal{M}(\overline{\Omega})} \leq 10d^{2-\frac{2\theta}{\pi}}$$

and

$$\int_{\overline{\Omega}} P(w)\mathrm{d}v(w) = P'(1), \quad P \in \mathcal{P}^{\leq d}(\overline{\Omega}). \tag{5.10}$$

Since $\widehat{T}(\mathbf{s}) = 0$ when $|\mathbf{s}| > d$, we use (5.10), Jensen inequality and Theorem 3.1 to derive

$$\begin{aligned}
\|\mathcal{L}(T)\|_{L_p} &= \Big\| \sum_{\mathbf{s}\in\{0,1,2,3\}^n, |\mathbf{s}|\leq d} |\mathbf{s}|\widehat{T}(\mathbf{s})\sigma_{\mathbf{s}} \Big\|_{L_p} \\
&\overset{(5.10)}{=} \Big\| \int_{\overline{\Omega}} \sum_{\mathbf{s}\in\{0,1,2,3\}^n, |\mathbf{s}|\leq d} w^{|\mathbf{s}|}\widehat{T}(\mathbf{s})\sigma_{\mathbf{s}} \, \mathrm{d}v(w) \Big\|_{L_p} \\
&= \Big\| \int_{\overline{\Omega}} w^{\mathcal{L}}(T) \, \mathrm{d}v(w) \Big\|_{L_p} \\
&\leq \int_{\overline{\Omega}} \|w^{\mathcal{L}}(T)\|_{L_p} \mathrm{d}|v|(w) \\
&\leq M_p \int_{\overline{\Omega}} \|T\|_{L_p} \mathrm{d}|v|(w) \\
&\leq 10M_p d^{2-\frac{2\theta}{\pi}} \|T\|_{L_p}.
\end{aligned}$$

The proof is complete. □

Next, we aim to apply the three-lines lemma to derive a bound for $\mathcal{L}^k$ when acting on low-degree polynomial. For each $p \in (1,\infty)$ and $\epsilon > 0$, we put

$$p_\epsilon \coloneqq \begin{cases} p+\epsilon, & \text{if } p \in [2,\infty); \\ p-\epsilon, & \text{if } 1 < p < 2, \ \epsilon \in (0, p-1). \end{cases}$$

THEOREM 5.7. *Let $d \in [n]$, and let $p \in (1,\infty)$, $\theta \in (0,1)$ be such that*

$$\frac{1}{p} = \frac{\theta}{p_\epsilon} + \frac{1-\theta}{2}, \quad \epsilon \in (0,\infty).$$

There exists a constant $K_{p,\epsilon} > 0$ (depending only on p and $\epsilon > 0$) such that for each $k \in \mathbb{N}$ we have

$$\big\|\mathcal{L}^k(T)\big\|_{L_p} \leq K_{p,\epsilon}^k d^k \|T\|_{L_2}^{1-\theta} \|T\|_{L_{p_\epsilon}}^{\theta}, \quad T \in \mathcal{P}^{\leq d}(\mathbb{M}_{2^n}). \tag{5.11}$$

The proof of Theorem 5.7 is a consequence of the three-lines lemma and the following noncommutative Cowling inequality regarding the bound of imaginary Riesz potential.

THEOREM 5.8 ([**JW17**]). *Suppose that Δ is the infinitesimal generator of the Markov semigroup $\{e^{-t\Delta}\}_{t\geq 0}$ acting on a noncommutative probability space. For each $1 < p < \infty$, there exists a universal constant $K > 0$ such that*

$$\left\|\Delta^{iz}\right\|_{L_p\to L_p} \leq \frac{Kp^2}{p-1}(1+|z|^3\log^2(|z|))^{|1/p-1/2|}\exp\{\pi|1/p-1/2||z|\}, \quad z\in\mathbb{R}.$$

We now proceed to prove Theorem 5.7.

PROOF OF THEOREM 5.7. For each $z\in\mathbb{C}$ and $T\in\mathbb{M}_{2^n}$, we write

$$\mathcal{L}^z(T) = \sum_{\mathbf{s}\in\{0,1,2,3\}^n} \widehat{T}(\mathbf{s})|\mathbf{s}|^z\sigma_{\mathbf{s}}.$$

Since the summation above is finite, it follows that $\mathcal{L}^z(T):\mathbb{C}\to\mathbb{M}_{2^n}$ is analytic.

In order to derive (5.11), it suffices to show that there exists a constant $K_{p,\epsilon}>0$ such that

$$\left|\langle\mathcal{L}^k(T),G\rangle\right| \leq K_{p,\epsilon}^k d^k\|T\|_{L_2}^{1-\theta}\|T\|_{L_{p_\epsilon}}^{\theta}, \tag{5.12}$$

for all $G\in L_{p'}(\mathbb{M}_{2^n})$ with $\|G\|_{L_{p'}}=1$ where p' is the conjugate index of p, i.e., $\frac{1}{p}+\frac{1}{p'}=1$. We now apply the three-lines lemma to prove (5.12). For any $G\in L_{p'}(\mathbb{M}_{2^n})$ with $\|G\|_{L_{p'}}=1$, we define $\psi: S:=\{z\in\mathbb{C}: 0\leq\Re(z)\leq 1\}\to\mathbb{C}$ as follows

$$\psi(z) := \exp\left\{z^2\pi k|1/p_\epsilon-1/2|/(1-\theta)\right\}\left\langle\mathcal{L}^{(1-z)k/(1-\theta)}(T),G_z^*\right\rangle,$$

where $G_z := u(g)|G|^{\frac{p'(1-z)}{2}+\frac{p'z}{p'_\epsilon}}$ and $u(G)$ is the partial isometry coming from the polar decomposition $G=u(G)|G|$. Following from the spectral decomposition of $|G|$, it is clear that G_z is analytic, and hence ψ is an analytic function on the open strip $\{z\in\mathbb{C}:0<\Re(z)<1\}$ and continuous on closed unit strip S. It is clear that $G_\theta=G$. Moreover, notice that $u(G)^*u(G)$ is an orthogonal projection onto range$(|G|)$, it follows that

$$|G_z| = \begin{cases} |G|^{\frac{p'}{2}}, & \text{if } \Re(z)=0,\\ |G|^{\frac{p'}{p'_\epsilon}}, & \Re(z)=1. \end{cases} \tag{5.13}$$

We first consider the case $z\in S$ with $\Re(z)=0$. In this case, we note that $\|G_z\|_2=1$ and $\|\mathcal{L}^{-\frac{zk}{1-\theta}}\|=1$ for each $z\in S$. By the Cauchy-Schwarz inequality, we have

$$\begin{aligned} |\psi(z)| &= \left|\exp\left\{z^2\pi k|1/p_\epsilon-1/2|/(1-\theta)\right\}\left\langle\mathcal{L}^{(1-z)k/(1-\theta)}(T),G_z^*\right\rangle\right| \\ &= \left|\left\langle\mathcal{L}^{-\frac{zk}{1-\theta}}\left(\mathcal{L}^{\frac{k}{1-\theta}}(T)\right),G_z^*\right\rangle\right| \\ &\leq \left\|\mathcal{L}^{\frac{k}{1-\theta}}(T)\right\|_{L_2} \leq d^{\frac{k}{1-\theta}}\|T\|_{L_2}, \end{aligned} \tag{5.14}$$

where we used (5.9) in the last line.

We now turn to consider the case $z \in S$ with $\Re(z) = 1$. Let $z = 1 + iv$ with $v \in \mathbb{R}$. Applying Theorem 5.8, we get

$$\begin{aligned} |\psi(z)| &= \left|\exp\left\{z^2\pi k|1/p_\epsilon - 1/2|/(1-\theta)\right\}\left\langle \mathcal{L}^{(1-z)k/(1-\theta)}(T), G_z^*\right\rangle\right| \\ &\leq \exp\left\{(1-v^2)\pi k|1/p_\epsilon - 1/2|/(1-\theta)\right\}\left\|\mathcal{L}^{ikv/(1-\theta)}(T)\right\|_{L_{p_\epsilon}} \\ &\leq K_{p_\epsilon}\left(1+\left(\frac{k|v|}{1-\theta}\right)^3\log^2\left(\frac{k|v|}{1-\theta}\right)\right)^{|1/p_\epsilon-1/2|} \\ &\quad\times\exp\left\{(1+|v|-v^2)\pi k|1/p_\epsilon-1/2|/(1-\theta)\right\}\|T\|_{L_{p_\epsilon}}. \end{aligned} \tag{5.15}$$

Here we used the Hölder inequality and the fact $\|G_z\|_{p'_\epsilon} = 1$. Note that $1 + t^3\log^2(t) \leq 6e^t$ for all $t \geq 0$. Therefore,

$$\begin{aligned} |\psi(z)| &\leq K_{p_\epsilon}\exp\left\{\frac{\pi k|1/p_\epsilon - 1/2|\left(1+(1+\frac{1}{\pi})|v|-v^2\right)}{1-\theta}\right\}\|T\|_{L_{p_\varepsilon}} \\ &\leq K_{p_\epsilon}\exp\left\{\frac{\pi k|1/p_\epsilon - 1/2|\left(\frac{4\pi^2+(1+\pi)^2}{4\pi^2}\right)}{1-\theta}\right\}\|T\|_{L_{p_\epsilon}}. \end{aligned} \tag{5.16}$$

Combining (5.14) and (5.16) with the three-lines lemma [**Gra14**, Lemma 1.3.5], we get that

$$|\psi(\theta)| \leq K_{p,\epsilon}d^k\exp\left\{\frac{\pi\theta k|1/p_\epsilon - 1/2|\left(\frac{4\pi^2+(1+\pi)^2}{4\pi^2}\right)}{1-\theta}\right\}\|T\|_{L_2}^{1-\theta}\|T\|_{L_{p_\epsilon}}^{\theta},$$

where $\theta = \frac{(2-p)p_\epsilon}{(2-p_\epsilon)p}$. The proof is complete. □

COROLLARY 5.9. *Let $d \in [n]$, $p \in (1,\infty)$ and $\epsilon \in (0,\infty)$. There exists a constant $K_{p,\epsilon} > 0$ (depending only on p and $\epsilon > 0$) such that*

$$\|\mathcal{L}(T)\|_{L_p} \leq K_{p,\epsilon}d\|T\|_{L_{p+\epsilon}}, \quad T \in \mathcal{P}^{\leq d}(\mathbb{M}_{2^n}).$$

PROOF. **Case 1:** $p \in [2,\infty)$. The assertion follows from Theorem 5.7 with $k = 1$ and the fact $\|T\|_{L_2} \leq \|T\|_{L_{p+\epsilon}}$.

Case 2: $p \in (1,2)$. By duality, we have

$$\begin{aligned} \|\mathcal{L}(T)\|_{L_p} &= \sup_{\|G\|_{L_{p'}}\leq 1}|\langle\mathcal{L}(T),G\rangle| \\ &= \sup_{\|G\|_{L_{p'}}\leq 1}|\langle T,\mathcal{L}(G)\rangle| \\ &\leq \|T\|_{L_{p+\epsilon}}\|\mathcal{L}(G)\|_{L_{(p+\epsilon)'}}. \end{aligned} \tag{5.17}$$

One can choose $\epsilon > 0$ small enough so that $p < p+\epsilon < 2$ and $(p+\epsilon)' + \epsilon < p'$. Applying the result of **Case 1**, to (5.17), it follows that

$$\|\mathcal{L}(T)\|_{L_p} \leq K_{p,\epsilon}d\|T\|_{L_{p+\epsilon}}\|G\|_{L_{(p+\epsilon)'+\epsilon}} \leq K_{p,\epsilon}d\|T\|_{L_{p+\epsilon}}\|G\|_{L_{p'}} \leq K_{p,\epsilon}d\|T\|_{L_{p+\epsilon}}.$$

This finishes the proof. □

We have a further consequence of Theorem 5.7.

COROLLARY 5.10. *Let $d \in [n]$, $p \in (1,\infty)$, and let $T \in \mathcal{P}^{\leq d}(\mathbb{M}_{2^n})$. If T is an unitary or a projection operator, then there exists a constant $K_p > 0$ such that*

$$\|\mathcal{L}(T)\|_{L_p} \leq K_p d \|T\|_{L_p}.$$

PROOF. Note that, for any $r \in (0,\infty)$, $\|T\|_{L_r} = 1$ whenever $T \in \mathbb{M}_{2^n}$ is unitary. The assertion follows immediately from Theorem 5.7 (with $k = 1$ and any given $\epsilon > 0$).

Now let us assume that T is a projection, that is, T is self-adjoint and $T^2 = T$. In this case, we have

$$\|T\|_{L_2}^2 = \|T\|_{L_p}^p = \|T\|_{L_{p_\epsilon}}^{p_\epsilon}, \quad \forall \epsilon > 0.$$

Consequently,

$$\|T\|_{L_2}^{1-\theta} \|T\|_{L_{p_\epsilon}}^{\theta} = \|T\|_{L_p}^{(1-\theta)\frac{p}{2}} \|T\|_{L_p}^{\theta \frac{p}{p_\epsilon}} = \|T\|_{L_p}.$$

Then the desired assertion for projection T follows immediately from Theorem 5.7 (with $k = 1$ and any given $\epsilon > 0$). □

REMARK 5.11. The last corollary means that quantum version of Conjecture 1.4 holds for unitary or projection elements which are of at most of degree d. Its commutative version can be found in Domelevo et al. [**DDF$^+$25**].

5.3. Quantum Naor's inequality and Bernstein-Markov type inequalities for gradients

This section is devoted to deriving gradient estimate inequality and the corresponding Riesz transform-type inequality in quantum Boolean cubes. To begin with, for $T \in \mathbb{M}_{2^n}$, let

$$\|\nabla(T)\|_{L_p} := \left\|(\mathbf{d}_j(T))_{j=1}^n\right\|_{L_p(\mathbb{M}_{2^n};\ell_2^{cr})},$$

where the norm given in the right-hand side comes from the noncommutative Khintchine inequality (see more details in Lemma 2.5). We first establish the gradient estimates presented in Theorem 5.12 and Theorem 5.14, which are quantum analogues of the results due to Naor, Eskenazis and Ivanisvili (see for example [**EI20b**, Proposition 43] and [**EI20b**, Lemma 44]), and then we prove Bernstein-Markov type inequalities for gradient in Theorem 5.16.

THEOREM 5.12. *For each $p \in (1,\infty)$ and each $T \in L_p(\mathbb{M}_{2^n})$, there exists a constant $K_p > 0$ depending only on p such that the following inequality holds*

$$\mathbb{E}_\delta \left\| \sum_{j=1}^n \delta_j \mathbf{d}_j(e^{-t\mathcal{L}}(T)) \right\|_{L_p} \leq \frac{K_p}{(e^{\gamma_p t}-1)^{1/\gamma_p}} \|T\|_{L_p}, \quad \forall t > 0, \tag{5.18}$$

where $\delta = \{\delta_j\}_{j=1}^n \in \{-1,1\}^n$ and $\gamma_p = \min\{p,2\}$.

PROOF. Fix $t > 0$, and let $p \in [2,\infty)$ and $T \in \mathbb{M}_{2^n}$. Recall that the following quantum (dual) Buser-type inequality was recently proved in [**JLLZ24**, Lemma 5.6] (see also its proof there with changing $e^{2t}-1$ to $2t$):

$$\left\| \left(\sum_{j=1}^n |\mathbf{d}_j(e^{-t\mathcal{L}}(T))|^2 \right)^{1/2} \right\|_{L_p} \leq \frac{1}{\sqrt{e^{2t}-1}} \|T\|_{L_p}, \quad 2 \leq p \leq \infty. \tag{5.19}$$

Then, by the noncommutative Khintchine inequality (Lemma 2.5), we have

$$\mathbb{E}_\delta \left\| \sum_{j=1}^n \delta_j \mathbf{d}_j(e^{-t\mathcal{L}}(T)) \right\|_{L_p} \leq C\sqrt{p}\|\nabla(e^{-t\mathcal{L}}(T))\|_{L_p} \leq \frac{K_p}{\sqrt{e^{2t}-1}}\|T\|_{L_p}. \tag{5.20}$$

It remains to deal with the case $p \in (1,2]$. We divide its proof into three steps.

Step I. For each $T \in \mathbb{M}_{2^n}$ and $t > 0$, we write

$$T = \sum_{\mathbf{s}\in\{0,1,2,3\}^n} \widehat{T}(\mathbf{s})\sigma_{\mathbf{s}},$$

and define $H_t : L_p(\mathbb{M}_{2^n}) \to L_p(\{-1,1\}^n, L_p(\mathbb{M}_{2^n}))$ by setting

$$H_t(T)(\delta) \coloneqq \sum_{\mathbf{s}\in\{0,1,2,3\}^n} \widehat{T}(\mathbf{s})\sigma_{\mathbf{s}} \prod_{j\in supp\,(\mathbf{s})} (e^{-t} + (\frac{1-e^{-t}}{3})\delta_j), \tag{5.21}$$

where $\delta = \{\delta_j\}_{j=1}^n \in \{-1,1\}^n$.

Recall that for each $A \subset [n]$, the Walsh function $w_A : \{-1,1\}^n \to \{-1,1\}$ is given by $w_A(\delta) = \prod_{j\in A}\delta_j$, $\delta = \{\delta_j\}_{j=1}^n \in \{-1,1\}^n$. By the definition of $H_t(T)$,

$$H_t(T) = \sum_{\mathbf{s}\in\{0,1,2,3\}^n} \sum_{B\subseteq supp\,(\mathbf{s})} e^{-t|B|}(\frac{1-e^{-t}}{3})^{|\mathbf{s}|-|B|}\widehat{T}(\mathbf{s})\sigma_{\mathbf{s}} w_{supp\,(\mathbf{s})\setminus B}. \tag{5.22}$$

We now make the connection between the term $\sum_{j=1}^n \delta_j \mathbf{d}_j\left(e^{-t\mathcal{L}}(T)\right)$ with $H_t(T)$ as follows. Due to (5.22), it is clear that

(5.23)

$$\begin{aligned}
\left(\mathrm{Id}_{\mathbb{M}_{2^n}} \otimes \mathrm{Rad}^{=1}_{\{-1,1\}^n}\right)(H_t(T)) &= \sum_{j=1}^n \delta_j \left(\mathrm{Id}_{\mathbb{M}_{2^n}} \otimes \mathbb{E}_\delta\right)(\delta_j H_t(T)) \\
&= \sum_{j=1}^n \delta_j \sum_{\mathbf{s}\in\{0,1,2,3\}^n} \sum_{B\subseteq supp\,(\mathbf{s})} \widehat{T}(\mathbf{s})\sigma_{\mathbf{s}} e^{-t|B|}(\frac{1-e^{-t}}{3})^{|\mathbf{s}|-|B|}\mathbb{E}(\delta_j w_{supp\,(\mathbf{s})\setminus B}) \\
&= e^t(\frac{1-e^{-t}}{3})\sum_{j=1}^n \delta_j \sum_{\substack{\mathbf{s}\in\{0,1,2,3\}^n \\ s_j\neq 0}} e^{-t|\mathbf{s}|}\widehat{T}(\mathbf{s})\sigma_{\mathbf{s}} \\
&= \frac{e^t-1}{3}\sum_{j=1}^n \delta_j \mathbf{d}_j\left(e^{-t\mathcal{L}}(T)\right).
\end{aligned}$$

Here, $\mathrm{Rad}^{=1}_{\{-1,1\}^n}$ is the classical Rademacher projection (see (3.3) or [**Pis82b**]). Since $\mathbb{E}_\delta w_A = 0$ whenever $A \neq \varnothing$, only the term $B = supp\,(\mathbf{s})$ survives after taking expectation. Therefore,

$$\left(\mathrm{Id}_{\mathbb{M}_{2^n}} \otimes \mathbb{E}_\delta\right)(H_t(T)) = \sum_{\mathbf{s}} e^{-t|\mathbf{s}|}\widehat{T}(\mathbf{s})\sigma_{\mathbf{s}} = e^{-t\mathcal{L}}(T). \tag{5.24}$$

Step II. For each $t > 0$ and $T \in \mathbb{M}_{2^n}$, let

$$S_t(T) \coloneqq H_t(T) - e^{-t\mathcal{L}}(T).$$

Note that

$$\|S_t(T)\|_{L_1} \leq \|H_t(T)\|_{L_1} + \|e^{-t\mathcal{L}}(T)\|_{L_1} \leq \|H_t(T)\|_{L_1} + \|T\|_{L_1}.$$

To get an upper bound for $\|S_t(T)\|_{L_1}$, it remains to estimate the term $\|H_t(T)\|_{L_1}$. For $-1/3 \le \rho \le 1$, define the single-qubit Pauli channel $D_\rho : \mathbb{M}_2 \to \mathbb{M}_2$ by

$$D_\rho(T) \coloneqq \frac{1+3\rho}{4} T + \frac{1-\rho}{4} \sum_{\ell=1}^{3} \sigma_\ell T \sigma_\ell.$$

The coefficients in the preceding expression are nonnegative and their sum is equal to one. Thus, D_ρ is completely positive and trace-preserving. Moreover, by (2.1),

$$D_\rho(\sigma_0) = \sigma_0, \qquad D_\rho(\sigma_\ell) = \rho\sigma_\ell, \quad \ell = 1,2,3.$$

For $\delta \in -1,1^n$, set

$$\rho_j(\delta) \coloneqq e^{-t} + \frac{1-e^{-t}}{3}\delta_j.$$

we have

$$-\frac{1}{3} \le \rho_j(\delta) \le 1.$$

By the definition of H_t,

$$H_t(T)(\delta) = \left(D_{\rho_1(\delta)} \otimes \cdots \otimes D_{\rho_n(\delta)}\right)(T).$$

The tensor product in the preceding identity is completely positive and trace-preserving and is therefore contractive on L_1. Consequently,

$$\|H_t(T)\|_{L_1(\Omega_n; L_1(\mathbb{M}_{2^n}))} = \mathbb{E}_\delta \|H_t(T)(\delta)\|_{L_1} \le \|T\|_{L_1}.$$

Since $e^{-t\mathcal{L}}$ is also contractive on L_1, it follows that $\|S_t(T)\|_{L_1} \le 2\|T\|_{L_1}$.

Since $\{\sigma_{\mathbf{s}} \otimes w_A\}_{\mathbf{s}\in\{0,1,2,3\}^n, A\subseteq[n]}$ forms an orthonormal basis in $L_2(\mathbb{M}_{2^n}) \otimes L_2(\{-1,1\}^n)$, it follows from (5.22) that

$$\begin{aligned}
\|S_t(T)\|_{L_2}^2 &= \left\|H_t(T) - e^{-t\mathcal{L}}(T)\right\|_{L_2}^2 \\
&= \left\| \sum_{\mathbf{s}\in\{0,1,2,3\}^n} \sum_{B\subsetneq supp\,(\mathbf{s})} e^{-t|B|} \left(\frac{1-e^{-t}}{3}\right)^{|\mathbf{s}|-|B|} \widehat{T}(\mathbf{s}) \sigma_{\mathbf{s}} w_{supp\,(\mathbf{s})\setminus B} \right\|_{L_2}^2 \\
&= \sum_{\mathbf{s}\in\{0,1,2,3\}^n} \sum_{B\subsetneq supp\,(\mathbf{s})} |\widehat{T}(\mathbf{s})|^2 e^{-2t|B|} \left(\frac{1-e^{-t}}{3}\right)^{2(|\mathbf{s}|-|B|)} \\
&= \sum_{\mathbf{s}\in\{0,1,2,3\}^n} |\widehat{T}(\mathbf{s})|^2 \left(\sum_{l=0}^{|\mathbf{s}|-1} \binom{|\mathbf{s}|}{l} e^{-2tl} \left(\frac{1-e^{-t}}{3}\right)^{2(|\mathbf{s}|-l)} \right).
\end{aligned}$$

Note that for each $\mathbf{s} \in \{0,1,2,3\}^n$,

$$\begin{aligned}
\sum_{l=0}^{|\mathbf{s}|-1} \binom{|\mathbf{s}|}{l} e^{-2tl} \left(\frac{1-e^{-t}}{3}\right)^{2(|\mathbf{s}|-l)} &= \left(e^{-2t} + \left(\frac{1-e^{-t}}{3}\right)^2\right)^{|\mathbf{s}|} - e^{-2t|\mathbf{s}|} \\
&\le \left(e^{-2t} + (1-e^{-t})^2\right)^{|\mathbf{s}|} - e^{-2t|\mathbf{s}|} \le 1 - e^{-t},
\end{aligned}$$

where we used the elementary fact $(u^2+(1-u)^2)^l \le u^{2l}+(1-u)$ whenever $u \in (0,1)$ and $l \in \mathbb{N}$. Indeed, it suffices to note that

$$\begin{aligned}\sum_{j=1}^{l}\binom{l}{j}u^{2(l-j)}(1-u)^{2j} &\le (1-u)\left(\sum_{j=1}^{l}\binom{l}{j}u^{2(l-j)}(1-u)^{j}\right)\\ &\le (1-u)\left(\sum_{j=0}^{l}\binom{l}{j}u^{l-j}(1-u)^{j}\right) = 1-u.\end{aligned}$$

Now we conclude from the above argument that

$$\|S_t(T)\|_{L_2} \le (1-e^{-t})^{1/2}\|T\|_{L_2}.$$

For general $p \in (1,2)$, we choose $\theta \in (0,1)$ such that $\frac{1}{p} = \frac{\theta}{1} + \frac{1-\theta}{2}$ and apply the complex interpolation theorem (see e.g. [**BL76**, Theorem 1.1.1]) to get that

$$\|S_t\|_{L_p \to L_p} \le 2^{\frac{2}{p}-1}(1-e^{-t})^{1-\frac{1}{p}}.$$

Step III. For general $p \in (1,2]$ and $\mathfrak{T} \in L_p(\{-1,1\}^n; L_p(\mathbb{M}_{2^n}))$, we have

$$\left\|\left(\mathrm{Id}_{\mathbb{M}_{2^n}} \otimes \mathrm{Rad}^{=1}_{\{-1,1\}^n}\right)(\mathfrak{T})\right\|_{L_p} \le C_p\|\mathfrak{T}\|_{L_p}.$$

Therefore, combining the previous estimates with (5.23) and taking the expectation with respect to $\{\delta_j\}_{j=1}^n$ yields that

$$\begin{aligned}\mathbb{E}_\delta\left\|\sum_{j=1}^{n}\delta_j \mathbf{d}_j(e^{-t\mathcal{L}}(T))\right\|_{L_p} &= \frac{3}{e^t-1}\mathbb{E}_\delta\left\|\left(\mathrm{Id}_{\mathbb{M}_{2^n}} \otimes \mathrm{Rad}^{=1}_{\{-1,1\}^n}\right)(H_t(T))\right\|_{L_p}\\ &= \frac{3}{e^t-1}\mathbb{E}_\delta\left\|\left(\mathrm{Id}_{\mathbb{M}_{2^n}} \otimes \mathrm{Rad}^{=1}_{\{-1,1\}^n}\right)(S_t(T))\right\|_{L_p}\\ &\le \frac{3C_p}{(e^t-1)}\|S_t(T)\|_{L_p}\\ &\le \frac{3C_p 2^{\frac{2}{p}-1}(1-e^{-t})^{1-\frac{1}{p}}}{(e^t-1)}\|T\|_{L_p}\\ &\le K_p\frac{1}{(e^{tp}-1)^{1/p}}\|T\|_{L_p},\end{aligned} \tag{5.25}$$

where the last inequality follows from

$$\frac{(1-e^{-t})^{1-\frac{1}{p}}}{e^t-1} \le \frac{p^{1/p}}{(e^{pt}-1)^{1/p}}.$$

The proof is complete. □

Remark 5.13. For each $p \in (1,\infty]$, there exists a constant $K_p > 0$ depending only on p such that

$$\|\nabla(e^{-t\mathcal{L}}(T))\|_{L_p} \le \begin{cases}\frac{K_p}{(e^{pt}-1)^{1/p}}\|T\|_{L_p}, & 1<p<2,\\ \frac{1}{\sqrt{e^{2t}-1}}\|T\|_{L_p}, & 2\le p\le\infty,\end{cases} \quad \forall\, t>0. \tag{5.26}$$

Proof. In the case of $2 \le p \le \infty$, we can obtain it directly from (5.19). Applying the noncommutative Khintchine inequality Lemma 2.5 to the left-hand

side of (5.18) of Theorem 5.12 yields a constant $K_p > 0$ depending only on $p \in (1,2)$ such that

$$\left\|\nabla(e^{-t\mathcal{L}}(T))\right\|_{L_p} \leq \frac{K_p}{(e^{tp}-1)^{1/p}}\|T\|_{L_p}, \quad \text{for all } t>0,$$

which is the desired inequality. □

The following result, a noncommutative analogy of Naor's Riesz transform inequality [**EI20b**, Proposition 43], can be deduced from Theorem 5.12 and the functional calculus of the number operator.

THEOREM 5.14. *For each $p \in (1,\infty)$ and $\epsilon \in (\max\{0, \frac{1}{2}-\frac{1}{p}\}, 1-\frac{1}{p})$, there exists a constant $K_p > 0$ depending only on p such that for each $n \in \mathbb{N}$, we have*

$$\|\nabla(T)\|_{L_p} \leq \left(\frac{K_p}{\epsilon}\right)\left\|\mathcal{L}^{\frac{1}{p}+\epsilon}(T)\right\|_{L_p}, \quad \forall\ T \in L_p(\mathbb{M}_{2^n}). \tag{5.27}$$

PROOF. By the functional calculus of $\mathcal{L}$, we have

$$\mathcal{L}^{-(\frac{1}{p}+\epsilon)} = \frac{1}{\Gamma(\frac{1}{p}+\epsilon)}\int_0^\infty t^{\frac{1}{p}+\epsilon-1}e^{-t\mathcal{L}}dt. \tag{5.28}$$

By Lemma 2.5, the Kahane-Khintchine inequality (see for instance [**Pis16**]) and (5.28), we have

$$\begin{aligned}\left\|\nabla\mathcal{L}^{-(\frac{1}{p}+\epsilon)}(T)\right\|_{L_p} &\leq A_p^{(1)}\mathbb{E}_\delta\left\|\sum_{j=1}^n \delta_j \mathbf{d}_j\left(\mathcal{L}^{-(\frac{1}{p}+\epsilon)}(T)\right)\right\|_{L_p}\\ &\leq A_p^{(2)}\int_0^\infty t^{\frac{1}{p}+\epsilon-1}\mathbb{E}_\delta\left\|\sum_{j=1}^n \delta_j \mathbf{d}_j(e^{-t\mathcal{L}}(T))\right\|_{L_p}dt,\end{aligned} \tag{5.29}$$

which, together with Theorem 5.12 for $1 < p \leq 2$, implies

$$\left\|\nabla\mathcal{L}^{-(\frac{1}{p}+\epsilon)}(T)\right\|_{L_p} \leq A_p^{(3)}\|T\|_{L_p}\int_0^\infty \frac{t^{\frac{1}{p}+\epsilon-1}}{(e^{tp}-1)^{1/p}}dt.$$

Note that (see [**EI20b**, eq. (214)])

$$\int_0^\infty \frac{t^{\frac{1}{p}+\epsilon-1}}{(e^{tp}-1)^{1/p}}dt = O(\frac{1}{\epsilon}), \quad p \in (1,2].$$

The desired inequality for $p \in (1,2]$ follows.

As for the case $p \in (2,\infty)$, combining (5.29) and Theorem 5.12 for $2 < p < \infty$, we have

$$\left\|\nabla\mathcal{L}^{-(\frac{1}{p}+\epsilon)}(T)\right\|_{L_p} \leq A_p^{(4)}\|T\|_{L_p}\int_0^\infty \frac{t^{\frac{1}{p}+\epsilon-1}}{\sqrt{e^{2t}-1}}dt.$$

Hence, it suffices to check

$$I := \int_0^\infty \frac{t^{\frac{1}{p}+\epsilon-1}}{\sqrt{e^{2t}-1}}dt < \infty.$$

We write

$$I = \int_0^1 \frac{t^{\frac{1}{p}+\epsilon-1}}{\sqrt{e^{2t}-1}}dt + \int_1^\infty \frac{t^{\frac{1}{p}+\epsilon-1}}{\sqrt{e^{2t}-1}}dt =: I_1 + I_2.$$

For I_1, as $t \to 0^+$, we have $e^{2t} - 1 \approx 2t$, the integrand becomes:

$$\frac{t^{\frac{1}{p}+\epsilon-1}}{\sqrt{e^{2t}-1}} \approx \frac{t^{\frac{1}{p}+\epsilon-1}}{\sqrt{2t}} = \frac{t^{\frac{1}{p}-\frac{1}{2}+\epsilon-1}}{\sqrt{2}}.$$

Thus, the integrand behaves like $t^{\frac{1}{p}-\frac{1}{2}+\epsilon-1}$, and the integral converges since $\epsilon > \frac{1}{2} - \frac{1}{p}$. The exact result is:

$$\int_0^1 \frac{t^{\frac{1}{p}-\frac{1}{2}+\epsilon-1}}{\sqrt{2}} dt = \frac{1}{\sqrt{2}} \int_0^1 t^{\frac{1}{p}-\frac{1}{2}+\epsilon-1} dt = \frac{1}{\sqrt{2}} \left[\frac{t^{\frac{1}{p}-\frac{1}{2}+\epsilon}}{\frac{1}{p}-\frac{1}{2}+\epsilon} \right]_0^1 =: \frac{K_p}{\epsilon}.$$

In the second part I_2, e^{pt} grows very rapidly as $t \to \infty$, so the denominator $\sqrt{e^{2t}-1} \approx e^t$. The integrand becomes:

$$\frac{t^{\frac{1}{p}+\epsilon-1}}{\sqrt{e^{2t}-1}} \approx t^{\frac{1}{p}+\epsilon-1} e^{-t}.$$

Due to the exponential decay e^{-t}, the integral always converges as $t \to \infty$. The exact result is:

$$\int_1^\infty t^{\frac{1}{p}+\epsilon-1} e^{-t} dt = \Gamma\left(\frac{1}{p}+\epsilon, 1\right),$$

where $\Gamma(a, x)$ is the incomplete Gamma function. Combining the results of the two parts, we obtain the desired result. □

The following result is motivated by [**O'D14**, Theorem 9.22].

LEMMA 5.15. *For every $T \in \mathcal{P}^{\leq d}(\mathbb{M}_{2^n})$, we have*

$$\|T\|_{L_2} \leq e^d \|T\|_{L_1}.$$

PROOF. For $\epsilon > 0$, and let $0 < \theta < 1$ be the solution of $\frac{1}{2} = \frac{\theta}{1} + \frac{1-\theta}{2+\epsilon}$, that is $\theta = \frac{\epsilon}{2(1+\epsilon)}$. Applying Hölder's inequality, we get

$$\|T\|_{L_2} \leq \|T\|_{L_{2+\epsilon}}^{1-\theta} \|T\|_{L_1}^{\theta}. \tag{5.30}$$

In addition, for each $2 \leq q < \infty$ and $T \in \mathcal{P}^{\leq d}(\mathbb{M}_{2^n})$, we have

$$\|T\|_{L_q} \leq \sqrt{q-1}^d \|T\|_{L_2}. \tag{5.31}$$

Indeed, using Lemma 2.1, we have

$$\|T\|_{L_q}^2 = \|e^{-\log\sqrt{q-1}}(e^{\log\sqrt{q-1}}(T))\|_{L_q}^2 \leq \|e^{\log\sqrt{q-1}}(T)\|_{L_2}^2$$

and

$$\|e^{\log\sqrt{q-1}}(T)\|_{L_2}^2 = \sum_{k=0}^{d} \sum_{|\mathbf{s}|=k} (q-1)^k |\widehat{T}(\mathbf{s})|^2 \leq (q-1)^d \|T\|_{L_2}^2.$$

The inequality (5.31) now follows from the last two inequalities. Then, by (5.30) and (5.31) with $q = 2 + \epsilon$, we get

$$\|T\|_{L_2} \leq (1+\epsilon)^{\frac{d(1-\theta)}{2}} \|T\|_{L_2}^{1-\theta} \|T\|_{L_1}^{\theta},$$

which implies

$$\|T\|_{L_2} \leq (1+\epsilon)^{\frac{d(1-\theta)}{2\theta}} \|T\|_{L_1} = \left((1+\epsilon)^{\frac{1}{\epsilon}+\frac{1}{2}}\right)^d \|T\|_{L_1}.$$

Since $\lim_{\epsilon\to 0}(1+\epsilon)^{\frac{1}{\epsilon}+\frac{1}{2}} = e$, the desired result follows. □

Now we can prove the following quantum Bernstein–Markov type inequality for the gradient as a consequence of Theorem 5.14 and Theorem 5.5.

THEOREM 5.16. *Let $n \geq 1$, $d \in [n]$, $1 \leq p \leq \infty$ and $\theta_p \in (0, \frac{\pi}{2}]$ as Theorem 3.1. Then there exist $\alpha_p \in [1,2)$ and $C_p \in (0,\infty)$ such that, for every $T \in \mathcal{P}^{\leq d}(\mathbb{M}_{2^n})$, we have*

$$\|\nabla(T)\|_{L_p} \leq \begin{cases} C_p d^{\frac{2}{p}-\frac{\theta_p}{p\pi}} \log(d+1)\|T\|_{L_p}, & 1 < p < \infty; \\ 2d\|T\|_{L_p}, & p = \infty; \\ e^d\sqrt{d}\|T\|_{L_p} & p = 1. \end{cases}$$

PROOF. For each $1 < p < \infty$ and $\epsilon \in (\max\{0, \frac{1}{2} - \frac{1}{p}\}, 1 - \frac{1}{p})$, by Theorem 5.14 and Lemma 4.18 with $\beta = \frac{1}{p} + \epsilon$, we get

$$\|\nabla(T)\|_{L_p} \leq \frac{K_p}{\epsilon}\|\mathcal{L}^{\frac{1}{p}+\epsilon}(T)\|_{L_p} \leq \frac{4K_p}{\epsilon}\|\mathcal{L}(T)\|_{L_p}^{\frac{1}{p}+\epsilon}\|T\|_{L_p}^{1-(\frac{1}{p}+\epsilon)}.$$

Using Theorem 5.5, there exist $\alpha_p = 2 - \frac{\theta_p}{\pi} \in [1,2)$ (where $\theta_p \in (0, \frac{\pi}{2}]$ is as in Theorem 3.1) and $c_p > 0$ such that

$$\|\nabla(T)\|_{L_p} \leq \left(\frac{4c_pK_p}{\epsilon}\right) d^{(2-\frac{\theta_p}{\pi})(\frac{1}{p}+\epsilon)}\|T\|_{L_p}.$$

Choosing $\epsilon = \frac{1}{\log(d+1)}$, we have

$$\begin{aligned} \|\nabla(T)\|_{L_p} &\leq 4c_pK_p d^{\frac{2}{p}-\frac{\theta_p}{p\pi}} d^{(2-\frac{\theta_p}{\pi})\frac{1}{\log(d+1)}} \log(d+1)\|T\|_{L_p} \\ &\leq 4c_pK_p d^{\frac{2}{p}-\frac{\theta_p}{p\pi}} d^{\frac{2}{\log(d+1)}} \log(d+1)\|T\|_{L_p} \\ &\leq C_p d^{\frac{2}{p}-\frac{\theta_p}{p\pi}} \log(d+1)\|T\|_{L_p}, \end{aligned}$$

where the last inequality follows from $d^{\frac{2}{\log(d+1)}} \leq e^2$.

For the case $p = \infty$, by the quantum (dual) Buser inequality (5.19) and Theorem 5.1 with $T_d(e^t) \leq e^{td^2}$, we have

$$\|\nabla(T)\|_{L_\infty} \leq \frac{1}{\sqrt{e^{2t}-1}}\|e^{t\mathcal{L}}(T)\|_{L_\infty} \leq \frac{e^{td^2}}{\sqrt{e^{2t}-1}}\|T\|_{L_\infty}.$$

Taking $t = \frac{1}{d^2}$ and by the inequality $\sqrt{e^{2t}-1} \geq \sqrt{2t}$, we obtain

$$\|\nabla(T)\|_{L_\infty} \leq \frac{e}{\sqrt{2t}}\|T\|_{L_\infty} \leq 2d\|T\|_{L_\infty}.$$

For the final case where $p = 1$, the result follows from orthogonality, $\widehat{T}(\mathbf{s}) = 0$ whenever $|\mathbf{s}| > d$ and Lemma 5.15. Indeed, it is clear that

$$\begin{aligned} \|\nabla(T)\|_{L_1} \leq \|\nabla(T)\|_{L_2} &\leq \left\| \sum_{j=1}^n |d_j(T)|^2 \right\|_{L_1}^{\frac{1}{2}} \\ &\leq \left(\sum_{j=1}^n \||d_j(T)|^2\|_{L_1} \right)^{\frac{1}{2}} \leq \left(\sum_{j=1}^n \||d_j(T)|\|_{L_2}^2 \right)^{\frac{1}{2}}, \end{aligned}$$

which, together with Lemma 5.15, implies

$$\|\nabla(T)\|_{L_1} \leq \left(\sum_{\substack{\mathbf{s}\in\{0,1,2,3\}^n \\ |\mathbf{s}|\leq d}} |\mathbf{s}||\widehat{T}(\mathbf{s})|^2 \right)^{\frac{1}{2}} \leq \sqrt{d}\|T\|_{L_2} \leq e^d\sqrt{d}\|T\|_{L_1}.$$

□

REMARK 5.17. Theorem 5.16 can be traced back to [**EI20b**, Theorem 15] and [**EI20b**, Proposition 46]; however, for $2 \leq p < \infty$, [**EI20b**, Theorem 15] was proved by using Lust-Piquard's Riesz transform inequality ([**ELP08a, LP98**]), and here we apply the quantum (dual) Buser inequality established recently in [**JLLZ24**].

CHAPTER 6

Covariance quantum Poincaré and quantum Talagrand-type inequalities in tail spaces

This chapter develops Talagrand and covariance inequalities for quantum observables whose Pauli–Fourier spectrum is supported on the tail space $\mathcal{T}^{\geq d}(\mathbb{M}_{2^n})$. The main point is that the lower spectral cutoff degree d yields specific decay property under the quantum Ornstein–Uhlenbeck semigroup and therefore strengthens the corresponding influence-based estimates.

We first establish tail Talagrand inequalities for $1 \leq p < 2$. In particular, the geometric influence estimate obtained from Theorem 6.1 has a full logarithmic denominator and improves the dependence on small influences in the corresponding result of [**RWZ24**]. We then derive tail-sensitive covariance Poincaré and covariance Talagrand inequalities. These results are obtained within a common semigroup framework and make the improvement arising from spectral localization explicit.

6.1. Quantum tail Talagrand inequalities

For $j \in [n]$, $1 \leq p < \infty$, and $T \in \mathbb{M}_{2^n}$, define the j-th L_p-influence of T by

$$\mathrm{Inf}_j^p(T) \coloneqq \|\mathbf{d}_j(T)\|_{L_p}^p.$$

The case $p = 1$ is commonly referred to as the *geometric influence*; see, for example, [**RWZ24**]. When $p = 2$, we write

$$\mathrm{Inf}_j(T) \coloneqq \mathrm{Inf}_j^2(T), \qquad \mathrm{Inf}(T) \coloneqq \sum_{j=1}^n \mathrm{Inf}_j(T).$$

By (2.6) and Parseval's identity,

$$\mathrm{Inf}_j(T) = \sum_{\substack{\mathbf{s}\in\{0,1,2,3\}^n \\ s_j \neq 0}} |\widehat{T}(\mathbf{s})|^2.$$

We use the variance convention

$$\mathrm{Var}(T) \coloneqq \mathrm{tr}(|T|^2) - |\mathrm{tr}(T)|^2 = \|T - \mathrm{tr}(T)\mathbf{1}\|_{L_2}^2.$$

The next result is tail-version of the *Talagrand inequality*, which quantifies the variance decay property for T whose Fourier support is contained in levels at least d.

THEOREM 6.1. *Let $p \in [1, 2)$ and $d \in [n]$. For every $T \in \mathcal{T}^{\geq d}(\mathbb{M}_{2^n})$ with $\|T\|_{L_\infty} \leq 1$, we have*

$$\mathrm{Var}(T) \leq \left(\frac{4}{2-p}\right) \sum_{j=1}^n \frac{\|d_j(T)\|_{L_p}^p}{d + \log\left(1/\|d_j(T)\|_{L_p}^p\right)}.$$

To prove Theorem 6.1, we begin by recalling some elementary estimates. The first is the spectral decay on tail spaces.

LEMMA 6.2. *For every $T \in \mathcal{T}^{\geq d}(\mathbb{M}_{2^n})$, we have*

$$\|e^{-t\mathcal{L}}(T)\|_{L_2} \leq e^{-td}\|T\|_{L_2}.$$

The second estimate is a scalar integral bound that will retain both the tail parameter and the logarithmic influence term.

LEMMA 6.3. *For every $d \geq 1$ and $\lambda \geq 0$,*

$$\int_0^\infty e^{-2dt-\lambda \tanh(t)}\,\mathrm{d}t \leq \frac{1}{d+\lambda}.$$

PROOF. Set $y = \tanh(t) = \frac{e^t-e^{-t}}{e^t+e^{-t}}$. Since

$$e^{-2t} = \frac{1-y}{1+y} \qquad \text{and} \qquad \mathrm{d}t = \frac{\mathrm{d}y}{1-y^2},$$

changing the variable the integral equals

$$\int_0^1 \frac{(1-y)^{d-1}}{(1+y)^{d+1}} e^{-\lambda y}\,\mathrm{d}y.$$

For $0 \leq y < 1$, let

$$F_d(y) \coloneqq dy + (d-1)\log(1-y) - (d+1)\log(1+y).$$

It is easy to see that $F_d(0) = 0$ and

$$F_d'(y) = \frac{2y - d(1+y^2)}{1-y^2} \leq 0,$$

where we used $d \geq 1$ and $2y \leq 1+y^2$. Hence

$$\frac{(1-y)^{d-1}}{(1+y)^{d+1}} \leq e^{-dy}.$$

It follows that

$$\int_0^\infty e^{-2dt-\lambda \tanh t}\,\mathrm{d}t \leq \int_0^1 e^{-(d+\lambda)y}\,\mathrm{d}y \leq \frac{1}{d+\lambda}.$$

□

PROOF OF THEOREM 6.1. Let $d \in [n]$ and $T \in \mathcal{T}^{\geq d}$. By Lemma 6.5 and Lemma 6.2, we have

$$\begin{aligned}
\mathrm{Var}(T) &= \sum_{j=1}^n \int_0^\infty \mathrm{tr}((e^{-t\mathcal{L}}\mathbf{d}_j(T))^* \cdot \mathbf{d}_j(T))\mathrm{d}t \\
&= 4\sum_{j=1}^n \int_0^\infty \mathrm{tr}((e^{-2t\mathcal{L}}\mathbf{d}_j(T))^* \cdot e^{-2t\mathcal{L}}\mathbf{d}_j(T))\mathrm{d}t \\
&= 4\sum_{j=1}^n \int_0^\infty \left\|e^{-2t\mathcal{L}}(\mathbf{d}_j(T))\right\|_{L_2}^2 \mathrm{d}t \qquad (6.1) \\
&\leq 4\sum_{j=1}^n \int_0^\infty e^{-2td}\|e^{-t\mathcal{L}}(\mathbf{d}_j(T))\|_{L_2}^2 \mathrm{d}t \\
&\leq 4\sum_{j=1}^n \int_0^\infty e^{-2td}\|\mathbf{d}_j(T)\|_{L_{1+e^{-2t}}}^2 \mathrm{d}t,
\end{aligned}$$

where the last inequality is due to the (optimal) hypercontractivity given in Theorem 2.1. Hence, to get the desired inequality it suffices to estimate

$$\int_0^\infty e^{-2td}\|\mathbf{d}_j(T)\|_{L_{1+e^{-2t}}}^2 \mathrm{d}t, \quad j \in [n]. \qquad (6.2)$$

Assume first that $1 < p < 2$. Denote that $X_j \coloneqq \mathbf{d}_j(T)$ for each $j \in [n]$. Put

$$a_k = \frac{k}{2}\log\Big(\frac{1}{p-1}\Big), \qquad k \geq 0,$$

and, for a fixed j, set

$$I_j \coloneqq \int_0^\infty e^{-2td}\|X_j\|_{L_{1+e^{-2t}}}^2 \,\mathrm{d}t.$$

Writing the integral as a sum over the intervals $[a_k, a_{k+1}]$ and translating each interval to $[0, a_1]$, we obtain

$$\begin{aligned}
I_j &= \sum_{k=0}^\infty (p-1)^{kd} \int_0^{a_1} e^{-2ud}\|X_j\|_{L_{1+(p-1)^k e^{-2u}}}^2 \,\mathrm{d}u \\
&\leq \sum_{k=0}^\infty (p-1)^{k} \int_0^{a_1} e^{-2ud}\|X_j\|_{L_{1+e^{-2u}}}^2 \,\mathrm{d}u \\
&= \frac{1}{2-p}\int_0^{a_1} e^{-2ud}\|X_j\|_{L_{1+e^{-2u}}}^2 \,\mathrm{d}u.
\end{aligned}$$

Here we used $d \geq 1$ and the monotonicity of the L_r-norms with respect to r for the normalized trace. By Lemma 2.3, $\|X_j\|_{L_\infty} \leq 1$ for each $j \in [n]$. Set

$$b_j = \|X_j\|_{L_p}^p \qquad \text{and} \qquad \lambda_j = \log(1/b_j), \quad j \in [n]$$

whenever $X_j \neq 0$. For $0 \leq u \leq a_1$, the exponent $r(u) = 1 + e^{-2u}$ belongs to $[p, 2]$. Interpolation between L_p and L_∞ therefore gives

$$\|X_j\|_{L_{r(u)}}^2 \leq b_j^{2/r(u)} = b_j^{1+\tanh(u)} = b_j e^{-\lambda_j \tanh(u)}.$$

Consequently, Lemma 6.3 yields

$$I_j \le \frac{b_j}{(2-p)(d+\lambda_j)} = \left(\frac{1}{2-p}\right) \frac{\|X_j\|_{L_p}^p}{d + \log\left(1/\|X_j\|_{L_p}^p\right)}. \tag{6.3}$$

Combining this estimate with (6.1) proves the claim for $1 < p < 2$.

It remains to consider $p = 1$. In this case interpolation between L_1 and L_∞ applies for every $t \ge 0$. If $a_j = \|X_j\|_{L_1} \ne 0$ and $\mu_j = \log(1/a_j)$, then

$$\|X_j\|_{L_{1+e^{-2t}}}^2 \le a_j^{2/(1+e^{-2t})} = a_j e^{-\mu_j \tanh(t)}.$$

Combining Lemma 6.3 and (6.1), we obtain

$$\mathrm{Var}(T) \le 4 \sum_{j=1}^n \frac{\|\mathbf{d}_j(T)\|_{L_1}}{d + \log\big(1/\|\mathbf{d}_j(T)\|_{L_1}\big)},$$

which is the asserted estimate at $p = 1$. □

A direct consequence of Theorem 6.1 is the following geometric influence Talagrand inequality, which was established in [**RWZ24**, Theorem 3.2] and serves as the key ingredient for quantum KKL-type inequality invoking geometric influence; see [**RWZ24**, Theorem 3.9].

COROLLARY 6.4. *For each $T \in \mathbb{M}_{2^n}$ with $\|T\|_{L_\infty} \le 1$, we have*

$$\mathrm{Var}(T) \le 4 \sum_{j=1}^n \frac{\|\mathbf{d}_j(T)\|_{L_1}}{1 + \log\left(1/\|\mathbf{d}_j(T)\|_{L_1}\right)}.$$

PROOF. Choosing $p = 1$ and $d = 1$ in Theorem 6.1 yields the desired result. □

We conclude this section by comparing Theorem 6.1 with the geometric-influence inequality obtained in [**RWZ24**, Theorem 3.2]. The latter asserts that, for every $A \in \mathbb{M}_{2^n}$ with $\|A\|_{L_\infty} \le 1$,

$$\mathrm{Var}(A) \le C \sum_{j=1}^n \frac{\|\mathbf{d}_j(A)\|_{L_1}\big(1 + \|\mathbf{d}_j(A)\|_{L_1}\big)}{\big[1 + \log^+\big(1/\|\mathbf{d}_j(A)\|_{L_1}\big)\big]^{1/2}},$$

where $C > 0$ is universal constant. By contrast, the case $p = d = 1$ of Theorem 6.1 gives

$$\mathrm{Var}(A) \le 8 \sum_{j=1}^n \frac{\|\mathbf{d}_j(A)\|_{L_1}}{1 + \log\big(1/\|\mathbf{d}_j(A)\|_{L_1}\big)}.$$

Since $\|\mathbf{d}_j(A)\|_{L_1} \le 1$, the latter estimate improves the dependence on each individual geometric influence: it replaces the square-root logarithmic denominator in [**RWZ24**] by a full logarithmic denominator and removes the additional factor $1 + \|\mathbf{d}_j(A)\|_{L_1}$. In particular, writing $a_j = \|\mathbf{d}_j(A)\|_{L_1}$, one has, for $0 < a_j \le 1$,

$$\frac{a_j}{1 + \log(1/a_j)} \le \frac{a_j(1+a_j)}{\big[1 + \log^+(1/a_j)\big]^{1/2}}.$$

More generally, Theorem 6.1 provides a family of estimates for every $1 \le p < 2$ and retains the lower Fourier-level parameter d through the denominator $d + \log\big(1/\|\mathbf{d}_j(T)\|_{L_p}\big)$. Thus the estimate becomes stronger when the Fourier support of T is pushed farther into the tail space.

6.2. Covariance Poincaré and Talagrand-type inequalities

This section aims to derive covariance estimates for observables whose Pauli–Fourier spectra are supported on the tail space $\mathcal{T}^{\geq d}(\mathbb{M}_{2^n})$. More precisely, covariance Poincaré inequalities and covariance Talagrand-type inequalities will be established via semigroup tools. The semigroup representation below reduces covariance to coordinatewise pairings. For non-Hilbertian norms, we combine this representation with the heat smoothing estimate of Theorem 4.1; in Hilbertian case, the exact spectral decay yields the sharper factor d^{-1}.

For $T_1, T_2 \in \mathbb{M}_{2^n}$, the covariance of For T_1, T_2 is defined by

$$\mathrm{Cov}(T_1, T_2) = \mathrm{tr}((T_1 - \mathrm{tr}(T_1))^*)(T_2 - \mathrm{tr}(T_2)) = \mathrm{tr}(T_1^* T_2) - \overline{\mathrm{tr}(T_1)}\mathrm{tr}(T_2). \tag{6.4}$$

Thus, $\mathrm{Cov}(T, T) = \mathrm{Var}(T)$ for each $T \in \mathbb{M}_{2^n}$.

LEMMA 6.5. *For $T_1, T_2 \in \mathbb{M}_{2^n}$, we have*

$$\mathrm{Cov}(T_1, T_2) = \sum_{j=1}^{n} \int_0^\infty \mathrm{tr}[((e^{-t\mathcal{L}}\mathbf{d}_j(T_1)))^* \mathbf{d}_j(T_2)]dt.$$

PROOF. For $T_1, T_2 \in \mathbb{M}_{2^n}$, according to the definition (6.4) and the interpolation of semigroups, we have

$$\begin{aligned} \mathrm{Cov}(T_1, T_2) &= \mathrm{tr}(T_1^* T_2) - \overline{\mathrm{tr}(T_1)}\mathrm{tr}(T_2) \\ &= -\int_0^\infty \mathrm{tr}((\partial_t e^{-t\mathcal{L}}(T_1))^* T_2)\ \mathrm{d}t, \end{aligned} \tag{6.5}$$

where we used $\lim_{t\to 0^+} e^{-t\mathcal{L}}(T) = T$ and $\lim_{t\to\infty} e^{-t\mathcal{L}}(T) = \mathrm{tr}(T)$. Since for each $j \in [n]$ and each $T \in \mathbb{M}_{2^n}$, we have

$$\begin{aligned} \partial_t e^{-t\mathcal{L}}(T) &= \partial_t \left(\sum_{\mathbf{s}\in\{0,1,2,3\}^n} e^{-t|\mathbf{s}|}\widehat{T}(\mathbf{s})\sigma_{\mathbf{s}} \right) \\ &= -\sum_{\mathbf{s}\in\{0,1,2,3\}^n} |\mathbf{s}| e^{-t|\mathbf{s}|}\widehat{T}(|\mathbf{s}|)\sigma_{\mathbf{s}} \\ &= -\sum_{j=1}^{n} \sum_{\substack{\mathbf{s}\in\{0,1,2,3\}^n \\ j\in\mathrm{supp}(\mathbf{s})}} e^{-t|\mathbf{s}|}\widehat{T}(\mathbf{s})\sigma_{\mathbf{s}} \\ &= -\sum_{j=1}^{n} \mathbf{d}_j e^{-t\mathcal{L}}(T). \end{aligned}$$

Combining the above equation with Lemma 2.2 and the orthogonality of $\{\sigma_{\mathbf{s}}\}_{\mathbf{s}\in\{0,1,2,3\}^n}$ in $L_2(\mathbb{M}_{2^n})$, we obtain that

$$\begin{aligned} \mathrm{Cov}(T_1, T_2) &= \sum_{j=1}^{n} \int_0^\infty \mathrm{tr}((\mathbf{d}_j e^{-t\mathcal{L}}(T_1))^* T_2)dt \\ &= \sum_{j=1}^{n} \int_0^\infty \mathrm{tr}[(e^{-t\mathcal{L}}\mathbf{d}_j(T_1))^* \mathbf{d}_j(T_2)]dt. \end{aligned}$$

The proof is complete. □

The following lemma provides consequences of tail heat smoothing that will be used here and in the next section. The first estimate is a dimension-free exponential decay estimate on the first tail space (i.e., tail space with $d = 1$); the second is an estimate on the d-tail space.

LEMMA 6.6. *Fix $1 < p < \infty$, and let $C_p, c_p > 0$ and $A_p \geq 1$ be the constants in Theorem 4.1. Put $\alpha_p := A_p^{-1}$.*

(i) *There exist constants $B_r \geq 1$ and $b_p > 0$, depending only on p, such that*

$$\|e^{-t\mathcal{L}}T\|_{L_p} \leq B_p e^{-b_p t}\|T\|_{L_p}, \qquad t \geq 0, \tag{6.6}$$

whenever $\operatorname{tr}(T) = 0$.

(ii) *There exists a constant $K_p > 0$, depending only on p, such that*

$$\int_0^\infty \|e^{-t\mathcal{L}}T\|_{L_p}\,\mathrm{d}t \leq K_p d^{-\alpha_p}\|T\|_{L_p} \tag{6.7}$$

for every $d \in [n]$ and every $T \in \mathcal{T}^{\geq d}(\mathbb{M}_{2^n})$.

PROOF. If $\operatorname{tr}(T) = 0$, that is, $T \in \mathcal{T}^{\geq 1}(\mathbb{M}_{2^n})$. For $0 \leq t \leq 1$, contractivity gives $\|e^{-t\mathcal{L}}T\|_{L_p} \leq \|T\|_{L_p}$, whereas for $t \geq 1$, Theorem 4.1 with $d = 1$ gives

$$\|e^{-t\mathcal{L}}T\|_{L_p} \leq C_p e^{-c_p t}\|T\|_{L_p}.$$

Thus (6.6) holds, for example, with $b_p = c_p$ and $B_p = \max\{e^{c_p}, C_p\}$.

For $T \in \mathcal{T}^{\geq d}(\mathbb{M}_{2^n})$, combining $A_p \geq 1$ with Theorem 4.1 yields

$$\begin{aligned}\int_0^\infty \|e^{-t\mathcal{L}}T\|_{L_p}\,\mathrm{d}t &\leq C_p\|T\|_{L_p}\left(\int_0^1 e^{-c_p dt^{A_p}}\,\mathrm{d}t + \int_1^\infty e^{-c_p dt}\,\mathrm{d}t\right)\\ &\leq C_p\|T\|_{L_p}\left(\frac{\Gamma(\alpha_p)}{(c_p d)^{\alpha_p}} + \frac{e^{-c_p d}}{c_p d}\right).\end{aligned}$$

Since $d^{-1} \leq d^{-\alpha_p}$ for $d \geq 1$ and $0 < \alpha_p \leq 1$, it follows that there exists a constant $K_p > 0$ depending only on p such that

$$\frac{\Gamma(\alpha_p)}{(c_p d)^{\alpha_p}} + \frac{e^{-c_p d}}{c_p d} \leq K_p d^{-\alpha_p},$$

which yields the desired estimate. □

THEOREM 6.7. *Let $d \in [n]$, $2 < p < \infty$, and let $q = p/(p-1)$. With α_p and K_p as in Lemma 6.6, one has*

$$\big|\operatorname{Cov}(T_1, T_2)\big| \leq K_p d^{-\alpha_p}\sum_{j=1}^n \big(\operatorname{Inf}_j^p(T_1)\big)^{1/p}\big(\operatorname{Inf}_j^q(T_2)\big)^{1/q} \tag{6.8}$$

for all $T_1, T_2 \in \mathcal{T}^{\geq d}(\mathbb{M}_{2^n})$.

PROOF. For each $j \in [n]$, the operator $\mathbf{d}_j(T_1)$ still belongs to $\mathcal{T}^{\geq d}(\mathbb{M}_{2^n})$. Lemma 6.5, Hölder's inequality, and (6.7) therefore give

$$\begin{aligned}\big|\operatorname{Cov}(T_1, T_2)\big| &\leq \sum_{j=1}^n \int_0^\infty \|e^{-t\mathcal{L}}\mathbf{d}_j(T_1)\|_{L_p}\|\mathbf{d}_j(T_2)\|_{L_q}\,\mathrm{d}t\\ &\leq K_p d^{-\alpha_p}\sum_{j=1}^n \|\mathbf{d}_j(T_1)\|_{L_p}\|\mathbf{d}_j(T_2)\|_{L_q}.\end{aligned}$$

The conclusion follows from the definition of the L_p-influences. □

In the Hilbertian case, we apply Parseval's identity to give the optimal dependence on the lower Fourier level.

COROLLARY 6.8. *For every $d \in [n]$ and all T_1, $T_2 \in \mathcal{T}^{\geq d}(\mathbb{M}_{2^n})$,*

$$\big|\mathrm{Cov}(T_1, T_2)\big| \leq \left(\frac{1}{d}\right) \sum_{j=1}^{n} \sqrt{\mathrm{Inf}_j(T_1)\mathrm{Inf}_j(T_2)}. \tag{6.9}$$

In particular,

$$\mathrm{Var}(T) \leq \left(\frac{1}{d}\right) \mathrm{Inf}(T), \quad T \in \mathcal{T}^{\geq d}(\mathbb{M}_{2^n}).$$

PROOF. By Lemmas 6.5 and 6.2,

$$\begin{aligned}\big|\mathrm{Cov}(T_1, T_2)\big| &\leq \sum_{j=1}^{n} \int_0^\infty \|e^{-t\mathcal{L}}\mathbf{d}_j(T_1)\|_{L_2}\|\mathbf{d}_j(T_2)\|_{L_2}\,\mathrm{d}t \\ &\leq \sum_{j=1}^{n} \int_0^\infty e^{-dt}\,\mathrm{d}t\,\|\mathbf{d}_j(T_1)\|_{L_2}\|\mathbf{d}_j(T_2)\|_{L_2},\end{aligned}$$

which is (6.9). Taking $T_1 = T_2 = T$ gives the variance estimate. □

The covariance Poincaré inequalities stated previously control the bilinear form by the sizes of the coordinate derivatives. The purpose of the rest of the section is to refine those estimates into Talagrand-type inequalities invoking L_1 and L_p influences. The next theorem is a covariance form of Talagrand's influence inequality. It may be viewed as a logarithmic refinement of the covariance Poincaré estimate at level $d = 1$.

THEOREM 6.9. *For every $1 < p, q < \infty$ with $\frac{1}{p} + \frac{1}{q} = 1$, there exist a constant C_p such that for any T_1, and $T_1 \in \mathbb{M}_{2^n}$, we have*

$$\big|\mathrm{Cov}(T_1, T_2)\big| \leq C_p \sum_{j=1}^{n} \frac{\|\mathbf{d}_j(T_1)\|_{L_p}\|\mathbf{d}_j(T_2)\|_{L_q}}{\sqrt{\left(1 + \log\left(\frac{\|\mathbf{d}_j(T_1)\|_{L_p}}{\|\mathbf{d}_j(T_1)\|_{L_1}}\right)\right)\left(1 + \log\left(\frac{\|\mathbf{d}_j(T_2)\|_{L_q}}{\|\mathbf{d}_j(T_2)\|_{L_1}}\right)\right)}} \tag{6.10}$$

PROOF. By conjugate symmetry of covariance, after interchanging T_1 and T_2 if necessary, it suffices to consider $2 \leq p < \infty$ and $1 < q \leq 2$. For $j \in [n]$, set

$$X_j = \mathbf{d}_j(T_1), \qquad Y_j = \mathbf{d}_j(T_2),$$

and it is clear that both X_j and Y_j are trace zero. Let $B_p, b_p > 0$ be the constants in (6.6), and put $a_p = 2/b_p$. Applying Lemma 6.5, changing variables according to $t = (a_p + 2)u$, and using self-adjointness of the semigroup, we obtain

$$\begin{aligned}\big|\mathrm{Cov}(T_1, T_2)\big| &\leq (a_p + 2)\sum_{j=1}^{n} \int_0^\infty \big|\mathrm{tr}\big((e^{-a_p u\mathcal{L}}e^{-u\mathcal{L}}X_j)^* e^{-u\mathcal{L}}Y_j\big)\big|\,\mathrm{d}u \\ &\leq (a_p + 2)B_p \sum_{j=1}^{n} \int_0^\infty e^{-2u}\|e^{-u\mathcal{L}}X_j\|_{L_p}\|e^{-u\mathcal{L}}Y_j\|_{L_q}\,\mathrm{d}u.\end{aligned}$$

Indeed, the second inequality follows from Hölder's inequality and (6.6), applied to the trace-zero operator $e^{-u\mathcal{L}}X_j$ for the time $a_p u$.

Define

$$p(u) = 1 + (p-1)e^{-2u}, \qquad q(u) = 1 + (q-1)e^{-2u}.$$

Hypercontractivity gives

$$\|e^{-u\mathcal{L}}X_j\|_{L_p} \le \|X_j\|_{L_{p(u)}}, \qquad \|e^{-u\mathcal{L}}Y_j\|_{L_q} \le \|Y_j\|_{L_{q(u)}}.$$

Interpolation between L_1 and L_p, and between L_1 and L_q, respectively, yields

$$\|X_j\|_{L_{p(u)}} \le \|X_j\|_{L_1}^{\theta_p(u)} \|X_j\|_{L_p}^{1-\theta_p(u)},$$
$$\|Y_j\|_{L_{q(u)}} \le \|Y_j\|_{L_1}^{\theta_q(u)} \|Y_j\|_{L_q}^{1-\theta_q(u)},$$

where

$$\frac{1}{p(u)} = \theta_p(u) + \frac{1-\theta_p(u)}{p}, \qquad \frac{1}{q(u)} = \theta_q(u) + \frac{1-\theta_q(u)}{q}.$$

Fix a summand for which X_j and Y_j are nonzero, and write

$$x_j = \frac{\|X_j\|_{L_1}}{\|X_j\|_{L_p}}, \qquad y_j = \frac{\|Y_j\|_{L_1}}{\|Y_j\|_{L_q}}.$$

Since the trace is normalized, $0 < x_j, y_j \le 1$. The preceding estimates and the Cauchy–Schwarz inequality reduce the remaining integral to $A_j K_j$, where

$$A_j^2 = \int_0^\infty e^{-2u} x_j^{2\theta_p(u)} \,\mathrm{d}u, \qquad K_j^2 = \int_0^\infty e^{-2u} y_j^{2\theta_q(u)} \,\mathrm{d}u.$$

Changing variables from u to $\nu = p(u)$ gives

$$A_j^2 = \frac{1}{2(p-1)} \int_1^p x_j^{2\theta_p(\nu)} \,\mathrm{d}\nu, \qquad \theta_p(\nu) = \frac{p-\nu}{\nu(p-1)}.$$

A second change of variables, $s = 2\theta_p(\nu)$, yields

$$A_j^2 \le \frac{p}{4} \int_0^2 x_j^s \,\mathrm{d}s \le \left(\frac{p}{2}\right) \frac{1}{1+\log(1/x_j)}.$$

Here the last inequality follows from the elementary estimate

$$\int_0^2 x^s \,\mathrm{d}s = \frac{1-x^2}{\log(1/x)} \le \frac{2}{1+\log(1/x)}, \quad 0 < x \le 1,$$

with the value at $x = 1$ understood by continuity. The same argument gives

$$K_j^2 \le \left(\frac{q}{2}\right) \frac{1}{1+\log(1/y_j)}.$$

Substitution proves (6.10), for example with

$$C_p = \frac{(a_p+2)B_p\sqrt{pq}}{2}.$$

□

The preceding theorem does not yet record the location of the Fourier support. In L_2, the exact spectral decay on $\mathcal{T}^{\ge d}(\mathbb{M}_{2^n})$ can be inserted before the hypercontractive step. The next theorem provides a *covariance Talagrand L_1-L_2 inequality for tail spaces*, which captures the decay property depending the lower Fourier level.

THEOREM 6.10. *For each $n \in \mathbb{N}$, $d \in [n]$, and T_1, $T_2 \in \mathbb{M}_{2^n}$, we have*

$$|\mathrm{Cov}(T_1, T_2)| \le 4 \sum_{j=1}^n \frac{\|\mathbf{d}_j(T_1)\|_{L_2} \|\mathbf{d}_j(T_2)\|_{L_2}}{\sqrt{\left(d + \log\left(\frac{\|\mathbf{d}_j(T_1)\|_{L_2}}{\|\mathbf{d}_j(T_1)\|_{L_1}}\right)\right)\left(d + \log\left(\frac{\|\mathbf{d}_j(T_2)\|_{L_2}}{\|\mathbf{d}_j(T_2)\|_{L_1}}\right)\right)}} \tag{6.11}$$

PROOF. Set $X_j = \mathbf{d}_j(T_1)$ and $Y_j = \mathbf{d}_j(T_2)$. By Lemma 6.5, the change of variables $t = 4u$, and self-adjointness of the semigroup give

$$\begin{aligned}\bigl|\mathrm{Cov}(T_1,T_2)\bigr| &\le 4\sum_{j=1}^{n}\int_0^\infty \bigl|\mathrm{tr}\bigl((e^{-3u\mathcal{L}}X_j)^* e^{-u\mathcal{L}}Y_j\bigr)\bigr|\,\mathrm{d}u \\ &\le 4\sum_{j=1}^{n}\int_0^\infty e^{-2du}\|e^{-u\mathcal{L}}X_j\|_{L_2}\|e^{-u\mathcal{L}}Y_j\|_{L_2}\,\mathrm{d}u.\end{aligned}$$

In the second line we used Lemma 6.2 in the form

$$\|e^{-3u\mathcal{L}}X_j\|_{L_2} = \|e^{-2u\mathcal{L}}(e^{-u\mathcal{L}}X_j)\|_{L_2} \le e^{-2du}\|e^{-u\mathcal{L}}X_j\|_{L_2}.$$

Hypercontractivity and interpolation between $L_1(\mathbb{M}_{2^n})$ and $L_2(\mathbb{M}_{2^n})$ imply

$$\|e^{-u\mathcal{L}}X_j\|_{L_2} \le \|X_j\|_{L_{1+e^{-2u}}} \le \|X_j\|_{L_2}\left(\frac{\|X_j\|_{L_1}}{\|X_j\|_{L_2}}\right)^{\tanh(u)},$$

and the same estimate holds for Y_j. For a nonzero derivative, put

$$\lambda_{1,j} = \log\left(\frac{\|X_j\|_{L_2}}{\|X_j\|_{L_1}}\right), \qquad \lambda_{2,j} = \log\left(\frac{\|Y_j\|_{L_2}}{\|Y_j\|_{L_1}}\right).$$

The Cauchy–Schwarz inequality therefore gives

$$\bigl|\mathrm{Cov}(T_1,T_2)\bigr| \le 4\sum_{j=1}^{n}\|X_j\|_{L_2}\|Y_j\|_{L_2} I_{1,j}^{1/2} I_{2,j}^{1/2},$$

where

$$I_{i,j} = \int_0^\infty e^{-2du-2\lambda_{i,j}\tanh(u)}\,\mathrm{d}u, \qquad i \in \{1,2\}.$$

Recall that Lemma 6.3 yields

$$I_{i,j} \le \frac{1}{d+2\lambda_{i,j}} \le \frac{1}{d+\lambda_{i,j}}.$$

Substituting these bounds yields (6.11). □

Taking the two observables to be equal gives the following tail Talagrand L_1-L_2 inequality.

COROLLARY 6.11. *For each $d \in [n]$ and $T \in \mathbb{M}_{2^n}$, we have*

$$\|T\|_{L_2}^2 \le 4\sum_{j=1}^{n}\frac{\|\mathbf{d}_j(T)\|_{L_2}^2}{d+\log\left(\|\mathbf{d}_j(T)\|_{L_2}/\|\mathbf{d}_j(T)\|_{L_1}\right)}.$$

We conclude this section with the following *covariance Talagrand L_1-L_p inequality for tail spaces*, which is the most general covariance inequality established in this chapter.

THEOREM 6.12. *Let $1 < p < \infty$, let $q = p/(p-1)$, and set $r = \max\{p,q\}$. Let $A_r \ge 1$ be the exponent in Theorem 4.1, and put $\alpha_r = A_r^{-1}$. There exists a constant $C_p > 0$, depending only on p, such that*

$$\bigl|\mathrm{Cov}(T_1,T_2)\bigr| \le C_p\sum_{j=1}^{n}\frac{\|\mathbf{d}_j(T_1)\|_{L_p}\|\mathbf{d}_j(T_2)\|_{L_q}}{\sqrt{\left(d^{\alpha_r}+\log\left(\frac{\|\mathbf{d}_j(T_1)\|_{L_p}}{\|\mathbf{d}_j(T_1)\|_{L_1}}\right)\right)\left(d^{\alpha_r}+\log\left(\frac{\|\mathbf{d}_j(T_2)\|_{L_q}}{\|\mathbf{d}_j(T_2)\|_{L_1}}\right)\right)}}$$

for all $d \in [n]$ and all T_1, $T_2 \in \mathcal{T}^{\ge d}(\mathbb{M}_{2^n})$.

PROOF. By the symmetry of conjugation, it is enough to treat the case $p = r \geq 2$ and $1 < q \leq 2$. Let $C_p^{(0)}, c_p > 0$ and $A_p \geq 1$ be constants for which Theorem 4.1 holds. Set $X_j = \mathbf{d}_j(T_1)$ and $Y_j = \mathbf{d}_j(T_2)$. After the change of variables $t = 3u$, Lemma 6.5 and self-adjointness of the semigroup yield

$$\big|\mathrm{Cov}(T_1, T_2)\big| \leq 3 \sum_{j=1}^{n} \int_0^\infty \|e^{-2u\mathcal{L}} X_j\|_{L_p} \|e^{-u\mathcal{L}} Y_j\|_{L_q} \, \mathrm{d}u.$$

Since $e^{-u\mathcal{L}}(X_j) \in \mathcal{T}^{\geq d}$, Theorem 4.1, applied for the time u, gives

$$\|e^{-2u\mathcal{L}} X_j\|_{L_p} \leq C_p^{(0)} e^{-c_p d \min\{u, u^{A_p}\}} \|e^{-u\mathcal{L}} X_j\|_{L_p}.$$

Consequently,

$$\big|\mathrm{Cov}(T_1, T_2)\big| \leq C_p^{(1)} \sum_{j=1}^{n} \|X_j\|_{L_p} \|Y_j\|_{L_q} \int_0^\infty K_d(u) x_j^{\theta_p(u)} y_j^{\theta_q(u)} \, \mathrm{d}u, \tag{6.12}$$

where

$$K_d(u) = e^{-c_p d \min\{u, u^{A_p}\}}, \qquad x_j = \frac{\|X_j\|_{L_1}}{\|X_j\|_{L_p}}, \qquad y_j = \frac{\|Y_j\|_{L_1}}{\|Y_j\|_{L_q}},$$

and

$$\theta_s(u) = \frac{1 - e^{-2u}}{1 + (s-1)e^{-2u}}, \qquad s \in \{p, q\}.$$

Here we used the same hypercontractive and interpolation estimates as in the proof of Theorem 6.9.

We now claim the following estimate needed to bound the integral in (6.12). Fix $1 < s < \infty$, $A \geq 1$, and $c > 0$. There exists $D_{s,A,c} > 0$ such that, for every $d \geq 1$ and $\lambda \geq 0$,

$$\int_0^\infty e^{-cd \min\{u, u^A\} - 2\lambda\theta_s(u)} \, \mathrm{d}u \leq \frac{D_{s,A,c}}{d^{1/A} + \lambda}. \tag{6.13}$$

Indeed, with $\kappa_s = (1 - e^{-2})/s$, one has

$$\theta_s(u) \geq \kappa_s \min\{u, 1\}, \qquad u \geq 0.$$

Writing $R = d^{1/A}$ and splitting the integral at $u = 1$, we obtain

$$\int_0^1 e^{-cdu^A - 2\kappa_s\lambda u} \, \mathrm{d}u \leq \min\left\{C_{A,c} R^{-1}, (2\kappa_s\lambda)^{-1}\right\} \leq \frac{C_{s,A,c}}{R + \lambda},$$

where the second term in the minimum is omitted when $\lambda = 0$. Moreover, by

$$\int_1^\infty e^{-cdu - 2\kappa_s\lambda} \, \mathrm{d}u \leq \frac{C_{s,c}}{d(1+\lambda)} \leq \frac{C_{s,c}}{R + \lambda}.$$

the desired inequality (6.13) follows.

Put

$$\lambda_{1,j} \coloneqq \log(1/x_j), \qquad \lambda_{2,j} \coloneqq \log(1/y_j).$$

The Cauchy–Schwarz inequality and (6.13), first with $s = p$ and then with $s = q$, give

$$\begin{aligned}
&\int_0^\infty K_d(u) x_j^{\theta_p(u)} y_j^{\theta_q(u)} \, \mathrm{d}u \\
&\leq \left(\int_0^\infty K_d(u) x_j^{2\theta_p(u)} \, \mathrm{d}u \right)^{1/2} \left(\int_0^\infty K_d(u) y_j^{2\theta_q(u)} \, \mathrm{d}u \right)^{1/2} \\
&\leq \frac{C_p}{\sqrt{(d^{\alpha_p} + \lambda_{1,j})\,(d^{\alpha_p} + \lambda_{2,j})}}.
\end{aligned}$$

Substituting this estimate into (6.12) proves the desired inequality. If $p < 2$, the same argument is applied after interchanging the two observables; then $r = q$ and the exponent is $\alpha_r = A_r^{-1}$. □

Remark 6.13. Theorems 6.10 and 6.12 distinguish two features of tail covariance estimates. In L_2, spectral orthogonality produces the exact lower-level parameter d; for general conjugate exponents, the verified dimension-free gain is governed by the heat-smoothing scale d^{α_r}.

CHAPTER 7

Quantum correlation inequalities with lower bound

A function $f : \{-1,1\}^n \to \mathbb{R}$ is said monotone increasing if $f(x) \le f(y)$ whenever $x = (x_j)_{j=1}^n \le y = (y_j)_{j=1}^n$, $x, y \in \{-1,1\}^n$. Here $x \le y$ means that $x_j \le y_j$ for all $j \in [n]$. Recall that $f : \{-1,1\}^n \to \mathbb{R}$ is motonone increasing if and only if

$$w_{\{j\}} D_j(f) \ge 0, \quad \forall j \in [n]. \tag{7.1}$$

The remarkable Harris-Kleitman correlation inequality ([**Har60, Kle66**]) states that for monotone increasing functions $f, g : \{-1,1\}^n \to \{-1,1\}$, their correlation is always positive, that is,

$$\mathrm{Cov}(f,g) := \mathbb{E}(fg) - \mathbb{E}(f)\mathbb{E}(g) \ge 0. \tag{7.2}$$

Talagrand [**Tal96**] improved the Harris–Kleitman inequality via establishing the following quantitative inequality: for monotone increasing functions $f, g : \{-1,1\}^n \to \{-1,1\}$, there exists an absolute constant $C > 0$ such that

$$\mathrm{Cov}(f,g) := \mathbb{E}(fg) - \mathbb{E}(f)\mathbb{E}(g) \ge C\Psi\Big(\sum_{j\in[n]} \|D_j(f)\|_{L_1} \|D_j(g)\|_{L_1} \Big), \tag{7.3}$$

where $\Psi(t) = t/\log(\frac{e}{t})$. Talagrand's method and results inspired a number of important works in combinatorics and probability theory. An alternative correlation lower bound was recently proved by Keller, Mossel, and Sen in [**KMS14**]. For monotone increasing functions $f, g : \{-1,1\}^n \to \{-1,1\}$, there exists an absolute constant $K > 0$ such that

$$\mathrm{Cov}(f,g) \ge K \sum_{j=1}^n \frac{\|D_j(f)\|_{L_1} \|D_j(g)\|_{L_1}}{\sqrt{\log(e/\|D_j(f)\|_{L_1}) \log(e/\|D_j(g)\|_{L_1})}}. \tag{7.4}$$

Note that the lower bounds in (7.3) and (7.4) are uncomparable. To combine the advantages of (7.3) and (7.4), Kalai, Keller and Mossel [**KKM16**] studied stronger correlation lower bounds for monotone increasing functions under additional natural regularity or symmetry conditions. Recently, the famous Chvátal's conjecture was found by Friedgut et al. [**FKKK18**] to be equivalent to a certain strengthening of Talagrand's correlation inequality. Motivated by this conjecture, Eldan [**Eld22**] improved (7.3) and (7.4) when one of the monotone increasing functions is so-called "antipodal". We also refer the reader to [**CC25**] for recent developments on this problem.

In this chapter, we aim to extend the above results (7.2), (7.3), and (7.4) to the quantum Boolean cube setting. The main results of this chapter are Theorem 7.5, Theorem 7.10 and Theorem 7.17.

7.1. Quantum monotonically increasing function and Harris-Kleitman inequality

We first introduce the definition of monotone elements in $\mathbb{M}_{2^n}$. To this end, for each $j \in [n]$ and $\ell \in \{1,2,3\}$, we define

$$\mathbf{d}_{j,\ell}(T) \coloneqq \sum_{\substack{\mathbf{s}\in\{0,1,2,3\}^n \\ s_j=\ell}} \widehat{T}(\mathbf{s})\sigma_{\mathbf{s}},$$

$$\partial_{j,\ell}(T) \coloneqq \sigma_{\ell e_j}\mathbf{d}_{j,\ell}(T),$$

and

$$\partial_j(T) \coloneqq \sum_{\ell=1}^{3} \partial_{j,\ell}(T).$$

Here, $e_j = (0,0,\cdots,1,0,0)$, and 1 only appears in the j-th position. Similarly, $\ell e_j = (0,0,\cdots,\ell,0,0)$ and ℓ only appears in the j-th position.

Definition 7.1. An element $T \in \mathbb{M}_{2^n}$ is said to be monotonically increasing if

$$\partial_{j,\ell}(T) \geq 0, \quad \text{for each } j \in [n], \ell \in \{1,2,3\}.$$

This definition is of course motivated by (7.1). We here provide a basic example of monotonically increasing elements in $\mathbb{M}_{2^n}$.

Example 7.2. Let $T \in \mathbb{M}_{2^n}$ and assume that the Fourier coefficients of T satisfy $\widehat{T}(\mathbf{s}) \in \mathbb{R}$ and

$$\widehat{T}(\ell e_j) \geq \sum_{\ell\in\{1,2,3\}} \sum_{\substack{\mathbf{s}_j=\ell \\ |\mathbf{s}|\geq 2}} |\widehat{T}(\mathbf{s})|.$$

Then T is monotonically increasing.

Proof. According to the choice of T, it follows that T is self-adjoint. For each $j \in [n]$ and $\ell \in \{1,2,3\}$, to show $\partial_{j,\ell}(T) \geq 0$, it suffices to verify that $\mathrm{tr}[p\cdot\partial_{j,\ell}(T)] \geq 0$ for each projection $p \in \mathcal{P}(\mathbb{M}_{2^n})$. Indeed, we have

$$\begin{aligned}
\mathrm{tr}[p\cdot\partial_{j,\ell}(T)] &= \sum_{\mathbf{s}_j=\ell} \widehat{T}(\mathbf{s})\mathrm{tr}(p\cdot\sigma_{\ell e_j}\sigma_{\mathbf{s}}) \\
&= \widehat{T}(\ell e_j)\mathrm{tr}(p) + \sum_{\substack{\mathbf{s}_j=\ell \\ |\mathbf{s}|\geq 2}} \widehat{T}(\mathbf{s})\mathrm{tr}(p\cdot\sigma_{\ell e_j}\sigma_{\mathbf{s}}) \\
&\geq \widehat{T}(\ell e_j)\mathrm{tr}(p) - \sum_{\substack{\mathbf{s}_j=\ell \\ |\mathbf{s}|\geq 2}} |\widehat{T}(\mathbf{s})|\cdot|\mathrm{tr}(p\cdot\sigma_{\ell e_j}\sigma_{\mathbf{s}})| \\
&\geq \big(\widehat{T}(\ell e_j) - \sum_{\substack{\mathbf{s}_j=\ell \\ |\mathbf{s}|\geq 2}} |\widehat{T}(\mathbf{s})|\big)\mathrm{tr}(p) \geq 0,
\end{aligned}$$

where we used the fact $|\mathrm{tr}(p\cdot\sigma_{\ell e_j}\sigma_{\mathbf{s}})| \leq \mathrm{tr}(p)$ and the assumption. The desired assertion follows. □

We collect useful properties of the partial differential operators $\partial_{j,\ell}$ and ∂_j.

Lemma 7.3. *Let $T_1, T_2 \in \mathbb{M}_{2^n}$.*

(i) *For each $j \in [n]$ and $1 \leq p \leq \infty$, $\|\partial_j(T)\|_{L_p} \leq 3\|\mathbf{d}_j(T)\|_{L_p}$.*

(ii) *For all $j \in [n]$ and $\ell \in \{1,2,3\}$, we have*

$$\sigma_{\ell e_j} e^{-t\mathcal{L}} \mathbf{d}_{j,\ell}(T) = \partial_{j,\ell}\left(e^{-t\mathcal{L}}(T)\right) = e^{-t} e^{-t\mathcal{L}}\left(\partial_{j,\ell}(T)\right).$$

(iii) *We always have the following covariance identity*

$$\mathrm{Cov}(T_1, T_2) = \sum_{j=1}^{n}\sum_{\ell=1}^{3}\int_0^\infty e^{-t}\mathrm{tr}\left(\partial_{j,\ell}(T_1)^* \cdot e^{-t\mathcal{L}}\left(\partial_{j,\ell}(T_2)\right)\right) dt. \tag{7.5}$$

Proof. (i) Fix $j \in [n]$. It is clear that

$$\|\mathbf{d}_{j,\ell}(T)\|_{L_p} = \left\|\mathcal{E}_{\mathbb{M}_{2^n}\setminus\{\sigma_{me_j}: m\neq\ell\}}\mathbf{d}_j(T)\right\|_{L_p} \leq \|\mathbf{d}_j(T)\|_{L_p} \qquad \ell = \{1,2,3\}.$$

The definition of conditional expectation is referred to Chapter 8.1. By the contractivity of conditional expectations on L_p, we have

$$\|\partial_j(T)\|_{L_p} \leq \sum_{\ell=1}^{3}\|\sigma_{\ell e_j}\mathbf{d}_{j,\ell}(T)\|_{L_p} \leq 3\|\mathbf{d}_j(T)\|_{L_p}.$$

(ii) The first equality is obvious. We here only show the second equality. Fix $j \in [n]$ and $\ell \in \{1,2,3\}$. According to the definitions, we write

$$\partial_{j,\ell}\left(e^{-t\mathcal{L}}(T)\right) = \partial_{j,\ell}\left(\sum_{\mathbf{s}\in\{0,1,2,3\}^n}\widehat{T}(\mathbf{s})e^{-t|\mathbf{s}|}\sigma_{\mathbf{s}}\right) = \sum_{\mathbf{s}_j=\ell}\widehat{T}(\mathbf{s})e^{-t|\mathbf{s}|}\sigma_{\mathbf{s}\ominus\ell e_j} \tag{7.6}$$

and

$$e^{-t\mathcal{L}}\left(\partial_{j,\ell}(T)\right) = \sum_{\mathbf{s}_j=\ell}\widehat{T}(\mathbf{s})e^{-t(|\mathbf{s}|-1)}\sigma_{\mathbf{s}\ominus\ell e_j}. \tag{7.7}$$

Comparing (7.6) and (7.7) yields the desired result.

(iii) Combining Lemma 6.5 and item (ii) above, we have

$$\begin{aligned}
\mathrm{Cov}(T_1,T_2) &= \sum_{j=1}^{n}\int_0^\infty \mathrm{tr}\left(\mathbf{d}_j(T_1)^* \cdot \mathbf{d}_j(e^{-t\mathcal{L}}(T_2))\right) dt \\
&= \sum_{j=1}^{n}\sum_{\ell=1}^{3}\int_0^\infty \mathrm{tr}\left(\mathbf{d}_{j,\ell}(T_1)^* \cdot \mathbf{d}_{j,\ell}\left(e^{-t\mathcal{L}}(T_2)\right)\right) dt \\
&= \sum_{j=1}^{n}\sum_{\ell=1}^{3}\int_0^\infty \mathrm{tr}\left(\partial_{j,\ell}(T_1)^* \cdot \partial_{j,\ell}\left(e^{-t\mathcal{L}}(T_2)\right)\right) dt \\
&= \sum_{j=1}^{n}\sum_{\ell=1}^{3}\int_0^\infty e^{-t}\mathrm{tr}\left(\partial_{j,\ell}(T_1)^* \cdot e^{-t\mathcal{L}}\left(\partial_{j,\ell}(T_2)\right)\right) dt.
\end{aligned}$$

The proof is complete. □

Useful properties of monotonically increasing elements in $\mathbb{M}_{2^n}$ are stated below.

Lemma 7.4. *Let $T \in \mathbb{M}_{2^n}$ be monotonically increasing.*

(i) *For each $j \in [n]$, $\|\mathbf{d}_j(T)\|_{L_1} \leq \|\partial_j(T)\|_{L_1}$.*

(ii) *For each $j \in [n]$ and $\ell \in \{1,2,3\}$,*

$$\|\mathbf{d}_{j,\ell}(T)\|_{L_1} = \widehat{T}(\ell e_j).$$

(iii) *For each* $j \in [n]$ *and* $\ell \in \{1,2,3\}$,

$$\partial_{j,\ell}\left(e^{-t\mathcal{L}}(T)\right) \geq 0,$$

that is, $e^{-t\mathcal{L}}(T)$ *is also monotonically increasing.*

PROOF. (i) Since T is monotonically increasing, it follows that $\partial_{j,\ell}(T) \geq 0$ for all $j \in [n]$ and $\ell \in \{1,2,3\}$. Then, for each $j \in [n]$, we have

$$\begin{aligned}\|\partial_j(T)\|_{L_1} &= \sum_{\ell=1}^{3} \operatorname{tr}\left(\partial_{j,\ell}(T)\right) = \sum_{\ell=1}^{3} \|\mathbf{d}_{j,\ell}(T)\|_{L_1} \\ &\geq \left\| \sum_{\ell=1}^{3} \mathbf{d}_{j,\ell}(T) \right\|_{L_1} = \|\mathbf{d}_j(T)\|_{L_1}.\end{aligned}$$

(ii) Since $\|\cdot\|_{L_1}$ is unitary variant, for each $j \in [n]$ and $\ell \in \{1,2,3\}$, we have

$$\begin{aligned}\|\mathbf{d}_{j,\ell}(T)\|_{L_1} &= \|\sigma_{\ell e_j}\mathbf{d}_{j,\ell}(T)\|_{L_1} = \|\partial_{j,\ell}(T)\|_{L_1} \\ &= \operatorname{tr}(\partial_{j,\ell}(T)) = \widehat{T}(\ell e_j),\end{aligned}$$

where we used the fact that $T \in \mathbb{M}_{2^n}$ is monotonically increasing in the third equality, and the last equality is due to orthogonality of the quantum Walsh series.

(iii) By Lemma 7.3 (ii) and the fact that $e^{-t\mathcal{L}}$ is positive preserving, it follows that

$$\partial_{j,\ell}\left(e^{-t\mathcal{L}}(T)\right) = e^{-t}e^{-t\mathcal{L}}\left(\partial_{j,\ell}(T)\right) \geq 0, \quad \forall j \in [n], \ell \in \{1,2,3\}.$$

□

Now we are ready to prove quantum version of the Harris–Kleitman inequality (7.2).

THEOREM 7.5. *Let* $T_1, T_2 \in \mathbb{M}_{2^n}$ *be monotonically increasing operators. Then*

$$\operatorname{Cov}(T_1, T_2) \geq 0.$$

PROOF. For $t > 0$, set $h(t) = \operatorname{tr}(e^{-t\mathcal{L}}(T_1)^* T_2)$. Then we find that

$$h(0) = \operatorname{tr}(T_1^* T_2) \quad \text{and} \quad h(\infty) = \operatorname{tr}(T_1)^* \operatorname{tr}(T_2).$$

By the definition of $h(t)$ and Lemma 7.4 (iii), we get

$$\begin{aligned}h'(t) &= -\operatorname{tr}((\mathcal{L}e^{-t\mathcal{L}}(T_1))^* T_2) \\ &= -\sum_{j=1}^{n} \operatorname{tr}(e^{-t\mathcal{L}}\mathbf{d}_j(T_1)^*\mathbf{d}_j(T_2)) \\ &= -\sum_{j=1}^{n}\sum_{\ell=1}^{3} \operatorname{tr}(e^{-t\mathcal{L}}\mathbf{d}_{j,\ell}(T_1)^*\mathbf{d}_{j,\ell}(T_2)) \\ &= -\sum_{j=1}^{n}\sum_{\ell=1}^{3} \operatorname{tr}((\sigma_{\ell e_j}e^{-t\mathcal{L}}\mathbf{d}_{j,\ell}(T_1))^*\mathbf{d}_{j,\ell}(T_2)\sigma_{\ell e_j}) \\ &= -\sum_{j=1}^{n}\sum_{\ell=1}^{3} e^{-t}\operatorname{tr}\Big(e^{-t\mathcal{L}}(\partial_{j,\ell}(T_1))^*\partial_{j,\ell}(T_2)\Big) \\ &\leq 0,\end{aligned}$$

where the last equality is due to Lemma 7.3 (ii). Hence, the function h is decreasing, and consequently, we have

$$\mathrm{Cov}(T_1, T_2) = h(0) - h(\infty) \geq 0.$$

The proof is complete. □

7.2. The Talagrand correlation inequality

We next aim to establish quantum version of the Talagrand quantitative correlation inequality (7.3). Given $T_1, T_2 \in \mathbb{M}_{2^n}$, $d \in [n]$ and $J \subset [n]$, we set

$$\mathcal{W}_d(T_1, T_2) = \sum_{|\mathbf{s}|=d} \widehat{T_1}(\mathbf{s})\widehat{T_2}(\mathbf{s}),$$

and

$$\mathcal{W}_{1,J}(T_1, T_2) = \sum_{j\in J}\sum_{\ell=1}^{3} \widehat{T_1}(\ell e_j)\widehat{T_2}(\ell e_j).$$

If $T_1 = T_2 = T$, the preceding symbols will be abbreviated as $\mathcal{W}_d(T)$, $\mathcal{W}_{1,J}(T)$, respectively.

For each T_1, $T_2 \in \mathbb{M}_{2^n}$ with $\max\{\|T_1\|_{L_\infty}, \|T_2\|_{L_\infty}\} \leq 1$, it is clear that $|\mathcal{W}_d(T_1, T_2)| \leq 1$ for each $d \in [n]$. Indeed, by the Cauchy-Schwarz inequality we have

$$\begin{aligned} |\mathcal{W}_d(T_1, T_2)| &= \left| \sum_{|\mathbf{s}|=d} \widehat{T_1}(\mathbf{s})\widehat{T_2}(\mathbf{s}) \right| \\ &\leq \Big(\sum_{|\mathbf{s}|=d} |\widehat{T_1}(\mathbf{s})|^2 \Big)^{1/2} \Big(\sum_{|\mathbf{s}|=d} |\widehat{T_2}(\mathbf{s})|^2 \Big)^{1/2} \\ &\leq \|T_1\|_{L_2}\|T_2\|_{L_2} \leq \|T_1\|_{L_\infty}\|T_2\|_{L_\infty} \leq 1. \end{aligned} \tag{7.8}$$

Next, we will establish Talagrand's lower bound correlation inequality for the covariance between monotonically increasing elements $T_1, T_2 \in \mathbb{M}_{2^n}$. Due to the following counterexample, the lower bound we obtain is of the $\mathcal{W}_1(T_1, T_2)$-form rather than $M(T_1, T_2)$-form lower bound, where

$$M(T_1, T_2) := \sum_{j=1}^{n} \|\mathbf{d}_j(T_1)\|_{L_1}\|\mathbf{d}_j(T_2)\|_{L_1}.$$

Example 7.6. Let $T_1 = \sum_{j=1}^n \sigma_{e_j}$ and $T_2 = \sum_{j=1}^n \sigma_{2e_j}$. Then we have $\mathrm{Cov}(T_1, T_2) = 0$, but $M(T_1, T_2) > 0$.

Proof. It is clear that both T_1 and T_2 are monotonically increasing. By the definition of the covariance and for any $i \in \{1, 2, 3\}$, $\mathrm{tr}(\sigma_i) = 0$, we have

$$\mathrm{Cov}(T_1, T_2) = \mathrm{tr}(T_1T_2) - \mathrm{tr}(T_1)\mathrm{tr}(T_2) = 0.$$

In addition,

$$M(T_1, T_2) = \sum_{j=1}^{n} \|d_j(T_1)\|_{L_1}\|d_j(T_2)\|_{L_1} = \sum_{j=1}^{n} \|\sigma_{e_j}\|_{L_1}\|\sigma_{2e_j}\|_{L_1} = n > 0.$$

Therefore, we cannot obtain the following correlation inequality

$$\mathrm{Cov}(T_1, T_2) \geq cM(T_1, T_2) \cdot \left(\log \left(\frac{e}{M(T_1, T_2)} \right) \right)^{-1}.$$

□

For each $j \in [n]$ and any $f : \{-1, 1\}^n \to \mathbb{R}$, the classical framework involves only a single Fourier coefficient $\widehat{f}(\{j\})$. For each $j \in [n]$ and $T \in \mathbb{M}_{2^n}$, whereas the noncommutative setting yields three first-order Fourier coefficients $\widehat{T}(\ell e_j), \ell \in \{1, 2, 3\}$. This fundamental difference directly leads to the fact that there is no inherent relation between $M(T_1, T_2)$ and $\mathcal{W}_1(T_1, T_2)$, when T_1, T_2 are monotone increasing matrices. Motivated by the role of structural conditions in classical inequalities [**KKM16**, Definition 1.6], we introduce the following similarity condition in the noncommutative setting.

Definition 7.7. Let $T_1, T_2 \in \mathbb{M}_{2^n}$ be monotonically increasing. We say T_1, T_2 satisfy the similarity condition if for each $j \in [n]$, there exists a constant $c_j \in (0, \infty)$ such that

$$\frac{\widehat{T_1}(e_j)}{\widehat{T_2}(e_j)} = \frac{\widehat{T_1}(2e_j)}{\widehat{T_2}(2e_j)} = \frac{\widehat{T_1}(3e_j)}{\widehat{T_2}(3e_j)} = c_j. \tag{7.9}$$

Remark 7.8. The above similarity condition holds automatically in the classical setting. Indeed, for each $j \in [n]$ and any functions $f, g : \{-1, 1\}^n \to \mathbb{R}$, the pointwise Fourier coefficient reduces to a single term, and it suffices to set $c_j = \frac{\widehat{f}(\{j\})}{\widehat{g}(\{j\})}$, which in effect means that no additional condition is imposed.

Remark 7.9. If we replace the above condition (7.9) with the following weak similarity condition

$$\max_{\ell \in \{1,2,3\}} \left\{ \frac{\widehat{T_1}(\ell e_j)}{\widehat{T_2}(\ell e_j)} \right\} \leq c_j \min_{\ell \in \{1,2,3\}} \left\{ \frac{\widehat{T_1}(\ell e_j)}{\widehat{T_2}(\ell e_j)} \right\}, \quad c_j > 0, \tag{7.10}$$

all the corresponding results (like Theorem 7.10, Proposition 7.14 and Proposition 7.16) remain valid.

This section is devoted to establishing the following *quantum Talagrand correlation inequality*. Note that it follows from Lemma 7.4 (ii) that

$$\mathcal{W}_1(T_1, T_2) = \sum_{j=1}^{n} \sum_{\ell=1}^{3} \|\mathbf{d}_{j,\ell}(T_1)\|_{L_1} \|\mathbf{d}_{j,\ell}(T_2)\|_{L_1} \tag{7.11}$$

provided both $T_1, T_2 \in \mathbb{M}_{2^n}$ are monotonically increasing.

Theorem 7.10 (quantum Talagrand correlation inequality). *Suppose that monotonically increasing elements $T_1, T_2 \in \mathbb{M}_{2^n}$ satisfy condition* (7.9) *and*

$$\max\{\|T_1\|_{L_\infty}, \|T_2\|_{L_\infty}\} \leq 1.$$

Then exists a universal constant $K > 0$ such that

$$\mathrm{Cov}(T_1, T_2) \geq K \frac{\mathcal{W}_1(T_1, T_2)}{\log\left(e/\mathcal{W}_1(T_1, T_2)\right)}. \tag{7.12}$$

Before proving Theorem 7.10, we need following auxiliary lemmas regarding the basic properties related to the partial differential operators and Rademacher projections (see Chapter 3).

LEMMA 7.11. *Let T_1, $T_2 \in \mathbb{M}_{2^n}$. For each $j \in [n]$ and $\ell \in \{1,2,3\}$, we have*

$$\langle \mathbf{d}_{j,\ell}(T_1), \mathbf{d}_{j,\ell}(T_2) \rangle = \langle \partial_{j,\ell}(T_1), \partial_{j,\ell}(T_2) \rangle ,$$

and consequently,

$$\langle \mathbf{d}_{j}(T_1), \mathbf{d}_{j}(T_2) \rangle = \sum_{\ell=1}^{3} \langle \partial_{j,\ell}(T_1), \partial_{j,\ell}(T_2) \rangle . \tag{7.13}$$

PROOF. We write

$$\begin{aligned} \langle \mathbf{d}_{j,\ell}(T_1), \mathbf{d}_{j,\ell}(T_2) \rangle &= \operatorname{tr} \left(\mathbf{d}_{j,\ell}(T_1)^* \mathbf{d}_{j,\ell}(T_2) \right) \\ &= \operatorname{tr} \left(\left(\sigma_{\ell e_j} \mathbf{d}_{j,\ell}(T_1) \right)^* \left(\sigma_{\ell e_j} \mathbf{d}_{j,\ell}(T_2) \right) \right) \\ &= \langle \partial_{j,\ell}(T_1), \partial_{j,\ell}(T_2) \rangle , \end{aligned}$$

which completes the proof. □

LEMMA 7.12. *If $T \in \mathbb{M}_{2^n}$ is monotonically increasing, then, for each $j \in [n]$ and $\ell \in \{1,2,3\}$, we have*

$$\partial_{j,\ell} \left(\mathrm{Rad}^{=1}(T) \right) = \| \partial_{j,\ell}(T) \|_{L_1} = \widehat{T}(\ell e_j).$$

PROOF. Fix $j \in [n]$ and $\ell \in \{1,2,3\}$. Since $T \in \mathbb{M}_{2^n}$ is monotonically increasing, it follows that

$$\| \partial_{j,\ell}(T) \|_{L_1} = \operatorname{tr}(\sigma_{\ell e_j} d_{j,\ell}(T)) = \sum_{\mathbf{s}:\mathbf{s}_j = \ell} \widehat{T}(\mathbf{s}) \operatorname{tr}(\sigma_{\ell e_j} \sigma_{\mathbf{s}}) = \widehat{T}(\ell e_j).$$

On the other hand, for a given $j \in [n]$ and $\ell \in \{1,2,3\}$, we note that

$$\partial_{j,\ell} \left(\mathrm{Rad}^{=1}(T) \right) = \sum_{|\mathbf{s}|=1} \widehat{T}(\mathbf{s}) \partial_{j,\ell}(\sigma_{\mathbf{s}}) = \widehat{T}(\ell e_j).$$

The desired result follows. □

LEMMA 7.13. *Suppose that $T_1, T_2 \in \mathbb{M}_{2^n}$ are monotonically increasing. Then the function*

$$H(\xi) \coloneqq \operatorname{Cov} \left(e^{-\xi \mathcal{L}}(T_1), T_2 \right), \quad \xi \geq 0$$

is decreasing.

PROOF. By Lemma 6.5, we have

$$\operatorname{Cov} \left(e^{-\xi \mathcal{L}}(T_1), T_2 \right) = \sum_{j=1}^{n} \int_0^\infty \left\langle \mathbf{d}_j e^{-\xi \mathcal{L}}(T_1), \mathbf{d}_j e^{-t \mathcal{L}}(T_2) \right\rangle dt.$$

Hence, by the fact that $\{\mathbf{d}_j\}_{j=1}^n$ are orthogonal projections (7.13) and Lemma 7.3 (ii), we have

$$\begin{aligned}
\frac{d}{d\xi}H(\xi) &= -\sum_{j=1}^{n}\int_0^\infty \left\langle \mathcal{L}\mathbf{d}_j e^{-\xi\mathcal{L}}(T_1), \mathbf{d}_j e^{-t\mathcal{L}}(T_2)\right\rangle dt \\
&= -\sum_{j=1}^{n}\int_0^\infty \left\langle \mathbf{d}_j e^{-\xi\mathcal{L}}(T_1), \mathbf{d}_j e^{-t\mathcal{L}}(T_2)\right\rangle dt \\
&= -\sum_{j=1}^{n}\sum_{\ell=1}^{3}\int_0^\infty \left\langle \mathbf{d}_{j,\ell} e^{-\xi\mathcal{L}}(T_1), \mathbf{d}_{j,\ell} e^{-t\mathcal{L}}(T_2)\right\rangle dt \qquad (7.14)\\
&= -\sum_{j=1}^{n}\sum_{\ell=1}^{3}\int_0^\infty \left\langle \partial_{j,\ell} e^{-\xi\mathcal{L}}(T_1), \partial_{j,\ell} e^{-t\mathcal{L}}(T_2)\right\rangle dt \\
&= -\sum_{j=1}^{n}\sum_{\ell=1}^{3}\int_0^\infty e^{-(\xi+t)}\left\langle e^{-\xi\mathcal{L}}\partial_{j,\ell}(T_1), e^{-t\mathcal{L}}\partial_{j,\ell}(T_2)\right\rangle dt \le 0,
\end{aligned}$$

where the last inequality is due to the fact that $T_1, T_2 \in \mathbb{M}_{2^n}$ are monotonically increasing and the semigroup $\{e^{-t\mathcal{L}}\}_{t\ge 0}$ is positive preserving. □

To prove Theorem 7.10, we need to prove the following proposition. We will provide its proof after we prove Theorem 7.10.

PROPOSITION 7.14. *For each $2 \le d \in [n]$, suppose that monotonically increasing elements T_1 and $T_2 \in \mathbb{M}_{2^n}$ satisfy condition* (7.9)*, $\max\{\|T_1\|_{L_\infty}, \|T_2\|_{L_\infty}\} \le 1$ and $\mathcal{W}_1(T_1,T_2) \le e^{-2(d-1)}$. Then we have*

$$|\mathcal{W}_d(T_1,T_2)| \le \frac{26e}{d}\Big(\frac{4e}{d-1}\Big)^{d-1}\mathcal{W}_1(T_1,T_2)\Big(\log(\frac{d}{\mathcal{W}_1(T_1,T_2)})\Big)^{d-1}.$$

We here provide the proof of Theorem 7.10.

PROOF OF THEOREM 7.10. Take monotonically increasing elements T_1, $T_2 \in \mathbb{M}_{2^n}$. Choose specific $\xi > 0$ such that

$$\rho := e^{-\xi} = \frac{\epsilon}{\log(e/\mathcal{W}_1(T_1,T_2))}$$

with $0 < \epsilon < \frac{1}{208e^2}$. According to Lemma 7.13,

$$\mathrm{Cov}(T_1,T_2) \ge \mathrm{Cov}\left(e^{-\xi\mathcal{L}}(T_1), T_2\right).$$

Hence, it suffices to estimate $\mathrm{Cov}\left(e^{-\xi\mathcal{L}}(T_1), T_2\right)$. By Lemma 6.5, we have

$$\begin{aligned}
\mathrm{Cov}\left(e^{-\xi\mathcal{L}}(T_1), T_2\right) &= \sum_{j=1}^{n}\int_0^\infty \left\langle \mathbf{d}_j e^{-\xi\mathcal{L}}(T_1), e^{-t\mathcal{L}}\mathbf{d}_j(T_2)\right\rangle dt \\
&= \sum_{j=1}^{n}\int_0^\infty \left\langle \mathbf{d}_j\left(e^{-\xi\mathcal{L}}\left(\mathrm{Rad}^{=1}(T_1)\right)\right), e^{-t\mathcal{L}}\left(\mathbf{d}_j(\mathrm{Rad}^{=1}(T_2))\right)\right\rangle dt \\
&\quad + \sum_{j=1}^{n}\int_0^\infty \left\langle \mathbf{d}_j\left(e^{-\xi\mathcal{L}}\left(\mathrm{Rad}^{\ge 2}(T_1)\right)\right), e^{-t\mathcal{L}}\left(\mathbf{d}_j(\mathrm{Rad}^{\ge 2}(T_2))\right)\right\rangle dt \\
&:= \mathbf{A} + \mathbf{B}.
\end{aligned}$$

We claim that

$$\mathbf{A} = \rho \mathcal{W}_1(T_1, T_2) \tag{7.15}$$

and

$$\mathbf{B} \leq \frac{\rho}{2} \mathcal{W}_1(T_1, T_2). \tag{7.16}$$

Once these claims have been proved, we deduce that

$$\mathrm{Cov}(T_1, T_2) \geq \mathrm{Cov}\left(e^{-\xi \mathcal{L}}(T_1), T_2\right) \geq \big(\frac{3\epsilon}{2}\big) \frac{\mathcal{W}_1(T_1, T_2)}{\log\Big(e/\mathcal{W}_1(T_1, T_2)\Big)},$$

which is the desired inequality.

We next prove the claims (7.15) and (7.16) respectively.

Estimate of A. For the term $\mathbf{A}$, by Lemma 7.11 and Lemma 7.12, we have

$$\begin{aligned}
\mathbf{A} &= \sum_{j=1}^{n} \int_0^{\infty} \left\langle \mathbf{d}_j \left(e^{-\xi \mathcal{L}} \left(\mathrm{Rad}^{=1}(T_1)\right)\right), e^{-t\mathcal{L}} \left(\mathbf{d}_j(\mathrm{Rad}^{=1}(T_2))\right)\right\rangle dt \\
&= \sum_{j=1}^{n} e^{-\xi} \left(\int_0^{\infty} e^{-t} dt\right) \left\langle \mathbf{d}_j \mathrm{Rad}^{=1}(T_1), \mathbf{d}_j \mathrm{Rad}^{=1}(T_2)\right\rangle \\
&= \sum_{j=1}^{n} \sum_{\ell=1}^{3} e^{-\xi} \left\langle \mathbf{d}_{j,\ell} \mathrm{Rad}^{=1}(T_1), \mathbf{d}_{j,\ell} \mathrm{Rad}^{=1}(T_2)\right\rangle \\
&= \sum_{j=1}^{n} \sum_{\ell=1}^{3} e^{-\xi} \left\langle \partial_{j,\ell} \mathrm{Rad}^{=1}(T_1), \partial_{j,\ell} \mathrm{Rad}^{=1}(T_2)\right\rangle \\
&= e^{-\xi} \mathcal{W}_1(T_1, T_2) = \rho \mathcal{W}_1(T_1, T_2).
\end{aligned}$$

This verifies the claim (7.15).

Estimate of B. For the term $\mathbf{B}$, by the orthogonality of $\{\sigma_{\mathbf{s}}\}_{\mathbf{s}}$, we have

$$\begin{aligned}
\mathbf{B} &= \sum_{j=1}^{n} \int_0^{\infty} \left\langle \mathbf{d}_j \mathrm{Rad}^{\geq 2} \left(e^{-\xi \mathcal{L}}(T_1)\right), e^{-t\mathcal{L}} \left(\mathbf{d}_j \mathrm{Rad}^{\geq 2}(T_2)\right)\right\rangle dt \\
&= \sum_{j=1}^{n} \sum_{\substack{|\mathbf{s}| \geq 2 \\ \mathbf{s}_j \neq 0}} \sum_{\substack{|\mathbf{w}| \geq 2 \\ \mathbf{w}_j \neq 0}} \int_0^{\infty} e^{-\xi|\mathbf{s}|} e^{-t|\mathbf{w}|} \widehat{T_1}(\mathbf{s}) \widehat{T_2}(\mathbf{w}) \langle \sigma_{\mathbf{s}}, \sigma_{\mathbf{w}} \rangle dt \\
&= \sum_{j=1}^{n} \sum_{\substack{|\mathbf{s}| \geq 2 \\ \mathbf{s}_j \neq 0}} e^{-\xi|\mathbf{s}|} \left(\int_0^{\infty} e^{-t|\mathbf{s}|} dt\right) \widehat{T_1}(\mathbf{s}) \widehat{T_2}(\mathbf{s}) \\
&= \sum_{j=1}^{n} \sum_{\substack{|\mathbf{s}| \geq 2 \\ \mathbf{s}_j \neq 0}} \frac{e^{-\xi|\mathbf{s}|}}{|\mathbf{s}|} \widehat{T_1}(\mathbf{s}) \widehat{T_2}(\mathbf{s}).
\end{aligned}$$

After interchanging the order of summation, we write

$$\begin{aligned}\mathbf{B} &= \sum_{|\mathbf{s}|\geq 2} |\mathbf{s}| \frac{e^{-\xi|\mathbf{s}|}}{|\mathbf{s}|} \widehat{T_1}(\mathbf{s})\widehat{T_2}(\mathbf{s}) \\ &= \sum_{\ell:=|\mathbf{s}|=2}^{n} \rho^{\ell} \mathcal{W}_{\ell}(T_1, T_2) \\ &= \sum_{\ell=2}^{\frac{1}{2}\log\left(\frac{e}{\mathcal{W}_1(T_1,T_2)}\right)} \rho^{\ell}\mathcal{W}_{\ell}(T_1,T_2) + \sum_{\ell>\frac{1}{2}\log\left(\frac{e}{\mathcal{W}_1(T_1,T_2)}\right)}^{n} \rho^{\ell}\mathcal{W}_{\ell}(T_1,T_2) \\ &\coloneqq \mathbf{B_I} + \mathbf{B_{II}},\end{aligned} \tag{7.17}$$

where $\mathcal{W}_{\ell}(T_1,T_2) \coloneqq \sum_{|\mathbf{s}|=\ell} \widehat{T_1}(\mathbf{s})\widehat{T_2}(\mathbf{s})$.

We assume from now on that $\frac{1}{2}\log\left(\frac{e}{\mathcal{W}_1(T_1,T_2)}\right) \geq 2$, otherwise the desired inequality is trivial. By this assumption, we have

$$0 < \mathcal{W}_1(T_1,T_2) \leq e^{-3}. \tag{7.18}$$

We proceed to estimate $\mathbf{B_I}$. In this case, $2 \leq \ell \leq \frac{1}{2}\log\left(\frac{e}{\mathcal{W}_1(T_1,T_2)}\right)$, and hence, $\mathcal{W}_1(T_1,T_2) \leq e^{-2\ell+1} \leq e^{-2(\ell-1)}$. Note that for each $\ell \geq 2$, we have

$$\left(\frac{4e}{\ell-1}\right)\log\left(\frac{\ell}{\mathcal{W}_1(T_1,T_2)}\right) \leq 4e\cdot\log\left(\frac{e}{\mathcal{W}_1(T_1,T_2)}\right). \tag{7.19}$$

Indeed, rearranging (7.19), this is equivalent to

$$\ell\mathcal{W}_1(T_1,T_2)^{\ell-2} \leq e^{\ell-1},$$

which is trivial since by (7.18), we have

$$\ell\mathcal{W}_1(T_1,T_2)^{\ell-2} \leq \ell e^{-3(\ell-2)} \leq \ell \leq e^{\ell-1}, \quad \ell \geq 2.$$

Combining Proposition 7.14 and (7.19), we get

$$\begin{aligned}|\mathbf{B_I}| &\leq \sum_{\ell=2}^{\frac{1}{2}\log\left(\frac{e}{\mathcal{W}_1(T_1,T_2)}\right)} \rho^{\ell}\cdot\frac{26e}{\ell}\left(\frac{4e}{\ell-1}\right)^{\ell-1}\mathcal{W}_1(T_1,T_2)\left(\log\left(\frac{\ell}{\mathcal{W}_1(T_1,T_2)}\right)\right)^{\ell-1} \\ &\leq \sum_{\ell=2}^{\frac{1}{2}\log\left(\frac{e}{\mathcal{W}_1(T_1,T_2)}\right)} \rho^{\ell}\cdot\frac{26e}{\ell}(4e)^{\ell-1}\left(\log\left(\frac{e}{\mathcal{W}_1(T_1,T_2)}\right)\right)^{\ell-1}\mathcal{W}_1(T_1,T_2) \\ &= \rho\mathcal{W}_1(T_1,T_2)\sum_{\ell=2}^{\frac{1}{2}\log\left(\frac{e}{\mathcal{W}_1(T_1,T_2)}\right)} \epsilon^{\ell-1}\left(\frac{26e}{\ell}\right)(4e)^{\ell-1}.\end{aligned}$$

Note that for $0 < \epsilon < \frac{1}{208e^2}$, we have

$$\epsilon^{\ell-1}\left(\frac{26e}{\ell}\right)(4e)^{\ell-1} \leq 2^{-\ell}, \quad \ell \geq 2.$$

Thus, we further have

$$|\mathbf{B_I}| \leq \rho \mathcal{W}_1(T_1,T_2) \sum_{\ell=2}^{\frac{1}{2}\log\left(\frac{e}{\mathcal{W}_1(T_1,T_2)}\right)} 2^{-\ell}. \tag{7.20}$$

It remains to estimate $\mathbf{B_{II}}$. Note here that (see also (7.8))

$$\begin{aligned} |\mathbf{B_{II}}| &\leq \sum_{\ell > \frac{1}{2}\log\left(\frac{e}{\mathcal{W}_1(T_1,T_2)}\right)} \rho^\ell |\mathcal{W}_\ell(T_1,T_2)| \\ &\leq \sum_{\ell > \frac{1}{2}\log\left(\frac{e}{\mathcal{W}_1(T_1,T_2)}\right)} \rho^\ell \\ &= \rho \sum_{\ell > \frac{1}{2}\log\left(\frac{e}{\mathcal{W}_1(T_1,T_2)}\right)} \left(\frac{\epsilon^{\ell-1}}{\left(\log \frac{e}{\mathcal{W}_1(T_1,T_2)}\right)^{\ell-1}} \right). \end{aligned} \tag{7.21}$$

We now show that

$$\left(\log \frac{e}{\mathcal{W}_1(T_1,T_2)}\right)^{-(\ell-1)} \leq e^2 \mathcal{W}_1(T_1,T_2), \tag{7.22}$$

whenever $\ell > \frac{1}{2}\log\left(\frac{e}{\mathcal{W}_1(T_1,T_2)}\right)$ and $0 < \mathcal{W}_1(T_1,T_2) \leq e^{-3}$. Indeed, changing the variable $\mathcal{W}_1(T_1,T_2) \mapsto e^{-u}$ with $u \geq 3$, (7.22) is equivalent to

$$1 \leq e^{2-u}(1+u)^{\ell-1},$$

whenever $u \geq 3$ and $\ell > \frac{u+1}{2} \geq 2$. Since $e^{2-u}(1+u)^{\frac{u-1}{2}} \leq e^{2-u}(1+u)^{\ell-1}$ for $\ell - 1 \geq \frac{u-1}{2}$. It now remains to show that

$$1 \leq e^{2-u}(1+u)^{\frac{u-1}{2}}, \tag{7.23}$$

for $u \geq 3$. Rearranging (7.23) yields that the desired inequality is equivalent to

$$u - \left(\frac{u-1}{2}\right)\log(1+u) \leq 2, \tag{7.24}$$

for $u \geq 3$. We define $F(u) \coloneqq u - \left(\frac{u-1}{2}\right)\log(1+u)$. We now show that $F(u) \leq 2$ for $u \geq 3$. For $3 \leq u \leq 4$, we have

$$F(u) \leq u - \frac{u-1}{2}\log 4 \leq 4 - \frac{3}{2}\log 4 < 2.$$

For $u \geq 4$, we compute

$$F'(u) = \frac{1}{2} - \frac{1}{2}\log(1+u) + \frac{1}{1+u}$$

and

$$F''(u) = -\frac{1}{2(1+u)} - \frac{1}{(1+u)^2} < 0.$$

Since $F'(4) < 0$, it follows that F is decreasing on $[4,\infty)$. Hence,

$$F(u) \leq F(4) < 2, \qquad u \geq 4.$$

This proves (7.24), and therefore (7.22). Substituting (7.22) to (7.21) yields that

$$\begin{aligned}|\mathbf{B_{II}}| &\le e^2\mathcal{W}_1(T_1,T_2)\rho \sum_{\ell>\frac{1}{2}\log\left(\frac{e}{\mathcal{W}_1(T_1,T_2)}\right)} \epsilon^{\ell-1} \\ &\le \rho\mathcal{W}_1(T_1,T_2) \sum_{\ell>\frac{1}{2}\log\left(\frac{e}{\mathcal{W}_1(T_1,T_2)}\right)} 2^{-\ell},\end{aligned} \tag{7.25}$$

where we used the observation

$$e^2\epsilon^{\ell-1} \le 2^{-\ell}, \quad \text{for} \quad \ell > \frac{1}{2}\log\left(\frac{e}{\mathcal{W}_1(T_1,T_2)}\right) \ge 2.$$

Indeed, this observation holds for $\ell = 2$, since

$$e^2\epsilon < \frac{1}{208} < \frac{1}{4},$$

and the general case follows by induction because $\epsilon < \frac{1}{208e^2} < 1/2$. Combining (7.25) and (7.20), we arrive at (7.16). The proof of the theorem is complete. □

It remains to prove Proposition 7.14. To this end, we first establish another useful result in Proposition 7.16. For each $T_1, T_2 \in \mathbb{M}_{2^n}$, $J \subset [n]$, $d \ge 2$ and $t > 0$, set

$$L_{J,d,t}(T_1,T_2) = \left\{ j \in J : \sum_{\substack{supp\,(\mathbf{s})\subset J^c \\ |\mathbf{s}|=d-1,\alpha\in\{1,2,3\}}} \left|\widehat{T_1}(\mathbf{s}\oplus\alpha e_j)\widehat{T_2}(\mathbf{s}\oplus\alpha e_j)\right| > t\sum_{\ell=1}^{3}\left|\widehat{T_1}(\ell e_j)\widehat{T_2}(\ell e_j)\right| \right\}.$$

If T_1, T_2 are monotonically increasing operators and satisfy condition (7.9), then, by Lemma 7.4 (ii) and by (7.9), for each $\ell \in \{1,2,3\}$ and $j \in [n]$, $\widehat{T_1}(\ell e_j), \widehat{T_2}(\ell e_j) \ge 0$, and we have

$$\sum_{\ell=1}^{3}\widehat{T_1}(\ell e_j)\widehat{T_2}(\ell e_j) = \big(\sum_{\ell=1}^{3}\widehat{T_1}(\ell e_j)^2\big)^{\frac{1}{2}} \cdot \big(\sum_{\ell=1}^{3}\widehat{T_2}(\ell e_j)^2\big)^{\frac{1}{2}},$$

and consequently, by the Cauchy–Schwarz inequality, we have

$$L_{J,d,t}(T_1,T_2) \subset L_{J,d,t}(T_1,T_1) \cup L_{J,d,t}(T_2,T_2). \tag{7.26}$$

We need the following lemma.

LEMMA 7.15. *Let $T \in \mathbb{M}_{2^n}$ be monotonically increasing with $\|T\|_{L_\infty} \le 1$. For each $J \subset [n]$, $d \ge 2$ and $t \ge 12(4e)^{d-1}$, we have*

$$\sum_{j\in L_{J,d,t}(T,T)}\sum_{\ell=1}^{3}\widehat{T}(\ell e_j)^2 \le \frac{5e}{3}\big(\frac{t}{12}\big)^{-\frac{1}{d-1}}\exp\left\{-\frac{(d-1)(\frac{t}{12})^{\frac{1}{d-1}}}{2e}\right\}. \tag{7.27}$$

PROOF. Take monotonically increasing $T \in \mathbb{M}_{2^n}$, $J \subset [n]$, $d \ge 2$ and $t \ge 12(4e)^{d-1}$. During the proof, let us write L_t^T instead of $L_{J,d,t}(T,T)$. Applying Lemma 7.4 (ii), for each $j \in [n]$, we have

$$\|\mathbf{d}_j(T)\|_{L_1} \le \sum_{\ell=1}^{3}\|\mathbf{d}_{j,\ell}(T)\|_{L_1} = \sum_{\ell=1}^{3}\widehat{T}(\ell e_j),$$

which further implies

$$\|\mathbf{d}_j(T)\|_{L_1}^2 \leq 3\sum_{\ell=1}^{3}\widehat{T}(\ell e_j)^2.$$

Thus, we have

$$\begin{aligned}
&\sum_{j\in J}\sum_{\substack{supp\,(\mathbf{s})\subset J^c\\ |\mathbf{s}|=d-1,\alpha\in\{1,2,3\}}}|\widehat{T}(\mathbf{s}\oplus\alpha e_j)|^2\\
&\qquad\leq 2t_0^2\sum_{j\in J}\|\mathbf{d}_j(T)\|_{L_1}^2+10et_0^{2-\frac{2}{d-1}}\exp\left\{-\frac{(d-1)t_0^{\frac{2}{d-1}}}{2e}\right\}\\
&\qquad\leq 6t_0^2\sum_{j\in J}\sum_{\ell=1}^{3}\widehat{T}(\ell e_j)^2+10et_0^{2-\frac{2}{d-1}}\exp\left\{-\frac{(d-1)t_0^{\frac{2}{d-1}}}{2e}\right\},
\end{aligned}$$

where the first inequality is taken from the proof of [**JLLZ24**, Proposition 5.2] (provided $t_0 \geq (4e)^{\frac{d-1}{2}}$). According to the definition of L_t^T, taking $t_0 = \frac{\sqrt{t}}{2\sqrt{3}}$, it follows that

$$\begin{aligned}
t\sum_{j\in L_t^T}\sum_{\ell=1}^{3}\widehat{T}(le_j)^2 &\leq \sum_{j\in L_t^T}\sum_{\substack{supp\,(\mathbf{s})\subset J^c\\ |\mathbf{s}|=d-1,\alpha\in\{1,2,3\}}}|\widehat{T}(\mathbf{s}\oplus\alpha e_j)|^2\\
&\leq \frac{t}{2}\sum_{j\in J}\sum_{\ell=1}^{3}\widehat{T}(\ell e_j)^2+10e(\frac{t}{12})^{1-\frac{1}{d-1}}\exp\left\{-\frac{(d-1)(\frac{t}{12})^{\frac{1}{d-1}}}{2e}\right\}.
\end{aligned}$$

From the above inequality, by subtracting the first term on the right-hand side from the left-hand side and subsequently dividing both sides by $\frac{t}{2}$, we have

$$\begin{aligned}
\sum_{j\in L_t^T}\sum_{\ell=1}^{3}\widehat{T}(\ell e_j)^2 &\leq \sum_{j\in L_t^T}\sum_{\substack{supp\,(\mathbf{s})\subset J^c\\ |\mathbf{s}|=d-1,\alpha\in\{1,2,3\}}}|\widehat{T}(\mathbf{s}\oplus\alpha e_j)|^2\\
&\leq \frac{2}{t}\cdot 10e(\frac{t}{12})^{1-\frac{1}{d-1}}\exp\left\{-\frac{(d-1)(\frac{t}{12})^{\frac{1}{d-1}}}{2e}\right\}\\
&= \frac{5e}{3}(\frac{t}{12})^{-\frac{1}{d-1}}\exp\left\{-\frac{(d-1)(\frac{t}{12})^{\frac{1}{d-1}}}{2e}\right\}.
\end{aligned}$$

The proof is complete. □

PROPOSITION 7.16. *Let $T_1, T_2 \in \mathbb{M}_{2^n}$ be monotonically increasing operators satisfying the condition* (7.9)*, $\max\{\|T_1\|_{L_\infty}, \|T_2\|_{L_\infty}\} \leq 1$ and $\mathcal{W}_1(T_1,T_2) \leq e^{-2(d-1)}$. Then, for each $J \subset [n]$ and $d \geq 2$, we have*

$$\sum_{j\in J}\sum_{\substack{supp\,(\mathbf{s})\subset J^c\\ |\mathbf{s}|=d-1,\alpha\in\{1,2,3\}}}\left|\widehat{T_1}(\mathbf{s}\oplus\alpha e_j)\widehat{T_2}(\mathbf{s}\oplus\alpha e_j)\right| \leq 26\left(\frac{4e}{d-1}\right)^{d-1}\mathcal{W}_{1,J}(T_1,T_2)\left(\log\left(\frac{1}{\mathcal{W}_{1,J}(T_1,T_2)}\right)\right)^{d-1}.$$

PROOF. We divide the proof into three steps. We write for convenience L_t (resp. $L_t^{T_1}$, $L_t^{T_2}$) instead of $L_{J,d,t}(T_1,T_2)$ (resp. $L_{J,d,t}(T_1,T_1)$, $L_{J,d,t}(T_2,T_2)$).

Step I. For each $j \in J$, we define the following function on J by setting

$$f_1(j) = \sum_{\ell=1}^{3} \widehat{T_1}(\ell e_j)\widehat{T_2}(\ell e_j).$$

We further introduce the function $f_2(\cdot)$ on J by setting

$$\sum_{\substack{supp\,(\mathbf{s})\subset J^c \\ |\mathbf{s}|=d-1,\alpha\in\{1,2,3\}}} \left|\widehat{T_1}(\mathbf{s}\oplus\alpha e_j)\widehat{T_2}(\mathbf{s}\oplus\alpha e_j)\right| = f_1(j)f_2(j).$$

By [**KK13**, Lemma 11], we get

$$\begin{aligned}\sum_{j\in J}\sum_{\substack{supp\,(\mathbf{s})\subset J^c \\ |\mathbf{s}|=d-1,\alpha\in\{1,2,3\}}} \left|\widehat{T_1}(\mathbf{s}\oplus\alpha e_j)\widehat{T_2}(\mathbf{s}\oplus\alpha e_j)\right| &= \sum_{j\in J} f_1(j)f_2(j) \\ &= \int_0^\infty \big(\mathbb{E}_J\big(f_1(j)\cdot\chi_{\{j:f_2(j)>t\}}(j)\big)\big)dt = \int_0^\infty \sum_{j\in L_t}\sum_{\ell=1}^{3}\widehat{T_1}(\ell e_j)\widehat{T_2}(\ell e_j)dt \\ &= \int_0^{t_1} \sum_{j\in L_t}\sum_{\ell=1}^{3}\widehat{T_1}(\ell e_j)\widehat{T_2}(\ell e_j)dt + \int_{t_1}^\infty \sum_{j\in L_t}\sum_{\ell=1}^{3}\widehat{T_1}(\ell e_j)\widehat{T_2}(\ell e_j)dt \\ &:= Y_{1,j} + Y_{2,j},\end{aligned}$$

where t_1 is chosen to satisfy

$$\mathcal{W}_{1,J}(T_1,T_2) = \exp\{-\frac{(d-1)(\frac{t_1}{12})^{\frac{1}{d-1}}}{4e}\}. \tag{7.28}$$

Note that the assumption $\mathcal{W}_{1,J}(T_1,T_2) \le \mathcal{W}_1(T_1,T_2) \le e^{-2(d-1)}$ and the above equation imply that

$$t_1 \ge 12(8e)^{d-1} \quad \text{and} \quad t_1 = 12(\tfrac{4e}{d-1})^{d-1}(\log\tfrac{1}{\mathcal{W}_{1,J}(T_1,T_2)})^{d-1}.$$

Then,

$$Y_{1,j} \le t_1\mathcal{W}_{1,J}(T_1,T_2) = 12(\frac{4e}{d-1})^{d-1}\mathcal{W}_{1,J}(T_1,T_2)\left(\log\frac{1}{\mathcal{W}_{1,J}(T_1,T_2)}\right)^{d-1}. \tag{7.29}$$

Step II. Since $T_1, T_2 \in \mathbb{M}_{2^n}$ are monotonically increasing and satisfy the condition (7.9), we can apply (7.26) to get

$$\sum_{j\in L_t}\sum_{\ell=1}^{3}\widehat{T_1}(\ell e_j)\widehat{T_2}(\ell e_j) \le \sum_{j\in L_t^{T_1}}\sum_{\ell=1}^{3}\widehat{T_1}(\ell e_j)\widehat{T_2}(\ell e_j) + \sum_{j\in L_t^{T_2}}\sum_{\ell=1}^{3}\widehat{T_1}(\ell e_j)\widehat{T_2}(\ell e_j). \tag{7.30}$$

By Cauchy-Schwarz inequality and (7.8), we can get $\mathcal{W}_1(T_2) \le 1$ and

$$\begin{aligned}\sum_{j\in L_t^{T_1}}\sum_{\ell=1}^{3}\widehat{T_1}(\ell e_j)\widehat{T_2}(\ell e_j) &\le \Big(\sum_{j\in L_t^{T_1}}\sum_{\ell=1}^{3}\widehat{T_1}(\ell e_j)^2\Big)^{\frac{1}{2}}\Big(\sum_{j\in L_t^{T_1}}\sum_{\ell=1}^{3}\widehat{T_2}(\ell e_j)^2\Big)^{\frac{1}{2}} \\ &\le \Big(\sum_{j\in L_t^{T_1}}\sum_{\ell=1}^{3}\widehat{T_1}(\ell e_j)^2\Big)^{\frac{1}{2}}.\end{aligned}$$

Similarly,

$$\sum_{j\in L_t^{T_2}}\sum_{\ell=1}^{3}\widehat{T_1}(\ell e_j)\widehat{T_2}(\ell e_j)\le \big(\sum_{j\in L_t^{T_2}}\sum_{\ell=1}^{3}\widehat{T_2}(\ell e_j)^2\big)^{\frac{1}{2}}.$$

Hence, combining the above argument, (7.30) and (7.27), for all $t\ge 12(4e)^{d-1}$, we have

$$\begin{aligned}\sum_{j\in L_t}\sum_{\ell=1}^{3}\widehat{T_1}(\ell e_j)\widehat{T_2}(\ell e_j)&\le \big(\sum_{j\in L_t^{T_1}}\sum_{\ell=1}^{3}\widehat{T_1}(\ell e_j)^2\big)^{\frac{1}{2}}+\big(\sum_{j\in L_t^{T_2}}\sum_{\ell=1}^{3}\widehat{T_2}(\ell e_j)^2\big)^{\frac{1}{2}}\\ &\le 2\cdot(\frac{5e}{3})^{\frac{1}{2}}(\frac{t}{12})^{-\frac{1}{2(d-1)}}\exp\left\{-\frac{(d-1)\left(\frac{t}{12}\right)^{\frac{1}{d-1}}}{4e}\right\}.\end{aligned}\tag{7.31}$$

Step III. For $Y_{2,j}$ and all $t\ge t_1\ge 12(8e)^{d-1}\ge 12(4e)^{d-1}$, by the above estimate (7.31), we have

$$Y_{2,j}\le\int_{t_1}^{\infty}2\cdot\left(\frac{5e}{3}\right)^{\frac{1}{2}}\left(\frac{t}{12}\right)^{-\frac{1}{2(d-1)}}\exp\left\{-\frac{(d-1)\left(\frac{t}{12}\right)^{\frac{1}{d-1}}}{4e}\right\}dt.$$

To estimate the above inequality, we first apply a change of variables, setting

$$s=\frac{(d-1)\left(\frac{t}{12}\right)^{\frac{1}{d-1}}}{4e},\tag{7.32}$$

obviously, $s\ge 2(d-1)$, and we obtain

$$Y_{2,j}=2\times 12\cdot\left(\frac{5e}{3}\right)^{\frac{1}{2}}\left(\frac{4e}{d-1}\right)^{d-\frac{3}{2}}(d-1)\int_{s=s_1}^{\infty}s^{d-\frac{5}{2}}e^{-s}ds,$$

where s_1 is derived by substituting t_1 into (7.32). If we denote $\varphi(s)=s^{d-\frac{5}{2}}e^{-s}$, then $\varphi(s)$ is decreasing and $\frac{\varphi(s+1)}{\varphi(s)}\le e^{-\frac{1}{2}}$ on $s\ge 2(d-1)$ and $d\ge 2$, and we have

$$\int_{s=s_1}^{\infty}s^{d-\frac{5}{2}}e^{-s}ds\le\frac{1}{1-e^{-1/2}}s_1^{d-\frac{5}{2}}e^{-s_1}.$$

It therefore follows that

$$\begin{aligned}Y_{2,j}&\le 2\times 12\cdot\left(\frac{5e}{3}\right)^{\frac{1}{2}}\times 4e\cdot\frac{1}{1-e^{-1/2}}\cdot\left(\frac{t_1}{12}\right)^{1-\frac{3}{2(d-1)}}\cdot\exp\left\{-\frac{(d-1)(\frac{t_1}{12})^{\frac{1}{d-1}}}{4e}\right\}\\ &\le 2\times 12\cdot\left(\frac{5e}{3}\right)^{\frac{1}{2}}\times 4e\cdot\frac{1}{1-e^{-1/2}}\cdot(8e)^{-3/2}\left(\frac{t_1}{12}\right)\cdot\exp\left\{-\frac{(d-1)(\frac{t_1}{12})^{\frac{1}{d-1}}}{4e}\right\}\\ &\le 14\left(\frac{4e}{d-1}\right)^{d-1}\mathcal{W}_{1,J}(T_1,T_2)\left(\log\frac{1}{\mathcal{W}_{1,J}(T_1,T_2)}\right)^{d-1},\end{aligned}$$

where the last inequality follows from the choice of t_1 in (7.28). Combining (7.29) with the above estimate, we finish the proof. □

We now prove Proposition 7.14.

PROOF OF PROPOSITION 7.14. For each $\mathbf{s} \in \{0,1,2,3\}^n$ with $|\mathbf{s}| = d$, we know the probability of J such that $\mathbf{s} = \mathbf{u} \oplus \alpha e_j$ with $j \in J$ and $supp\,(\mathbf{u}) \subseteq J^c$ is $\binom{d}{1} \cdot \frac{1}{d}(1-\frac{1}{d})^{d-1} = (1-\frac{1}{d})^{d-1} \geq \frac{1}{e}$. Hence,

$$\begin{aligned}
&\mathbb{E}_J \left(\sum_{\substack{supp\,(\mathbf{s}) \subseteq J^c, |\mathbf{s}| = d-1 \\ j \in J, \alpha \in \{1,2,3\}}} \left| \widehat{T_1}\,(\mathbf{s} \oplus \alpha e_j)\, \widehat{T_2}\,(\mathbf{s} \oplus \alpha e_j) \right| \right) \\
&= \left(1 - \frac{1}{d}\right)^{d-1} \sum_{\substack{\mathbf{s} \in \{0,1,2,3\}^n, \\ |\mathbf{s}| = d}} \left| \widehat{T_1}(\mathbf{s}) \widehat{T_2}(\mathbf{s}) \right| \qquad (7.33) \\
&\geq \frac{1}{e} \sum_{\substack{\mathbf{s} \in \{0,1,2,3\}^n, \\ |\mathbf{s}| = d}} \left| \widehat{T_1}(\mathbf{s}) \widehat{T_2}(\mathbf{s}) \right| \geq \frac{1}{e} |\mathcal{W}_d(T_1, T_2)|.
\end{aligned}$$

Secondly, Proposition 7.16, together with the assumption $\mathcal{W}_1(T_1, T_2) \leq e^{-2(d-1)}$ and (7.33), yields that

$$|\mathcal{W}_d(T_1, T_2)| \leq e \mathbb{E}_J \left[26 \left(\frac{4e}{d-1} \right)^{d-1} \mathcal{W}_{1,J}(T_1, T_2) \left(\log \left(\frac{1}{\mathcal{W}_{1,J}(T_1, T_2)} \right) \right)^{d-1} \right].$$

Note that $x \mapsto x\big(\log(\frac{1}{x})\big)^{d-1}$ is a concave function on $(0, e^{-(d-1)})$. Then it follows from the above inequality and the Jensen inequality that

$$\begin{aligned}
|\mathcal{W}_d(T_1, T_2)| &\leq 26e \left(\frac{4e}{d-1} \right)^{d-1} \left(\mathbb{E}_J \left(\mathcal{W}_{1,J}(T_1, T_2) \right) \right) \left(\log \left(\frac{1}{\mathbb{E}_J(\mathcal{W}_{1,J}(T_1, T_2))} \right) \right)^{d-1} \\
&= 26e \left(\frac{4e}{d-1} \right)^{d-1} \left(\frac{\mathcal{W}_1(T_1, T_2)}{d} \right) \left(\log \left(\frac{d}{\mathcal{W}_1(T_1, T_2)} \right) \right)^{d-1},
\end{aligned}$$

where we used $\mathbb{E}_J(\mathbb{1}_J(j)) = \frac{1}{d}$ in the last equality. Hence, the desired inequality follows. □

7.3. Quantum KMS-correlation inequalities

In this section, we aim to establish the quantum analogy of the Keller-Mossel-Sen correlation inequality (7.4). Recall the *quantum reverse hypercontractivity inequality* as follows: Let $\{e^{-t\mathcal{L}}\}_{t \geq 0}$ be the quantum Ornstein-Uhlenbeck semigroup acting on $L_p(\mathbb{M}_{2^n})$, then the following reverse hypercontractivity of $\{e^{-t\mathcal{L}}\}_{t \geq 0}$ holds

$$\mathrm{tr}\left(|T| e^{-t\mathcal{L}}(|S|) \right) \geq \|T\|_{L_p} \|S\|_{L_q}, \qquad (7.34)$$

whenever $e^{-2t} \leq (1-p)(1-q)$ with $(p,q) \in (0,1) \times (0,1)$. For further details, we refer the reader to [**BDR20, CKMT15**].

THEOREM 7.17. *Let monotonically increasing T_1, $T_2 \in \mathbb{M}_{2^n}$ satisfy*

$$\max\{\|T_1\|_{L_\infty}, \|T_2\|_{L_\infty}\} \leq 1.$$

There exists a universal constant $K > 0$ such that the following inequality holds

$$\mathrm{Cov}(T_1, T_2) \geq K \sum_{j=1}^n \frac{\|\mathbf{d}_j(T_1)\|_{L_1} \|\mathbf{d}_j(T_2)\|_{L_1}}{\sqrt{\log(e/\|\mathbf{d}_j(T_1)\|_{L_1}) \log(e/\|\mathbf{d}_j(T_2)\|_{L_1})}}. \qquad (7.35)$$

The following lemma motivated by Keller, Mossel and Sen ([**KMS14**]).

LEMMA 7.18. *Let $T \in \mathbb{M}_{2^n}$ with $\|T\|_{L_\infty} \leq 1$, and $0 < p < \infty$.*

(i) *For each $p \geq 1$, we have*

$$\left\|\partial_j\left(e^{-t\mathcal{L}}(T)\right)\right\|_{L_p} \leq 3^{\frac{p-1}{p}} t^{-(p-1)/2p} e^{-t/p} \|\partial_j(T)\|_{L_1}^{1/p}, \quad t > 0.$$

(ii) *If T is monotonically increasing, then the following holds for every $0 < p < 1$,*

$$\left\|\partial_j\left(e^{-t\mathcal{L}}(T)\right)\right\|_{L_p} \geq 3^{\frac{p-1}{p}} t^{(1-p)/2p} e^{-t/p} \|\partial_j(T)\|_{L_1}^{1/p}, \quad t > 0.$$

PROOF. (i) Fix $p \geq 1$. Note that

$$\|\mathbf{d}_j(e^{-t\mathcal{L}}(T))\|_{L_\infty} \leq t^{-1/2}\|T\|_{L_\infty} \leq t^{-1/2}$$

by the Buser inequality (5.19) with $p = \infty$, the fact that $e^{2t} - 1 \geq t$ for $t > 0$ and the assumption that $\|T\|_{L_\infty} \leq 1$. Then by the above estimate and Lemma 7.3 (i), we have

$$\|\partial_j(e^{-t\mathcal{L}}(T))\|_{L_\infty} \leq 3\|\mathbf{d}_j(e^{-t\mathcal{L}}(T))\|_{L_\infty} \leq 3t^{-1/2}. \tag{7.36}$$

By this inequality and Lemma 7.3 (i) (ii), we get

$$\begin{aligned}
\left\|\partial_j\left(e^{-t\mathcal{L}}(T)\right)\right\|_{L_p}^p &= \mathrm{tr}(|\partial_j(e^{-t\mathcal{L}}(T))|^{p-1}|\partial_j(e^{-t\mathcal{L}}(T))|) \\
&\leq (3t^{-1/2})^{p-1}\|\partial_j(e^{-t\mathcal{L}}(T)\|_{L_1} \\
&= 3^{p-1}t^{-(p-1)/2}e^{-t}\|e^{-t\mathcal{L}}(\partial_j(T))\|_{L_1} \\
&\leq 3^{p-1}t^{-(p-1)/2}e^{-t}\|\partial_j(T)\|_{L_1}, \quad t > 0.
\end{aligned}$$

We finish the proof of (i).

(ii) Fix $0 < p < 1$ and $t > 0$. Since T is monotonically increasing, it follows that

$$\|\partial_j(e^{-t\mathcal{L}}(T))\|_{L_1} = e^{-t}\mathrm{tr}\left(e^{-t\mathcal{L}}\left(\partial_j(T)\right)\right) = e^{-t}\|\partial_j(T)\|_{L_1},$$

where the first equality is due to Lemma 7.3 (ii) and Lemma 7.4 (iii), and the second equality is due to that $e^{-t\mathcal{L}}$ is trace preserving. As $0 < p < 1$, by the elementary inequality and (7.36),

$$\|\partial_j(e^{-t\mathcal{L}}(T))\|_{L_1} \leq \|\partial_j(e^{-t\mathcal{L}}(T))\|_{L_p}^p \|\partial_j(e^{-t\mathcal{L}}(T))\|_{L_\infty}^{1-p} \leq 3^{1-p}t^{-\frac{1-p}{2}}\|\partial_j(e^{-t\mathcal{L}}(T))\|_{L_p}^p.$$

Thus, combining the above argument, we further have

$$\|\partial_j(T)\|_{L_1} = e^t\|\partial_j(e^{-t\mathcal{L}}(T))\|_{L_1} \leq 3^{1-p}e^t t^{-\frac{1-p}{2}}\|\partial_j(e^{-t\mathcal{L}}(T))\|_{L_p}^p.$$

This verifies item (ii). □

The lemma below was stated in [**KMS14**] without proof, we here provide its proof for the sake of completeness.

LEMMA 7.19. *For each $t, s \in (0,1)$, we have*

$$\frac{t^2 + 2st + 1}{1 - s^2} = \inf_{\substack{s^2 \leq (1-p)(1-q) \\ (p,q)\in(0,1)^2}} \frac{t^2}{p} + \frac{1}{q}.$$

PROOF. On the one hand, for the range $s^2 \leq (1-p)(1-q)$ with $(s,p,q) \in (0,1)^3$, since the function $s \mapsto \frac{t^2+2st+1}{1-s^2}$ is increasing, it follows that

$$\frac{t^2 + 2st + 1}{1 - s^2} \leq \frac{t^2 + 2\sqrt{(1-p)(1-q)}t + 1}{1 - (1-p)(1-q)} = \frac{t^2 + 2\sqrt{(1-p)(1-q)}t + 1}{p + q - pq}. \tag{7.37}$$

We claim here that

$$\frac{t^2+2\sqrt{(1-p)(1-q)}t+1}{p+q-pq} \le \frac{t^2}{p}+\frac{1}{q}, \quad \text{for } (p,q)\in(0,1)^2. \tag{7.38}$$

Indeed, rearranging (7.38) yields that

$$q^2(1-p)t^2-2pq\sqrt{(1-p)(1-q)}t+p^2(1-q)=\left(q\sqrt{1-p}t-p\sqrt{1-q}\right)^2\ge 0,$$

which verifies the claim. Combining (7.37) with (7.38) entails that

$$\inf_{\substack{s^2\le(1-p)(1-q)\\(p,q)\in(0,1)^2}} \frac{t^2}{p}+\frac{1}{q} \ge \frac{t^2+2st+1}{1-s^2}.$$

On the other hand, for given $(s,t)\in(0,1)^2$, we define

$$p_0 \coloneqq \frac{t(1-s^2)}{s+t} \quad \text{and} \quad q_0 \coloneqq \frac{1-p_0-s^2}{1-p_0}$$

It is clear that $(p_0,q_0)\in(0,1)^2$ with $s^2=(1-p_0)(1-q_0)$. Moreover, substituting the fact $s^2=(1-p_0)(1-q_0)$ into $F(s)\coloneqq\frac{t^2+2st+1}{1-s^2}$ yields that

$$F([(1-p_0)(1-q_0)]^{\frac{1}{2}})=\frac{t^2}{p_0}+\frac{1}{q_0}\ge \inf_{\substack{s^2\le(1-p)(1-q)\\(p,q)\in(0,1)^2}} \frac{t^2}{p}+\frac{1}{q}.$$

Combining the two inequalities completes the proof. □

Now we are ready to prove Theorem 7.17.

Proof of Theorem 7.17. By Lemma 2.3, it follows from the assumption that from

$$\max\{\|\mathbf{d}_j(T_1)\|_{L_\infty},\|\mathbf{d}_j(T_2)\|_{L_\infty}\}\le 1. \tag{7.39}$$

for each $j\in[n]$. Applying the representation of covariance (7.5) and Lemma 7.3 (ii), we have

$$\begin{aligned}
\mathrm{Cov}(T_1,T_2) &= \sum_{j=1}^n \int_0^\infty e^{-t}\mathrm{tr}\left(\partial_j(T_1)e^{-t\mathcal{L}}\left(\partial_j(T_2)\right)\right)dt\\
&\ge \sum_{j=1}^n \int_1^\infty e^{-(t-1)}\mathrm{tr}\left[\partial_j\left(e^{-\mathcal{L}/2}(T_1)\right)\cdot e^{-(t-1)\mathcal{L}}\left(\partial_j\left(e^{-\mathcal{L}/2}(T_2)\right)\right)\right]dt\\
&= \sum_{j=1}^n \int_0^\infty e^{-t}\mathrm{tr}\left[\partial_j\left(e^{-\mathcal{L}/2}(T_1)\right)\cdot e^{-t\mathcal{L}}\left(\partial_j\left(e^{-\mathcal{L}/2}(T_2)\right)\right)\right]dt.
\end{aligned}$$

Changing the variable $e^{-t}\mapsto s$, we further have

$$\begin{aligned}
\mathrm{Cov}(T_1,T_2) &\ge \sum_{j=1}^n \int_0^1 \mathrm{tr}\left[\partial_j\left(e^{-\mathcal{L}/2}(T_1)\right)\cdot e^{\log(s)\mathcal{L}}\left(\partial_j\left(e^{-\mathcal{L}/2}(T_2)\right)\right)\right]ds\\
&= \sum_{j=1}^n \int_0^1 A^j(s)ds.
\end{aligned} \tag{7.40}$$

It suffices to estimate the integral $\int_0^1 A^j(s)ds$. For each $j \in [n]$ and $s \in (0,1)$, by (7.34) and Lemma 7.18, we have

$$\begin{aligned} A^j(s) &\geq \left\| \partial_j \left(e^{-\mathcal{L}/2}(T_1) \right) \right\|_{L_p} \cdot \left\| \partial_j \left(e^{-\mathcal{L}/2}(T_2) \right) \right\|_{L_q} \\ &\geq 2 \cdot 3^{\frac{p-1}{p}+\frac{q-1}{q}} \cdot (2e)^{-(\frac{1}{2p}+\frac{1}{2q})} \|\partial_j(T_1)\|_{L_1}^{1/p} \cdot \|\partial_j(T_2)\|_{L_1}^{1/q}, \end{aligned} \tag{7.41}$$

for each $(p,q) \in (0,1)^2$ with $s^2 \leq (1-p)(1-q)$. Applying Lemma 7.4 (i) to (7.41) yields that,

$$A^j(s) \geq 18 \cdot 3^{-(\frac{1}{p}+\frac{1}{q})} \cdot (2e)^{-(\frac{1}{2p}+\frac{1}{2q})} \|\mathbf{d}_j(T_1)\|_{L_1}^{1/p} \cdot \|\mathbf{d}_j(T_2)\|_{L_1}^{1/q}. \tag{7.42}$$

For each $j \in [n]$, take a_j, $b_j \in (0,\infty)$ such that

$$\frac{1}{3\cdot\sqrt{2e}} \|\mathbf{d}_j(T_1)\|_{L_1} = \exp\left\{ -\frac{a_j^2}{2} \right\}, \quad \frac{1}{3\cdot\sqrt{2e}} \|\mathbf{d}_j(T_2)\|_{L_1} = \exp\left\{ -\frac{b_j^2}{2} \right\}. \tag{7.43}$$

Then (7.42) turns into

$$\begin{aligned} A^j(s) &\geq 18 \sup_{\substack{s^2 \leq (1-p)(1-q) \\ (p,q)\in(0,1)^2}} \exp\left\{ -\frac{1}{2}\left(\frac{a_j^2}{p} + \frac{b_j^2}{q} \right) \right\} \\ &= 18 \exp\left\{ -\frac{1}{2}\left(\inf_{\substack{s^2 \leq (1-p)(1-q) \\ (p,q)\in(0,1)^2}} \frac{a_j^2}{p} + \frac{b_j^2}{q} \right) \right\}. \end{aligned} \tag{7.44}$$

Without loss os generality, by symmetry, we assume from now on that a_j^2 and b_j^2 are non-zeros and $\frac{a_j^2}{b_j^2} \in (0,1)$. Applying Lemma 7.19 with $t = a_j/b_j$ to (7.44) entails that

$$A^j(s) \geq 18 \exp\left\{ -\frac{1}{2}\left(\frac{a_j^2 + 2sa_jb_j + b_j^2}{1-s^2} \right) \right\}, \text{for all } s \in (0,1). \tag{7.45}$$

Since $F(s) \coloneqq \exp\left\{ -\frac{1}{2}\left(\frac{a_j^2+2sa_jb_j+b_j^2}{1-s^2} \right) \right\}$ is decreasing on $s \in (0,1)$, we have

$$\begin{aligned} \int_0^1 A^j(s)ds &\geq 18 \int_0^1 \exp\left\{ -\frac{1}{2}\left(\frac{a_j^2 + 2sa_jb_j + b_j^2}{1-s^2} \right) \right\} ds \\ &\geq 18\epsilon \exp\left\{ -\frac{1}{2}\left(\frac{a_j^2 + 2\epsilon a_jb_j + b_j^2}{1-\epsilon^2} \right) \right\}, \end{aligned} \tag{7.46}$$

for every $\epsilon \in (0,1)$ and $j \in [n]$. By (7.43) and (7.39), it follows that

$$\min\{a_j, b_j\} \geq (\log(18e))^{1/2}, \quad j \in [n],$$

and hence, $\frac{1}{a_jb_j} \in (0,1)$ for each $j \in [n]$. We further have

(7.47)
$$\sup_{\epsilon\in(0,1)} \epsilon \exp\left\{-\frac{1}{2}\left(\frac{a_j^2+2\epsilon a_jb_j+b_j^2}{1-\epsilon^2}\right)\right\} \geq \liminf_{\epsilon\to\frac{1}{a_jb_j}} \epsilon\exp\left\{-\frac{1}{2}\left(\frac{a_j^2+2\epsilon a_jb_j+b_j^2}{1-\epsilon^2}\right)\right\}$$
$$= \frac{1}{a_jb_j}\exp\left\{-\frac{1}{2}\left(\frac{a_j^2+2+b_j^2}{1-\frac{1}{a_j^2b_j^2}}\right)\right\}.$$

We now claim that there exists a universal constant $\delta > 0$ such that

$$\frac{a_j^2+2+b_j^2}{1-\frac{1}{a_j^2b_j^2}} \leq a_j^2+b_j^2+\delta, \tag{7.48}$$

provided that $a_j,\, b_j \geq (\log(18e))^{1/2}$. Rearranging (7.48), it remains to show that there exists a universal constant $\delta > 0$ such that

$$\left(\frac{ab}{ab-1}\right)\left(2+\frac{1}{a}+\frac{1}{b}\right) \leq \delta, \quad \text{when } a=a_j^2,\ b=b_j^2 \geq \log(18e).$$

Since $ab \geq \frac{4}{3}$ for $a,\, b \geq \log(18e)$, it follows that $ab-1 \geq ab/4$. Hence, we can choose $\delta = 12 \geq 8\left(1+\frac{1}{\log(18e)}\right)$ such that the following holds

$$\left(\frac{ab}{ab-1}\right)\left(2+\frac{1}{a}+\frac{1}{b}\right) \leq 4\left(2+\frac{1}{a}+\frac{1}{b}\right) \leq 8\left(1+\frac{1}{\log(18e)}\right) \leq 12.$$

Combining $\delta = 12$, (7.48), (7.47) and (7.43) yields that

$$\begin{aligned}\int_0^1 A^j(s)ds &\geq \frac{18}{a_jb_je^6}\exp\left\{-\frac{a_j^2+b_j^2}{2}\right\} \\ &\geq \frac{1}{4e^7}\frac{\|\mathbf{d}_j(T_1)\|_{L_1}\|\mathbf{d}_j(T_2)\|_{L_1}}{\sqrt{\log(e/\|\mathbf{d}_j(T_1)\|_{L_1})\log(e/\|\mathbf{d}_j(T_2)\|_{L_1})}}.\end{aligned} \tag{7.49}$$

Indeed, for each $j \in [n]$, it follows from (7.43) that

$$\exp\left\{-\frac{a_j^2+b_j^2}{2}\right\} = \frac{\|\mathbf{d}_j(T_1)\|_{L_1}\|\mathbf{d}_j(T_2)\|_{L_1}}{18e}.$$

Moreover, since $\|\mathbf{d}_j(T_i)\|_{L_1} \leq 1$ and $3\sqrt{2e} < e^2$, we have

$$a_j^2 = 2\log\left(\frac{3\sqrt{2e}}{\|\mathbf{d}_j(T_1)\|_{L_1}}\right) \leq 4\log\left(\frac{e}{\|\mathbf{d}_j(T_1)\|_{L_1}}\right),$$

and similarly,

$$b_j^2 \leq 4\log\left(\frac{e}{\|\mathbf{d}_j(T_2)\|_{L_1}}\right).$$

which proves the last inequality in (7.49). The proof is complete.

□

CHAPTER 8

Quantum BKS inequality and its inverse

This chapter aims to investigate the quantum BKS-inequality in detail. In a seminal paper, Benjamini, Kalai and Schramm [**BKS99**] (classical BKS theorem) proved that if the sum of the squares of the influences of f is close to zero, then f must be noise sensitive. In [**KK13**], Keller and Kindler obtained a quantitative version of the BKS theorem. It was shown in [**KMS14**] that the quantitative BKS theorem also holds for the standard Gaussian measure on $\mathbb{R}^n$ with respect to geometric influences.

Recently, a quantum quantitative BKS theorem has been obtained by Blecher, Gao and Xu [**BGX25**] via a *random restriction technique* (see also [**JLLZ24**] for the random restriction technique used there to derive a quantum Eldan-Gross inequality). In this chapter, we will combine the semigroup tools with the noncommutative martingales to provide a simpler proof of the quantum BKS-inequality in Theorem 8.5. Our proof is actually motivated by [**JLZ25**]. We prefer to provide all the details for the reader's convenience. Moreover, inspired by the semigroup approach by Keller, Mossel and Sen [**KMS14**], we obtain an *inverse* quantum BKS-inequality.

8.1. The quantum BKS-inequality

We first recall some basic facts on noncommutative martingale theory. Recall that conditional expectation in the noncommutative setting could be defined as follows. Let $\mathcal{B}$ be a $*$-subalgebra of $\mathbb{M}_{2^n}$. There exists a linear operator $\mathcal{E}_{\mathcal{B}} : \mathbb{M}_{2^n} \to \mathcal{B}$ satisfying the following conditions: for every $T \in \mathbb{M}_{2^n}$

(i) $\mathcal{E}_{\mathcal{B}}$ is a normal contractive positive projection from $\mathbb{M}_{2^n}$ onto $\mathcal{B}$;
(ii) $\mathcal{E}_{\mathcal{B}}(aTb) = a\mathcal{E}_{\mathcal{B}}(T)b$ provided $a, b \in \mathcal{B}$;
(iii) $\tau(\mathcal{E}_{\mathcal{B}}(T)) = \tau(T)$.

The operator $\mathcal{E}_{\mathcal{B}}$ given as above is called the conditional expectation with respect to $\mathcal{B}$. By (i) and (iii), one can deduce via interpolation that $\mathcal{E}_B$ is a contraction on $L_p(\mathbb{M}_{2^n})$ for all $1 \le p \le \infty$.

For every $\mathbf{s} \in \{0,1,2,3\}^n$, we denote that $\max(\mathbf{s}) \coloneqq \max\{k \in [n] : s_k \neq 0\}$ and let $\mathcal{M}_j$ be the $*$-subalgebra generated by $\{\sigma_{\mathbf{s}}\}_{\max(\mathbf{s}) \le j}$. Since $(\mathcal{M}_j)_{j=1}^n$ forms an increasing sequence of $*$-subalgebras, it follows that, for each $T \in \mathbb{M}_{2^n}$, $(\mathcal{E}_{\mathcal{M}_j}(T))_{j=1}^n$ forms a noncommutative martingale (see e.g. [**PX97**] for its definition), and the related martingale difference is given as follows: $\mathcal{D}_1(T) \coloneqq \mathcal{E}_{\mathcal{M}_1}(T)$ and

$$\mathcal{D}_j(T) \coloneqq \mathcal{E}_{\mathcal{M}_j}(T) - \mathcal{E}_{\mathcal{M}_{j-1}}(T), \quad 2 \le j \le n.$$

The next elementary lemma contains necessary properties of conditional expectations.

LEMMA 8.1. *Let* $T = \sum_{\mathbf{s} \in \{0,1,2,3\}^n} \widehat{T}(\mathbf{s})\sigma_{\mathbf{s}} \in \mathbb{M}_{2^n}$.

(i) *For each $j \in [n]$, we have $\mathcal{E}_{\mathcal{M}_j}(T) = \sum_{\max(\mathbf{s})\leq j} \widehat{T}(\mathbf{s})\sigma_{\mathbf{s}}$ and*

$$\mathcal{D}_j(T) = \sum_{\max(\mathbf{s})=j} \widehat{T}(\mathbf{s})\sigma_{\mathbf{s}}. \tag{8.1}$$

(ii) *We have*

$$\|T\|_{L_2}^2 = \sum_{j=1}^{n} \|\mathcal{D}_j(T)\|_{L_2}^2. \tag{8.2}$$

Proof. Since $\mathcal{M}_j$ is a $*$-subalgebra of $\mathbb{M}_{2^n}$ generated by $\{\sigma_{\mathbf{s}}\}_{\max(\mathbf{s})\leq j}$, by the orthonormality of stabilizer operators, we have

$$\mathcal{E}_{\mathcal{M}_j}(\sigma_{\mathbf{s}}) = \begin{cases} \sigma_{\mathbf{s}}, \text{ if } \max(\mathbf{s}) \leq j \\ 0, \text{ otherwise.} \end{cases}$$

Hence the claim follows easily from the orthnormality of the martingale differences $\{\mathcal{D}_j(T)\}_{j=1}^n$. □

Lemma 8.2. *Let $\mathcal{M}_{[n]\setminus\{j\}}$ be the $*$-subalgebra of $\mathbb{M}_{2^n}$ generated by $\{\sigma_{\mathbf{s}}\}_{s_j=0}$. Then for $T = \sum_{\mathbf{s}\in\{0,1,2,3\}^n} \widehat{T}(\mathbf{s})\sigma_{\mathbf{s}} \in \mathbb{M}_{2^n}$ and $j \in [n]$, we have*

$$\mathcal{E}_{\mathcal{M}_{[n]\setminus\{j\}}}(T) = \sum_{s_j=0} \widehat{T}(\mathbf{s})\sigma_{\mathbf{s}}.$$

Consequently, the following holds

$$\mathbf{d}_j(T) = T - \mathcal{E}_{\mathcal{M}_{[n]\setminus\{j\}}}(T).$$

Proof. The proof is trivial, which is left to the reader. □

Corollary 8.3. *For $T \in \mathbb{M}_{2^n}$ and each $j \in [n]$, we have*

$$\mathcal{D}_j(T) = \mathcal{E}_{\mathcal{M}_j}(\mathbf{d}_j(T)).$$

Consequently, $\|\mathcal{D}_j(T)\|_{L_p} \leq \|\mathbf{d}_j(T)\|_{L_p}$ for every $1 \leq p \leq \infty$.

Proof. By the last lemma, we have

$$\begin{aligned} \mathcal{E}_{\mathcal{M}_j}(\mathbf{d}_j(T)) &= \mathcal{E}_{\mathcal{M}_j}(T) - \mathcal{E}_{\mathcal{M}_j}\left(\sum_{s_j=0} \widehat{T}(\mathbf{s})\sigma_{\mathbf{s}}\right) \\ &= \mathcal{E}_{\mathcal{M}_j}(T) - \sum_{\max(\mathbf{s})\leq j-1} \widehat{T}(\mathbf{s})\sigma_{\mathbf{s}} \\ &= \mathcal{E}_{\mathcal{M}_j}(T) - \mathcal{E}_{\mathcal{M}_{j-1}}(T) = \mathcal{D}_j(T), \end{aligned}$$

where we used Lemma 8.1 (i) in the third line. Since every conditional expectation is a contraction on $L_p(\mathbb{M}_{2^n})$ for $1 \leq p \leq \infty$, the desired inequality follows immediately. □

Lemma 8.4. *For each $T \in \mathbb{M}_{2^n}$ with $\mathrm{tr}(T) = 0$, we have*

$$\sum_{j=1}^{n} \|\mathbf{d}_j(T)\|_{L_2}^2 = \sum_{j=1}^{n}\sum_{i=1}^{n} \|\mathbf{d}_j(\mathcal{D}_i(T))\|_{L_2}^2.$$

PROOF. Clearly, by the martingale decomposition of T and $\mathrm{tr}(T)=0$, we have $T=\sum_{i=1}^n \mathcal{D}_i(T)$. Hence, the orthonormality of martingale differences yields that

$$\|\mathbf{d}_j(T)\|_{L_2}^2=\sum_{i=1}^n \|\mathbf{d}_j(\mathcal{D}_i(T))\|_{L_2}^2,\ \forall\ j\in[n].$$

Summing over $j\in[n]$ entails the desired identity. □

We now arrive to state and prove the quantum BKS-inequality as follows.

THEOREM 8.5. *Suppose that $T\in\mathbb{M}_{2^n}$ with $\|T\|_{L_\infty}\le 1$ and $T^*=T$. Then*

$$\mathrm{Var}\left(e^{-t\mathcal{L}}(T)\right)\le\Big(\sum_{j=1}^n\|\mathbf{d}_j(T)\|_{L_1}^2\Big)^{\tanh(t)},\quad t\ge 0.$$

PROOF. We assume without loss of generality that $\mathrm{tr}(T)=0$. It is not hard to see that $e^{-t\mathcal{L}}(\mathcal{D}_j(T))=\mathcal{D}_j(e^{-t\mathcal{L}}(T))$ for each $j\in[n]$. Indeed, by Lemma 8.1 (i), we have

$$e^{-t\mathcal{L}}\left(\mathcal{D}_j(T)\right)=e^{-t\mathcal{L}}\Big(\sum_{\max(\mathbf{s})=j}\widehat{T}(\mathbf{s})\sigma_{\mathbf{s}}\Big)=\sum_{\max(\mathbf{s})=j}\widehat{T}(\mathbf{s})e^{-t|\mathbf{s}|}\sigma_{\mathbf{s}}=\mathcal{D}_j(e^{-t\mathcal{L}}(T)).$$

Thus, for each $t\ge 0$, by (8.2) and the hypercontractivity (see Lemma 2.1), we obtain

$$\begin{aligned}
\mathrm{Var}\big(e^{-t\mathcal{L}}(T)\big)&\le\sum_{j=1}^n\big\|e^{-t\mathcal{L}}\big(\mathcal{D}_j(T)\big)\big\|_{L_2}^2\le\sum_{j=1}^n\|\mathcal{D}_j(T)\|_{L_{1+e^{-2t}}}^2\\
&\le\sum_{j=1}^n\|\mathcal{D}_j(T)\|_{L_2}^{2(1-\tanh(t))}\cdot\|\mathcal{D}_j(T)\|_{L_1}^{2\tanh(t)}\\
&\le\Big(\sum_{j=1}^n\|\mathcal{D}_j(T)\|_{L_2}^2\Big)^{1-\tanh(t)}\Big(\sum_{j=1}^n\|\mathcal{D}_j(T)\|_{L_1}^2\Big)^{\tanh(t)}\\
&=\|T\|_{L_2}^{2(1-\tanh(t))}\Big(\sum_{j=1}^n\|\mathcal{D}_j(T)\|_{L_1}^2\Big)^{\tanh(t)}\le\Big(\sum_{j=1}^n\|\mathbf{d}_j(T)\|_{L_1}^2\Big)^{\tanh(t)}.
\end{aligned}$$

Here, in the second inequality, we used the Hölder inequality with parameters $\frac{1}{1+e^{-2t}}=\frac{\tanh(t)}{1}+\frac{1-\tanh(t)}{2}$, and used both the assumption $\|T\|_{L_\infty}\le 1$ and Corollary 8.3 in the last inequality. □

REMARK 8.6. It is not hard to see

$$\frac{1}{2}(1-e^{-2t})\le\tanh(t)\le 1-e^{-2t},\quad\forall t>0.$$

Recall that the classical quantitative BKS inequality proved in [**KK13**, Theorem 4] states that for some $c>0.234$,

$$\mathrm{Var}\big(e^{-t\mathcal{L}}(f)\big)\le 20\Big(\sum_{j=1}^n\|d_j(T)\|_{L_1(\{-1,1\}^n)}^2\Big)^{c(1-e^{-2t})},\quad t\ge 0.$$

It seems that our quantum BKS inequality is better, more or less, since we find that $c\in[1/2,1]$ is good enough for the purpose. Additionally, our proof is simpler than the one given in [**KK13**].

REMARK 8.7. The classical quantitative BKS inequality [**KK13**, Theorem 4] was recently extended into the quantum setting via the some quantum random restriction method by Blecher et al. [**BGX25**]. Our proof of Theorem 8.5 is simpler than [**BGX25**].

8.2. The inverse quantum BKS-inequality

THEOREM 8.8. *For each $t > 0$ and monotone increasing $T \in \mathbb{M}_{2^n}$ defined as in Lemma* 7.1*, we have*

$$\mathrm{Var}\left(e^{-t\mathcal{L}}(T)\right) \geq e^{-2t} \sum_{j=1}^{n} \|\mathbf{d}_j(T)\|_{L_1}^2. \tag{8.3}$$

PROOF. For a monotone increasing element $T \in \mathbb{M}_{2^n}$, we apply the covariance identity (7.5) in Lemma 7.4 (iii) to get

$$\mathrm{Var}(T) = \sum_{j=1}^{n} \int_0^\infty e^{-t} \mathrm{tr}\left(\partial_j(T) \cdot e^{-t\mathcal{L}}\left(\partial_j(T)\right)\right) dt.$$

For each $j \in [n]$, we define

$$h_j(t) \coloneqq \mathrm{tr}\left(\partial_j(T) \cdot e^{-t\mathcal{L}}\left(\partial_j(T)\right)\right), \quad \text{for } t > 0.$$

It is clear that

$$\begin{aligned}
\frac{d}{dt} h_j(t) &= -\mathrm{tr}\left(\partial_j(T) \cdot \mathcal{L} e^{-t\mathcal{L}}\left(\partial_j(T)\right)\right) \\
&= -\sum_{l=1}^{n} \mathrm{tr}\left(\mathbf{d}_l \partial_j(T) \cdot \mathbf{d}_l e^{-t\mathcal{L}}\left(\partial_j(T)\right)\right) \\
&= -\sum_{l=1}^{n} \left\| e^{-t\mathcal{L}/2} \mathbf{d}_l \partial_j(T) \right\|_{L_2}^2 \leq 0.
\end{aligned}$$

This means that $h_j(t)$ is a decreasing function on $t > 0$. Hence,

$$\begin{aligned}
\mathrm{Var}(T) &= \sum_{j=1}^{n} \int_0^\infty e^{-t} h_j(t) dt \\
&\geq \sum_{j=1}^{n} \int_0^\infty e^{-t} \lim_{t\to\infty} h_j(t) dt \qquad (8.4) \\
&= \sum_{j=1}^{n} \mathrm{tr}\left(\partial_j(T)\right)^2 \cdot \left(\int_0^\infty e^{-t} dt \right) = \sum_{j=1}^{n} \|\partial_j(T)\|_{L_1}^2,
\end{aligned}$$

where we used T is monotone increasing in the last line.

Now, for each $t > 0$ note here that $e^{-t\mathcal{L}}(T)$ is still monotone increasing (see Lemma 7.4 (iii)), substituting $e^{-t\mathcal{L}}(f)$ to (8.4) and Lemma 7.3 (ii) entail that

$$\begin{aligned}\mathrm{Var}\left(e^{-t\mathcal{L}}(f)\right) &\geq \sum_{j=1}^{n} \left\|\partial_j\left(e^{-t\mathcal{L}}(f)\right)\right\|_{L_1}^2 \\ &= e^{-2t}\sum_{j=1}^{n} \mathrm{tr}\left(e^{-t\mathcal{L}}\left(\partial_j(f)\right)\right)^2 \\ &= e^{-2t}\sum_{j=1}^{n} \mathrm{tr}\left(\partial_j(f)\right)^2 \\ &= e^{-2t}\sum_{j=1}^{n} \|\partial_j(f)\|_{L_1}^2 \geq e^{-2t}\sum_{j=1}^{n} \|\mathbf{d}_j(f)\|_{L_1}^2,\end{aligned}$$

where the second equality is due to that $e^{-t\mathcal{L}}$ is trace preserving, and we used Lemma 7.4 (i) in the last inequality. □

THEOREM 8.9. *Let $T \in \mathbb{M}_{2^n}$ be monotonically increasing operator. Then*

$$\mathrm{Var}(T) \geq \sum_{j=1}^{n} \|\mathbf{d}_j(T)\|_{L_1(\mathbb{M}_{2^n})}^2.$$

PROOF. Combining (8.4) with Lemma 7.4 (i) yields the expected result. □

CHAPTER 9

Quantum Bohnenblust–Hille inequality and approximation of polynomials from tail spaces

In Chapter 9.1, we improve the constant of the quantum Bohnenblust–Hille inequality, which is the central technical result behind recent progress in learning bounded local Hamiltonians and observables [**HCP23, VZ24b**]. In Chapter 9.2, we study L_1 distance of polynomials on the quantum Boolean from tail spaces. This study relies on the quantum Bohnenblust–Hille inequality. The reduction method used in this chapter is also useful in next chapter concerning quantum learning problems.

9.1. Quantum Bohnenblust–Hille inequality

In [**Ble01**], Blei established Bohnenblust–Hille inequalities for polynomials on Boolean cubes with dimension-free constants: for any function $f \in \mathcal{P}^{\leq d}(\{-1,1\}^n)$, we have the dimension-free estimate

$$\left(\sum_{|A|\leq d} |\widehat{f}(A)|^{\frac{2d}{d+1}}\right)^{\frac{d+1}{2d}} \leq C_d \|f\|_\infty. \tag{9.1}$$

Denoting $\mathrm{BH}_{\{\pm 1\}}^{\leq d}$ the best constant such that (9.1) holds. Defant et al. [**DMP19**] proved that the dimension-free Bohnenblust–Hille constants grow at most subexponentially in the degree, that is, $\mathrm{BH}_{\{\pm 1\}}^{\leq d} \leq C^{\sqrt{d \log d}}$ with $C > 1$ being a universal constant. Recently, this Bohnenblust–Hille inequality has found great applications in learning bounded low-degree functions on Boolean cubes [**EI22**], and in the L_p distance estimate of polynomials on the Boolean from Walsh tail spaces [**EZ24**].

An analog of (9.1) on the quantum Boolean $\mathbb{M}_{2^n}$ was proved by Huang-Chen-Preskill [**HCP23**]. For any operator $T \in \mathbb{M}_{2^n}$ of degree at most $d \in [n]$, one has

$$\left(\sum_{|\mathbf{s}|\leq d} |\widehat{T}(\mathbf{s})|^{\frac{2d}{d+1}}\right)^{\frac{d+1}{2d}} \leq \mathrm{BH}_{\mathbb{M}_2}^{\leq d} \|T\|,$$

where $\mathrm{BH}_{\mathbb{M}_2}^{\leq d} < \infty$ denotes the best constant. Clearly, $\mathrm{BH}_{\{\pm 1\}}^{\leq d} \leq \mathrm{BH}_{\mathbb{M}_2}^{\leq d}$. Huang-Chen-Preskill [**HCP23**] showed that

$$\mathrm{BH}_{\mathbb{M}_2}^{\leq d} \leq d^{O(d)}.$$

Later, Volberg and Zhang [**VZ24b**, Theorem 1.2] improved this constant to be

$$\mathrm{BH}_{\mathbb{M}_2}^{\leq d} \leq 3^d \mathrm{BH}_{\{\pm 1\}}^{\leq d}$$

via a reduction method. In the very recent work [**BSVZ25**], Becker et al. further improved this to be

$$\mathrm{BH}_{\mathbb{M}_2}^{\leq d} \leq \sqrt{3}^{d+1}\mathrm{BH}_{\{\pm 1\}}^{\leq d}.$$

Our main contribution here is to show that

$$\mathrm{BH}_{\mathbb{M}_2}^{\leq d} \leq \sqrt{3}^{d}\mathrm{BH}_{\{\pm 1\}}^{\leq d},$$

and we will also show that this is optimal in some sense (see Remark 9.3 below). Our method draws inspiration from the reduction method developed by Volberg and Zhang [**VZ24b**].

THEOREM 9.1. *The following inequality holds for every $T \in \mathcal{P}(\mathbb{M}_{2^n})$,*

$$\left(\sum_{\substack{\mathbf{s}\in\{0,1,2,3\}^n \\ |\mathbf{s}|\leq d}} |\widehat{T}(\mathbf{s})|^{\frac{2d}{d+1}}\right)^{\frac{d+1}{2d}} \leq 3^{d/2}\mathrm{BH}_{\{\pm 1\}}^{\leq d}\|T\|_{L_\infty(\mathbb{M}_{2^n})}.$$

To begin with, we shall present a useful lemma. Write

$$\varepsilon = \big(\varepsilon_1^{(1)},\dots,\varepsilon_n^{(1)},\varepsilon_1^{(2)},\dots,\varepsilon_n^{(2)},\varepsilon_1^{(3)},\dots,\varepsilon_n^{(3)}\big) \in \{-1,1\}^{3n}.$$

For each $1 \leq j \leq n$, define $\rho_j : \{-1,1\}^3 \to \mathbb{M}_2(\mathbb{C})$ by setting

$$\rho_j(\varepsilon) := \mathbf{1}_2 + \frac{\varepsilon_j^{(1)}\sigma_1 + \varepsilon_j^{(2)}\sigma_2 + \varepsilon_j^{(3)}\sigma_3}{\sqrt{3}} \in \mathbb{M}_2(\mathbb{C}), \tag{9.2}$$

and define $\rho : \{-1,1\}^{3n} \to \mathbb{M}_{2^n}(\mathbb{C})$ by setting

$$\rho := \rho_1 \otimes \cdots \otimes \rho_n. \tag{9.3}$$

LEMMA 9.2. *Let ρ be as in* (9.3)*. Given $d \in [n]$, let $T \in \mathcal{P}^{\leq d}(\mathbb{M}_{2^n})$ and define the function $f_T : \{-1,1\}^{3n} \to \mathbb{C}$ by setting*

$$f_T(\varepsilon) := \mathrm{tr}\big(T\,\rho(\varepsilon)\big), \quad \varepsilon \in \{-1,1\}^{3n}.$$

We have the following properties.

(i) *The matrix ρ is a density matrix, that is, $\rho(\varepsilon) \geq 0$ and $\mathrm{tr}(\rho(\varepsilon)) = 1$ for each $\varepsilon \in \{-1,1\}^{3n}$.*
(ii) *The function f_T is at most of degree d, that is, $f_T \in \mathcal{P}^{\leq d}(\{-1,1\}^{3n})$.*
(iii) *We have $\{\widehat{f_T}(A)\}_{A\subset[3n]} = \{\frac{\widehat{T}(\mathbf{s})}{3^{|\mathbf{s}|/2}}\}_{\mathbf{s}\in\{0,1,2,3\}^n}$ and*

$$\left(\sum_{|A|\leq d} |\widehat{f_T}(A)|^{\frac{2d}{d+1}}\right)^{\frac{d+1}{2d}} = \left(\sum_{\substack{\mathbf{s}\in\{0,1,2,3\}^n \\ |\mathbf{s}|\leq d}} \left|\frac{\widehat{T}(\mathbf{s})}{3^{|\mathbf{s}|/2}}\right|^{\frac{2d}{d+1}}\right)^{\frac{d+1}{2d}}. \tag{9.4}$$

PROOF. (i) It is easy to verify via (2.1) that for each $1 \leq j \leq n$,

$$\frac{\varepsilon_j^{(1)}\sigma_1 + \varepsilon_j^{(2)}\sigma_2 + \varepsilon_j^{(3)}\sigma_3}{\sqrt{3}}$$

is hermitian and unitary. Hence, for each $\varepsilon \in \{-1,1\}^{3n}$, the eigenvalues of $\rho_j(\varepsilon)$ are 0 and 2, and therefore $\rho_j \geq 0$. Also note that

$$\mathrm{tr}_2(\rho_j(\varepsilon)) = \mathrm{tr}_2\Big(\mathbf{1}_2 + \frac{\varepsilon_j^{(1)}\sigma_1 + \varepsilon_j^{(2)}\sigma_2 + \varepsilon_j^{(3)}\sigma_3}{\sqrt{3}}\Big) = 1, \quad 1 \leq j \leq n.$$

Consequently, for each ε,

$$\rho \geq 0, \qquad \operatorname{tr}(\rho(\varepsilon)) = \prod_{j=1}^{n} \operatorname{tr}_2(\rho_j(\varepsilon)) = 1. \tag{9.5}$$

(ii) It is not hard to see that for each $\varepsilon \in \{-1,1\}^{3n}$

$$\begin{aligned} f_T(\varepsilon) &= \sum_{\substack{\mathbf{s}\in\{0,1,2,3\}^n \\ |\mathbf{s}|\leq d}} \widehat{T}(\mathbf{s}) \langle \rho(\varepsilon), \sigma_{\mathbf{s}} \rangle \\ &= \sum_{\substack{\mathbf{s}\in\{0,1,2,3\}^n \\ |\mathbf{s}|\leq d}} \widehat{T}(\mathbf{s}) \prod_{j\in\operatorname{supp}(\mathbf{s})} \langle \rho_j(\varepsilon), \sigma_{s_j} \rangle \\ &= \sum_{\substack{\mathbf{s}\in\{0,1,2,3\}^n \\ |\mathbf{s}|\leq d}} \left(\frac{\widehat{T}(\mathbf{s})}{3^{|\mathbf{s}|/2}} \right) \prod_{j\in\operatorname{supp}(\mathbf{s})} \varepsilon_j^{(s_j)}, \end{aligned} \tag{9.6}$$

where the second equality is due to (9.5) and the last equality is due to the fact $\operatorname{tr}(\sigma_j) = 0$ for $j \in \{1,2,3\}$. The desired assertion follows.

(iii) The formula (9.4) follows from (9.6) directly. □

Now we turn to prove Theorem 9.1.

Proof of Theorem 9.1. Let $T \in \mathcal{P}^{\leq d}(\mathbb{M}_{2^n})$, and let $f_T : \{-1,1\}^{3n} \to \mathbb{C}$ be as in Lemma 9.2. By (9.5) and Hölder's inequality, for each $\varepsilon \in \{-1,1\}^{3n}$, we have

$$\begin{aligned} |f_T(\varepsilon)| &= |\operatorname{tr}(T\rho(\varepsilon))| \leq \|T\|_{L_\infty(\mathbb{M}_{2^n})} \|\rho(\varepsilon)\|_{L_1(\mathbb{M}_{2^n})} \\ &= \|T\|_{L_\infty(\mathbb{M}_{2^n})} \operatorname{tr}(\rho(\varepsilon)) = \|T\|_{L_\infty(\mathbb{M}_{2^n})}, \end{aligned} \tag{9.7}$$

and consequently,

$$\|f_T\|_\infty \leq \|T\|_{L_\infty(\mathbb{M}_{2^n})}. \tag{9.8}$$

Applying the classical Bohnenblust–Hille inequality (9.1) to the function $f_T : \{-1,1\}^{3n} \to \mathbb{C}$ defined in Lemma 9.2, we have

$$\left(\sum_{|A|\leq d} |\widehat{f_T}(A)|^{\frac{2d}{d+1}} \right)^{\frac{d+1}{2d}} \leq \mathrm{BH}_{\{\pm1\}}^{\leq d} \|f_T\|_\infty \leq \mathrm{BH}_{\{\pm1\}}^{\leq d} \|T\|_{L_\infty(\mathbb{M}_{2^n})}.$$

Thus, by Lemma 9.2 (iii), we have

$$\left(\sum_{\substack{\mathbf{s}\in\{0,1,2,3\}^n \\ |\mathbf{s}|\leq d}} \left| \frac{\widehat{T}(\mathbf{s})}{3^{|\mathbf{s}|/2}} \right|^{\frac{2d}{d+1}} \right)^{\frac{d+1}{2d}} \leq \mathrm{BH}_{\{\pm1\}}^{\leq d} \|T\|_{L_\infty(\mathbb{M}_{2^n})}.$$

We can deduce the desired quantum Bohnenblust–Hille inequality whenever we note that $3^{-d/2} \leq 3^{-|\mathbf{s}|/2}$ for all $\mathbf{s} \in \{0,1,2,3\}^n$ with $|\mathbf{s}| \leq d$. The proof is complete. □

Remark 9.3. We now explain that the factor $3^{d/2}$ should be optimal in the case $d = 1$. In fact, we shall show that there exists T such that inequality in Theorem

9.1 can be a equality. Let

$$T \coloneqq \sum_{l=1}^{3} \sigma_{le_1} = \begin{pmatrix} 1 & 1-i \\ 1+i & -1 \end{pmatrix} \otimes \mathbf{1}_2 \otimes \cdots \otimes \mathbf{1}_2.$$

It is not hard to see that $\|T\|_{L_\infty(\mathbb{M}_{2^n})} = 3^{1/2}$ and

$$\sum_{\substack{\mathbf{s}\in\{0,1,2,3\}^n \\ |\mathbf{s}|=1}} |\widehat{T}(\mathbf{s})| = 3.$$

Recall that $\mathrm{BH}^{\leq 1}_{\{\pm 1\}} = 1$. Combing above, we verify the assertion.

Remark 9.4. Our proof of the quantum Bohnenblust–Hille inequality Theorem 9.1 essentially builds upon the reduction method introduced by Volberg and Zhang [**VZ24b**]. However, by judiciously choosing the density matrix ρ and the reduction function f_T here, we are able to obtain improved constants.

In [**VZ24b**], the authors applied the quantum Bohnenblust–Hille inequality to study quantum junta theorem and learning problems of low degree quantum observables. We also study quantum learning problems in Chapter 10 but with different arguments motivated by [**EIS23**].

In Chapter 9.2 below, motivated by [**EZ24**], we shall provide a new and interesting application of the quantum Bohnenblust–Hille inequality.

9.2. Approximation of polynomials from quantum tail spaces

In [**Ole17**], Oleszkiewicz estimated the L_1-distance of the Rademacher sum $f = \sum_{|A|=1} w_A$ from the d-th tail space $\mathcal{T}^{\geq d}(\{-1,1\}^n)$.

Theorem 9.5 (Oleszkiewicz). *There exist universal constants C_1 and C_2 such that for every Rademacher sums $n \in \mathbb{N}$ and every $f \in \mathcal{P}^{=1}(\{-1,1\}^n)$, the following inequality holds*

$$C_1 K(f, m, \ell_2^n, \ell_\infty^n) \leq \inf_{g\in\mathcal{T}^{>m}(\{-1,1\}^n)} \|f-g\|_{L_1} \leq C_2 K(f, m, \ell_2^n, \ell_\infty^n),$$

where $1 \leq m \leq n$.

The K-functional used here is given later. This is a restatement of initial Oleszkiewicz's theorem in terms of the K-functional, which was observed by Eskenazis and Zhang [**EZ24**]. As pointed out by Oleszkiewicz, the method of [**Ole17**] does not appear to extend beyond Rademacher sums. Recently, using a different method by employing the Bohnenblust–Hille inequality (9.1) and a class bound of Figiel on the Rademacher projection of polynomials, Eskenazis and Zhang [**EZ24**] established various bounds for the L_p distance of polynomials on the Boolean from Walsh tail spaces, extending some of Oleszkiewicz's results for Rademacher sums. In this section, we aim to establish counterpart results in [**EZ24**] for the quantum Boolean.

We now recall basic concept like K-functional in interpolation theory. For two compatible Banach spaces X and Y, define norms $\|\cdot\|_{X\cap Y}$ and $\|\cdot\|_{X+Y}$ on $X\cap Y$ and $X+Y$, respectively as follows

$$\|f\|_{X\cap Y} \coloneqq \max\{\|f\|_X, \|f\|_Y\}, \quad \forall\ f \in X\cap Y,$$

and

$$\|f\|_{X+Y} \coloneqq \inf\{\|g\|_X + \|h\|_Y : f = g + h,\ g \in X,\ h \in Y\}.$$

More generally, for each $t > 0$, we define

$$\|f\|_{X\cap tY} \coloneqq \max\{\|f\|_X, t\|f\|_Y\}, \quad \forall\ f \in X \cap Y,$$

and the K-functional as follows

$$K(f,t,X,Y) = \|f\|_{X+tY} \coloneqq \inf\{\|g\|_X + t\|h\|_Y : f = g + h,\ g \in X,\ h \in Y\}.$$

It is well-known that for two compatible Banach spaces X and Y, we have

$$(X \cap tY, \|\cdot\|_{X\cap tY})^* = (X^* + t^{-1}Y^*, \|\cdot\|_{X^*+t^{-1}Y^*}), \quad \forall\ t > 0.$$

We refer to the monograph [**BL76**] for more details.

We first derive a quantum version of the Figiel inequality (see e.g. [**EI22**]). Suppose that $T_d(x) = \sum_{\ell=0}^d c(d,\ell)x^\ell$ is the Chebyshev polynomial of the first kind which is characterized by $T_d(\cos(x)) = \cos(dx)$ for all $x \in \mathbb{R}$. We now define for simplicity

$$\tilde{c}(d,\ell) \coloneqq \begin{cases} \frac{T_d^{(\ell)}(0)}{\ell!}, & \text{if } d-\ell \text{ is even,} \\ \frac{T_{d-1}^{(\ell)}(0)}{\ell!}, & \text{if } d-\ell \text{ is odd.} \end{cases}$$

Recall the following well-known Markov inequality. Let h be a polynomial on $[-1,1]$ with degree d. Then for each positive integer $\ell \le d$ we have

$$|h^{(\ell)}(0)| \le \begin{cases} |T_d^{(\ell)}(0)|\|h\|_{L_\infty}, & \text{if } d-\ell \text{ is even,} \\ |T_{d-1}^{(\ell)}(0)|\|h\|_{L_\infty}, & \text{if } d-\ell \text{ is odd.} \end{cases} \tag{9.9}$$

LEMMA 9.6. *Suppose that $T \in \mathcal{P}^{\le d}(\mathbb{M}_{2^n})$. Then, for each $\ell \in \mathbb{N}$ with $\ell \le d$, we have*

$$\left\|\mathrm{Rad}^{=\ell}(T)\right\|_{L_\infty} \le \tilde{c}(d,\ell)\|T\|_{L_\infty}.$$

PROOF. Without loss of generality, we write $T = \sum_{\mathbf{s}\in\{0,1,2,3\}^n, |\mathbf{s}|\le d} \widehat{T}(\mathbf{s})\sigma_{\mathbf{s}}$. It follows that the function defined by

$$h_T(t) \coloneqq \sum_{\substack{\mathbf{s}\in\{0,1,2,3\}^n \\ |\mathbf{s}|\le d}} \widehat{T}(\mathbf{s})\sigma_{\mathbf{s}} t^{|\mathbf{s}|}, \quad \forall\ t \in [-1,1],$$

satisfies $\sup_{t\in[-1,1]} \|h_T(t)\|_{L_\infty} \le \|T\|_{L_\infty}$. For each $\xi,\ \eta \in L_2(\mathbb{M}_{2^n})$ with $\|\xi\|_{L_2} = \|\eta\|_{L_2} = 1$, we define

$$h_T(t)[\xi,\eta] \coloneqq \langle \xi, h_T(t)\eta\rangle.$$

It is clear that

$$\sup\{h_T(t)[\xi,\eta] : t \in [-1,1], \|\xi\|_{L_2} = \|\eta\|_{L_2} = 1\} \le \|T\|_{L_\infty},$$

and $h_T(\cdot)[\xi,\eta]$ is of degree d on $[-1,1]$. Hence, by (9.9), we get that

$$|h_T^{(\ell)}(0)[\xi,\eta]| \le \begin{cases} |T_d^{(\ell)}(0)|\|T\|_{L_\infty}, & \text{if } d-\ell \text{ is even} \\ |T_{d-1}^{(\ell)}(0)|\|T\|_{L_\infty}, & \text{if } d-\ell \text{ is odd,} \end{cases} \tag{9.10}$$

for every $\xi,\ \eta \in L_2(\mathbb{M}_{2^n})$. By the definition of Rademacher projection $\mathrm{Rad}^{=\ell}$, it follows that for each $\xi,\ \eta \in L_2(\mathbb{M}_{2^n})$ with $\|\xi\|_{L_2} = \|\eta\|_{L_2} = 1$ we have

$$|\langle \xi, \mathrm{Rad}^{=\ell}(T)\eta\rangle| = \Big|\sum_{\substack{\mathbf{s}\in\{0,1,2,3\}^n \\ |\mathbf{s}|=\ell}} \widehat{T}(\mathbf{s})\langle \xi, \sigma_{\mathbf{s}}\eta\rangle\Big| = \Big|\frac{h^{(\ell)}(0)[\xi,\eta]}{\ell!}\Big|. \tag{9.11}$$

Combining with (9.10) and (9.11) and taking the supremum over ξ and η in the unit sphere of $L_2(\mathbb{M}_{2^n})$ yields the desired result. □

For each k, $d \in [n]$ with $d < k$, we define

$$\mathcal{P}_{(d,k]}(\mathbb{M}_{2^n}) \coloneqq \left\{ T \in \mathbb{M}_{2^n} : T = \sum_{\substack{\mathbf{s}\in\{0,1,2,3\}^n \\ d<|\mathbf{s}|\le k}} \widehat{T}(\mathbf{s})\sigma_{\mathbf{s}} \right\}.$$

In what follows, we shall write $\mathrm{Rad}^{\le d} = \sum_{\ell\le d}\mathrm{Rad}^{=\ell}$, $d \in [n]$. Given $T \in \mathbb{M}_{2^n}$, denote by $\widehat{T}$ the vector of Fourier coefficients of T.

LEMMA 9.7. *For each $d < k \le n$, let $N_d \coloneqq \sum_{l=0}^{d} 3^l\binom{n}{l}$. Then the following inequality holds for every $T \in \mathcal{P}^{\le k}(\mathbb{M}_{2^n})$,*

$$\max\left\{ \left\|\widehat{\mathrm{Rad}^{\le d}(T)}\right\|_{\ell_2^{N_d}}, z(k,d)^{-1}\left\|\widehat{\mathrm{Rad}^{\le d}(T)}\right\|_{\ell_{2d/(d+1)}^{N_d}} \right\} \le \inf_{G\in\mathcal{P}_{(d,k]}(\mathbb{M}_{2^n})} \|T-G\|_{L_\infty},$$

where $z(k,d) \coloneqq \mathrm{BH}_{\mathbb{M}_2}^{\le d}\sum_{l=0}^{d}|\tilde{c}(k,l)|$ and $\mathrm{BH}_{\mathbb{M}_2}^{\le d} \le 3^{d/2}\mathrm{BH}_{\{\pm1\}}^{\le d}$ is the constant coming from the quantum Bohnenblust–Hille inequality.

PROOF. Take $T \in \mathcal{P}^{\le k}(\mathbb{M}_{2^n})$. It follows from the Parseval identity that

$$\left\|\widehat{\mathrm{Rad}^{\le d}(T)}\right\|_{\ell_2^{N_d}} = \left\|\mathrm{Rad}^{\le d}(T)\right\|_{L_2} \le \|T\|_{L_2} \le \|T\|_{L_\infty}.$$

On the other hand, by the quantum Bohnenblust–Hille inequality (Theorem 9.1), we have

$$\begin{aligned}\left\|\widehat{\mathrm{Rad}^{\le d}(T)}\right\|_{\ell_{2d/(d+1)}^{N_d}} &\le \mathrm{BH}_{\mathbb{M}_2}^{\le d}\left\|\mathrm{Rad}^{\le d}(T)\right\|_{L_\infty} \\ &\le \mathrm{BH}_{\mathbb{M}_2}^{\le d}\sum_{l=0}^{d}\left\|\mathrm{Rad}^{=l}(T)\right\|_{L_\infty} \le \mathrm{BH}_{\mathbb{M}_2}^{\le d}\sum_{l=0}^{d}|\tilde{c}(k,l)|\|T\|_{L_\infty},\end{aligned}$$

where we used Lemma 9.6 in the last inequality.

Therefore, the Rademacher projection $\widehat{\mathrm{Rad}^{\le d}}$ is a linear operator which is bounded from $(\mathcal{P}^{\le k}(\mathbb{M}_{2^n}), \|\cdot\|_{L_\infty})$ to $\ell_2^{N_d} \cap z(k,d)^{-1}\ell_{2d/(d+1)}^{N_d}$. Note here that $\ker(\mathrm{Rad}^{\le d}) = \mathcal{P}_{(d,k]}(\mathbb{M}_{2^n})$. Hence, the operator $\widehat{\mathrm{Rad}^{\le d}}$ induces an bounded operator (which is still denoted by $\widehat{\mathrm{Rad}^{\le d}}$) from the quotient space $\mathcal{P}^{\le k}(\mathbb{M}_{2^n})/\mathcal{P}_{(d,k]}(\mathbb{M}_{2^n})$ to $\ell_2^{N_d} \cap z(k,d)^{-1}\ell_{2d/(d+1)}^{N_d}$ with the norm less than 1. and the desired inequality follows directly from the definition of the quotient norm. The proof is complete. □

We now apply Lemma 9.7 to study the L_1-distance of the at most d-degree polynomials from the quantum tail spaces, which is a quantum analogy of the approximate theorem of Eskenazis and Zhang [**EZ24**, Theorem 3].

THEOREM 9.8. *For any integers $n \ge k \ge d$, there exists a constant $c_d > 0$ depending only on d such that, for each $T \in \mathcal{P}^{\le d}(\mathbb{M}_{2^n})$, we have*

$$\|T\|_{L_1} \le K\left(\widehat{T}, c_d k^d, \ell_2^{N_d}, \ell_{2d/(d-1)}^{N_d}\right),$$

where $N_d \coloneqq \sum_{l=0}^{d} 3^l \binom{n}{l}$. *Consequently, we have*

$$\inf_{G\in\mathcal{T}^{>k}(\mathbb{M}_{2^n})} \|T-G\|_{L_1} \leq K(\widehat{T}, c_d k^d, \ell_2^{N_d}, \ell_{2d/(d-1)}^{N_d}). \tag{9.12}$$

Proof. The proof follows from an easy duality argument of Lemma 9.7 as follows. Taking the adjoint of the operator $\widehat{\mathrm{Rad}^{\leq d}}$, we get that $\widehat{\mathrm{Rad}^{\leq d}}^*$ is an operator from $\ell_2^{N_d} + z(k,d)\ell_{2d/(d-1)}^{N_d}$ to $\ker(\widehat{\mathrm{Rad}^{\leq d}}(\mathbb{M}_{2^n}))^\perp$ with norm less than 1.

On the other hand, it is easy to see that

$$\ker(\widehat{\mathrm{Rad}^{\leq d}}(\mathbb{M}_{2^n}))^\perp = \mathcal{P}^{\leq d}(\mathbb{M}_{2^n}) \oplus \mathcal{P}^{>k}(\mathbb{M}_{2^n}),$$

If this is the case, the second desired inequality follows directly. and

$$\widehat{\mathrm{Rad}^{\leq d}}^* \left(\{x_{\mathbf{s}}\}_{\substack{\mathbf{s}\in\{0,1,2,3\}^n \\ |\mathbf{s}|\leq d}} \right) = \sum_{\substack{\mathbf{s}\in\{0,1,2,3\}^n \\ |\mathbf{s}|\leq d}} x_{\mathbf{s}}\sigma_{\mathbf{s}}.$$

Hence,

$$\|T\|_{L_1} \leq K(\widehat{T}, z(k,d), \ell_2^{N_d}, \ell_{2d/(d-1)}^{N_d}).$$

To get the desired bound, it suffices to note that

$$z(k,d) = \mathrm{BH}_{\mathbb{M}_2}^{\leq d} \sum_{l=0}^{d} |\tilde{c}(k,l)| \leq \mathrm{BH}_{\mathbb{M}_2}^{\leq d} \sum_{l=0}^{d} \frac{k^l}{l!} \leq e\mathrm{BH}_{\mathbb{M}_2}^{\leq d} k^d.$$

Now we turn to show (9.12). We assume this inequality does not hold. Then, there exist $T \in \mathcal{P}^{\leq d}(\mathbb{M}_{2^n})$ and $\alpha \in (0,\infty)$ such that for some $d \leq k \in [n]$,

$$A \coloneqq \inf_{G\in\mathcal{T}^{>k}(\mathbb{M}_{2^n})} \|T-G\|_{L_1} > \alpha > K(\widehat{T}, c_d k^d, \ell_2^{N_d}, \ell_{2d/(d-1)}^{N_d}).$$

Note that we can always take $G \in \mathcal{T}^{>k}(\mathbb{M}_{2^n})$ such that $\|G\|_{L_1} < \frac{A-\alpha}{2}$. Then

$$\begin{aligned} \|T\|_{L_1} &\geq \|T-G\|_{L_1} - \|G\|_{L_1} \\ &\geq A - \frac{A-\alpha}{2} \\ &> A - (A-\alpha) = \alpha, \end{aligned}$$

which contradicts the first assertion of the theorem. □

Even in the commutative setting, Theorem 9.8 is far from optimal; see [**EZ24**, Theorem 3] and its related discussion there. However, in the case $d = 1$, Theorem 9.8 is optimal. We show this by establishing the quantum version of the Oleszkiewicz inequality (Theorem 9.5 above).

Theorem 9.9. *There exist universal constant* $C_1, C_2 > 0$ *that*

$$C_1^{-1} K\left(\widehat{T}, m, \ell_2^{3n}, \ell_\infty^{3n}\right) \leq \inf_{G\in\mathcal{T}^{>m}(\mathbb{M}_{2^n})} \|G-T\|_{L_1} \leq C_2 K\left(\widehat{T}, m, \ell_2^{3n}, \ell_\infty^{3n}\right),$$

for $T \in \mathcal{P}^{=1}(\mathbb{M}_{2^n})$ *and* $1 \leq m \leq 3n$.

Proof. The right-hand side inequality follows from Theorem 9.8. It suffices to show the left-hand side inequality. We here apply the reduction method used in the proof of Theorem 9.1. Let ρ_j be as in (9.2) and define

$$\gamma \coloneqq \frac{\rho_1}{2} \otimes \frac{\rho_2}{2} \cdots \otimes \frac{\rho_n}{2}.$$

It was already shown in the proof of Lemma 9.2 that for each $j \in [n]$ and $\varepsilon \in \{-1,1\}^{3n}$, the eigenvalues of $\rho_j(\varepsilon)$ are 0 and 2. Thus, it follows that for each $\varepsilon \in \{-1,1\}^{3n}$, $\|\gamma(\varepsilon)\|_{L_\infty} = 1$.

For $T \in \mathcal{P}^{=1}(\mathbb{M}_n)$ and $G \in \mathcal{T}^{>m}(\mathbb{M}_n)$, we define two functions on $\{-1,1\}^{3n}$ as

$$f(\cdot) \coloneqq \langle \gamma(\cdot), T\rangle, \quad \text{and} \quad g(\cdot) \coloneqq \langle \gamma(\cdot), G\rangle.$$

The same reason as to (9.6), we have $f \in \mathcal{P}^{=1}(\{-1,1\}^{3n})$ and $g \in \mathcal{T}^{>m}(\{-1,1\}^{3n})$. Moreover, we have

$$\{\widehat{T}(\mathbf{s})\}_{\mathbf{s}} = 2\sqrt{3}\{\widehat{f}(A)\}_A.$$

Applying the Oleszkiewicz inequality (i.e., Theorem 9.5) to the functions f and g, it gives that

$$\begin{aligned} K\left(\widehat{T}, m, \ell_2^{3n}, \ell_\infty^{3n}\right) &= 2\sqrt{3}K\left(f, m, \ell_2^{3n}, \ell_\infty^{3n}\right) \\ &\le C_1^{-1}\mathbb{E}|f-g| \\ &= C_1^{-1}\mathbb{E}\left(|\langle \gamma(\varepsilon), T-G\rangle|\right) \\ &\le C_1^{-1}\mathbb{E}\left(\|\gamma(\varepsilon)\|_{L_\infty}\right)\|T-G\|_{L_1} \\ &= C_1^{-1}\|T-G\|_{L_1}, \end{aligned}$$

where the expectation is taken over $\{-1,1\}^{3n}$. Since $G \in \mathcal{T}^{>m}(\mathbb{M}_{2^n})$ is arbitrary, the desired result follows. □

Particularly, they obtained a more accurate estimate for *symmetric polynomials* in [**EZ24**, Theorem 4]. Inspired by this, we introduce the quantum analogies of *symmetric polynomials* as follows.

Definition 9.10. For each $l \in [n]$, we say

$$E_l \coloneqq \sum_{\substack{\mathbf{s}\in\{0,1,2,3\}^n \\ |\mathbf{s}|=l}} \sigma_{\mathbf{s}}$$

is the *quantum elementary symmetric polynomial of order* l. We say $T \in \mathbb{M}_{2^n}$ is a *quantum symmetric polynomial of degree at most* $d \in [n]$ if T is given by $T \coloneqq \sum_{l=0}^d \alpha_l E_l$ for some sequence $\{\alpha_l\}_l \subset \mathbb{C}$.

We first collect a basic property for each quantum elementary symmetric polynomial of order l. To this end, for each $j \in [n]$ and $A \subseteq [n]$, we define

$$u_j \coloneqq \frac{1}{\sqrt{3}}\sum_{l=1}^3 \sigma_{le_j} \in \mathbb{M}_{2^n} \tag{9.13}$$

and

$$u_A \coloneqq \prod_{j\in A} u_j, \quad u_\emptyset \coloneqq \mathbf{1}. \tag{9.14}$$

Note here that $u_iu_j = u_ju_i$ for each $i, j \in [n]$ and $\{u_A\}_{A\subseteq[n]}$ forms an orthonormal sequence in $L_2(\mathbb{M}_{2^n})$. Indeed, for each $A, B \subseteq [n]$ with $A \neq B$, we have

$$\|u_A\|_{L_2}^2 = \prod_{j\in A}\|u_j\|_{L_2}^2 = 1, \tag{9.15}$$

and we assume without loss of generality that $i \in A \setminus B$, then

$$\langle u_A, u_B\rangle = \mathrm{tr}_2(u_i)\cdot\langle u_{A\setminus\{i\}}, u_B\rangle = 0. \tag{9.16}$$

Let $p \coloneqq \prod_{j=1}^{n}(\mathbf{1}+u_j)$ be the *Riesz product*. Then it is clear that

$$p = \sum_{A\subseteq[n]} u_A. \tag{9.17}$$

Since each $\mathbf{1}+u_j$ is nonnegative and $\{1+u_j\}_{j\in[n]}$ are commutative, it follows from that p is positive. Thus

$$\|p\|_{L_1} = \operatorname{tr}\left(\sum_{A\subseteq[n]} u_A\right) = \operatorname{tr}(u_\emptyset) = 1. \tag{9.18}$$

LEMMA 9.11. *Let E_l be a elementary symmetric polynomials for some $l \in [n]$. Then*

$$E_l = 3^{l/2} \sum_{\substack{A\subseteq[n]\\|A|=l}} u_A.$$

PROOF. It is clear that

$$E_l = \sum_{\substack{\mathbf{s}\in\{0,1,2,3\}^n\\|\mathbf{s}|=l}} \sigma_{\mathbf{s}} = \sum_{\substack{A\subseteq[n]\\|A|=l}} \sum_{\substack{\mathbf{s}\in\{0,1,2,3\}^n\\ \operatorname{supp}(\mathbf{s})=A}} \sigma_{\mathbf{s}} = 3^{l/2} \sum_{\substack{A\subseteq[n]\\|A|=l}} u_A.$$

□

THEOREM 9.12. *Let $n \geq k \geq d$ be integers. Then for each symmetric polynomial T of degree at most d, we have*

$$\|T\|_{L_1} \leq \sum_{l=0}^{d} 3^{l/2}|\alpha_l| \cdot |\tilde{c}(k,l)|, \quad T = \sum_{l=0}^{d} \alpha_l E_l. \tag{9.19}$$

PROOF. Since every symmetric polynomial is a linear combination of elementary symmetric polynomials, by the triangle inequality it suffices to bound $\|E_l\|_{L_1}$ for each $l \in [d]$. Take $G \in L_\infty(\mathbb{M}_{2^n})$ with $\|G\|_{L_\infty} \leq 1$. By Lemma 9.11, we have

$$\operatorname{tr}(E_l \cdot G) = 3^{l/2} \sum_{\substack{A\subseteq[n]\\|A|=l}} \langle u_A, G\rangle = 3^{l/2} \left\langle \sum_{A\subseteq[n]} u_A, \operatorname{Rad}^{=l}(G)\right\rangle = 3^{l/2} \left\langle p, \operatorname{Rad}^{=l}(G)\right\rangle,$$

where p is given as in (9.17). Hence, for arbitrary $G \in L_\infty(\mathbb{M}_n)$ with $\|G\|_{L_\infty} \leq 1$, we have

$$|\operatorname{tr}(E_l \cdot G)| \leq 3^{l/2}\|p\|_{L_1}\|\operatorname{Rad}^{=l}(G)\|_{L_\infty} \leq 3^{l/2}|\tilde{c}(k,l)|,$$

where we used the Figiel inequality (i.e., Lemma 9.6) and (9.18) in the last line. This implies

$$\|E_l\|_{L_1} \leq 3^{l/2}|\tilde{c}(k,l)|$$

for each $l \in [d]$. It now remains to apply the triangle inequality to conclude that

$$\|T\|_{L_1} \leq \sum_{l=0}^{d} 3^{l/2}|\alpha_l| \cdot \|E_l\|_{L_1} \leq \sum_{l=0}^{d} 3^{l/2}|\alpha_l| \cdot |\tilde{c}(k,l)|.$$

The proof of this theorem is complete. □

REMARK 9.13. In the classical setting of Boolean cubes, let f_l be the l-th elementary symmetric polynomial, defined by

$$f_l(\varepsilon) \coloneqq \sum_{\substack{A\subseteq[n]\\|A|=l}} w_A(\varepsilon), \quad \varepsilon \in \{-1,1\}^n, \text{ with } l \in [n].$$

The upper bound for symmetric polynomials with respect to L_1 norm is due to Esekenazis and Zhang (see [**EZ24**, Theorem 4]), which states that $\|f_l\|_{L_1} \leq |\tilde{c}(k,l)|$ for each $n \geq k \geq l$. By contrast, in the quantum counterpart, the factor $3^{l/2}$ appearing on the right hand side of (9.19) is necessary, as shown in the elementary example below. For the case $n = l = 1$, we have

$$e_1 = \sigma_1 + \sigma_2 + \sigma_3 = \begin{pmatrix} 1 & 1-i \\ 1+i & -1 \end{pmatrix}.$$

Hence, $\|e_1\|_{L_1} = 3^{1/2}$. For this same case, the Chebyshev polynomial of the first kind of degree 1 is $T_1(x) = x$ for all $x \in \mathbb{R}$, which gives $|\tilde{c}(1,1)| = 1$. This show the necessity of the desired factor in (9.19).

We now conclude this section with the following lower estimate of quantum analogy of the L_1-distance of elementary symmetric polynomials from tail spaces.

THEOREM 9.14. *Let E_d be the quantum elementary symmetric polynomial of order d. For integers $n \geq k \geq d$, there exists a positive function $\varepsilon(n,k,d)$ with $\lim_{n\to\infty} \varepsilon(n,k,d) = 0$ such that*

$$3^{d/2}\left(|\tilde{c}(k,d)| - \varepsilon(n,k,d)\right) \leq \inf_{G\in\mathcal{T}^{>k}(\mathbb{M}_{2^n})} \|E_d - G\|_{L_1} \leq 3^{d/2}|\tilde{c}(k,d)|.$$

PROOF. Let T_k be the k-th Chebyshev polynomial of first kind, and we define

$$h_{n,k}(\varepsilon) \coloneqq T_k\left(\frac{1}{n}\sum_{j=1}^{n}\varepsilon_j\right), \quad \forall\, \varepsilon = \{\varepsilon_j\}_{j=1}^n \in \{-1,1\}^n.$$

It is clear that $\|h_{n,k}\|_{L_\infty} \leq \sup_{t\in[-1,1]} |T_k(t)| \leq 1$.

Since $h_{n,k}$ is a symmetric polynomial, it follows that

$$h_{n,k} = \sum_{j=0}^{k} \beta_{j,k,n} f_j, \tag{9.20}$$

where $\beta_{j,k,n} \in \mathbb{R}$ and f_l is the l-th elementary symmetric polynomial on $\{-1,1\}^n$. We now define $\xi_l \coloneqq \sum_{\substack{A\subseteq[n]\\|A|=l}} u_A$, where u_A is given as in (9.14) for each $A \subseteq [n]$. Furthermore, let

$$H_{n,k} \coloneqq \sum_{j=0}^{k} \beta_{j,k,n}\xi_j = \sum_{j=0}^{k} \frac{\beta_{j,k,n}}{3^{j/2}} E_j,$$

where we used Lemma 9.11. We denote that $\mathcal{F} \coloneqq \operatorname{span}\{u_A : A \subseteq [n]\} \subseteq \mathbb{M}_{2^n}$, and define $\Gamma : L_\infty(\{-1,1\}^n) \to \mathcal{F}$ by

$$f = \sum_{A\subseteq[n]} \widehat{f}(A) w_A \mapsto \sum_{A\subseteq[n]} \widehat{f}(A) u_A.$$

From the fact that $\{u_A\}_{A\subseteq[n]}$ commute with each other, it is easy to see that Γ is a trace preserving $*$-homomorphism. Thus, we have

$$\|f\|_{L_{2m}} = \|\Gamma(f)\|_{L_{2m}}, \quad m \in \mathbb{N}. \tag{9.21}$$

Letting $m \to \infty$ on both sides of (9.21) yields that

$$\|f\|_{L_\infty} = \|\Gamma(f)\|_{L_\infty}, \quad f \in L_\infty(\{-1,1\}^n).$$

Due to $\Gamma(h_{n,k}) = H_{n,k}$, we have $\|h_{n,k}\|_{L_\infty} = \|H_{n,k}\|_{L_\infty} = 1$.

Case I. If $k-d$ is even, we define

$$\varphi_{d,k,n} \coloneqq \sum_{\substack{0\le j\le d\\ d-j \text{ is even}}} \operatorname{sign}(\beta_{j,k,n}) E_j.$$

By Theorem 9.12, it follows that

$$\|\varphi_{d,k,n}\|_{L_1} \le \sum_{\substack{0\le j\le d\\ d-j \text{ is even}}} 3^{j/2} |\tilde{c}(k,j)|. \tag{9.22}$$

For any $G \in \mathcal{T}^{>k}(\mathbb{M}_{2^n})$, it follows from $\|H_{k,n}\|_{L_\infty} \le 1$ that

$$\begin{aligned}
\|\varphi_{d,k,n} - G\|_{L_1} &\ge \operatorname{tr}(\varphi_{d,k,n} H_{k,n}) \\
&= \sum_{\substack{0\le d\le j\\ d-j \text{ is even}}} \frac{|\beta_{j,k,n}|}{3^{j/2}} \|E_j\|_{L_2}^2 \\
&= \sum_{\substack{0\le j\le d\\ d-j \text{ is even}}} 3^{j/2} |\beta_{j,k,n}| \binom{n}{j} \\
&= \sum_{\substack{0\le j\le d\\ d-j \text{ is even}}} 3^{j/2} |\beta_{j,k,n}| f_j(1,\dots,1).
\end{aligned} \tag{9.23}$$

Since (9.23) holds for arbitrary $G \in \mathcal{T}^{>k}(\mathbb{M}_{2^n})$, it yields

$$\inf_{G\in\mathcal{T}^{>k}(\mathbb{M}_{2^n})} \|\varphi_{d,k,n} - G\|_{L_1} \ge \sum_{\substack{0\le j\le d\\ d-j \text{ is even}}} 3^{j/2} |\beta_{j,k,n}| f_j(1,\dots,1). \tag{9.24}$$

It is clear that $E_d = \operatorname{sign}(\beta_{d,k,n})(\varphi_{d,k,n} - \varphi_{d-2,k,n})$ with $\varphi_{-1,k,n} = \varphi_{0,k,n} \equiv 0$. By the triangle inequality and (9.22) we get that

$$\begin{aligned}
&\inf_{G\in\mathcal{T}^{>k}(\mathbb{M}_{2^n})} \|E_d - G\|_{L_1} \\
&\quad\geq \inf_{G\in\mathcal{T}^{>k}(\mathbb{M}_{2^n})} \|\varphi_{d,k,n} - G\|_{L_1} - \inf_{G\in\mathcal{T}^{>k}(\mathbb{M}_{2^n})} \|\varphi_{d-2,k,n} - G\|_{L_1} \\
&\quad\geq \sum_{\substack{0\leq j\leq d\\ d-j \text{ is even}}} 3^{j/2}|\beta_{j,k,n}| f_j(1,\dots,1) - \sum_{\substack{0\leq j\leq d-2\\ d-2-j \text{ is even}}} 3^{j/2}|\tilde{c}(k,j)| \\
&\quad\geq \sum_{\substack{0\leq j\leq d,\\ d-j \text{ is even}}} 3^{j/2}|\tilde{c}(k,j)| - \sum_{\substack{0\leq j\leq d\\ d-j \text{ is even}}} \frac{2(k+1)!3^{j/2}\max_{l\geq j}|\tilde{c}(k,l)|}{n} \\
&\qquad - \sum_{\substack{0\leq j\leq d-2\\ d-2-j \text{ is even}}} 3^{j/2}|\tilde{c}(k,j)| \\
&\quad\geq 3^{d/2}\left(|\tilde{c}(k,d)| - \frac{2d(k+1)!\max_{k\geq l\geq 0}|\tilde{c}(k,l)|}{n}\right),
\end{aligned}$$

where we used [**IRR**$^+$**21**, Lemma 27] in the third inequality. This completes the proof for the case $k-d$ is even.

Case II. If $k-d$ is odd, we need to reduce the approach to the even case by considering $k-d-1$. Indeed, we define $\varphi_{-1,k-1,n} = \varphi_{0,k-1,n} \equiv 0$ and

$$E_d = \operatorname{sign}(\beta_{d,k-1,n})(\varphi_{d,k-1,n} - \varphi_{d-2,k-1,n}) \quad 0 \leq d < k.$$

Recall that

$$|\tilde{c}(k,d)| = \begin{cases} \frac{|T_k^{(d)}(0)|}{d!}, & \text{if } k-d \text{ is even,} \\ \frac{|T_{k-1}^{(d)}(0)|}{d!}, & \text{if } k-d \text{ is odd.} \end{cases} \tag{9.25}$$

Hence,

$$\begin{aligned}
\inf_{G\in\mathcal{T}^{>k}(\mathbb{M}_{2^n})} \|E_d - G\|_{L_1} &\geq \inf_{G\in\mathcal{T}^{>k}(\mathbb{M}_{2^n})} \|\varphi_{d,k-1,n} - G\|_{L_1} - \inf_{G\in\mathcal{T}^{>k}(\mathbb{M}_{2^n})} \|\varphi_{d-2,k-1,n} - G\|_{L_1} \\
&\geq 3^{d/2}\left(|\tilde{c}(k,d)| - \frac{2d(k+1)!\max_{k\geq l\geq 0}|\tilde{c}(k,l)|}{n}\right),
\end{aligned}$$

where we used [**IRR**$^+$**21**, Lemma 27] and (9.25) in the second inequality. The proof is complete. □

CHAPTER 10

Learning quantum observables

In this chapter, we have two main objectives. The first is to investigate the optimal asymptotics of the quantum random query model in quantum learning theory, and the second is to study quantum learning problems of related types. To describe the quantum learning problem, let us first review the classical one. Let $\mathcal{F}$ be a class of functions on $\{-1,1\}^n$. Given $f \in \mathcal{F}$ and access to examples $(X_1, f(X_1)), \ldots, (X_Q, f(X_Q))$, the goal is to algorithmically construct a hypothesis function $h : \{-1,1\}^n \to \mathbb{R}$ which effectively approximates f. Different access models to examples give rise to concrete versions of the learning problem. The two most standard models are the query model and random query model which are stated as follows

PROBLEM 10.1 (Query model). *Let $\epsilon \in (0,1)$. Determine the least number $Q = Q_{\{\pm 1\}}(\mathcal{F}, \epsilon)$ such that for any $f \in \mathcal{F}$ together with Q queries*

$$(X_1, f(X_1)), \ldots, (X_Q, f(X_Q)),$$

one can construct a function $h : \{-1,1\}^n \to \mathbb{R}$ such that

$$\|h - f\|_{L_2}^2 \leq \varepsilon.$$

PROBLEM 10.2 (Random query model). *Let $\epsilon, \delta \in (0,1)$. Determine the least number $Q_r = Q_{\{\pm 1\},r}(\mathcal{F}, \epsilon, \delta)$ such that for any $f \in \mathcal{F}$ together with Q random queries*

$$(X_1, f(X_1)), \ldots, (X_Q, f(X_Q)),$$

one can construct a random function $h : \{-1,1\}^n \to \mathbb{R}$ such that

$$\|h - f\|_{L_2}^2 \leq \varepsilon$$

with probability at least $1 - \delta$.

The least number Q (resp. Q_r) of examples required to solve the learning problem in each case is called the query complexity of the model. The query complexity of learning problems has been studied extensively for various classes $\mathcal{F}$ of functions on the discrete Boolean $\{-1,1\}^n$. Set

$$\mathcal{F}_{n,d} \stackrel{\text{def}}{=} \{f : \{-1,1\}^n \to [-1,1] : f \text{ has degree at most } d\}. \tag{10.1}$$

In [**EI22**], by applying the Bohnenblust–Hille inequality (9.1), Eskenazis and Ivanisvili, proved that

$$Q_{\{\pm 1\},r}(\mathcal{F}_{n,d}, \epsilon, \delta) \leq c_d \frac{1}{\epsilon^{d+1}} \log\left(\frac{n}{\delta}\right), \tag{10.2}$$

where c_d depends on the Bohnenblust–Hille constant $\mathrm{BH}_{\{\pm 1\}}^{\leq d}$. This improves significantly the previous work studied in [**LMN93b**] and [**IRR**$^+$**21**]. Recently, Eskenazis, Ivanisvili, and Streck [**EIS23**] established the sharpness as of the above upper

bound (10.2) as $n \to \infty$. More precisely, they showed that there exists constant $c_{d,\epsilon,\delta}$ such that

$$Q_{\{\pm 1\},r}(\mathcal{F}_{n,d}, \epsilon, \delta) \geq c_{d,\epsilon,\delta} \log(n). \tag{10.3}$$

In the same paper, Eskenazis, Ivanisvili, and Streck [**EIS23**] also established a more robust version of the upper bound (10.2) that applies to different concept classes $\mathcal{F}$ which are not necessarily of bound degree.

Now suppose that we want to learn quantum observable T over n-qubits, i.e. $T \in \mathbb{M}_{2^n}$. Recall that a matrix $\rho \in \mathbb{M}_{2^n}$ is a density matrix whenever $\rho \geq 0$ and $\mathrm{tr}(\rho) = 1$. Let $\mathcal{S}$ be some set of density matrices in $\mathbb{M}_{2^n}$. The quantum learning problem is similar to the classical setting, however the queries are now replaced with

$$\mathrm{tr}[T\rho], \quad \rho \sim \mathcal{S},$$

where ρ are density matrices in $\mathbb{M}_{2^n}$. We elaborate on the concrete matters of quantum learning problems as follows.

PROBLEM 10.3 (Quantum query model). *Let $\mathcal{Q} \subset \mathbb{M}_{2^n}$ and let $\epsilon \in (0,1)$. Determine the least number $Q_{\mathbb{M}_2} = Q_{\mathbb{M}_2}(\mathcal{Q}, \epsilon)$ such that for any $T \in \mathcal{Q}$ together with $Q_{\mathbb{M}_2}$ queries*

$$(\rho^{(1)}, \mathrm{tr}(T\rho^{(1)})), \ldots, (\rho^{(Q_{\mathbb{M}_2})}, \mathrm{tr}(T\rho^{(Q_{\mathbb{M}_2})})), \quad \rho^{(k)} \in \mathcal{S},$$

one can construct another observable $\widetilde{T} \in \mathbb{M}_{2^n}$ such that

$$\|T - \widetilde{T}\|_{L_2}^2 \leq \epsilon.$$

PROBLEM 10.4 (Quantum random query model). *Let $\mathcal{Q} \subset \mathbb{M}_{2^n}$ and let $\epsilon, \delta \in (0,1)$. Determine the least number $Q_{\mathbb{M}_2,r} = Q_{\mathbb{M}_2,r}(\mathcal{Q}, \epsilon, \delta)$ such that for any $T \in \mathcal{Q}$ together with $Q_{\mathbb{M}_2,r}$ queries*

$$(\rho^{(1)}, \mathrm{tr}(T\rho^{(1)})), \ldots, (\rho^{(Q_{\mathbb{M}_2,r})}, \mathrm{tr}(T\rho^{(Q_{\mathbb{M}_2,r})})), \quad \rho^{(k)} \in \mathcal{S},$$

one can construct another 'random' observable $\widetilde{T} \in \mathbb{M}_{2^n}$ such that

$$\|T - \widetilde{T}\|_{L_2}^2 \leq \epsilon$$

with probability at least $1 - \delta$.

Recently, the learning problems similar to the upper bound (10.2) for low degree quantum observables were investigated by Volberg and Zhang [**VZ24b**] via the newly constructed quantum Bohnenblust–Hille inequality (see e.g. Theorem 9.1). For some $d \in [n]$, let

$$\mathcal{Q}_{n,d} := \{T = T^* \in \mathbb{M}_{2^n} : \|T\|_\infty \leq 1, T \in \mathcal{P}^{\leq d}(\mathbb{M}_{2^n})\}.$$

Volberg and Zhang's result states that for $\epsilon, \delta \in (0,1)$,

$$Q_{\mathbb{M}_2,r}(\mathcal{Q}_{n,d}, \epsilon, \delta) \leq \frac{C^{d\sqrt{d \log d}}}{\epsilon^{d+1}} \log(\frac{n}{\delta}) \tag{10.4}$$

for some large enough $C > 0$. Our first goal of the chapter is to show that (10.4) is asymptotically optimal as $n \to \infty$.

THEOREM 10.5. *For $\epsilon, \delta \in (0,1)$, we have*

$$Q_{\mathbb{M}_2,r}(\mathcal{Q}_{n,d}, \epsilon, \delta) \simeq_{d,\epsilon,\delta} \log(n).$$

This is the quantum analogue of (10.3). In Chapter 10.1, we provide its proof.

10.1. Asymptotic sharpness

The main objective here is to prove Theorem 10.5. Since we already know (10.4), it remains to show

$$Q_{\mathbb{M}_2,r}(\mathcal{Q}_{n,d},\epsilon,\delta) \geq c_{d,\epsilon,\delta}\log(n). \tag{10.5}$$

Let

$$\mathcal{B}_n := \{T \in \mathbb{M}_{2^n} : T = T^* = T^{-1}\}$$

and let

$$\mathcal{B}_{n,d} := \{T \in \mathcal{B}_n : T \in \mathcal{P}^{\leq d}(\mathbb{M}_{2^n})\}, \quad d \in [n].$$

As it is clear that $Q_{\mathbb{M}_2,r}(\mathcal{Q}_{n,d},\epsilon,\delta) \geq Q_{\mathbb{M}_2,r}(\mathcal{B}_{n,d},\epsilon,\delta)$, then (10.5) follows by combining Theorem 10.6 and Theorem 10.9 (or Theorem 10.10).

Proof of Theorem 10.5. The assertion follows by (10.4), Theorem 10.6 and Theorem 10.9 or Theorem 10.10. □

Now our primary remaining task is to establish Theorem 10.6 and Theorem 10.9. To proceed, let us recall the *packing number* of a metric space $(\mathcal{M},\wp)$ as follows. For any $\epsilon > 0$, the subset $\mathcal{N} \subseteq \mathcal{M}$ is said to be ϵ-packing, if for each ξ, $\xi' \in \mathcal{N}$, we have $\wp(\xi,\xi') > \epsilon$. For arbitrary $\epsilon > 0$, the packing number of the metric space $(\mathcal{M},\wp)$, denoted by $M(\mathcal{M},\wp,\epsilon)$, is defined by

$$M(\mathcal{M},\wp,\epsilon) := \sup\{|\mathcal{N}| : \mathcal{N} \subseteq \mathcal{M}, \text{ and } \mathcal{N} \text{ is an } \epsilon\text{-packing subset}\}.$$

The next result provides a lower bound for the (random) query complexity $Q_{\mathbb{M}_2,r}(\mathcal{Q}_{n,d},\epsilon,\delta)$ via the packing number.

Theorem 10.6. *Let $d \in [n]$. For every $\epsilon > 0$ and every $\delta \in (0,1)$,*

$$Q_{\mathbb{M}_2,r}(\mathcal{Q}_{n,d},\epsilon,\delta) \geq \log_2 M(\mathcal{Q}_{n,d},\|\cdot\|_2,2\sqrt{\epsilon}) + \log_2(1-\delta). \tag{10.6}$$

Proof. For simplicity, let

$$M := M(\mathcal{Q}_{n,d},\|\cdot\|_2,2\sqrt{\epsilon}).$$

By the definition of the packing number, we can take $A_1,\cdots,A_M \in \mathcal{Q}_{n,d}$ such that

$$\|A_i - A_j\|_{L_2} > 2\sqrt{\epsilon}, \quad i \neq j. \tag{10.7}$$

A random-example learning algorithm of length Q is a deterministic map

$$H : (\mathcal{S} \times \{-1,1\})^Q \to L_2(\mathbb{M}_{2^n},\mathrm{tr}).$$

According to the proof of (10.4) provided in [**VZ24b**], we can assume that the samples are generated by

$$\varepsilon^{(1)},\ldots,\varepsilon^{(Q)} \in \{-1,1\}^{3n}.$$

Define a probability measure on $\{-1,1\}$ with respect to some $A \in \mathcal{Q}_{n,d}$ and $\varepsilon \in \{-1,1\}^{3n}$ by setting

$$\mathbb{P}\big(Y = \pm 1 \mid \varepsilon, A\big) = \mathrm{tr}\left(\frac{1 \pm A}{2}\rho\big(\varepsilon\big)\right) = \frac{1 \pm \mathrm{tr}\big(A\rho(\varepsilon)\big)}{2}, \tag{10.8}$$

where ρ is defined as in (9.3).

In the sequel, we denote by

$$Q := Q_{\mathbb{M}_2,r}(\mathcal{Q}_{n,d},\epsilon,\delta)$$

the least $Q \in \mathbb{N}$ such that there exists such an algorithm satisfying for each $A \in \mathcal{Q}_{n,d}$ (10.9)

$$\mathbb{P}_{\varepsilon^{(1)},\dots,\varepsilon^{(Q)}\in\{-1,1\}^{3n}}\Big(\big\|H\big((\rho(\varepsilon^{(1)}),Y_1),\dots,(\rho(\varepsilon^{(Q)}),Y_Q)\big)-A\big\|_2^2\le\epsilon\Big)\ge 1-\delta.$$

Let

$$\widetilde{\varepsilon}=(\varepsilon^{(1)},\dots,\varepsilon^{(Q)})$$

be independent random vectors such that there exists such an algorithm satisfying $\rho(\varepsilon^{(t)}) \in \mathcal{S}, t = 1,\dots,Q$ and (10.9) for some learning algorithm H. For each $i \in \{1,\dots,M\}$ and each fixed $\widetilde{\varepsilon}$, let $\mathbb{P}_{i,\widetilde{\varepsilon}}$ denote the conditional distribution on $\{-1,1\}^Q$ of the measurement outcome string when the target observable is A_i. That is, by (10.8), we have

$$\mathbb{P}_{i,\widetilde{\varepsilon}}(Y)=\prod_{t=1}^{Q}\frac{1+Y_t\mathrm{tr}(A_i\rho(\varepsilon^{(t)}))}{2},\quad Y=(Y_1,\cdots,Y_Q)\in\{-1,1\}^Q.$$

It is clear that $\mathbb{P}_{i,\widetilde{\varepsilon}}$ is a probability measure on $\{-1,1\}^Q$. For each $i \in \{1,\dots,M\}$, define

$$E_i(\widetilde{\varepsilon}):=\{Y\in\{-1,1\}^Q:\|H\big((\rho(\varepsilon^{(1)}),Y_1),\dots,(\rho(\varepsilon^{(Q)}),Y_Q)\big)-A_i\|_2^2\le\epsilon\}.$$

For every fixed $\widetilde{\varepsilon}$, the sets $E_i(\widetilde{\varepsilon}),\dots,E_M(\widetilde{\varepsilon})$, are pairwise disjoint. Indeed, if $Y \in E_i(\widetilde{\varepsilon}) \cap E_j(\widetilde{\varepsilon})$ with $i \neq j$, then

$$\begin{aligned}\|A_i-A_j\|_2&\le\|A_i-H((\rho(\varepsilon^{(1)}),Y_1),\cdots,(\rho(\varepsilon^{(Q)}),Y_Q))\|_2\\&\quad+\|H((\rho(\varepsilon^{(1)}),Y_1),\cdots,(\rho(\varepsilon^{(Q)}),Y_Q))-A_j\|_2\le 2\sqrt{\varepsilon},\end{aligned}$$

contradicting the packing property (10.7). Hence, by (10.9), for each $i = 1,\dots,M$, we have

$$\begin{aligned}&\mathbb{E}_{\widetilde{\varepsilon}}\big[\mathbb{P}_{i,\widetilde{\varepsilon}}(E_i(\widetilde{\varepsilon}))\big]=\mathbb{E}_{\widetilde{\varepsilon}}\big[\mathbb{P}(Y\in E_i(\widetilde{\varepsilon})|\widetilde{\varepsilon},A_i)\big]\\&=\mathbb{P}_{\varepsilon^{(1)},\dots,\varepsilon^{(Q)}\in\{-1,1\}^{3n}}\Big(\big\|H\big((\rho(\varepsilon^{(1)}),Y_1),\dots,(\rho(\varepsilon^{(Q)}),Y_Q)\big)-A_i\big\|_2^2\le\epsilon\Big)\\&\ge 1-\delta.\end{aligned}$$

Summing over i, by the disjointness of $\{E_i(\widetilde{\varepsilon})\}_i$, we obtain

$$\begin{aligned}M(1-\delta)&\le\sum_{i=1}^{M}\mathbb{E}_{\widetilde{\varepsilon}}\big[\mathbb{P}_{i,\widetilde{\varepsilon}}(E_i(\widetilde{\varepsilon}))\big]\\&=\mathbb{E}_{\widetilde{\varepsilon}}\Big[\sum_{Y\in\{-1,1\}^Q}\sum_{i=1}^{M}\chi_{E_i(\widetilde{\varepsilon})}(Y)\mathbb{P}_{i,\widetilde{\varepsilon}}(Y)\Big]\\&\le\mathbb{E}_{\widetilde{\varepsilon}}\Big[\sum_{Y\in\{-1,1\}^Q}1\Big]=2^Q.\end{aligned}$$

Taking logarithms gives (10.6). The proof is complete. □

REMARK 10.7. Theorem 10.6 does not require $A^2 = I$, self-adjointness and contraction

$$\|A\|_{L_\infty}\le 1$$

suffices. Indeed, $\{-1,1\}^Q$ is not the range of A but of the outcomes Y produced by the two-outcome $\{\frac{1+A}{2},\frac{1-A}{2}\}$ in (10.8), which is valid for any self-adjoint A with $\|A\|_{L_\infty} \le 1$, this is exactly the condition $\frac{1\pm A}{2} \ge 0$, and satisfies $\mathbb{E}[Y_t] = \mathrm{tr}(A\rho(\varepsilon^{(t)}))$

regardless of whether A is Boolean. Consequently the proof gives (10.6) for all of $\mathcal{Q}_{n,d}$, not merely its Boolean subclass.

The classical counterpart of this observation is that the randomized bound in [**EIS23**, (20) of Proposition 8] already holds for the full class $\mathcal{F}_{n,d}$ of bounded (not necessarily $\{-1,1\}$-valued) functions, with no Boolean assumption: querying an oracle that, on input x, outputs $+1$ with probability $\frac{1+f(x)}{2}$ and -1 with probability $\frac{1-f(x)}{2}$ gives an unbiased $\{-1,1\}$-valued estimate of $f(x)$, and running the same counting argument with this randomized oracle in place of f itself recovers Proposition 8 (20) unchanged.

Now we turn to derive the packing bound for quantum functions. We need the following combinatorial lemma, which is analogous to [**EIS23**, Lemma 12]. For each m, $k \in \mathbb{N}$ with $k \le m$, let $\{0,1,2,3\}^m_{=k}$ be the set of all sequences with length k, more precisely,

$$\{0,1,2,3\}^m_{=k} \coloneqq \left\{ \eta \in \{0,1,2,3\}^m : \sum_{j=1}^m \operatorname{sign}(\eta(j)) = k \right\}.$$

We equip $\{0,1,2,3\}^m_{=k}$ with the uniform distribution μ, that is, $\mu(\eta) = \frac{1}{\binom{m}{k}3^k}$ for each $\eta \in \{0,1,2,3\}^m_{=k}$. For η_1 and $\eta_2 \in \{0,1,2,3\}^m_{=k}$, we define

$$|\eta_1 \cap \eta_2| \coloneqq |\{j \in [m] : \eta_1(j) = \eta_2(j) \neq 0\}|.$$

LEMMA 10.8. *For any m, $k \in \mathbb{N}$ with $k \le m$ and every $0 < \epsilon < \frac{1}{2}$, there exist at least $\{\eta_j\}_{j=1}^t \subseteq \{0,1,2,3\}^m_{=k}$ with $t \ge \left(\frac{3}{2k}\right)^k m^{(1-\epsilon)k} \left(\frac{m}{3m+2}\right)^{\epsilon k}$ such that*

$$|\eta_i \cap \eta_j| < (1-\epsilon)k, \quad i \neq j,\ i,\ j \in [t].$$

PROOF. We now provide a direct proof invoking the greedy algorithm. For a fixed $\eta \in \{0,1,2,3\}^m_{=k}$, we denote

$$D(\eta, (1-\epsilon)k) \coloneqq \{\zeta \in \{0,1,2,3\}^m_{=k} : |\eta \cap \zeta| \ge (1-\epsilon)k\}.$$

It is clear that

$$\begin{aligned} |D(\eta,(1-\epsilon)k)| &\le \sum_{j=\lceil (1-\varepsilon)k \rceil}^{k} \sum_{l=0}^{k-j} \binom{k}{j}\binom{k-j}{l}\binom{m-k}{k-j-l} 2^l 3^{k-j-l} \\ &= \sum_{r=0}^{\lfloor \epsilon k \rfloor} \sum_{l=0}^{r} \binom{k}{r}\binom{r}{l}\binom{m-k}{r-l} 2^l 3^{r-l}. \end{aligned} \tag{10.10}$$

By the fact $\binom{m-k}{r-l} \le m^{r-l}$, it follows that

$$\sum_{l=0}^{r} \binom{r}{l}\binom{m-k}{r-l} 2^l 3^{r-l} \le \sum_{l=0}^{r} \binom{r}{l} 2^l (3m)^{r-l} = (2+3m)^r. \tag{10.11}$$

Substituting (10.11) into (10.10) yields that

$$\begin{aligned} |D(\eta,(1-\epsilon)k))| &\le \sum_{r=0}^{\lfloor \epsilon k \rfloor} \binom{k}{r} (2+3m)^r \\ &\le (2+3m)^{\lfloor \epsilon k \rfloor} \sum_{r=0}^{k} \binom{k}{r} = 2^k (2+3m)^{\lfloor \epsilon k \rfloor}. \end{aligned} \tag{10.12}$$

We apply the greedy algorithm as follows. Pick an $\eta_1 \in \{0,1,2,3\}^m_{=k}$, and let $R_1 := \{0,1,2,3\}^m_{=k} \setminus D(\eta_1, (1-\epsilon)k)$. Having chosen $\eta_1, \ldots, \eta_i$ and formed R_i, pick $\eta_{i+1} \in R_i$ and let $R_{i+1} := R_i \setminus D(\eta_{i+1}, (1-\epsilon)k)$. Since $\{0,1,2,3\}^m_{=k}$ is a finite set, repeating the previous processes t-times turns the set $\{0,1,2,3\}^m_{=k}$ into an empty set, that means

$$\{0,1,2,3\}^m_{=k} = \bigcup_{i=1}^{t} D(\eta_i, (1-\epsilon)k).$$

By the fact

$$|\{0,1,2,3\}^m_{=k}| = \binom{m}{k} 3^k,$$

we obtain that

$$2^k (2+3m)^{\lfloor \epsilon k \rfloor} t \geq \binom{m}{k} 3^k \geq \left(\frac{3m}{k}\right)^k. \tag{10.13}$$

Rearranging (10.13) yields the desired result. □

We now turn to show a low bound for the packing number. Given $K \leq L \leq n$, we introduce a new symbol defined by setting

$$\begin{aligned} &\{0,1,2,3\}^{\geq L}_{=K} \\ &:= \left\{ \eta \in \{0,1,2,3\}^n : \eta(j) = 0, \text{ for } 1 \leq j \leq L-1, \text{ and } \sum_{l=1}^{n} \operatorname{sign}(\eta(j)) = K \right\}, \end{aligned}$$

where we order the set $\{0,1,2,3\}^{\geq L}_{=K}$ with the lexicography order. Hence, for each $\eta \in \{0,1,2,3\}^{\geq L}_{=K}$, there are $\{i_k\}_{1 \leq k \leq K}$ such that $L \leq i_k \leq K$ and $\eta(i_k) \in \{1,2,3\}$ for each $1 \leq k \leq K$.

THEOREM 10.9. *For any $\epsilon \in (0, \frac{1}{2})$ and n, $d \in \mathbb{N}$ with $2^d \leq n$, we have*

$$\log M(\mathcal{Q}_{n,d}, \|\cdot\|_{L_2}, \sqrt{2\epsilon}) \geq (1-\epsilon) 2^{d-1} \log(n) - d 2^{d-1}.$$

PROOF. The proof relies on an quantum analogy of decision tree. For each $j \in [n]$, we let

$$u_j := \sum_{l=1}^{3} \frac{\sigma_{l e_j}}{\sqrt{3}}. \tag{10.14}$$

Observe that for each $w = (w_1, \cdots, w_{d-1}) \in \{0,1\}^{d-1}$, $k(w) := 1 + \sum_{k=1}^{d-1} 2^{k-1} w_k$ runs over $\{1, 2, \cdots, 2^{d-1}\}$. For each $\eta \in \{0,1,2,3\}^{\geq d}_{=2^{d-1}}$, we define

$$\Gamma_\eta := \sum_{w \in \{0,1\}^{d-1}} \prod_{j=1}^{d-1} \left(\frac{\mathbf{1} + (2w_j - 1)u_j}{2} \right) \sigma_{\eta(i_{k(w)}) e_{i_{k(w)}}} \in \mathbb{M}_{2^n}.$$

We shall verify that $\Gamma_\eta \in \mathcal{Q}_{n,d}$. It is clear that Γ_η is of degree d. We can also verify that Γ_η is self-adjoint and unitary. For $w \in \{0,1\}^{d-1}$, we denote for simplicity

$$P_j(w) := \frac{\mathbf{1} + (2w_j - 1)u_j}{2}, \quad 1 \leq j \leq d-1$$

and

$$P(w) := \prod_{j=1}^{d-1} P_j(w).$$

Note that $\{u_j\}_{j=1}^n$ are unitaries. Then each $P_j(w)$ is a projection in $\mathbb{M}_{2^n}$. Moreover, since $d \le i_{k(w)} \le n$ for each w, it follows from the definition of P_j that P_j commutes with $\sigma_{\eta(i_{k(w)})e_{i_{k(w)}}}$ for all $1 \le j \le d-1$. Thus, it is easy to see that Γ_η is self-adjoint. We now show that

$$P(w)P(w') = \delta_{w,w'}P(w), \quad w,\ w' \in \{0,1\}^{d-1},$$

and

$$\sum_{w\in\{0,1\}^{d-1}} P(w) = \mathbf{1}.$$

Indeed, by the commuting of $\{u_j\}_{j=1}^n$, it follows that for arbitrary w, $w' \in \{0,1\}^{d-1}$, we have

$$\begin{aligned} P(w)P(w') &= \prod_{j=1}^{d-1} \frac{(\mathbf{1}+(2w_j-1)u_j)(\mathbf{1}+(2w'_j-1)u_j)}{4} \\ &= \prod_{j=1}^{d-1} \frac{\mathbf{1}+(2w_j-1)(2w'_j-1)+2(w_j+w'_j-1)u_j}{4} \\ &= \begin{cases} \prod_{j=1}^{d-1} \frac{1+(2w_j-1)u_j}{2}, & \text{if } w = w' \\ 0, & \text{otherwise.} \end{cases} \end{aligned}$$

Moreover, it is clear that

$$\begin{aligned} \sum_{w\in\{0,1\}^{d-1}} \prod_{j=1}^{d-1} P_j(w) &= \left(\frac{1-u_1}{2}+\frac{1+u_1}{2}\right) \sum_{v\in\{0,1\}^{d-2}} \prod_{j=2}^{d-1} \frac{1+(2v_j-1)u_j}{2} \\ &= \sum_{v\in\{0,1\}^{d-2}} \prod_{j=2}^{d-1} \frac{1+(2v_j-1)u_j}{2} = \mathbf{1}, \end{aligned}$$

where we applied the induction argument. Therefore, Γ_η is unitary by seeing that

$$\Gamma_\eta^2 = \left(\sum_{w\in\{0,1\}^{d-1}} P(w)\sigma_{\eta(i_{k(w)})e_{i_{k(w)}}}\right)^2 = \sum_{w\in\{0,1\}^{d-1}} P(w) = \mathbf{1}.$$

Applying Lemma 10.8 with parameters $k = 2^{d-1}$ and $m = n-d+1$, it follows that there exist $\{\eta_j\}_{j=1}^t$ with

$$t \ge \left(\frac{3}{2^d}\right)^{2^{d-1}} n^{(1-\epsilon)2^{d-1}} \left(\frac{n}{3n+2}\right)^{\epsilon 2^{d-1}}$$

such that $|\eta_i \cap \eta_j| < (1-\epsilon)2^{d-1}$. For any η, $\eta' \in \{\eta_j\}_{j=1}^t$, we have

$$\begin{aligned} \|\Gamma_\eta - \Gamma_{\eta'}\|_{L_2}^2 &= \sum_{w\in\{0,1\}^{d-1}} \|P(w)\|_{L_2}^2 \left\|\sigma_{\eta(i_{k(w)})e_{i_{k(w)}}} - \sigma_{\eta'(i_{k(w)})e_{i_{k(w)}}}\right\|_{L_2}^2 \\ &\ge 2^{2-d}|\{j\in\mathbb{N} : \eta_{i(j)} \ne \eta'_{i(j)}\}| \\ &\ge 2^{2-d}\left(2^{d-1} - |\eta\cap\eta'|\right) \ge 2\epsilon, \end{aligned} \tag{10.15}$$

where we used the fact

$$\|P(w)\|_{L_2}^2 = \prod_{j=1}^{d-1} \|P_j(w)\|_{L_2}^2 = \prod_{j=1}^{d-1} \operatorname{tr}\left(\frac{1+(2w_j-1)u_j}{2}\right) = \frac{1}{2^{d-1}}.$$

It follows from (10.15) that $\|\Gamma_\eta - \Gamma_{\eta'}\|_{L_2}^2 \geq 2\epsilon$ for each $i \neq j$. Therefore, we have

$$\log M(\mathcal{Q}_{n,d}, \|\cdot\|_{L_2}, \sqrt{2\epsilon}) \geq (1-\epsilon)2^{d-1}\log(n) - 2^{d-1}\log(2^d/3) - \epsilon 2^{d-1}\log\left(\frac{3n+2}{n}\right).$$

Since $2^{d-1}\log(3) \geq \epsilon 2^{d-1}\log(3+\frac{2}{n})$ for d, $n \in \mathbb{N}$ as $\epsilon \in (0,1/2)$, it follows that

$$\log M(\mathcal{Q}_{n,d}, \|\cdot\|_{L_2}, \sqrt{2\epsilon}) \geq (1-\epsilon)2^{d-1}\log(n) - d2^{d-1}.$$

This completes the proof. □

Modifying the proof of Theorem 10.9, we can provide an improved version of the lower bound for $\log M(\mathcal{Q}_{n,d}, \|\cdot\|_{L_2}, \sqrt{2\epsilon})$, whose construction is rather invoked.

THEOREM 10.10. *There exists a universal constant $C > 0$ such that the following holds. For every $0 < \epsilon < 1$ and every $n, d \in \mathbb{N}$ satisfying $n \geq \frac{3^d-1}{2}$ we have*

$$\log M\left(\mathcal{Q}_{n,d}, \sqrt{2\epsilon}, \|\cdot\|_{L_2}\right) \geq (1-\epsilon)3^{d-1}\log n - Cd3^{d-1}.$$

PROOF. Let $n, d \in \mathbb{N}$ be such that $n \geq \frac{3^d-1}{2}$. Set

$$K \coloneqq 3^{d-1}, \quad q \coloneqq 1 + 3 + \cdots + 3^{d-2} = \frac{3^{d-1}-1}{2}.$$

The integer q is the number of internal vertices of the tree of depth $d-1$, and K is the number of its leaves with

$$n \geq q + K = \frac{3^d - 1}{2}.$$

Let

$$\mathcal{V} \coloneqq \bigcup_{r=0}^{d-2} \{1,2,3\}^r,$$

where $[3]^0 = \{\emptyset\}$ for convenience. Let $\mathcal{B} : \mathcal{V} \to [q] \coloneqq \{1,2,3,\ldots,q\}$ be a fixed bijection. The set of leaves $\mathcal{L} \coloneqq \{1,2,3\}^{d-1}$ are indexed by the lexicographic order.

For $\alpha = (\alpha_1, \ldots, \alpha_{d-1}) \in \mathcal{L}$, let $r(\alpha) \in \{1, \ldots, K\}$ be the position of α in this lexicographic order. For $0 \leq s \leq d-2$, write

$$\alpha|_s \coloneqq (\alpha_1, \ldots, \alpha_s) \in [3]^s,$$

where $\alpha|_0 = \emptyset$. Define the quantum path corresponding to $\alpha \in \mathcal{L}$ by setting

$$\mathcal{P}_\alpha \coloneqq \prod_{s=0}^{d-2} \sigma_{\alpha_{s+1} e_{\mathcal{B}(\alpha|_s)}}.$$

Since $\mathcal{B}$ is a bijection, $\alpha|_0, \alpha|_1, \cdots, \alpha|_{d-2}$ are $d-1$ distinct elements of $\mathcal{V}$. For each $\eta \in \{0,1,2,3\}_{=K}^{\geq q+1}$, let

$$q < i_\eta(1) < i_\eta(2) < \cdots < i_\eta(K) \leq n$$

be the nonzero coordinates of η. We define the quantum decision tree associated with $\{0,1,2,3\}_{=K}^{\geq q+1}$ by the Walsh expansion

$$\Gamma_\eta \coloneqq 3^{-\frac{d-1}{2}} \sum_{\alpha\in[3]^{d-1}} \mathcal{P}_\alpha \cdot \sigma_{\eta(i_\eta(r(\alpha)))e_{i_\eta(r(\alpha))}}. \tag{10.16}$$

We first prove that $\Gamma_\eta \in \mathcal{Q}_{n,d}$. The fact that $\deg(\Gamma_\eta) \leq d$ follows from (10.16). Moreover, every $\mathcal{P}_\alpha$ and every $\sigma_{\eta(i_\eta(r(\alpha)))e_{i_\eta(r(\alpha))}}$ is self-adjoint, and their supports are disjoint. Thus, $\Gamma_\eta^* = \Gamma_\eta$.

It remains to prove that Γ_η is unitary, that is $\Gamma_\eta^2 = \mathbf{1}$. By the definition of Γ_η, we have

$$\begin{aligned}
\Gamma_\eta^2 &= 3^{-(d-1)} \sum_{\alpha,\beta\in[3]^{d-1}} \mathcal{P}_\alpha\mathcal{P}_\beta\sigma_{\eta(i_\eta(r(\alpha)))e_{i_\eta(r(\alpha))}}\sigma_{\eta(i_\eta(r(\beta)))e_{i_\eta(r(\beta))}} \\
&= 3^{-(d-1)} \sum_{\substack{\alpha,\beta\in[3]^{d-1}\\ \alpha=\beta}} \mathcal{P}_\alpha^2\sigma^2_{\eta(i_\eta(r(\alpha)))e_{i_\eta(r(\alpha))}} \\
&\quad + 3^{-(d-1)} \sum_{\substack{\alpha,\beta\in[3]^{d-1}\\ \alpha\neq\beta}} \mathcal{P}_\alpha\mathcal{P}_\beta\sigma_{\eta(i_\eta(r(\alpha)))e_{i_\eta(r(\alpha))}}\sigma_{\eta(i_\eta(r(\beta)))e_{i_\eta(r(\beta))}} \\
&= \mathbf{1} + 3^{-(d-1)} \sum_{\substack{\alpha,\beta\in[3]^{d-1}\\ \alpha\neq\beta}} \mathcal{P}_\alpha\mathcal{P}_\beta\sigma_{\eta(i_\eta(r(\alpha)))e_{i_\eta(r(\alpha))}}\sigma_{\eta(i_\eta(r(\beta)))e_{i_\eta(r(\beta))}}.
\end{aligned} \tag{10.17}$$

For $\alpha,\beta\in[3]^{d-1}$ with $\alpha\neq\beta$, let

$$\ell_0 \coloneqq \min\{\ell\in[d-1] : \alpha_\ell \neq \beta_\ell\}, \qquad s_0 \coloneqq \ell_0 - 1.$$

For $s < s_0$, the corresponding factors in $\mathcal{P}_\alpha$ and $\mathcal{P}_\beta$ coincide and cancel. At the first branching vertex,

$$\alpha|_{s_0} = \beta|_{s_0}, \qquad \alpha_{s_0+1} \neq \beta_{s_0+1},$$

so the two distinct Pauli matrices supported on the same coordinate $c(\alpha|_{s_0}) = c(\beta|_{s_0})$ anticommute. All the remaining factors commute, since they are supported on distinct coordinates. Therefore,

$$\mathcal{P}_\alpha\mathcal{P}_\beta\sigma_{\eta(i_\eta(r(\alpha)))e_{i_\eta(r(\alpha))}}\sigma_{\eta(i_\eta(r(\beta)))e_{i_\eta(r(\beta))}} = -\mathcal{P}_\beta\mathcal{P}_\alpha\sigma_{\eta(i_\eta(r(\beta)))e_{i_\eta(r(\beta))}}\sigma_{\eta(i_\eta(r(\alpha)))e_{i_\eta(r(\alpha))}}.$$

Pairing the terms indexed by (α,β) and (β,α) shows that all the off-diagonal terms in (10.17) cancel. Consequently,

$$\sum_{\substack{\alpha,\beta\in[3]^{d-1}\\ \alpha\neq\beta}} \mathcal{P}_\alpha\mathcal{P}_\beta\sigma_{\eta(i_\eta(r(\alpha)))e_{i_\eta(r(\alpha))}}\sigma_{\eta(i_\eta(r(\beta)))e_{i_\eta(r(\beta))}} = 0.$$

Thus, we deduce from (10.17) that Γ_η is unitary.

For $\eta,\zeta\in\{0,1,2,3\}_{=K}^{\geq q+1}$, we now turn to estimate $\|\Gamma_\eta - \Gamma_\zeta\|_{L_2}$. Recall here that

$$|\eta\cap\zeta| \coloneqq \#\left\{j\in\{q+1,\dots,n\} : \eta(j) = \zeta(j) \neq 0\right\},$$

then by $\|\Gamma_\eta - \Gamma_\zeta\|_{L_2}^2 = 2 - 2\langle\Gamma_\eta,\Gamma_\zeta\rangle_{L_2}$, it remains to estimate $\langle\Gamma_\eta,\Gamma_\zeta\rangle_{L_2}$ from below. We claim that

$$\langle\Gamma_\eta,\Gamma_\zeta\rangle_{L_2} \leq \frac{|\eta\cap\zeta|}{K}.$$

Indeed, by the Walsh expansion,

$$\langle \Gamma_\eta, \Gamma_\zeta \rangle_{L_2} = 3^{-(d-1)} \sum_{\alpha,\beta \in [3]^{d-1}} \langle \sigma_{\eta(i_\eta(r(\alpha)))e_{i_\eta(r(\alpha))}}, \sigma_{\zeta(i_\zeta(r(\beta)))e_{i_\zeta(r(\beta))}} \rangle \cdot \langle \mathcal{P}_\alpha, \mathcal{P}_\beta \rangle.$$

If $\alpha \neq \beta$, then by the construction of quantum path, it follows that at least one position of $\mathcal{P}_\alpha$ and $\mathcal{P}_\beta$ are different. Hence, by the orthogonality of $\{\sigma_{\mathbf{s}}\}_{\mathbf{s}\in\{0,1,2,3\}^n}$, it follows that $\langle \mathcal{P}_\alpha, \mathcal{P}_\beta \rangle = 0$.

If $\alpha = \beta$, then by $\|\mathcal{P}_\alpha\|_{L_2} = 1$ for each $\alpha \in \{1,2,3\}^{d-1}$, we have

$$\langle \Gamma_\eta, \Gamma_\zeta \rangle_{L_2} = \frac{1}{K} \sum_{\alpha \in [3]^{d-1}} \langle \sigma_{\eta(i_\eta(r(\alpha)))e_{i_\eta(r(\alpha))}}, \sigma_{\zeta(i_\zeta(r(\alpha)))e_{i_\zeta(r(\alpha))}} \rangle.$$

By the orthogonality of $\{\sigma_{\mathbf{s}}\}_{\mathbf{s}\in\{0,1,2,3\}^n}$ again, we have

$$\langle \sigma_{\eta(i_\eta(r(\alpha)))e_{i_\eta(r(\alpha))}}, \sigma_{\zeta(i_\zeta(r(\alpha)))e_{i_\zeta(r(\alpha))}} \rangle = \begin{cases} 1, & \text{if two index coincide,} \\ 0, & \text{otherwise.} \end{cases}$$

Since the number of leaves α for which $\sigma_{\eta(i_\eta(r(\alpha)))e_{i_\eta(r(\alpha))}} = \sigma_{\zeta(i_\zeta(r(\alpha)))e_{i_\zeta(r(\alpha))}}$ is at most $|\eta \cap \zeta|$. Hence

$$\langle \Gamma_\eta, \Gamma_\zeta \rangle_{L_2} \leq \frac{|\eta \cap \zeta|}{K},$$

which completes the proof of our claim.

Therefore, if

$$|\eta \cap \zeta| < (1-\epsilon)K,$$

then

$$\|\Gamma_\eta - \Gamma_\zeta\|_{L_2}^2 > 2 - 2(1-\epsilon) = 2\epsilon.$$

It remains to choose many such η. We shall apply Lemma 10.8 with $m = n - q$ and $k = K = 3^{d-1}$. Since $n \geq q + K$, we have $K \leq m$. Thus, there exists a family

$$\mathcal{E} \subseteq \{0,1,2,3\}^{\geq q+1}_{=K}$$

such that

$$|\eta \cap \zeta| < (1-\epsilon)K, \quad \eta \neq \zeta, \quad \eta, \zeta \in \mathcal{E},$$

and

$$\log |\mathcal{E}| \geq (1-\epsilon)K \log m - K \log K - CK.$$

As $n \geq q + K$ and $q = (K-1)/2$, we have

$$m = n - q \geq \frac{2}{3} n.$$

Hence,

$$\log m \geq \log n - \log \frac{3}{2}.$$

Consequently,

$$\begin{aligned} \log |\mathcal{E}| &\geq (1-\epsilon)K \log n - K \log K - CK \\ &\geq (1-\epsilon)3^{d-1} \log n - Cd3^{d-1}. \end{aligned}$$

For every distinct $\eta, \zeta \in \mathcal{E}$, the preceding distance estimate gives

$$\|\Gamma_\eta - \Gamma_\zeta\|_{L_2} > \sqrt{2\epsilon}.$$

Therefore $\{\Gamma_\eta : \eta \in \mathcal{E}\}$ is a $\sqrt{2\epsilon}$-packing of $\mathcal{Q}_{n,d}$ in L_2, and hence

$$\log M\left(\mathcal{Q}_{n,d}, \sqrt{2\epsilon}, \|\cdot\|_{L_2}\right) \geq \log |\mathcal{E}| \geq (1-\epsilon)3^{d-1} \log n - Cd3^{d-1}.$$

The proof is complete. □

In Theorem 10.6, we provide the lower bound for random query complexity. We also have similar estimate for query model. However, the upper estimate for this remains open even in the classical setting; see [**EIS23**, page 12].

A deterministic learning algorithm of length Q consists of a map

$$H : (\mathcal{S} \times \{-1, 1\})^Q \to L_2(\mathbb{M}_{2^n}, \mathrm{tr}),$$

together with an initial query state $\rho_1 \in \mathcal{D}_n$ and deterministic maps

$$\Phi_t : (\mathcal{S} \times \{-1, 1\})^{t-1} \to \mathcal{S}, \qquad t = 2, \ldots, Q,$$

where $\mathcal{S}$ denotes the set of density matrices on $\mathbb{M}_{2^n}$. We emphasize that "deterministic" here refers only to the maps $H, \Phi_2, \ldots, \Phi_Q$ themselves: given an unknown observable $A \in \mathcal{Q}_{n,d}$, the sequence of query states $\rho_1, \rho_2, \ldots, \rho_Q$ and the corresponding measurement outcomes $Y_1, \ldots, Y_Q$ are still random, since each Y_t is the outcome of a genuine quantum measurement on ρ_t. Given a realized outcome string $y = (y_1, \ldots, y_Q) \in \{-1, 1\}^Q$, we define recursively

$$\rho_1^y := \rho_1, \qquad \rho_t^y := \Phi_t\big((\rho_1^y, y_1), \ldots, (\rho_{t-1}^y, y_{t-1})\big), \qquad t = 2, \ldots, Q;$$

note that ρ_t^y depends only on the prefix $(y_1, \ldots, y_{t-1})$, and not on $y_t, \ldots, y_Q$. At stage t, given the query state ρ_t^y and the true observable A, the oracle returns a bit $Y_t \in \{-1, 1\}$ via the two-outcome measurement

$$\mathbb{P}(Y_t = \pm 1 \mid \rho_t^y, A) = \mathrm{tr}\left(\frac{1 \pm A}{2}\rho_t^y\right) = \frac{1 \pm \mathrm{tr}(A\rho_t^y)}{2}.$$

For $A \in \mathcal{B}_n$, we say that an outcome string $y \in \{-1, 1\}^Q$ is A-admissible if the transcript $(\rho_1^y, y_1), \ldots, (\rho_Q^y, y_Q)$ occurs with positive probability when the unknown observable is A, that is,

$$\prod_{t=1}^{Q} \frac{1 + y_t \,\mathrm{tr}(A\rho_t^y)}{2} > 0.$$

Because the outcome of each measurement is genuinely random, a fixed A may admit several distinct admissible outcome strings; the algorithm is required to succeed simultaneously on all of them. We denote by

$$Q_{\mathbb{M}_2}(\mathcal{Q}_{n,d}, \epsilon)$$

the least $Q \in \mathbb{N}$ for which there exists such a deterministic learning algorithm satisfying

$$\big\|H\big((\rho_1^y, y_1), \ldots, (\rho_Q^y, y_Q)\big) - A\big\|_{L_2}^2 \leq \varepsilon \tag{10.18}$$

for every $A \in \mathcal{Q}_{n,d}$ and every A-admissible outcome string $y \in \{-1, 1\}^Q$.

Remark 10.11. Let $d \in [n]$. For every $\epsilon > 0$, we have

$$Q_{\mathbb{M}_2}(\mathcal{Q}_{n,d}, \epsilon) \geq \log_2 M(\mathcal{Q}_{n,d}, \|\cdot\|_{L_2}, 2\sqrt{\epsilon}). \tag{10.19}$$

Proof. For simplicity, let

$$M := M(\mathcal{Q}_{n,d}, \|\cdot\|_2, 2\sqrt{\epsilon}),$$

and choose $A_1, \ldots, A_M \in \mathcal{Q}_{n,d}$ such that

$$\|A_i - A_j\|_2 > 2\sqrt{\epsilon} \qquad (i \neq j).$$

Denote $Q := Q_{\mathbb{M}_2}(\mathcal{Q}_{n,d}, \epsilon)$, and let H, ρ_1, and $\Phi_2, \dots, \Phi_Q$ be a deterministic learning algorithm of length Q satisfying the definition (10.18). For each $i \in \{1, \dots, M\}$, since A_i admits at least one admissible outcome string, fix one such string $y^{(i)} \in \{-1, 1\}^Q$, and consider the set

$$\Sigma := \{y^{(i)} : i = 1, \dots, M\} \subseteq \{-1, 1\}^Q.$$

We claim that $|\Sigma| = M$. Indeed, if $|\Sigma| < M$, then there exist $i \neq j$ such that

$$y^{(i)} = y^{(j)}.$$

Since ρ_t^y depends only on $(y_1, \dots, y_{t-1})$, the coincidence of the bit strings implies

$$\rho_t^{y^{(i)}} = \rho_t^{y^{(j)}}, \qquad t = 1, \dots, Q.$$

Hence the full data transcripts coincide, and the common output:

$$\widetilde{A} := H\big((\rho_1^{y^{(i)}}, y_1^{(i)}), \dots, (\rho_Q^{y^{(i)}}, y_Q^{(i)})\big) = H\big((\rho_1^{y^{(j)}}, y_1^{(j)}), \dots, (\rho_Q^{y^{(j)}}, y_Q^{(j)})\big)$$

satisfies

$$\|\widetilde{A} - A_i\|_2^2 \leq \varepsilon \qquad \text{and} \qquad \|\widetilde{A} - A_j\|_2^2 \leq \varepsilon,$$

since $y^{(i)}$ is A_i-admissible and $y^{(j)}$ is A_j-admissible. This contradicts the packing condition, since

$$\|A_i - A_j\|_2 \leq \|A_i - \widetilde{A}\|_2 + \|\widetilde{A} - A_j\|_2 \leq 2\sqrt{\varepsilon}.$$

This proves the claim. Since $\Sigma \subseteq \{-1, 1\}^Q$, we conclude that

$$M = |\Sigma| \leq 2^Q,$$

which yields (10.19). □

We end this section by the following exact learning result.

THEOREM 10.12. *Let $n \in \mathbb{N}$, $d \in \{1, \dots, n\}$ and $\delta \in (0, 1)$. Then*

$$Q_{\mathbb{M}_2}(\mathcal{Q}_{n,d}, 0) = \sum_{\ell=0}^{d} 3^\ell \binom{n}{\ell}. \tag{10.20}$$

Moreover, there exists some universal constant $C > 0$ such that

$$Q_{\mathbb{M}_2,r}(\mathcal{Q}_{n,d}, 0, \delta) \leq C\, d^2\, 6^d\, n^d\, \log\Big(\frac{n}{\delta}\Big). \tag{10.21}$$

PROOF. We now claim that $Q_{\mathbb{M}_2}(\mathcal{Q}_{n,d}, 0) \leq \sum_{\ell=0}^{d} 3^\ell \binom{n}{\ell}$. Indeed, let

$$A = \sum_{|\mathbf{s}| \leq d} \widehat{A}(\mathbf{s}) \sigma_{\mathbf{s}} \in \mathcal{Q}_{n,d}.$$

For $\mathbf{s} \neq 0$ with $|\mathbf{s}| \leq d$, define

$$\rho_{\mathbf{s}} := \mathbf{1} + \sigma_{\mathbf{s}}, \qquad \rho_0 := \mathbf{1}.$$

Since each $\sigma_{\mathbf{s}}$ is self-adjoint unitary, $\mathbf{1} + \sigma_{\mathbf{s}} \geq 0$, and therefore every $\rho_{\mathbf{s}}$ is a density matrix. Then

$$\operatorname{tr}(A\rho_0) = \operatorname{tr}(A) = \widehat{A}(0),$$

while for every $\mathbf{s} \neq 0$ with $|\mathbf{s}| \leq d$,

$$\operatorname{tr}(A\rho_{\mathbf{s}}) = \operatorname{tr}(A) + \operatorname{tr}(A\sigma_{\mathbf{s}}) = \widehat{A}(0) + \widehat{A}(\mathbf{s}).$$

It is clear that A has $\sum_{\ell=0}^{d} 3^{\ell}\binom{n}{\ell}$ Fourier coefficients as $|\mathbf{s}| \leq d$. Thus, according to the above correspondence, it suffices to take $\sum_{\ell=0}^{d} 3^{\ell}\binom{n}{\ell}$ query values

$$(\rho_0, \mathrm{tr}(A\rho_0)), \cdots (\rho_{\mathbf{s}}, \mathrm{tr}(A\rho_{\mathbf{s}})) \quad (0 < |\mathbf{s}| \leq d),$$

to recover A. Therefore A is exactly determined by $\sum_{\ell=0}^{d} 3^{\ell}\binom{n}{\ell}$ deterministic queries, so

$$Q_{\mathbb{M}_2}(\mathcal{Q}_{n,d}, 0) \leq \sum_{\ell=0}^{d} 3^{\ell}\binom{n}{\ell}.$$

It remains to show $Q_{\mathbb{M}_2}(\mathcal{Q}_{n,d}, 0) \geq \sum_{\ell=0}^{d} 3^{\ell}\binom{n}{\ell}$. We proceed by contradiction. Denote $Q \coloneqq Q_{\mathbb{M}_2}(\mathcal{Q}_{n,d}, 0)$and assume that $Q < \sum_{\ell=0}^{d} 3^{\ell}\binom{n}{\ell}$.Let $\{\rho_1, \ldots, \rho_Q\}$ be the adaptive density matrix queries. Now run the algorithm on the zero matrix 0. Consider the linear map

$$\Psi_0 : \mathcal{P}^{\leq d} \to \mathbb{R}^Q$$

defined by

$$\Psi_0(B) = \big(\mathrm{tr}(B\rho_1), \ldots, \mathrm{tr}(B\rho_Q)\big).$$

Since

$$\dim(\mathcal{P}^{\leq d}) = \sum_{\ell=0}^{d} 3^{\ell}\binom{n}{\ell} > Q = \dim(\mathbb{R}^Q),$$

then, $\ker\Psi_0 \neq \emptyset$. Choose $0 \neq B \in \ker \Psi_0$ and set

$$A \coloneqq \frac{B}{\|B\|_{\infty}} \in \mathcal{Q}_{n,d}.$$

Then

$$\mathrm{tr}(A\rho_j) = 0, \qquad \text{for each } j = 1, \ldots, Q.$$

Since all answers received by the algorithm on A are identical to those on 0. This is impossible, because exact learning requires the output to equal 0 on the 0 matrix and to equal $A \neq 0$ on the input A. Thus

$$Q \geq \sum_{\ell=0}^{d} 3^{\ell}\binom{n}{\ell},$$

which proves (10.20).

For the proof of the random example model (10.21). Set $\varepsilon^{(1)}, \ldots, \varepsilon^{(Q)} \in \{-1, 1\}^{3n}$, each $\varepsilon^{(t)} \in \{-1, 1\}^{3n}$ determines a density matrix $\rho(\varepsilon^{(t)})$ by (9.2) and (9.3). Define the operator

$$\Psi_{\varepsilon^{(1)}, \ldots, \varepsilon^{(Q)}} : \mathcal{Q}_{n,d} \to \mathbb{R}^Q$$

by

$$\Psi_{\varepsilon^{(1)}, \ldots, \varepsilon^{(Q)}}(A) \coloneqq \big(\mathrm{tr}(A\rho(\varepsilon^{(1)})), \ldots, \mathrm{tr}(A\rho(\varepsilon^{(Q)}))\big).$$

If this operator is injective, then the Q sample values uniquely determine the unknown $A \in \mathcal{Q}_{n,d}$, and one can reconstruct A exactly by solving a linear system for the Pauli coefficients $\{\widehat{A}(\mathbf{s}) : |\mathbf{s}| \leq d\}$. In order to prove the upper bound on the query complexity of the random example model without error, it suffices to show that if Q is large enough and $\varepsilon^{(1)}, \ldots, \varepsilon^{(Q)} \in \{-1, 1\}^{3n}$ are independent and uniformly distributed random vectors, then $\Psi_{\varepsilon^{(1)}, \ldots, \varepsilon^{(Q)}}$ is injective with high probability.

Fix points $\eta_1, \dots, \eta_q \in \{-1,1\}^{3n}$ and suppose that

$$\Psi_{\eta_1,\dots,\eta_q}$$

is not injective. Choose a nonzero matrix

$$B \in \ker \Psi_{\eta_1,\dots,\eta_q} \subseteq \mathcal{Q}_{n,d}.$$

Let ε be an independent uniform random vector on $\{-1,1\}^{3n}$. We have

$$\mathbb{P}\Big(\dim\ker \Psi_{\eta_1,\dots,\eta_q,\varepsilon} < \dim\ker \Psi_{\eta_1,\dots,\eta_q}\Big) \geq \mathbb{P}\big(\mathrm{tr}(B\rho(\varepsilon)) \neq 0\big) \geq 2^{-d}, \tag{10.22}$$

where the last inequality is due to [**O'D14**, Lemma 3.5].

Consider $Q > \sum_{\ell=0}^{d} 3^\ell \binom{n}{\ell} =: K$. Suppose that

$$\Psi_{\varepsilon^{(1)},\dots,\varepsilon^{(Q)}}$$

is not injective. Then its kernel has dimension at least 1, so along the chain

$$K \geq \dim\ker \Psi_{\varepsilon^{(1)}} \geq \dim\ker \Psi_{\varepsilon^{(1)},\varepsilon^{(2)}} \geq \cdots \geq \dim\ker \Psi_{\varepsilon^{(1)},\dots,\varepsilon^{(Q)}}$$

there can be at most $K-1$ strict drops. Hence at least $Q-K+1$ steps are equalities. By (10.22) and independence, this yields

$$\mathbb{P}\Big(\Psi_{\varepsilon^{(1)},\dots,\varepsilon^{(Q)}} \text{ is not injective}\Big) \leq \binom{Q}{Q-K+1}\big(1-2^{-d}\big)^{Q-K+1}.$$

Using

$$\binom{Q}{Q-K+1} = \binom{Q}{K-1} \leq Q^{K-1}, \qquad 1-2^{-d} \geq 1/2 \quad \text{for } d \geq 1,$$

we obtain

$$\mathbb{P}\Big(\Psi_{\varepsilon^{(1)},\dots,\varepsilon^{(Q)}} \text{ is not injective}\Big) \leq (2Q)^{K-1}\big(1-2^{-d}\big)^{Q} \leq \exp\Big((K-1)\log(2Q) - Q\,2^{-d}\Big),$$

the last inequality is due to the fact

$$1-2^{-d} \leq e^{-2^{-d}}.$$

Therefore, if

$$Q \geq C\,2^d K \log\Big(\frac{K}{\delta}\Big)$$

for a large enough universal constant $C > 0$, then

$$\mathbb{P}\Big(\Psi_{\varepsilon^{(1)},\dots,\varepsilon^{(Q)}} \text{ is injective}\Big) \geq 1-\delta.$$

Note that

$$K = \sum_{\ell=0}^{d} 3^\ell \binom{n}{\ell} \leq (d+1)3^d n^d.$$

Hence,

$$Q_{\mathbb{M}_2,r}(\mathcal{Q}_{n,d},0,\delta) \leq C\,2^d(d+1)3^d n^d \log\Big(\frac{(d+1)3^d n^d}{\delta}\Big).$$

Since $d \leq n$, one has

$$\log\Big(\frac{(d+1)3^d n^d}{\delta}\Big) \leq C\,d\,\log\Big(\frac{n}{\delta}\Big),$$

possibly after increasing C. Therefore

$$Q_{\mathbb{M}_2,r}(\mathcal{Q}_{n,d},0,\delta) \leq C\,d^2\,6^d\,n^d\,\log\Big(\frac{n}{\delta}\Big),$$

as claimed. The proof is complete. □

REMARK 10.13. Note that the number in (10.20) is different from its classical counterpart

$$Q_{\{\pm 1\}}(\mathcal{F}_{n,d}, 0) = \sum_{\ell=1}^{d} \binom{n}{\ell}$$

provided in [**EIS23**, Theorem 10]. The proof of deterministic identity (10.20) is actually simpler. In the classical case, one recovers the top Walsh layer from values on a Hamming ball via discrete derivatives. In the quantum setting, one can read off every low-degree Pauli coefficient directly by querying the density matrices

$$\rho_0 = \mathbf{1}, \qquad \rho_{\mathbf{s}} = \mathbf{1} + \sigma_{\mathbf{s}} \quad (1 \le |\mathbf{s}| \le d).$$

10.2. More interesting estimates

This section is devoted to investigating the following two fundamental objects:

(i) Deriving a broader upper bound for quantum learning without requiring the target quantum function to be of degree d;
(ii) Deriving the upper bound for the metric entropy of the class $(\mathcal{Q}_{n,d}, \|\cdot\|_{L_2})$.

The first main result in this section is now stated as follows.

THEOREM 10.14. *Let $\epsilon, \delta \in (0,1)$, $d \in [n]$ and $t, \eta \in [0,1)$, and let $A^* = A \in \mathbb{M}_{2^n}$ with $\|A\|_\infty \le 1$. Assume that there exists a set $\mathcal{T}(A) \subseteq \{0,1,2,3\}^n$ with $|\mathcal{T}(A)| \le m$, $m \in \mathbb{N}$, such that*

$$\sum_{|\mathbf{s}|>d} |\widehat{A}(\mathbf{s})|^2 \le t, \qquad \sum_{s \notin \mathcal{T}(A)} |\widehat{A}(\mathbf{s})|^2 \le \eta.$$

If

$$N \ge \Big\lceil \frac{18 \cdot 3^d m}{\varepsilon} \log\Big(\frac{2\sum_{\ell=0}^{d} 3^\ell \binom{n}{\ell}}{\delta}\Big)\Big\rceil,$$

then, from N independent random queries

$$\big(\rho(x^{(1)}), \mathrm{tr}(A\rho(x^{(1)}))\big), \ldots, \big(\rho(x^{(N)}), \mathrm{tr}(A\rho(x^{(N)}))\big), \qquad x^{(1)}, \ldots, x^{(N)} \in \{-1,1\}^{3n},$$

one can construct a random observable $\widetilde{A}$ such that

$$\|A - \widetilde{A}\|_2^2 \le \eta + t + \varepsilon$$

with probability at least $1 - \delta$.

PROOF. For $\mathbf{s} \in \{0,1,2,3\}^n$ with $|\mathbf{s}| \le d$, define the empirical coefficients

$$\alpha_{\mathbf{s}} := \frac{1}{N}\sum_{r=1}^{N} \mathrm{tr}(A\rho(x^{(r)}))\omega_S(x^{(r)}) = \frac{1}{N}\sum_{r=1}^{N} 3^{-|\mathbf{s}|/2}\widehat{A}(\mathbf{s})\omega_S(x^{(r)}), \quad x^{(r)} \in \{-1,1\}^{3n},$$

where S is determined by $supp\,(\mathbf{s})$, as defined in (9.6) within the proof of Lemma 9.2. Then, $\mathbb{E}[\alpha_{\mathbf{s}}] = 3^{-|\mathbf{s}|/2}\widehat{A}(\mathbf{s})$, where $\mathbb{E}$ is with respect to the uniform distribution. Since $\alpha_{\mathbf{s}}$ is an average of bounded i.i.d. random variables and $\|f_A\|_{L_\infty(\{-1,1\}^{3n})} \le \|A\|_{L_\infty} \le 1$, Hoeffding's inequality gives

$$\mathbb{P}\Big(|\alpha_{\mathbf{s}} - 3^{-|\mathbf{s}|/2}\widehat{A}(\mathbf{s})| > b\Big) \le 2e^{-Nb^2/2}, \qquad |\mathbf{s}| \le d.$$

Denote

$$G_b := \Big\{|\alpha_{\mathbf{s}} - 3^{-|\mathbf{s}|/2}\widehat{A}(\mathbf{s})| \le b \text{ for all } |\mathbf{s}| \le d\Big\}.$$

Using the union bound over $|\mathbf{s}| \le d$, we obtain that

$$\mathbb{P}(G_b) \ge 1 - 2\sum_{\ell=0}^{d} 3^\ell \binom{n}{\ell} e^{-Nb^2/2} \ge 1 - \delta,$$

provided

$$N \ge \frac{2}{b^2} \log\Big(\frac{2\sum_{\ell=0}^{d} 3^\ell \binom{n}{\ell}}{\delta}\Big).$$

Define

$$T_b := \{\mathbf{s} \in \{0,1,2,3\}^n : \ |\mathbf{s}| \le d, \ |\alpha_{\mathbf{s}}| \ge 2b\}, \qquad \widetilde{A}_b := \sum_{s \in T_b} 3^{|\mathbf{s}|/2} \alpha_{\mathbf{s}} \, \sigma_{\mathbf{s}}.$$

On the event G_b we have

$$\mathbf{s} \notin T_b \Longrightarrow |3^{-|\mathbf{s}|/2}\widehat{A}(\mathbf{s})| \le |\alpha_{\mathbf{s}}| + |\alpha_{\mathbf{s}} - 3^{-|\mathbf{s}|/2}\widehat{A}(\mathbf{s})| < 2b + b = 3b, \tag{10.23}$$

and

$$\mathbf{s} \in T_b \Longrightarrow |3^{-|\mathbf{s}|/2}\widehat{A}(\mathbf{s})| \ge |\alpha_{\mathbf{s}}| - |\alpha_{\mathbf{s}} - 3^{-|\mathbf{s}|/2}\widehat{A}(\mathbf{s})| \ge 2b - b = b. \tag{10.24}$$

Therefore,

$$\begin{aligned}
\|A - \widetilde{A}_b\|_2^2 &= \sum_{\mathbf{s} \in T_b} |\widehat{A}(\mathbf{s}) - 3^{|\mathbf{s}|/2}\alpha_{\mathbf{s}}|^2 + \sum_{\substack{|\mathbf{s}| \le d \\ \mathbf{s} \notin T_b}} |\widehat{A}(\mathbf{s})|^2 + \sum_{|\mathbf{s}| > d} |\widehat{A}(\mathbf{s})|^2 \\
&= \sum_{\mathbf{s} \in T_b} 3^{|\mathbf{s}|} |3^{-|\mathbf{s}|/2}\widehat{A}(\mathbf{s}) - \alpha_{\mathbf{s}}|^2 + \sum_{\substack{|\mathbf{s}| \le d \\ \mathbf{s} \notin T_b}} |\widehat{A}(\mathbf{s})|^2 + \sum_{|\mathbf{s}| > d} |\widehat{A}(\mathbf{s})|^2 \\
&= \sum_{\mathbf{s} \in T_b \cap \mathcal{T}(A)} 3^{|\mathbf{s}|} |\alpha_{\mathbf{s}} - 3^{-|\mathbf{s}|/2}\widehat{A}(\mathbf{s})|^2 + \sum_{\substack{|\mathbf{s}| \le d \\ \mathbf{s} \notin T_b, \ \mathbf{s} \in \mathcal{T}(A)}} |\widehat{A}(\mathbf{s})|^2 \\
&\quad + \sum_{\mathbf{s} \in T_b, \mathbf{s} \notin \mathcal{T}(A)} 3^{|\mathbf{s}|} |\alpha_{\mathbf{s}} - 3^{-|\mathbf{s}|/2}\widehat{A}(\mathbf{s})|^2 + \sum_{\substack{|\mathbf{s}| \le d \\ \mathbf{s} \notin T_b, \ \mathbf{s} \notin \mathcal{T}(A)}} |\widehat{A}(\mathbf{s})|^2 + \sum_{|\mathbf{s}| > d} |\widehat{A}(\mathbf{s})|^2.
\end{aligned}$$

On G_b, by (10.23), we have

$$\begin{aligned}
&\sum_{\mathbf{s} \in T_b \cap \mathcal{T}(A)} 3^{|\mathbf{s}|} |\alpha_{\mathbf{s}} - 3^{-|\mathbf{s}|/2}\widehat{A}(\mathbf{s})|^2 + \sum_{\substack{|\mathbf{s}| \le d \\ \mathbf{s} \notin T_b, \ \mathbf{s} \in \mathcal{T}(A)}} |\widehat{A}(\mathbf{s})|^2 \\
&= \sum_{s \in T_b \cap \mathcal{T}(A)} 3^{|\mathbf{s}|} |\alpha_{\mathbf{s}} - 3^{-|\mathbf{s}|/2}\widehat{A}(\mathbf{s})|^2 + \sum_{\substack{|\mathbf{s}| \le d \\ s \notin T_b, \ s \in \mathcal{T}(A)}} 3^{|\mathbf{s}|} |3^{-|\mathbf{s}|/2}\widehat{A}(\mathbf{s})|^2 \\
&\le 3^d 9b^2 \#\mathcal{T}(A) \le 3^d 9b^2 m.
\end{aligned}$$

And on G_b for $\mathbf{s} \in T_b$, by (10.24),

$$3^{|\mathbf{s}|} |\alpha_{\mathbf{s}} - 3^{-|\mathbf{s}|/2}\widehat{A}(\mathbf{s})|^2 \le 3^{|\mathbf{s}|} b^2 \le 3^{|\mathbf{s}|} |3^{-|\mathbf{s}|/2}\widehat{A}(\mathbf{s})|^2 = |\widehat{A}(\mathbf{s})|^2.$$

Therefore, the third and fourth terms are bounded by

$$\sum_{\mathbf{s} \notin \mathcal{T}(A)} |\widehat{A}(\mathbf{s})|^2 \le \eta,$$

while the last term is bounded by

$$\sum_{|\mathbf{s}|>d} |\widehat{A}(\mathbf{s})|^2 \leq t.$$

Thus

$$\|A - \widetilde{A}_b\|_2^2 \leq 3^d 9b^2 m + \eta + t = \varepsilon + \eta + t,$$

the last equality follows by choosing $b^2 = \varepsilon/(9 \cdot 3^d m)$, and this choice gives exactly

$$N \geq \left\lceil \frac{18 \cdot 3^d m}{\varepsilon} \log\Big(\frac{2\sum_{\ell=0}^{d} 3^\ell \binom{n}{\ell}}{\delta}\Big) \right\rceil.$$

The proof is complete. □

We observe in [**EIS23**] that the classical counterpart of Theorem 10.14 yields numerous applications concerning Fourier concentration. We skip applications of Theorem 10.14 herein, as they fall beyond the scope of the present paper to some extent.

Finally, we apply the reduction method used in the last section to prove the following upper bound for the ϵ-metric entropy of $(\mathcal{Q}_{n,d}, \|\cdot\|_{L_2})$. To proceed, let us recall the *covering number* of a metric space $(\mathcal{M}, d)$ as follows. For any $\epsilon > 0$, the subset $\mathcal{N} \subseteq \mathcal{M}$ is said to be ϵ-covering, if for each $\xi \in \mathcal{M}$, there exists some $\xi' \in \mathcal{N}$ with $d(\xi, \xi') \leq \epsilon$, and the covering number of the metric space $(\mathcal{M}, d)$, denoted by $N(\mathcal{M}, d, \epsilon)$, is defined by

$$N(\mathcal{M}, d, \epsilon) \coloneqq \inf\left\{|\mathcal{N}| : \mathcal{N} \subseteq \mathcal{M}, \text{ and } \mathcal{N} \text{ is an } \epsilon\text{-covering subset}\right\}.$$

In the classical setting, the corresponding result was proved (see [**EIS23**]) by using a deep junta theorem due to Dinur et al. [**DFKO07**]. Volberg and Zhang [**VZ24b**] established a noncommutative junta theorem only for operators that are exactly of degree at most d, corresponding to the special case in the classical framework. However, a noncommutative analogue of the more general junta theorem in [**DFKO07**] has not yet been proved in the quantum setting.

THEOREM 10.15. *For every $n, d \in \mathbb{N}$ and every $\epsilon > 0$, there exist a universal constant $C > 0$ and $\widetilde{\kappa}(d, \varepsilon) > 0$ depending only on d and ϵ such that*

$$\log_2 N\left(\mathcal{Q}_{n,d}, \|\cdot\|_{L_2}, \epsilon\right) \leq \frac{2^{Cd} 3^{2d}}{\epsilon^4} \log n + \widetilde{\kappa}(d, \epsilon).$$

PROOF. Recall that it was proved in [**EIS23**, Theorem 9] that there exists a universal constant $C_0 > 0$ and a function $\kappa(\cdot, \cdot)$ such that

$$\log_2 N(\mathcal{F}_{m,d}, \|\cdot\|_{L_2}, \eta) \leq \frac{2^{C_0 d}}{\eta^4} \log m + \kappa(d, \eta) \qquad m, d \in \mathbb{N},\ \eta \in (0, 1),$$

we claim that the following reduction holds

$$N\left(\mathcal{Q}_{n,d}, \|\cdot\|_{L_2}, \varepsilon\right) \leq N\left(\mathcal{F}_{3n,d}, \|\cdot\|_{L_2}, \frac{\epsilon}{2 \cdot 3^{d/2}}\right)$$

where $\mathcal{F}_{m,d}, \mathcal{F}_{3n,d}$ are defined as in (10.1). Once this claim has been proved, taking $m = 3n$ and $\eta = \epsilon/(2 \cdot 3^{d/2})$, combining the last two inequalities, we have

$$\log_2 N\ \left(\mathcal{Q}_{n,d}, \|\cdot\|_{L_2}, \epsilon\right) \leq 16 \cdot \frac{2^{C_0 d} 3^{2d}}{\epsilon^4} \log(3n) + \kappa\left(d, \frac{\epsilon}{2 \cdot 3^{d/2}}\right).$$

Since $16 \le 2^{4d}$ for every $d \ge 1$, we may increase the universal constant and absorb the harmless $\log 3$ term into the function depending only on (d, ϵ). Thus, for a possibly larger universal constant $C > 0$,

$$\log_2 N\ (\mathcal{Q}_{n,d}, \|\cdot\|_{L_2}, \epsilon) \le \frac{2^{Cd} 3^{2d}}{\epsilon^4} \log n + \widetilde{\kappa}(d, \epsilon),$$

where one may take for instance

$$\widetilde{\kappa}(d, \epsilon) := \kappa\left(d, \frac{\epsilon}{2 \cdot 3^{d/2}}\right) + 16 \cdot \frac{2^{C_0 d} 3^{2d}}{\epsilon^4} \log 3.$$

Now we turn to verify the claim. Let

$$\mathcal{G}_{n,d} := \{f_A :\ A \in \mathcal{Q}_{n,d}\} \subseteq \mathcal{F}_{3n,d},$$

where f_A is defined as in Lemma 9.2. We first compare the L_2 distance on $\mathcal{Q}_{n,d}$ with the L_2-distance on $\mathcal{G}_{n,d}$. If $A, B \in \mathcal{Q}_{n,d}$, then $\deg(A - B) \le d$, hence

$$\begin{aligned}
\|A - B\|_2^2 &= \sum_{|\mathbf{s}| \le d} |\widehat{A - B}(s)|^2 \\
&= \sum_{|S| \le d} 3^{|\mathbf{s}|} \left|\widehat{f_A - f_B}(S)\right|^2 \\
&\le 3^d \sum_{|S| \le d} \left|\widehat{f_A - f_B}(S)\right|^2 \\
&\le 3^d \|f_A - f_B\|_{L_2}^2.
\end{aligned}$$

Therefore

$$\|A - B\|_2 \le 3^{d/2} \|f_A - f_B\|_{L_2}. \tag{10.25}$$

Now set $\{h_1, \ldots, h_M\} \subseteq \mathcal{F}_{3n,d}$ be an $\epsilon/(2 \cdot 3^{d/2})$-cover of $\mathcal{F}_{3n,d}$ with

$$M = N\big(\mathcal{F}_{3n,d}, \|\cdot\|_{L_2}, \frac{\epsilon}{2 \cdot 3^{d/2}}\big).$$

Fix $A \in \mathcal{Q}_{n,d}$. Since $f_A \in \mathcal{G}_{n,d} \subseteq \mathcal{F}_{3n,d}$ and $\{h_j\}_{j=1}^M$ covers $\mathcal{F}_{3n,d}$, there exists j with

$$\|f_A - h_j\|_{L_2} \le \frac{\epsilon}{2 \cdot 3^{d/2}}.$$

In particular, we have $f_A \in \mathrm{Ball}_{L_2}(h_j, \epsilon/(2 \cdot 3^{d/2})) \cap \mathcal{G}_{n,d}$, so this intersection is automatically non-empty. Choose any point g_j from this intersection, and write $g_j = f_{A_j}$ for some $A_j \in \mathcal{Q}_{n,d}$. Then

$$\|g_j - h_j\|_{L_2} \le \frac{\epsilon}{2 \cdot 3^{d/2}}.$$

We claim that $\{A_j\}$ is an ε-cover of $(\mathcal{Q}_{n,d}, \|\cdot\|_{L_2})$. Indeed, let $A \in \mathcal{Q}_{n,d}$. By the triangle inequality,

$$\|f_A - g_j\|_{L_2} \le \|f_A - h_j\|_{L_2} + \|h_j - g_j\|_{L_2} \le 2\frac{\epsilon}{2 \cdot 3^{d/2}} = \frac{\varepsilon}{3^{d/2}}.$$

Applying (10.25),

$$\|A - A_j\|_2 \le 3^{d/2} \|f_A - g_j\|_{L_2} \le \varepsilon.$$

This proves

$$N\ (\mathcal{Q}_{n,d}, \|\cdot\|_2, \varepsilon) \le N\big(F_{3n,d}, \|\cdot\|_{L_2}, \frac{\varepsilon}{2 \cdot 3^{d/2}}\big).$$

The proof is complete. □

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